

MONICA NURSING STUDY CENTER

PRINCIPAL NCLEX STUDY GUIDE

COMPLETE CLASSES 1–20

Comprehensive Nursing Review • Clinical Systems • NCLEX Preparation

MONICA A. MAGARIÑO

Class 1: Ethical and Legal Principles, Management

Ethical Principles (P. 403)

- **Autonomy:** right to be informed and **self-determination**
- **Beneficence:** Duty **to do** good, to **prevent** and **remove** harm
- **Non-maleficence:** “Duty of **not doing** bad.” Do not harm
- **Justice:** Equitable distribution of resources and benefits, fairness
- **Veracity:** Obligation to tell the complete truth (**Do not hide nothing**)
- **Fidelity:** The duty to do what one has promised
- **Privacy Vs. Confidentiality**
- **Advocacy:** work for a patient, support patient rights.

Privacy: the right to keep personal information private. E.g., a CNA is looking into your belongings without authorization. Other: nurses taking pictures, sharing in social media,

Confidentiality rights of medical information (**HIPAA**): **you can know yours, your children, and from an adult for whom you received authorization. You must authorize disclosure of info. You can correct wrong info and request a copy.** E.g., nurses talking in the elevator about a clients diagnose. Others: IT issues like passwords, leaving open screens, etc.

Beneficence and Non-maleficence: “Same end, different paths”. Beneficence implies action to do good. Non maleficence implies restrain from doing (action of) harm. E.g., in the street, you stop the car to allow an elder pedestrian pass safely: non-maleficence; you get out the car and help the old person reach the sidewalk: beneficence.

Legal Issues (P 403)

- **Nurse practice act:** legal aspects of nursing: **Scope of Practice:** profession boundaries
- **American Nurse Association (state) (ANA):** professional organization: **Standards of Care:** acceptable nurse performance in patient care
 - **ANA code of ethic, “liability when performing duties, is Responsibility”**
- **State Board of Nursing:** apply **state** nursing act
- **Malpractice** = nonintentional negligence on nursing practice.
 - **Requirements for Negligence Lawsuit: Duty, Breach of duty (meter la pata), Damage, Causation.** (Needs to be present for be considered Malpractice)
 - Atropine administered to a client with glaucoma,*
 - Nurse rubbing the abdomen of a child with a Wilms’s tumor,*
 - Oxytocin initiated in late decelerations,*
 - Nurse turns coma client every 4 hours and develops pressure ulcer*
 - SCD initiated in a client with DVT*

**** Any time a patient sustains an injury after a fall, it is considered a form of Nurse Negligence ****

**** Placebos** are not allowed to give to any patient. Placebo is only authorized if the patient gives the consent during the research (Randomized Clinical Trial) ******

- **Good Samaritan Law (P 403)**
 - Out of hospital emergencies,
 - No gross negligence,
 - No abandonment,
 - No compensation

SBAR

S

Situation
 Briefly describe the situation.
 Give a succinct overview.

B

Background
 Briefly state pertinent history.
 What got us to this point?

A

Assessment
 Summarize the facts.
 What do you think is going on?

R

Recommendation
 What are you asking for?
 What needs to happen next?

**** First step patient identification**

S- the problem for communicate with other team member (acute change)

B- Past Hx

A- Objective data

R- recommendation

****Patient with DNR have a bracelet with that information****

Change of Shift Report: Don't report routines

- Client information: Name and age
- Medical plan:
 - Diagnoses, Assigned provider (MD), Allergies
 - Code status
 - Activity level (BR privileges)
 - Precautions (fall, bleeding, seizure)
 - Pertinent past medical-surgical history
 - Abnormal assessment findings
 - Tubes: IVs, catheters, PEGs, etc.
 - Abnormal labs
 - Important meds with potential adverse effects
 - Any test or procedure pending to carry out
- **Nursing plan:**
 - Referrals, teaching, Family and support system concerns
 - Discharge or transfer plan ([initiated on admissions process](#))

Documentation: Factual and Detailed

- **Correcting errors:** *draw a line*, enter correct information, initial and date. **Never use whiteout, no erase or scratch out an error**
- **Late entries:** *don't leave any blank space between documentation.*
- **Write 0.3 mg**, no .3 mg
- **Write 5 mg**, *no trailing zero is documented 5.0 mg*
- **Don't write U instead of Units.**
- **Don't use acronyms and abbreviations like QD, QID, TID, CHF, CKD, MgSO3**
- **Placebos are NOT documented as medication**

- If the doctor prescribes specific dose to a patient and the patient receive different dose, needs to be document (It is an error, but the nurse needs to document the dose administered)
- Do not use abbreviation for the name of the disease. Ex.: MS, write multiple sclerosis
- **Variance Report:** It is when the nurse made something different to the routine (is not expected to happen). This report is **going to the Risk Manager office**

Guidelines for documentation include:

- Use of permanent ink
- The use of only accepted terms and abbreviations
- Legible writing, accurate spelling, proper grammar, accurate dating and time of the entry
- The signature and title of the person who documented the entry, and a professional tone.
- If an error in documentation occurs, a thin line that does NOT obliterate the entry is drawn through the erroneous entry, the notation "Error" is written above the entry and the nurse signs this notation with their name and title.

Mandatory Incident Report:

- Child abuse < 18 years old
- Elderly abuse
- Human traffic
- Gunshot wound, dog bites
- STD (chlamydia, gonorrhea, syphilis, ...)
- Hepatitis C, bacterial meningitis (has vaccine), TB
- Use of drugs by staff members

NOTE: The purpose of report incidents, accidents, medical errors, and sentinel events is to prevent client injuries

**** Situations that require Incident Report:**

- Medication errors
- Complications for Dx or Tx
- Incorrect sponge count in surgery
- Failure to report change in pt condition
- Falls
- Client is burned
- Break in aseptic technique
- Client or family refuse Tx

STEPS:

- First safety of the patient (assessment, VS)
- Call the doctor and notify the incident and follow Dr order
- Document in the chart what happen factual and details
- Complete incident report form or variance report (include opinion and details what happen)
- Send to Risk Management

Infectious diseases of mandatory report: TB, STDs, Diseases with vaccine, Foodborne disease outbreaks, tropical diseases like malaria or dengue

INTENTIONAL TORTS (P 403) (Criminal Law)

- **Assault:** Threat of harmful or offensive contact (do not touch the patient but gives advice, treats)
- **Battery:** Intentional touching without other's consent (Ex. Given CPR for a pt with DNR)
- **False imprisonment**
Client is not allowed to leave facility (most important is inform the client about the eventual results of refusal care and leave)
Unauthorized restraining
- **Tarasoff Act:** duty to harm of threatened suicide or harm to others. (Ex: when pt tells the nurse his intention to hurt someone, needs to be reported)
- **Defamation**
Damages to reputation:
Verbal: **slander** (speech)
Written: **libel**
Tarasoff Act: duty to warn of threatened suicide or harm to others. (Ex. When the patient tells the nurse his intention to hurt someone, the RN needs report it)

Informed Consent: Self Determination Act

Requirements:

Adulthood and mental competence (Ex. Bipolar stable, schizophrenia stable, initial dementia)
Informed
Voluntary

Information given by the provider; consent obtained by the provider

Nurse: give document to sign and witnesses the client's voluntary signature

Incompetent to give consent:

- Under sedation drugs or alcohol, uncompensated psychosis
- Minors (<18 years old) except: (can do consent)
Emancipated minors (married, court designated, military)
"Non-emancipated mature minors": STDs, substance abuse, prenatal care, and contraceptive prescription, HIV, AIDs Tx.
Minors' women pregnant (No married), can give consent for her baby but not for herself, unless if she applies for any other exception.

Consent can be waived in emergencies **if it can't be obtained**

Consent should be given by client or by **authorized proxy**

Consent can be rescinded by competent client at any moment

Pregnant women after delivery cannot give own consent unless if she has another exception)

Advanced Directives (Self-Determination Act)

- **Living will for healthcare** (nurse cannot be a witness to a living will or advance directive)
 - List treatment that client chooses or refuse if becomes unable to make decisions and is terminally ill (CPR, mechanical ventilation, tube feeding, dialysis, antibiotics, comfort care, palliative care, organ and tissues donation for transplant, donation body for study)
 - Does not include financial decisions
- **Durable power of attorney**
 - Patient appoints health care proxy to carry out client's wishes.
 - Take decisions that are not already documented on living will
 - **The client "speaks" through the proxy**
- **Do not resuscitate (DNR) prescription** (Cardio and pulmonary resuscitation)

- Physician's written prescription (while the doctor does not place the order, the patient remain in full code)
- **LIMITED ONLY TO CPR**
- Order renewed at each admission
- Not abiding known DNR order: battery
- DNR form for change status to DO NOT EXIST, the only way is call HCP for made a new Rx to discontinue DNR
- Patients in Hospice are always DNR
- Assess patient understanding what DNR is

Multidisciplinary team:

- **Social worker:** patient problems, where go
- **Speech therapy:** patient problem with swallow, baby with cleft palate
- **Occupational therapy:** patient with problems of ADLs. The OT design spoon or special clothes for them)
- **Physical therapy:** rehabilitation, physical motion
- **Risk manager:** clarify missing information, explain what happened to client and family, refer pt to appropriate resources, care provided free of charge
- **Case manager:** coordinate, communicate, collaborate, resolve problems, follow for patient from admission to discharge
- **Case management:** coordinate, involve principles of CQI, promote professional practice

Restraints

Nursing Implications (P 402):

- Least restrictive restrain if all other measures were ineffective.
- **No PRN restraint orders**
- Renewed every 24 hours
- Assess and document neurovascular (capillary refill, pulses) and skin status (skin color, skin temperature) every 15-30' min
- Remove restraints at least every 2 hours: toileting, fluids, food, skin check, reposition, check bone prominence
- Half-bow safety knot tied to bed frame, not to side-rails
- Last option chemical restraints

Restrain types:

Tray board

Soft Wrist: patient increase agitated, cannot be redirected with distraction, keep trying to ankle.

Remove need medical devices.

Posey vest

Mummy restraint

Mittens: if the patient are treated to an immovable object, cannot remove it from their hands, it is for patients attempts to disconnect tubing or devices

Elbows: make difficult for a client to remove a medical device near the face or neck. Do not impede to removal of abdominal or urinary devices

Belt restraints is applied to patient waist, not chest, not abdomen

Waist: used for confused or impulsive patients who continually try to get out of bed or chair after repeated redirection.

Psychiatric Seclusion:

- Last resource
- Involuntary commitment:
 - Risk to self or others
 - Right for Habeas Corpus
 - Risk to refuse medication

Delegation

- **RN:**
 - Assessment (Head to toe, questions that you made for obtain a diagnostic, initial assessment when a patient arrive to emergency), diagnosis, care plan (independent interventions without ask the doctor and dependent intervention after consult with the doctor), teaching, evaluations, consultations
 - Admissions, transfers and discharges admission, telephone orders.
 - Central catheters, IV push, blood transfusions, chemo, IV anticoagulants
 - Unstable patients: “just received from... newly diagnosed...”
 - RN Float: stable patients only, cannot do teaching for new admission or discharge
- **LPN:**
 - Ongoing assessment of low risk/chronic patients (Ex. Before give med assess for any signs, VS, auscultate, always need to be delegated by RN, never will be a new admitted pt., or return from Sx, etc.)
 - Administration of oral, IM, SQ, topical medications
 - Maintain/regulate IV, non-IV push medications (NEVER can put an IV, only if this IV is passing by gravity never push)
 - NG or PEG tubes medication/feeding.
 - Reinforced teaching (remember the patient use the incentive in spirometer)
 - Insertion of Foleys (indwelling catheter)
 - Tracheotomy care, ostomy care
 - Traction and wound care
 - NOT PCA pump (patient requires assessment and teaching, pump with 2 doses of infusion)
- **UAP: (CNA)**
 - ADLs, feeding (no dysphagia), ambulation of stable patients, vital signs, urinary output, bladder scan, measuring I & O, skin barrier, tap water enemas, CPR.
 - No medications (no pills, not nice shampoo, not Abx cream)
 - Accu-check
 - UAP unlicensed assistant personal. Only interventions can be delegated
 - Reinforced teaching (remember the patient use the incentive in spirometer)
 - Report changes in the clients’ conditions to the RN
 - Provide comfort
 - Apply or removal physical restrains
 - Not allow to do mechanical modified to any patient (food processed with a machine like puree)
 - Report any intake fluid in once (oz) and cups
1 oz = 30 mL, 1 cup = 240 mL

Prioritization: ABCD alterations

MASLOW'S HIEARCHY OF BASIC HUMAN NEEDS



PAIN: PHYSIOLOGICAL: is **priority** if Chest pain, MI, compartment syndrome, burns, kidney stones, sickle cell crisis, cerebral hemorrhage (patient states “is the worse headache of my life”), aortic aneurysm (AAA, **pain on left side of back near spine, common in smoker elderly**)

- **Airway: (3 to 5 min to die)**
 - Stridor (epiglottitis, anaphylaxis),
 - Face burns,
 - Dislodged tracheotomy,
 - Dysphagia,
- **Breathing: (1 to 2 hours to die)**
 - Respiratory center depression (opioids, Mg sulfate, benzodiazepines),
 - Cervical spine lesions,
 - Muscle weakness (Myasthenia Gravis, G. Barre),
 - Chest trauma, pneumothorax, flail chest,
 - Pulmonary edema,
 - Chest tube and ventilator dysfunction,
- **Circulation:**
 - Chest pain,
 - Ventricular arrhythmias: Potassium, Ca and Mg, digoxin, thyroid,
 - Hypotension, shock, bleeding, dehydration,
 - Immediate postoperative and post procedures (Cath lab) clients,
- **Safety:**
- **Disability:** loss functions (amputation, CVA, retina detach, glaucoma, stroke)

Summarizing Prioritization

- Emergencies: ABC
- Lack of information to decide: Assessment (**use when do you don't know what is happen**)
- Acute Vs. Chronic
- Actual Problem Vs. Potential Problem
- Unexpected Vs. Expected
- **Disaster Triage**
 - **Red label, Emergency:** Go to ER (compartment syndrome, ABC)
 - **Yellow label, Urgency:** can wait 30 – 60 min without deterioration (appendix, fractures)
 - **Green label, non-Urgency:** Can wait more than 2 hours

- **Rapid Response Team vs Code Blue**

- **RRT:** respond in minutes
- **CB:** respond in seconds

Critical thinking Model categories:

- **Contextual awareness:** be alert what happen (values, relationships, environment)
- **Inquiry:** apply standards of ethical reasoning to your thinking
- **Analyzing assumptions**
- **Considering alternatives:** explore and imaging options to your situation
- **Reflecting:** questions, analyzing, reflect

Strategies...that work

- Read twice, if necessary, find key words, and **understand what is asked**.
- Carefully consider **each answer** option.
- **Eliminate** the options you are sure are wrong.
- Test how reasonable the answer is by posing **yourself or a loved one** in the situation
- Discard options **that deny the patient's feelings, concerns, or needs**.
- **"Always, never, none, all"**, are determinants rarely seen in health care.
- Identify **equally versus unique options**: Apple, apple, apple, pear.
- Identify the **umbrella option**: because it includes the others.
- Don't choose drug therapy as the first intervention in most of cases.
- Don't **"pass the bucket"** unless it is beyond RN scope of practice:
 - **Independent Interventions** → PRNs → Call MD.
- **Assessment first if it is needed** and there is a mix of assessment and interventions.
- Physical take priority over psychological needs. If all are physical needs, apply ABCs.
- Prioritize answers that focus on clients providing for **safety or reduce risk of injury**.

Class 2. Prevention

Quality Improvement and Evidence Based Practice

- Continuous Quality Improvement (CQI): a *client centered* continuous process involving everybody to produce good outcomes that meet standards (*safety*) and resolve problems. It follows the nursing process and is based in evidence. Focused team: group of 5-10 people
- Examples of quality indicators:
 - Reduction in medication-related adverse events
 - Reductions of bed sores and falls
 - Decreased number of urinary catheter infections
 - Reduced hospital readmissions
 - Decreased medication administration errors

Three Main Components of Evidence-Based Practice



Utilize the best external evidence



Draw on individual clinical expertise



Consider patient values & expectations

- **Primary prevention:** (Prevention of disease or injury before it occurs)
 - Health promotion and education, immunizations, hearing protection, body mechanics, folate in pregnancy, helmets, quitting smoking (Ex. Education for health, use of EPP for prevent to acquire any infection, worker using plugs in ears to prevent ear disease. Get the HPV vaccine to prevent the disease)
- **Secondary prevention:** (Do no prevent the disease, the patient has the disease but prevent the disease will complicated)

Early identification and treatment of **existing disease** to prevent disability and death:

 - Health **screening** for prostate, breast, cervical, and colon cancer; blood pressure and blood sugar screening, chemoprophylaxis TB, prophylactic antibiotics, losing weight, ASA in TIA, prenatal care, early treatment of acute diseases. (Ex, the mammography doesn't prevent breast cancer, but can detect if the patient has the early symptoms that can be treated or eradicate. Colon cancer detect with colonoscopy to check for polyps. Health fair for check HTN, cholesterol, blood sugar, screenings of HIV)
- **Tertiary prevention:**

Reducing the extent and severity of an injury or illness, to reduce disability, and restore or preserve function. (Tx of chronic disease)

 - Rehabilitation, treatment of chronic diseases, occupational therapy, physical therapy: stroke, heart failure, renal failure, diabetes mellitus, amputations.

Accident Prevention

- Medications and poisons in **locked cabinets**
- Prevent **aspiration**:
 - Peanut, hot dogs, carrots, popcorn, small toys, coins, batteries
- **Burns**: facing inward handles of pans on stove, **water heater 120°F or less** (children, DM)
- Falls in older:
 - Mandatory **screening of risk for fall** within the 4 hours of admission
 - Best predictor of risk for fall: **previous falls**
 - **Extrinsic factors**: **environment**
 - Elimination of **rugs and mats, electric cords across, pets, clutter free** (reguero)
 - Adequate illumination (night lights)
 - Nonskid soles
 - Grab bars in bathrooms, **use raised toilet**
 - Bed alarms, bedside table, and call light at client's reach
 - **Intrinsic factors**:
 - Age: ↓, muscle strength, balance, vision: exercise and assistive devices
 - Medication (sedation, orthostatic hypotension)
 - Incontinence, dementia
 - Bradycardia, syncope
- SIDS sudden infant death syndrome

- High risk for fall: older than 80, history of falls, live alone, has to climb stairs.

Riding Safely

- ✓ Ride in a straight line, single file.
- ✓ Go with the traffic flow. Ride on the right in the same direction as cars.
- ✓ Obey all traffic signs and signals.
- ✓ Ride with both hands on the handlebars except when signaling a turn or stop.
- ✓ Stop and look left-right-left for traffic before entering a street.
- ✓ Walk your bike across an intersection.
- ✓ Stay alert – use your eyes to look for things that could make you fall, like potholes, cracks, pebbles, or wet leaves.
- ✓ Stay alert – use your ears to listen for traffic. Don't wear earphones while riding.
- ✓ Watch for parked cars and cars pulling out or into parking spaces or driveways.
- ✓ Check your equipment. Make sure your bike tires are properly inflated and that the brakes work.

Don't forget your bicycle helmet. Wear it flat on your head and buckled!

Feet should touch the ground when sitting on the bicycle seat

Sudden infant death syndrome	
Risk factors	Protective factors
• Prone or side sleeping position • Tobacco smoke exposure before/after birth • Cosleeping in adult bed • Premature birth/low birth weight • Overheating during sleep • Soft sleep surface • Bumper pads, toys & loose pillows/blankets	• Supine sleeping position • Breast-feeding • Pacifier use during sleep • Up-to-date vaccinations • Appropriate clothing (eg, sleep sack) • Firm sleep surface • Removal of loose items from bed

Lead: *Old houses before 1978.*

Chelation therapy (BAL)

Children: brain damage

Adults: polyneuropathy

Crib safety: Slat's separation < 2 “



BAL (British anti-Lewisite) DMSA (Dimercaptosuccinic acid)

Environmental Safety

o Fire:

o **RACE (Rescue, Alarm, Confine, Extinguish)**

Evacuation order:

- o 1. Patients in immediate danger (visitors, patient can run or walk)
- o 2. Ambulatory patients
- o 3. Wheelchairs, isolates, cribs
- o 4. Bed bound patients (ICU patients)

o **Latex allergy = rubber allergy: (rubber is the same of latex)**

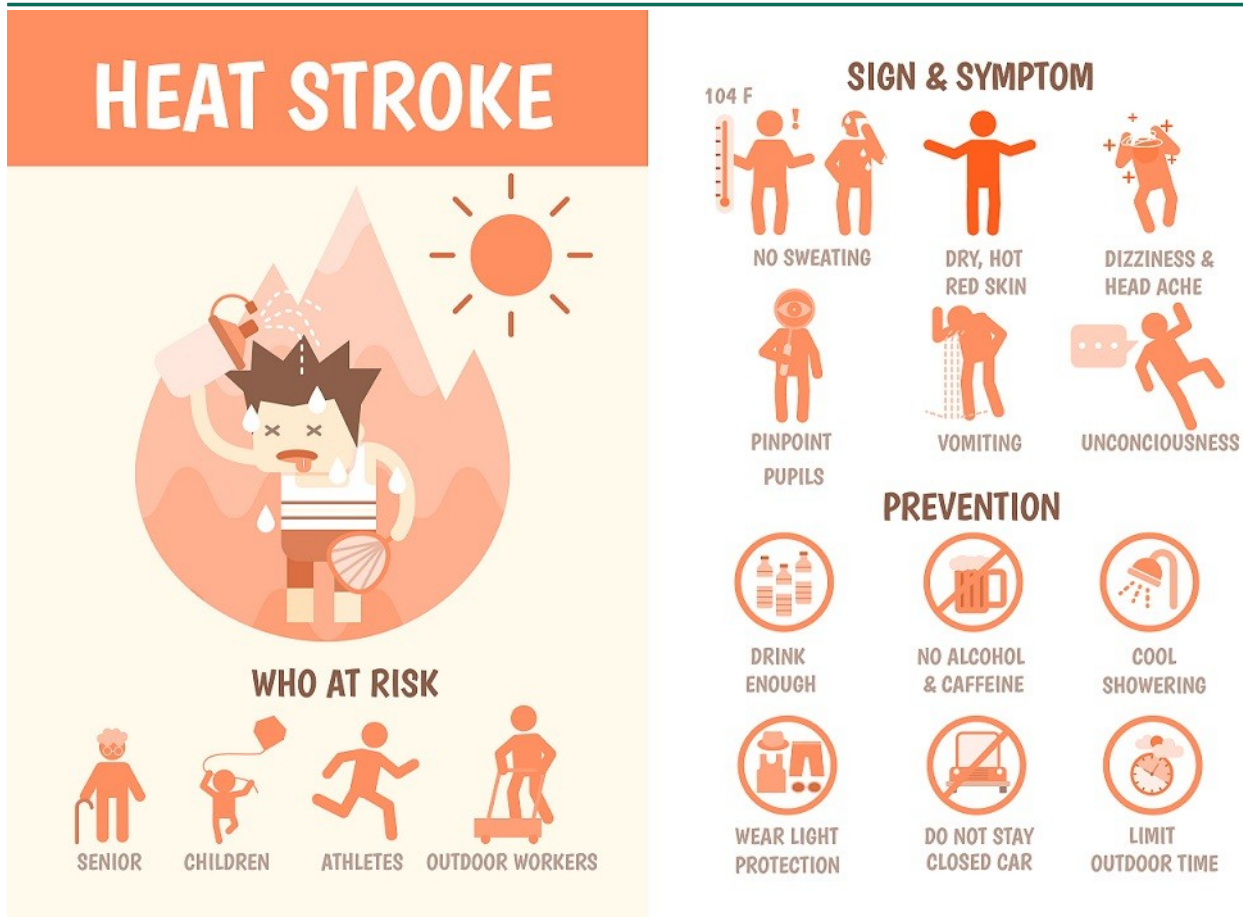
- o Allergy to avocado, banana, pineapple, kiwi, mango,
- o Spina bifida
- o Multiple surgeries

o **Lighting safety:**

- o Avoid standing on high places
- o Avoid trees during raining days
- o Avoid showering and pipes
- o Avoid using landlines telephones
- o If notice standing hair: drop to floor (because can be a ray close to you)

o **Home Safety during electrical storms (Thunderstorm)**

- o Stay away from windows
- o Do not use line telephones (home phone)
- o Do not bathe or shower
- o Avoid metal pipes



Heat stroke early symptoms: AMS, exhaustion, loss myoglobin on muscles

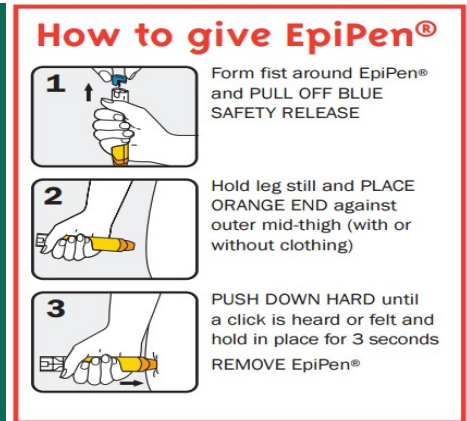
Anaphylaxis

- **Emergency order:**
 - Call for help, 911
 - Attend the patient
- **Symptoms:** angioedema, decrease BP, shock, hoarse voice, urticaria, stridor, wheezing, decrease O₂
- **Maintenance of airway**
- **First line drug:**
 - Epinephrine 1:1000 IM (inject in thigh)
- **Second line drugs after epinephrine:**
 - Oxygen
 - Albuterol inhaled
 - IV isotonic fluids
 - IV Glucocorticoids
 - IV Diphenhydramine

****Bee stinger, remove it with flat scrape**

Order for anaphylactic shock

1. Call for help (911)
2. Assist the patient



Bioterrorism: When to suspect?

- Sudden outbreak of uncommon disease in usually healthy people, coming from a limited area
- **Anthrax:**

Cutaneous, GI and respiratory: (flu-like syndrome → anthrax pneumonia (kill people))

No transmission person to person: **standard precautions**

Anthrax pneumonia if the patient arrives with the powder, give it a shower ASAP

Treatment: **ciprofloxacin or doxycycline for 60 days**. Vaccine available Doxycycline for 60 days

S/S: diarrhea with mucus and blood

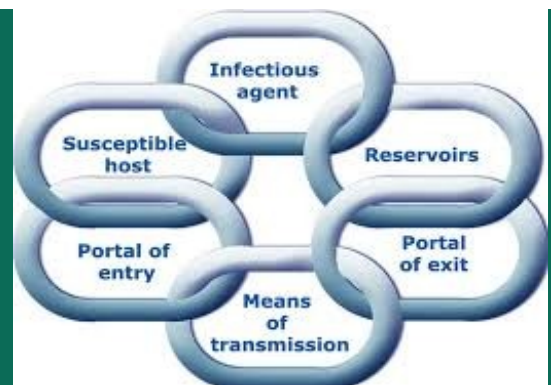
- **Plague (*Yersinia pestis*):**
 - Pneumonic plague: **droplets precautions. (Person to person transmission)**
 - **Bubonic plague:** (fleas), **standard precautions**
- **Tularemia:** Transmitted by **rabbits' ticks** (rabbit fever): **standard precautions**
- **Smallpox or monkeypox:** **Airborne precautions.**
- **Ebola:** Viral hemorrhagic fever: **contact and droplets, body fluids**
- **Botulism (*Clostridium botulinum*):** Ingestion of toxin **or in wounds**. Ptosis, diplopia, descending paralysis, apnea.

Many new cases of a disease not common in the area or age group

Decontamination out of area of exam, remove clothes without pulling over head, shower with water and soap, irrigate eyes

Chain of Infection

1. Infectious agent or pathogen
 2. Reservoir (place where the agent is located, Ex street dog with rabies)
 3. Portal of exit (Ex: mucosa nose and mouth)
 4. Mode of transmission
 5. Portal of entry
 6. Susceptible host
- Incidence: Newly cases
 - Prevalence: Sum of all cases that are alive at a given time



Transmission Based Precautions: Standard Precautions

- Bloodborne: **All patients**
- All body fluids **except sweat** are infectious
- Mucosa and broken skin are infectious
- **Hand hygiene** before and after touching patients
- **Gloves** added: secretions, mucosa or open skin, **body fluids (not use in intact skin)**.
- Eye, nose, mouth are portals of entry:
 - **Face shields**, or mask and **goggles**, if splash is expected
 - **Gown** should be used if splash
- Never recap needles, use sharp containers
- Solution of bleach for spills.
- Two different Standard Precautions' diseases **can be cohorted**:
 - E.g., Osteomyelitis and hepatitis B
 - Patient with shingle or herpes zoster with the lesion that you can cover.
 - *H. pyloric*, legionnaire's pneumonia, RMSF (Rocky Mountain Spotted Fever), Diarrhea by *E. coli*, Shigella or Salmonella in patient continent
 - Chemotherapy patients

Transmission Based Precautions: Contact Precautions

- Always: **gloves and gown**
 - **Surgical Mask only if respiratory MRSA or RSV with productive cough**
 - **If use gown and gloves, take out inside the room**
- **Dedicated equipment** like stethoscope and BP cuff.
- Double bagging
- Private room or can be cohorted: same infectious agent
- Door can be open
- Cover wounds for transportation
 - **Skin:**
 - **MDRO Infected wounds** (MRSA, VRSA, CRE, VRE)
 - Impetigo, abscesses, suppurative cellulitis, cutaneous diphtheria
 - Herpes zoster and herpes simple
 - Conjunctivitis,
 - **Scabies, pediculosis (24 hours cannot go to the school, after apply med can go)**
 - **Klebsiella (skin access)**
- Diarrheas: C. diff (Contact Enteric Precautions),** rotavirus, norovirus, **Diarrhea by E. coli, Shigella or Salmonella in patient incontinent (older patient with diaper, children)**
(Any other infectious diarrheas if incontinence)
 - **C. difficile produce Pseudomembranous colitis**
- **RSV (bronchiolitis) and parainfluenza**
- **Precautions:**
 - Wash hands with soap and water
 - Avoid use alcohol-base hand sanitizer
 - Disposable equipment (stethoscope)
 - Disinfect surface with bleach
 - Discontinue unnecessary antibiotic

Transmission Based Precautions: Droplets Precautions

- Regular **surgical mask (6 ft.)**
- Client wears surgical mask when out of isolation room.
- **If use gown and gloves, take out inside the room**
- Private room or cohort with same infection.
- Door can be open
- **Mnemonic:**
 - **S:** strep throat (24 hours), scarlet fever (**strawberry tongue**)
 - **P:** pneumonia (*pneumococcal, streptococcus, mycoplasma, pneumonic plague*), pertussis (*whooping cough*), parvovirus B19 (**produce 5th disease**)
 - **I:** influenza (*parainfluenza is contact...*)
 - **D:** pharyngeal diphtheria (*cutaneous diphtheria is contact...*)
 - **E:** epiglottitis
 - **R:** rubella (**Pregnant women less < 20 weeks, damage in fetus**)
 - **M:** mumps, meningitis: **only 24 hours for: meningococcus, (Neisseria meningitis bacteria)**
 - **AN:** adenovirus (**can produce pink eye**) (add **contact**) and rhinovirus (common cold)
- Meningitis for TB, gram negative, streptococcus pneumoniae, listeria, aseptic (viral): no isolation.

- **Adenovirus pneumonia:** droplets and contact, Adenovirus conjunctivitis: contact & droplets can produce pink eye.
- **Scarlet fever and acute glomerulonephritis** can produce rheumatic fever

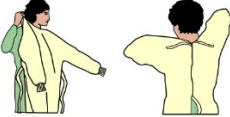
Transmission Based Precautions: Airborne Precautions

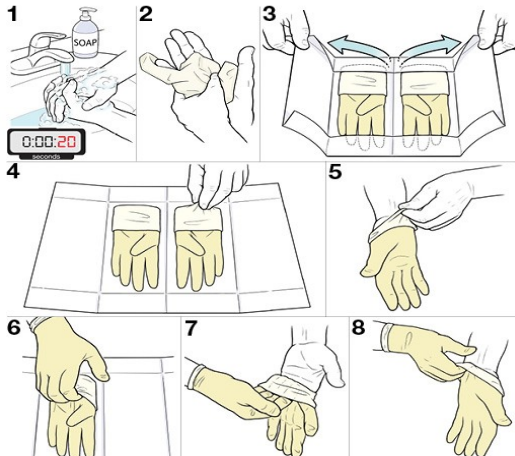
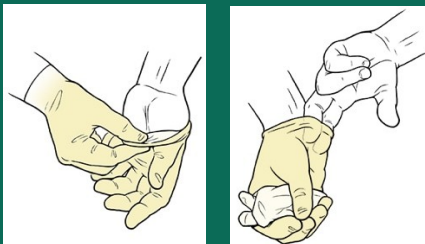
- Single room with **negative pressure**
- **Closed door:** *Social isolation, depression*
NI: room with windows
Can receive family visit with mask
TV
- **Particulate or “respirator” mask** (fitted N95)
 - Not required for measles/varicella if **immune staff**
- Client wears surgical mask when out of room
Measles (rubeola),
Chicken pox (varicella)
TB (tuberculosis)
Others: SARS, COVID-19, generalized or **disseminated** herpes zoster (add contact), avian influenza
- **NEMMONIC----- MY CHICKEN HEZ TB and SARS**

PPE

- **Wash Hands**
- **Donning PPE:**
 - Gown → Mask/Respirator → Goggle/shield → Gloves
- **Removing PPE:**
 - Gloves → Goggle/shield → Gown → Mask/Respirator
- **Wash Hands**

Note: everything is removed before leaving except the mask

CORRECT SEQUENCE FOR <i>DONNING</i> PERSONAL PROTECTIVE EQUIPMENT (PPE)	CORRECT SEQUENCE FOR <i>REMOVING</i> PERSONAL PROTECTIVE EQUIPMENT (PPE)
The type of PPE used will vary based on the level of precautions required; e.g., Standard and Contact, Droplet or Airborne Infection Isolation. Remove hand jewellery and tie back hair. Clean and dry hands thoroughly. GOWN / APRON Fully cover torso from neck to knees, arms to end of wrists, and wrap around the back Fasten in back of neck and waist  MASK OR RESPIRATOR Secure ties or elastic bands at middle of head and neck Fit flexible band to nose bridge Fit snug to face and below chin Fit-check respirator  GOGGLES OR FACE SHIELD If you wear glasses put them on. Place goggles or face shield over face and eyes and adjust to fit  GLOVES Extend to cover wrist   Capital & Coast District Health Board OPŌKO KI TE URU HAUORA Infection control January 2005. Developed using CDC Guidelines 2005	GLOVES Outside of gloves are contaminated—DO NOT TOUCH! Grasp outside of glove with opposite gloved hand; peel off Hold removed glove in gloved hand Slide fingers of ungloved hand under remaining glove at wrist Peel glove off over first glove Discard gloves in waste container Clean and dry your hands thoroughly  GOGGLES OR FACE SHIELD Outside of goggles or face shield are contaminated—DO NOT TOUCH! To remove, handle by head band or ear pieces Place in designated receptacle for reprocessing or in waste container Clean and dry your hands thoroughly  GOWN / APRON Gown front and sleeves are contaminated—DO NOT TOUCH! Unfasten ties Pull away from neck and shoulders, touching inside of gown only Turn gown inside out Fold or roll into a bundle and discard Clean and dry your hands thoroughly  MASK OR RESPIRATOR Front of mask/respirator is contaminated—DO NOT TOUCH! Grasp bottom, then top ties or elastics and remove Discard in waste container Clean and dry your hands thoroughly 

DONNING GLOVES	REMOVING GLOVES
	 Double Gloves Technique Sterile gloves are wearing in: IV, trachea, wound, bladder

Dominant hand first

Asepsis

- **Medical Asepsis (clean):**
 - Clean environment, no sterile
 - Hand washing, clean gloves, PPE, disinfectants
 - Hands are rinsed with fingers pointing down
 - E.g. Oral medications, vital signs, procedures in nose, mouth, vagina, rectum
- **Surgical Asepsis (sterile):**

- Sterile: no microorganisms
- Scrub. **Hands are rinsed with fingers pointing upward**, above elbows
- **Urinary tract, trachea, body cavities, IV therapy (TPN)**
- Opening sterile kits: check for torn open first fold away (cannot be broken or wet)
- Keep sterile field on visual attention all time, **do not turn your back**
- One inch (1-2.5 cm) margin around the edges are not sterile
- **Wet surface contaminates the sterile field** (capillary action)
- Gown is sterile only from waist to shoulders, **do not raise arms above the shoulders or below waist**

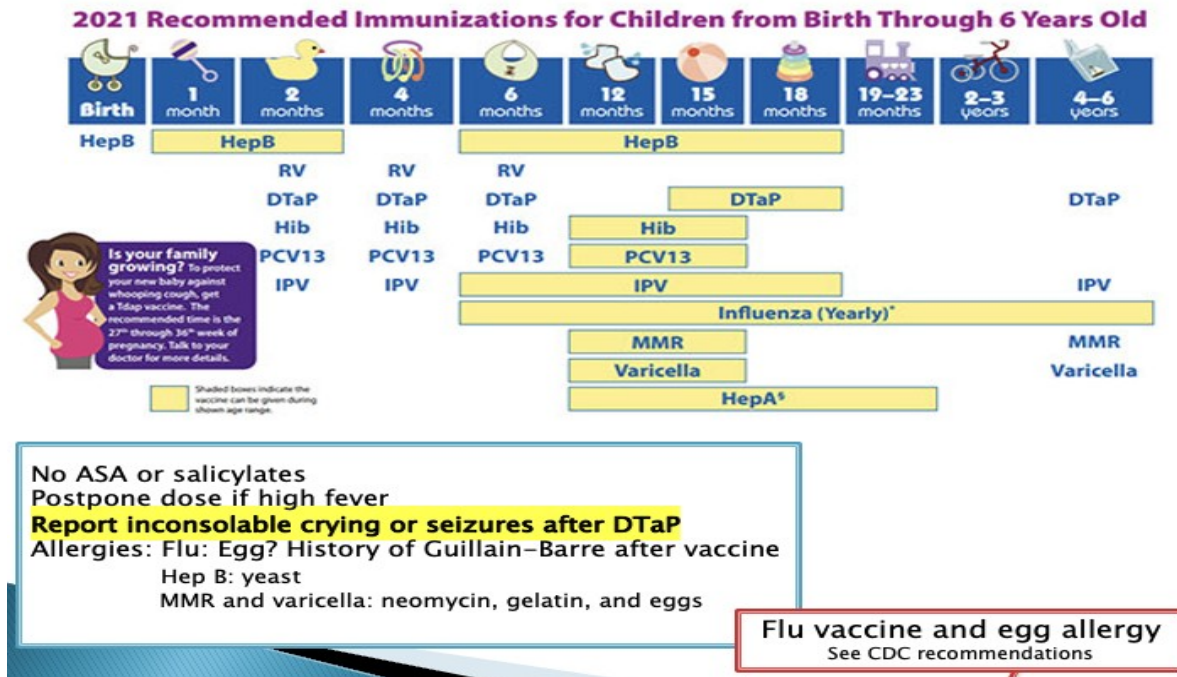
Immunizations

- **Vaccines (children)**
 - **Hepatitis B:**

First dose: at birth, two more doses (4 month and 6 month). IM applied in vastus lateralis
Mother HBsAg positive? HBIG. **If mother has Hep-B virus positive when the baby born, the newborn needs Hep-B and Immunoglobulin shot in legs**
Breastfeeding not contraindicated if mother has hepatitis B **because baby is immune after vaccine**
 - **At 6 to 8 weeks:**

RV (3 doses, PO),
DTaP (5 doses, IM, can produce encephalopathy: **inconsolable crying call HCP, bad reaction to pertussis**)
IPV (Salk, 4 doses, IM or SQ)
Hib (4 doses, IM),
PCV (4 doses, IM)
 - **At 6 months Rotavirus, is the only vaccine in drops**
 - **After first year: live virus**

MMR (2 doses, SQ),
Varicella (2 doses, SQ)
Women must wait 3 months for pregnancy
Recent **immunoglobulins or transfusions**: postpone vaccination around 8 months later
 - **Influenza (no live virus, can use in pregnant women):** yearly after 6 months, IM. **(Risk of Guillain Barre)**
 - IPV: inactivated polio virus
 - **Contraindications:**
 - Children or **adolescent** taking **salicylates (Syndrome Reye)**
 - Sever anaphylaxis with previous influenza vaccine (**doesn't include egg allergy**)
 - History of Guillain Barre
 - **Flu and pneumonia vaccines can be given the same day in adults**



Who should not get the flu vaccine?

1. Previous severe allergic reaction (anaphylaxis) to flu shot (any component).
2. People with history of Guillain Barre.
3. People with fever or feeling bad.
4. People taking aspirin.

Who should not get the **nasal flu** (live virus): same as before plus: pregnancy, immune depression, contact with immune depressed, children younger than 2 y or adult older than 50 y, people taking antiviral (Tamiflu)

NOTE: In the United States, **live virus vaccines** are routinely recommended for protection against **rotavirus, measles, mumps, rubella, and varicella**.

NOTE: Adult every year need Flu shot, every 10 years Tdap and when arrive to 65 years old PVC-15 and then PVC-23.

Infectious Diseases

- **Varicella (Chickenpox):**
 - **Airborne/contact until vesicles have crusted (6-7 days)**
 - Latent virus can reappear as h. zoster
 - Vaccine: live virus, after 1-year, **artificial active immunity**
 - **Aspirin and ibuprofen** should be avoided. **Aspirin** develop Reye Syndrome
 - **Trim nails** and bathe with uncooked oat
 - **Acyclovir** in severe cases (>12, pregnant, HIV)
- **Rubeola (Measles)**
 - Viral. **Airborne:** 4 days before/5 days after rash
 - **Koplik spot**
- **Rubella (German Measles)**
 - Viral. **Droplets.** MMR. Tender occipital nodes
 - **Highly teratogenic.** (**Warning for pregnant less than 18 weeks**)

- **Mumps**
 - Viral. **Droplets**. MMR
 - **Orchitis** and sterility.
- **Fifth Disease (Erythema Infectiosum)**
 - Erythema of face: **slapped cheek**
 - Human Parvovirus (HPV) B 19.
 - **Standard precautions, transmitted by blood, blood transfusion and respiratory secretions**
 - **Droplets precautions in immunosuppression.** No contagious after rash appears

Infectious Mononucleosis

- Epstein Barr Virus: Children, adolescents, young adults
- Symptoms:
 - Extreme **fatigue**
 - Sore throat
 - Enlarged lymph nodes
 - Hepatic-**splenomegaly**
- Transmission
 - Saliva, blood, body fluids
 - Standard precautions
- **Risks for complications:**
 - Splenic rupture
 - Avoid contact sports

Shingles: Herpes Zoster

- Reactivation of latent varicella-zoster virus
- **Pain** main problem (**Neuropathic pain**)
Gabapentin, amitriptyline, pregabalin, lidocaine patch, acyclovir can be prescribed
- Can affect V and VII cranial nerves
- Teratogenic
- Contact precautions
- Shingle vaccine: 50 years with history of varicella

Lyme Disease. Erythema Migrans (endemic)

- **Primary prevention: avoid tick bite**
- **Symptoms:**
 - Stage I: flu-like and bull's eye skin lesion: **7 days**
 - Stage II: Bell palsy, dysrhythmias
 - Stage III: arthritis, chronic fatigue: years
- Serology:
 - Test: Elisa and Western Blot.
 - **Positive after 4-6 weeks**
- Treatment:
 - Remove the ticks ASAP
 - Antibiotics:
 - IV penicillin
 - Tetracycline (doxycycline)

- **Interventions:** Primary prevention of Lyme Disease and other tick-borne diseases
 - Wear long sleeves shirt and long pants
 - Wear clear color cloths
 - Spray DEET (insect repellent) on cloths
 - Walk in the middle of trails avoiding bushes
 - Review your body for attached ticks and remove it
 - Check your accompany pets for attached ticks when returning from the hike or camping

Impetigo: Honey Colored Crust

- Contact Precautions: **hand washing**, avoid scratching
- Antibiotics
- Contact precautions until 48 hours after antibiotics.
- Risk for acute glomerulonephritis (Streptococcus)

Tinea (Ringworm)

- Ring-shaped lesions
 - Athlete foot (tinea pedis)
 - Onychomycosis (nail fungus)
 - Tinea corporis, cruris (**ingle**), pedis, **capitis**
- Prevention: **keep skin/feet dry**, wear shoes, use water shoes in public showers
- Treatment: **Azoles antimycotics** (ketoconazole, clotrimazole, terbinafine). (**Antibiotics are not effective**)
 - Azoles are hepatotoxic and inhibit liver metabolism of other drugs (toxicity)

Candidiasis

- Other names: thrush (oral), yeast (vagina)
- Risks:
 - Immunosuppression (AIDS, glucocorticoids)
 - Broad spectrum antibiotics
 - Uncontrolled diabetes
 - Infants (**thrush**)
- Treatment:
 - Nystatin: swish and swallow **for oral thrush**
 - Fluconazole

Pediculosis (lice)

- Permethrin: one application (10 min) and use comb to remove nits
- Re-treatment in 7 to 10 days for non-ovicides (permethrin)
- Nits are not contagious after application: send to school
- Contact precautions for only 24 hours
- Hot Wash: scarves, pillowcases, bedding, clothing, and towels
- Seal in plastic bags for 15 days not washable personal items
- Vacuum floor and furniture (no insecticides needed)
- Pets don't need to be treated (**dogs or cats**)
- Don't use lindane

Scabies

- Itching (more intense at night, hot weather)

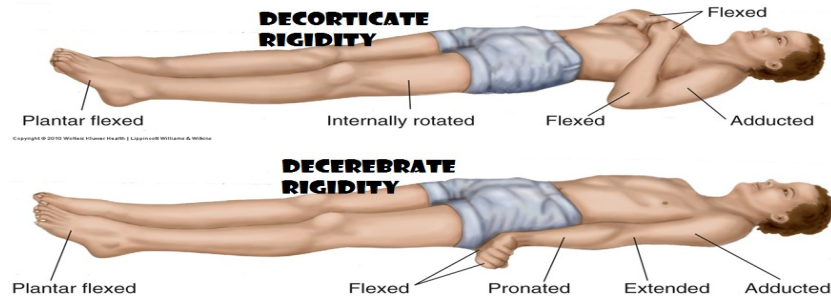
- Initial lesion in fingers and wrists
- Red wavy **burrows**
- Scabicide lotions: all body for 8-12 hours.
 - Can repeat in 7 – 10 days
 - **Permethrin**
 - Lindane **neurotoxic** (avoid)
 - Pruritus can persist after treatment
 - Not related to pets
 - Contact precautions

STI

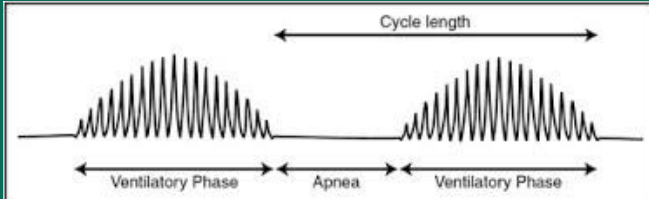
- **Chlamydia** (cervicitis, PID):
 - Can produce ectopic pregnancy, sterility for women
 - Use Condon, Tx for partner too
 - Doxycycline for 7 days. (Most used)
 - Azithromycin one dose (pregnancy).
- **Gonorrhea** (Urethritis):
 - In women produce too much discharge (not pain)
 - Ceftriaxone 500 mg if < 150 Kg, 1000 mg if > 150 Kg, IM one dose
- **Syphilis:** Treponema pallidum (Chancre, high contagious)
 - VDRL or RPR for screening.
 - TORCH, routine for every pregnant women
 - FTA-ABS for confirmation
 - Primary and secondary: Benzathine penicillin G 2.4 million units IM one dose
- For all:
 - Treat ASAP
 - Sexual abstention for 7 days, needs finish Tx
 - Treat partners (contact last 60 days)
 - Test for other STDs.

Class 3. Physical Examination

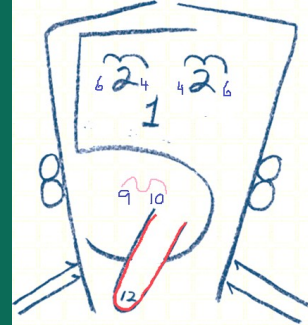
Glasgow Coma Score		
Eye Opening (E)	Verbal Response (V)	Motor Response (M)
4=Spontaneous 3=To voice 2=To pain 1=None	5=Normal conversation 4=Disoriented conversation 3=Words, but not coherent 2=No words.....only sounds 1=None	6=Normal 5=Localizes to pain 4=Withdraws to pain 3=Decorticate posture 2= <u>Decerebrate</u> 1=None
		Total = E+V+M



CHEYNE-STOKE RESPIRATION



Cranial Nerves



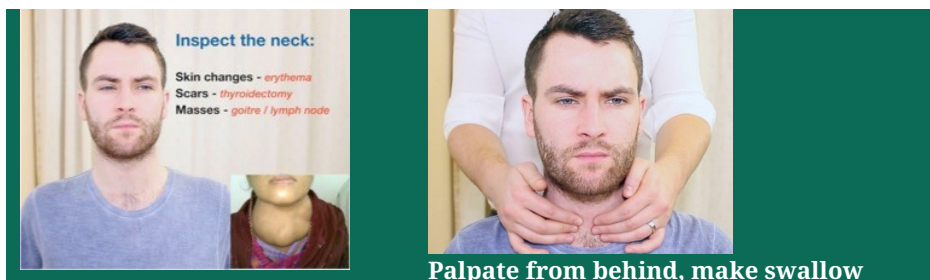
Cranial Nerves Assessment

Cranial Nerve	Responsibility and General Function
I Olfactory	Sense of smell
II Optic	Vision
III Oculomotor	Eye movement, pupillary constriction, accommodation
IV Trochlear	Movement of superior oblique muscle of eye
V Trigeminal	Sensation to the face, nasal and buccal mucosa, and teeth Muscles of mastication
VI Abducens	Movement of lateral rectus muscle of eye
VII Facial	Facial muscles and taste to anterior 2/3 of tongue
VIII Vestibulocochlear	Hearing, balance, and awareness of position
IX Glossopharyngeal	Sensation to the pharynx, posterior 1/3 of tongue, and tympanic membrane
X Vagus	Supplies parasympathetic fibers to viscera of chest, abdomen; motor fibers to the pharynx, larynx; sensory fibers to the ear, meninges, and viscera
XI Spinal Accessory	Supplies motor fibers to sternocleidomastoid and trapezius muscles
XII Hypoglossal	Supplies motor fibers to muscles of the tongue

NEUROLOGIC

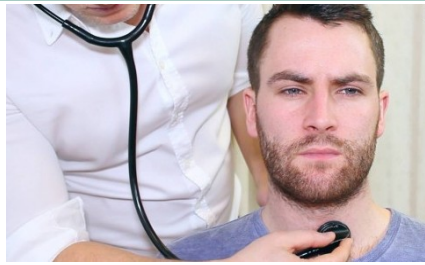
- Reflexes: DTR: Normal 2+
- Cerebellar: Romberg

Thyroid Assessment





No extension



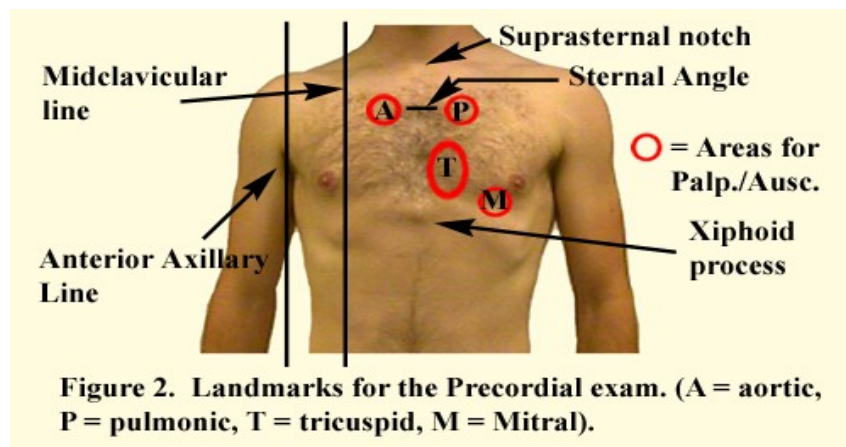
Bruit = hyperthyroidism

Pulse and BP Alterations

- **Apical-Radial Pulse deficit:**
 - Two persons counting in 1 minute:
 - One apical, another radial pulse rate
 - Any difference: EKG
 - Found in dysrhythmias: A. Fib
- **Pulsus Paradoxus:** Drop of systolic during inspiration causes disappearance of radial pulse in inspiration
- **Pulse Pressure:**
 - Systolic minus diastolic BP
 - E.g.: 130/80 = PP: 50 mmHg
 - **Neck Vein Distension Assessment**
 - HOB at 30-45 degree
- **Orthostatic Hypotension**
 - Lying, sitting and standing BP
 - Wait 3-5 minutes before each change
 - Drop 20 mmHg systolic (+ increased HR)
 - Can imply loss of >20% of volemia
 - Safety: Risk for fall

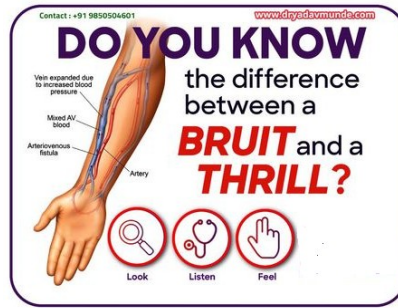
Areas of Cardiac Auscultation

PMI (Point Max impulse): V ICS/MCL



PMI ASSESSMENT

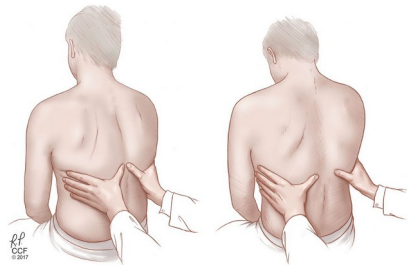
BRUIT Vs. THRILL



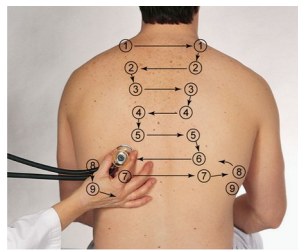
Bruit: turbulence, vibration, auscultated

Thrill: palpable vibration

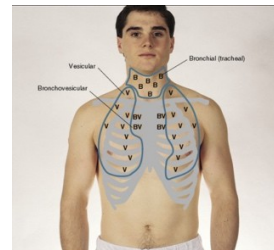
Lung Assessment: Inspection, palpation, percussion, auscultation



Lung expansion



Symmetrical



- rales or crackles: liquid or water in the alveoli
- abnormal: secretions in bronchi. NI: ask the patient deep-breath and cough
- if the patient is unconscious or intubated. NI: suctioning
- postural drainage position (rotate the patient with the head up)
- bronchoconstriction is produce by cranial nerve X or asthma, anaphylaxis, or bronchiolitis and is expected to listening **wheezes** during the expiration

Abdomen: Interview (know what happen), inspection, auscultation, percussion, palpation

Interview: pain, vomits, etc.?

Inspection: Cullen and Turner signs, distension, visible peristalsis, pulsations

Auscultation: bowel sounds, **bruits**

Percussion: shifting dullness (ascites)

Palpation: normal: liver span (inspiration-expiration), epigastric pulse

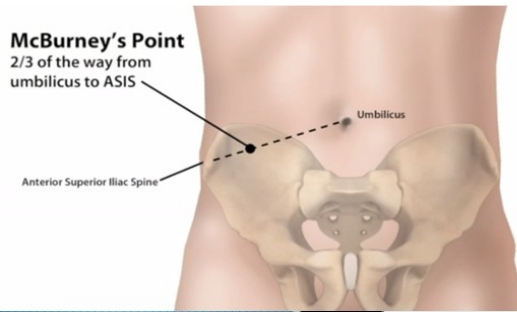
abnormal: pain, megalies, masses, rigidity, rebound tenderness (peritonitis), pulses below umbilicus,

Palpate symptomatic quadrant last.

Recognize which organs are in each abdominal quadrant

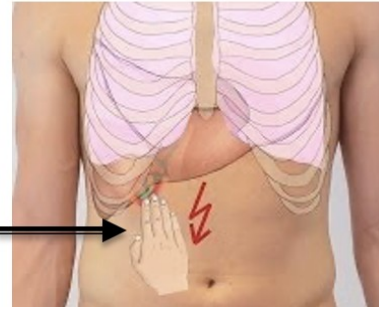
How to explore rebound tenderness

Appendicitis



Cholecystitis

Murphy's Sign
Interruption of
inspiration for pain



Infant Benchmarks

- **Vision:** See in black and white (NO color blind). Strabismus first 3 months.
- Visual fixation: 3-4 months (follow objects, see faces)
- Birth weight **doubles at 6 months and triples at 12 months**
- Anterior fontanel closes at **18 months**
- **Moro (startle), grasp, tonic neck and Galant reflexes disappear at 3-4 weeks**
 - **Babinski disappears by the first year**
- **Steady head control (no head lag) achieved at 3 months**
- **Rolling tummy to back at 4 months**
 - **Rolls in both directions (abdomen to back) by 6 months**
- Around **6 months old**, babies start to delightfully drop objects on purpose.
- **Sits supported at six months** and unsupported at **8-9 months**
- **Crawls** and make a **pincer** with **thumb and index at 9 months**
- **Holding a cup at 12 months**
- **Solitary play: Toys:** large toys, ball, rattle
- Breast or iron fortified formula milk until first year, solid food at 4-6 months
- **Pain:** Tachycardia, decreased O2 sat. Behavioral pain scales (FLACC, CRIES)
- **Separation anxiety:** starts between 6 to 8 months

PRIMITIVE REFLEXES



MORO REFLEX



TONIC NECK REFLEX



GALANT REFLEX



GRASP REFLEX



Step
reflex



BABINSKI

Faces Legs Activity Cry Consolability Revised Scale (FLACC-R)

- **Behavioral scale** used for children younger than 3 year that cannot report their pain, or in developmental delay. Also used in sedated children. **A score ≥ 3 requires pain management**

	Score 0	Score 1	Score 2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, uninterested	Frequent to constant quivering chin, clenched jaw
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking, or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting, back and forth, tense	Arched, rigid or jerking
Cry	No cry (awake or asleep)	Moans or whimpers; occasional complaint	Crying steadily, screams or sobs, frequent complaints
Consolability	Content, relaxed	Reassured by occasional touching, hugging or being talked to, distractible	Difficult to console or comfort

- **CRIES Scale:** Crying, Requires O2, Increased HR/FP, Expression, Sleepless.
A score ≥ 4 requires pain management
- **Self-report scale:** Children 4 or older can use the Face Pain Scale (**FPS-R, Wong-Baker**)

Wong-Baker FACES® Pain Rating Scale



- Numerical scale is better used after 8 years (concrete operational cognitive development)

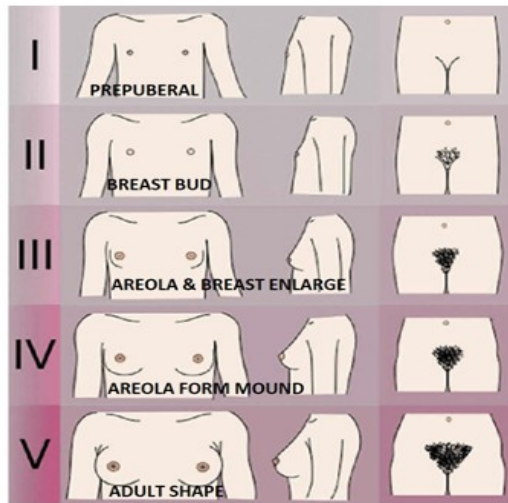
Toddler (1-2):

- Milk anemia (iron deficiency): Limit milk to 32 oz. per day
- Play: **parallel: push-pull toys. Avoid small pieces toys**
- **Walk by 18 months**
- **Throw a ball by 24 months**
- **Two-word sentences by 24 months**
- Goes up/downstairs holding by 24 months
- Hospitalization: **separation anxiety**

Preschool (3-5):

- **Copy a circle: 3 years**
- **Rides a tricycle: 3 years**
- **Stand one foot: 3 years**
- **Dress and undress: 3-4 year**
- **Hops one foot: 4 years**
- Transient **stuttering**
- Play: associative.
- **Imaginary friends**
- Allow to **play with medical equipment.**
- **Regression** as defense mechanism
- Think that death is reversible, but can experience **guilt**

TANNER STAGES OF DEVELOPMENT



Primary amenorrhea: failure to start menses within 3 years of Tanner Stage 2 or by age 15

- Stage 1: No glandular breast tissue palpable or areola (prepuberal)
- Stage 2: **Breast bud** palpable under the areola (**1st pubertal sign**)
- Stage 3: Breast tissue palpable outside areola
- Stage 4: Areola elevated above the contour of the breast (mound), forming a “double scoop” appearance
- Stage 5: Areolar mound recedes into single breast contour with areolar hyperpigmentation

Puberty in females begins with the development of breast buds under the areola, also known as thelarche, and represents entry into Tanner Stage 2. As puberty progresses, the glandular tissue of the breast increases in size and changes in contour. In females, thelarche is followed in 1 to 1.5 years by the onset of sexual hair (pubic and axillary), known as pubarche. Menarche, the onset of menses, arrives on average at age 12.5 years, regardless of ethnicity, following thelarche on average by 2.5 years (range 0.5 to 3 years).

Precocious puberty is defined as the onset of Tanner 2 secondary sexual characteristics before age 8 years in females or age 9 years in males if the continued progression of pubertal development occurs soon after. Delayed puberty should be considered if females have not reached Tanner 2 thelarche by age 13 years old or if males have not reached Tanner 2 gonad Arche by age 14 years. **Primary amenorrhea is defined as a failure to start menses within 3 years of Tanner Stage 2 (thelarche) or by age 15 years.**

Older Adult Normal Changes

- **Presbyopia:** reduce night vision (Ex: bright T-shirt for road workers at night)
- **Lentigo senile**, cherry angioma
- **Sunken eyes**, arcus senile
- **Presbycusis:** to high pitch sounds
- Decreased gustatory buds: salt cravings
- **Needs to sleep same time as younger.**
- ↓ Muscle, ↑ body fat: ↓ water, ↓ metabolic rate
- ↓ thirst: encourage to drink fluids
- ↓ hepatic and renal capacity to metabolize and excrete drugs
- **Barrel chest**, higher risk for pneumonia
- Atypical response to infections: hypothermia, AMS (confusion)
- Skin: **age spot**, **dryness**, pinna elongation, **Cherry angioma**
- **Increased incidence of cancer as age advances**
- Increased incidence of cancer as age advances
- Increased post-void residual
- Incontinence is not expected with aging

Class 4. Nutrition.

Pre-Albumin (15-30 mg/dL) and **Albumin** (3.5-5 g/dL): Prealbumin: Most sensitive for nutritional status (shorter half-life)

BMI: Normal 18.5-24.9, Weight changes with changes in water balance

Vitamins

Fat soluble	Water soluble
Vitamin A: - Liver, orange/green veggies, milk Night blindness. - Isotretinoin (acne) is a vitamin A derivate Vitamin A toxicity: ICP	Vitamin B1 (thiamine): - Pork, cereals - Wernicke-Korsakof (alcoholism)
Vitamin D (cholecalciferol): - Fortified milk, fish (salmon, cod, tuna), yolk - Calcium absorption Rickets and osteomalacia ↓CKD and sunlight deficiency. Vitamin D toxicity: hypercalcemia	Vitamin B3 (niacin): - Enriched breads ↓Cholesterol, ↑HDL-cholesterol - Nicotinic acid: Flushing. (ASA)
Vitamin E: - Vegetable oils, margarine - Antioxidant Inhibit coagulation in megadoses	Vitamin B6 (pyridoxine): - Pork, cereals, fortified bread - Neuritis (isoniazid)
Vitamin K (phytonadione) AquaMEPHYTON®): - Greens, liver - Synthesized in large intestine - Coagulation	Folic acid (Folate): - Green leafy vegetables, meat, liver, eggs - Megaloblastic anemia, neural tube defects
	Vitamin B12 (cyanocobalamin) - Animal protein - Pernicious anemia
	Vitamin C (ascorbic acid) - Citrus fruits Wound healing (together with Zinc) - Scurvy. Antioxidant

Therapeutic Diets

- **Clear liquid diet:**
 - No fiber: intestinal rest
 - Gelatin, broth, bouillon, popsicles, **coffee** (no cream), tea, apple/cranberry juice
- **Full liquid diet:**
 - Yogurt, pudding, ice cream, soups, opaque juices (**orange, peach, or pear juice**)

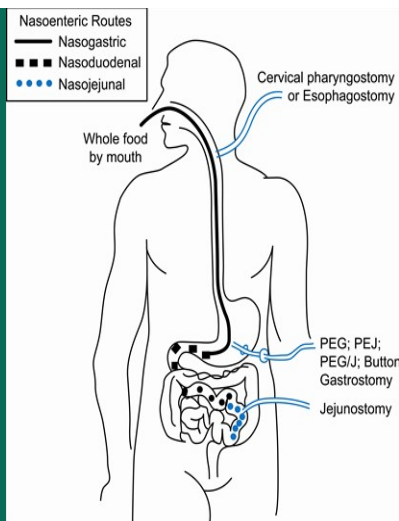
- **Mechanically Modified diet:**
 - Food processed with a machine (blender, puree)
 - **UAP is NOT allowed to do mechanical modified to any patient**
- **High/low protein diets**
 - Low: Liver/kidney diseases (with adequate CH/lipids for energy)
 - High: healing (burns, IBD, surgery), nephrotic syndrome, peritoneal dialysis
- **Gluten free (Celiac):**
 - **No: Barley, Rye, Oats, Wheat. (BROWN)**
- **High Potassium:**
 - Most fruits (apricot, avocado, cantaloupe, orange, tomato, banana, mango, papaya), dried fruits, skin of potatoes and sweet potatoes, beef green, meats, milk
- **Low Potassium:**
 - Apple, pear, peach, berries, grapes, watermelon, and canned fruits
 - Eggs, green beans, broccoli, kale, cabbage, lettuce, zucchini, yellow squash, carrots, celery
 - Avoid salt substitutes (KCl)
- **Low Sodium:**
 - **No:** process food, frozen foods, canned food (except fruits in syrup), soups, pickles, baked goods, commercial dressings
 - **Yes:** vegetables, white meats and fruits prepared from scratch
- **Tyramine restriction (MAOIs and migraine):**
 - Aged cheeses, red wine, beer, cured meats (bologna, salami), fermented food (tofu, sauerkraut), soy sauce, ripe bananas
- **High iron diet**
 - Meats, eggs yolks, green leafy vegetables, lentils, dried fruits, enriched breads and cereals, oatmeal
 - Orange juice has NO iron but helps to absorb it
- **High calcium diet**
 - Dairy, bony fishes (salmon, sardines), green vegetables (broccoli, collard green, spinach), tofu, fortified orange juice
- **Vegetarians**
 - **Vegans:** deficiency of B12 (animal food)
 - **Lacto and Lacto-ovo vegetarians:** no deficiency

Special Nutritional Needs

- **Pregnancy:**
 - Normal weight gain 25-35 lb.
 - Calories: pregnancy + 300 Kcal/day, breastfeeding + 500 Kcal/day
 - Increased iron and protein needs
 - Folate: 400 mcg/day
 - Avoid big fishes (mercury), cold cuts (listeria), raw vegetables (toxoplasma)
- **First year of life:**
 - Weight double at 6 months, triple at 12 months

- Avoid honey (botulism spores)
- No cow milk first year (breast milk or formula).
- **Whole** cow's milk after 1 year
- Water provided by the milk.
- Introduction of solid food (rice cereal): 4-6 months: one by one.
- Don't provide juices on bottle to prevent tooth decay
- **Risk for choking**

Gastro-Enteral Feeding



Gastrostomy (PEG):

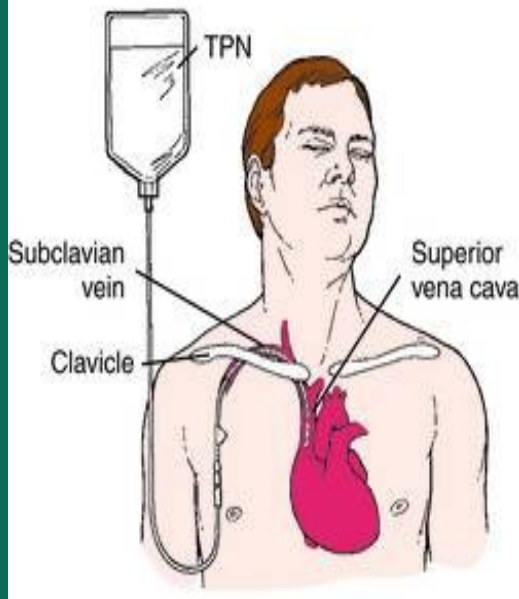
- bolus feeding

Jejunostomy (PEJ):

- continuous feeding

- **Indications:** Dysphagia or AMS
- **Contraindications**
 - Nonfunctional intestine
- **Complications:**
 - Risk for Aspiration
 - Diarrhea, electrolyte imbalance, hyperglycemia, pneumonia, peritonitis
- **Residual every 4 hours**
 - > 250 mL, prokinetic indicated
 - > 500 mL, hold feeding
- **Placement:**
 - X-rays
 - “NO” auscultation while injecting air
 - pH < 5, no supported by EBP
 - Change bag/tubing q 24 hours
 - Flush with water before/after bolus and meds administration
 - If the patients start coughing, remove the tube a little and do it again
- **Interventions:**
 - Measure tube from nose to ear and xiphoid process
 - Instruct the patient “tug your chin on chest” to prevent the tube goes to the trachea.
 - Salem Tube has 2 lumen (1 for suction and other for air)
 - HOB 30 degrees, right lateral after bolus

TPN



- **Indication:** Non-functional intestine (patients with IBD, intractable diarrhea, impaired absorption from GI, oral intake inadequate for > 7 days)
- **Central catheter.**
- **Sterile transparent dressing**
 - Contraindicated antibiotic ointment
- **IV pump required**
- **Change of tubing bag every 24 h**
- **Lipids piggybacked in TPN**
- **Do not add medications to TPN.**
- **Monitor:**
 - Hyperglycemia
 - Signs of Infections
 - Weight
 - I&O
 - Electrolytes
 - Hypokalemia (muscles cramps and spasms)
 - Adverse effect: tingling and numbness

Air embolism: churning heart murmur: Hallmark sign ^m = micron

Nutrition, Race, Culture, and Religion

- **Muslim Arabs:**
 - No pork
 - No alcohol
 - Fasting from dawn to sunset during Ramadan
- **Orthodox Jewish**
 - No pork, no shellfish, no rabbit
 - No combination of beef and dairy
- **Jehovah Witness:** No blood (blood sausage, raw meat)
- **Buddhism and Hinduism:** Vegetarians
- **Mormon:** No caffeine (coffee, tea, Red Bull), alcohol, tobacco
- **Seventh Day Adventist:** Ovo-Lacto-Vegetarian, occasional kosher lean meats, no alcohol, no caffeine, no pork, no shellfish, strict mealtime

DRUGS RELATED TO APPETITE

- **Appetite stimulants:**
 - Megestrol (Megace®): progesterone like (vaginal bleeding)
 - Oxandrolone: androgen like (hormonal side effects)
 - First gen anti-histaminic: cyproheptadine
 - Cannabinoids
- **Anorexigenic:**
 - Amphetamines: phentermine, benzphetamine, diethylpropion, methylphenidate

Addictive

Sympathomimetics (hypertension, arrhythmia, angina)
 Insomnia and nervousness

Indications and Precautions Using Herbs

- Inhibit coagulation (*warfarin, surgery*)
 - Garlic, Ginseng, Ginger and Ginkgo, Saw palmetto
 - Vitamin E, **Omega 3** (fish oil)
- Potentiate hypoglycemic drugs: Ginseng and **oolong tea**
- Potentiate SSRIs (serotonin syndrome) and MAOi: **St. John's Wort**
- Estrogen like: Black cohosh
- Stimulating of immunity: Echinacea: don't use in autoimmune diseases: SLE, RA, MS
- Antiemetic: Ginger
- Sedation: Valerian root: potentiate drugs that depress CNS
- Anticholinergic: Hyoscyamine
- Ephedra (**Ma Huang**)
 - Sympathomimetic: hypertension and tachycardia
 - Contraindicated with MAO inhibitors (phenelzine)
- Saw Palmetto: BPH

Foodborne Disease

- Foodborne illness is **caused by consuming contaminated foods or beverages**. Many different disease-causing microbes or pathogens can contaminate foods, so there are many different types of foodborne illnesses. Most foodborne diseases are infections caused by a variety of bacteria, viruses, and parasites.
- **NI:** wash the fruits and veggie before eating
 - Frozen meat can be re-frozen after thawing if still has crystal of ice
 - Hot leftovers must be stored in the refrigerator while still hot
 - Thaw frozen food in the refrigerator before cooking**

Insomnia

- Assessment of causes of insomnia: anxiety, pain, dyspnea, incontinence
- **Sleep hygiene:**
 - Keep your routine to sleep
 - Create a dark and cool environment
 - Avoid light and electronics before bed
 - Exercise early, not before going to bed
 - Avoid daytime naps
 - Avoid caffeine, nicotine, and alcohol
 - Light snack before bed: low fat **yogurt, milk, or cheese**
 - A warm bath can be helpful
 - Biofeedback** for anxiety
 - Auto-control, check your own pulse (meditation decrease pulse)
 - Use of physiologic parameter as mental control for relax.

Oxygen therapy

- Elevate HOB (**improve ventilation**)
- **Water based** lubrication of nares
- Humidifier if more than 4 L/min
 - Sterile water: change every 24 hours

- Drain condensation away from client
- Nasal cannula: precautions change weekly
 - Low flow, used in chronic airflow limitations, long-term oxygen
 - Cannot deliver higher than 40 % concentration, cannot be used in obstruction, headaches, dry mucous membrane, can dislodge
 - Deliver 1 to 6 L/min at a concentration 24 to 44 %
- Simple face mask:
 - Used in short-term/emergency
 - Deliver concentration of 35 to 50 %, required minimum 5 L/min
 - Monitor ABG, may humidity or use for aerosol Tx, protect bony prominence
- Non-rebreather mask:
 - **Bag stays at least ½ inflated at inspiration, no deflated**
 - Valves and flaps must move correctly
 - Used in deteriorated status who may need intubation
- Venturi Mask
 - Used in AR failure
 - High flow oxygen, available in 24 to 50 %
- Oxygen induced broncho-pulmonary dysplasia and retinopathy in premature (happen in children and is caused because they are receiving too much oxygen)



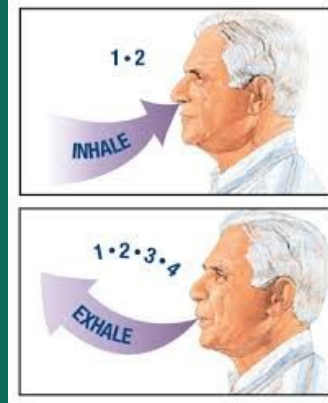
Green: Oxygen

Yellow: air

White: vacuum

Diaphragmatic Respiration with Pursed Lips

- Breathe in as if smelling a flower... Breathe out as if blowing a candle...
- Sit upright, place one hand on chest and other in abdomen
- Breathe in slowly through your nose so that your stomach moves out
- Use abdominal muscles to breathe out tightening your stomach, letting the hand on abdomen fall inward as you exhale through pursed lips
- Do not puff out cheeks
- Tuck lips to offer resistance to exhale
- Count until four when breathing out
- Benefits:
 - Reduction CO₂ in blood
 - Increase O₂ in blood



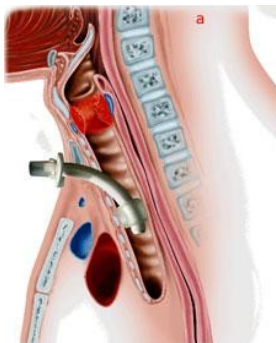
Incentive Spirometer



- Sitting or high HOB position
- After expiration, place mouthpiece inside mouth
- Inhale slowly and steady raising the flow indicator
- Remove mouthpiece and hold breath for 3 seconds
- Exhale slowly through pursed lips
- Then a short breath and cough
- Cough contraindicated in ICP, surgery of eye, ear or neck
- Encourage to repeat 5-10 times hourly

Managing Tubes and Catheters

- **Tracheostomy:**
 - Parts of tracheostomy tube
 - Complications:
 - Dislodgement:** Keep old tie until new one is on site
 - Tracheo-esophageal fistula/stenosis
 - Keep cuff pressure < 20 mmHg
 - Erosion innominate artery: Pulsatile tube, massive hemorrhage
 - Infection
 - Keep communication board.



Before 72 hours of surgical placement of trach: Manually ventilate with Ambu and rapid RT. After 72 h: extend neck, open with forceps, introduce obturator, then cannula, then remove obturator. Minimal leak technique: inflate to leak free pressure, then deflate slowly until a mild airflow is heard during peak insufflation

Question: What do you use for obtaining a sputum sample? [Use an In-line mucus trap.](#)

Suctioning

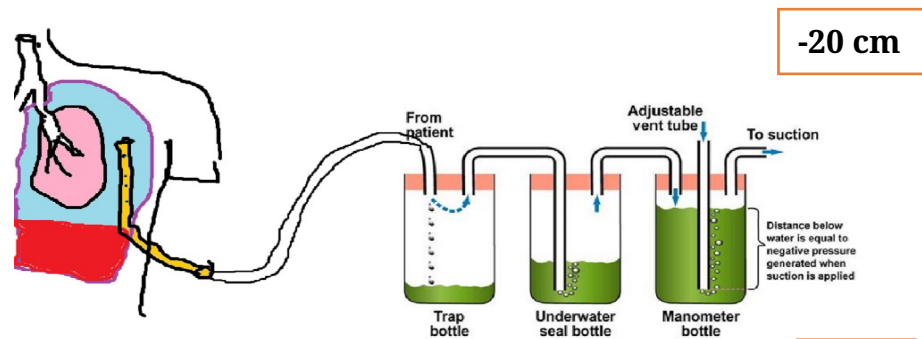
- Assessment to determine need for suctioning
- Position patient at 45-90 degrees and explain procedure
- Wall suction pressure: 100 – 120 mmHg (adults)
- Keep dominant hand and catheter sterile
- Hyper oxygenate before suction
- Insert catheter without suctioning [until resistance found.](#)
- Remove the catheter a little (1 cm) after resistance noted
- Start suctioning while going out
 - *Intermittent or continuous* suction with circular movement while withdrawing catheter
 - Limit suction time to 10 seconds
- Suction of non-sterile areas (mouth), after sterile trach suction
- Tracheostomy care: use only pre-cut gauze

Remove Bulb suction drain:

- Release the suction from the bulb
- Remove any sutures using a sterile suture remove kit
- Instruct the patient to breathe deeply and evenly
- Remove the drain using a continuous smooth motion
- Assess the exit site for irritation and drainage

Chest Tube

Used in patients with pleural effusion, pneumothorax, hemothorax, post-op cardiac/lungs Sx.



Collection chamber: [collect fluids/blood/air](#), observe for bleeding or outflow > **-2 cm** hour. If > 100 call HCP.

[Assess every 1 hour the first 8 hours, after that assess every 8 hours.](#)

Keep collection chamber **below chest level**. [The patient can ambulate with the chest tube.](#)

Water-seal chamber: Fluctuation (tidaling) and intermittent bubbling are OK

[Allow air to exit the pleura space and ensures it is not going back into the body](#)

[No fluctuation means lung re-expanded or kink](#)

[Excessive/continuous bubbling means air leaking](#)

Suction chamber: Continuous, gentle bubbling if suction connected. **If the suction is vigorous bubbling means suction too high**

Drainage system disconnected: submerge chest tube in water until reconnect

Chest tube dislodgement: cover with petroleum gauze and tape three sides, **call HCP**

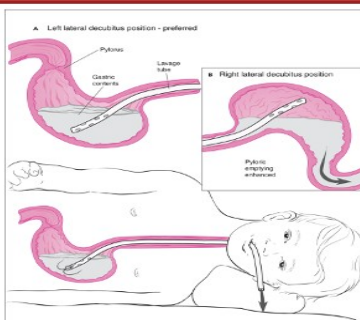
Sucking wound: when the air enters the thorax wall

NI: Keep a bottle of sterile water near the bed

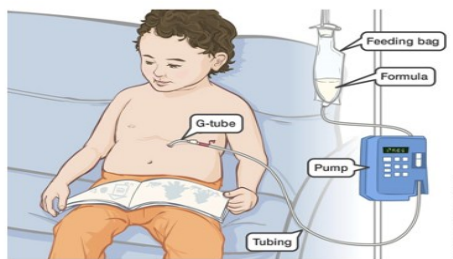
Naso-Gastric & Naso-Intestinal Tubes

- **Salem sump:** double lumen with a blue air vent
Keep vent above level of stomach and don't introduce liquids on it
- **Intestinal: Miller-Abbot and Cantor:** Do not secure with tape until final position is reached.
- **Insertion procedure:**
 - Obtain equipment and supplies, perform hand hygiene, apply clean gloves
 - Raise HOB to high Fowler
 - Assess nares and select a naris
 - Measure: **nostril-ear-xiphoid**, mark the limit
 - Lubricate tube with water-based lubricant
 - Instruct the patient to extend neck back slightly
 - Gently insert tube just past nasopharynx
 - Make patient **tuck chin on chest and take sips of water** (swallow) and advance tube to the mark point
 - Confirm position with X rays before initiating first feeding
 - Verify **placement and residual** before each successive feeding/medication: PH
 - Mouth care is priority (no glycerin swabs)
 - Flush 30 mL water before/after food or meds administered
 - NG tube (suction) not working client experiences nausea and distension
- If the client **can't swallow** the tube can be pass under **fluoroscopy**

Gastric Lavage: Left lateral Position



Enteral Feeding Position



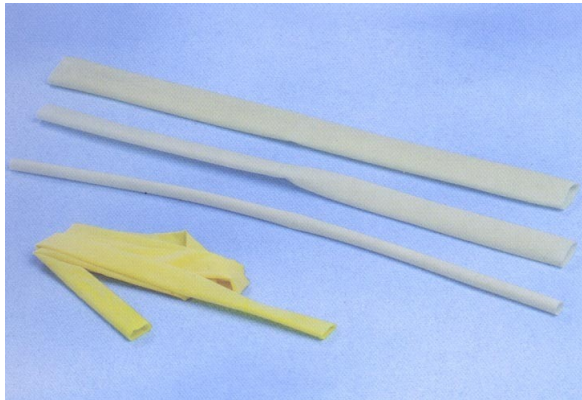
How to remove NG tube:

1. Disconnect from suction and remove tape
2. Inject 2-3 mL of air on tube
3. Pinch the tube with fingers
4. Ask the client to take a deep breath and bear down (Valsalva).
5. Pull out gently
6. Document procedure.



Drainages

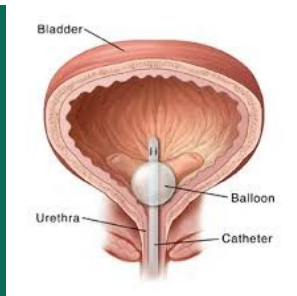
Jackson Pratt without clamp



Remove Bulb suction drain (JP)

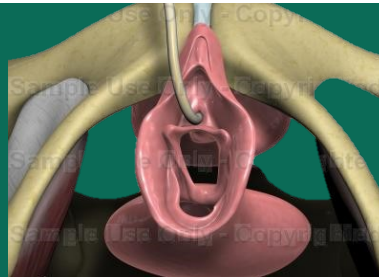
- Release the suction from the bulb
- Remove any sutures using a sterile suture remove kit
- Instruct to breathe deeply and evenly
- Remove the drain using a continuous, smooth motion
- Assess the exit site for irritation and drainage

Foley Insertion



Males: Advance the catheter 1 to 1.5 inch after urine starts flowing

Males: insert catheter positioning penis at 60-90 degrees. Remember return foreskin for uncircumcised



Females: if no urine flows after inserting the catheter, leave this catheter in place (vagina) and re-start procedure with other catheter

FOLEY INSERTION

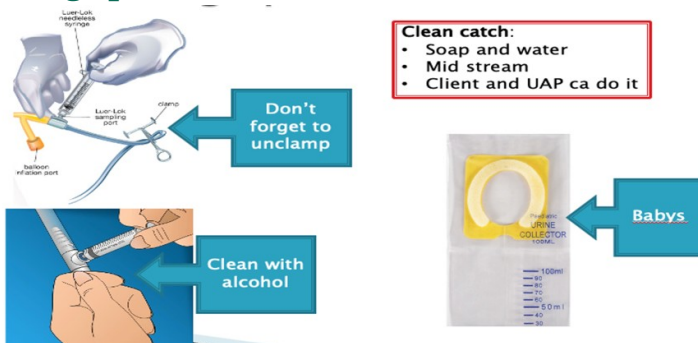
- Hand washing and sterile gloves
- Lubricate Foley and insert syringe with NS
- Separate labia with non-dominant hand
- Clean each labia front to back and clean meatus last
- Insert catheter until urine is visualized, advance one inch more
- Inflate the balloon with NS
- Anchor catheter on thigh
- Document

REMOVING FOLEY

- Protect linen with a pad and done clean gloves
- Empty balloon with syringe

- Ask client to take a breath and remove gently
- Clean the meatus
- Document

Taking Specimen of Urine



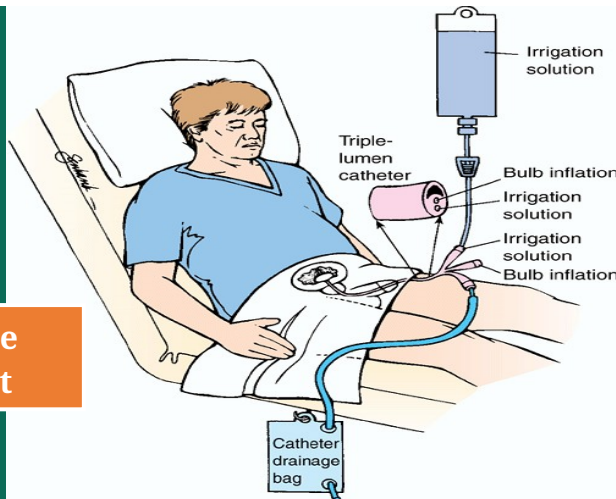
24-hour Urine Collection:

- Hour 0: urinate and discard urine
- Collect ALL ongoing urine
- Any spill requires start again
- Keep urine cold
- Hour 24: urinate and collect urine

Continuous Bladder Irrigation (CBI)

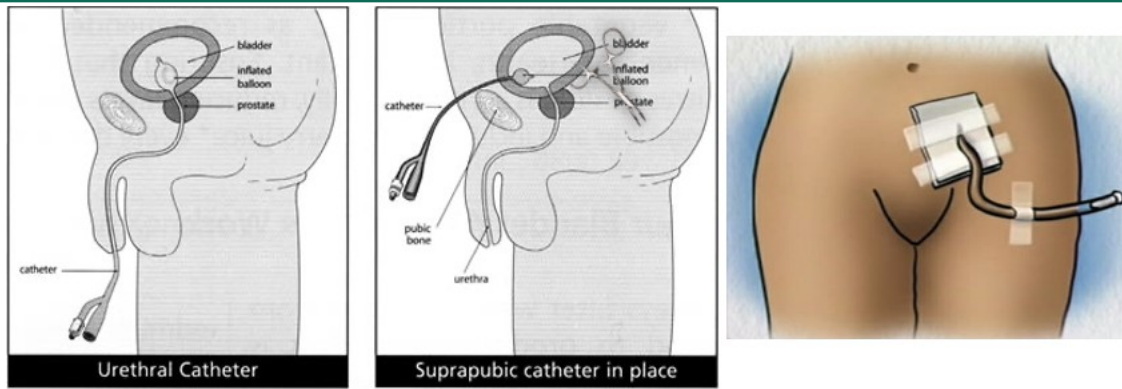
Irrigation: 1000 mL
 Drainage bag: 1500 mL
 Urinary output: ? mL
 Monitor LOC

leg should be kept straight

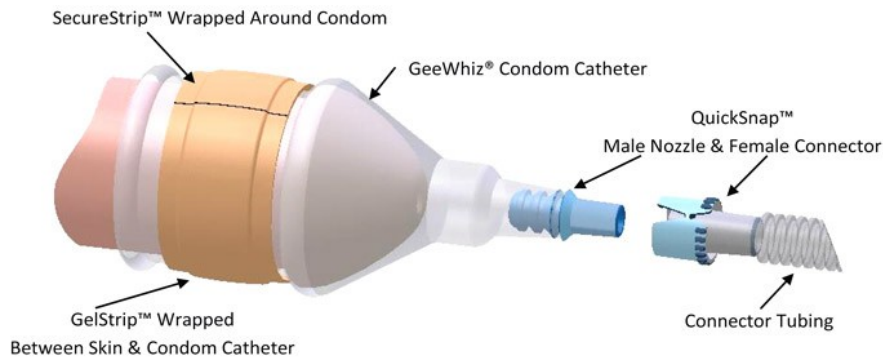


The intention of straight the leg is maintain the pressure in the prostate and have less bleeding

Suprapubic Catheter: Used for patients with swelling prostate and cannot pass the catheter through the urethra.



Condom Catheter: Can be performed by the patient or the UAP
Noninvasive alternative for incontinence management.



ENEMAS

- Sims left lateral, right knee flexed, **left leg needs to be extended**
- Lubricated tip introduced **3-4 inches**
- Bag 500-1000 ml warm solution
- Elevate no more than **12-18 inches or 1-1 ½** (**Patient can complaint pain if the bag is too high, put the bag a little down**)
- Lower or clamp if abdominal cramps
- Administer in **10-15 minutes**, **tell the UAP the enema cannot be admin too fast (< 5 min)**
- Hold liquid for **1-2 minutes**
- No more than 3 cleansing enemas (**more than 3 can produce Hyponatremia dilutional, call HCP for another solution**)
- Cleansing Vs. Retention enema (**Retention enema is used by patients with colon issues and the liquid used is kayexalate (Na+ polystyrene) or corticosteroid**)
- Tap water enema can be delegated
- Suppository: 1 to 1-1/2 "past sphincter and hold 15-20 minutes (**the suppository needs to be cold; use lubricate gloves**)

OSTOMY: POUCHING AN OSTOMY

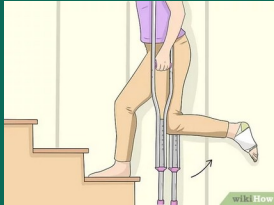
- Remove old pouch (standard precautions)
- Initial assessment (RN)
- Clean with warm water, pat dry (**it is normal when padding some blood comes out**)
- Detailed assessment of clean stoma
- Measure stoma (RN)
- Cut to fit pouch 1/8 inch larger (RN)
- Apply skin barrier around opening in pouch
- Apply skin barrier and pouch, press
- Fold bottom of open-end pouch and clamp.

- Evaluate drainage coming out.
- Scope of LPN or RN
- UAP can help the client trained to do it or can empty the bag and measure amount of outcome

Assistive Devices: Strength on Hands

- **Crutches:** (produce trauma on the axilla's)
 - **Tripod safe position:** 6 inches lateral, **6-8 inches forward**
 - **3 finger widths** between axilla and top of crutch
 - **Never lean** on your crutches
 - Elbows flexed at **30 degrees**

Upstairs: good leg goes first, weight sustained by crutches and weak leg. NEMONIC: Good leg Go to the sky



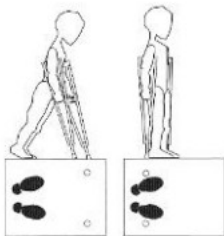
Downstairs: bad leg goes first after moving the crutches downstairs.

NEMONIC: Bad leg Go to evil first
Steps:

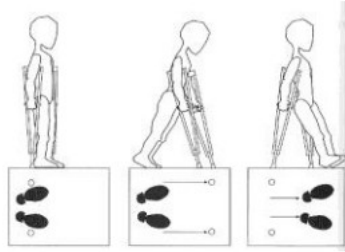
- Advance crutches
- Advance bad leg
- Advance good leg



- Gaits:
 - 2 points (one foot and one crutch (opposed sides) on floor while the other 2 advance on air)
 - 3 points (two crutches advance with weak leg while weight is supported by strong leg, then the strong leg advance swimming through)
 - 4 points (always two crutches and two feet are on floor, advancing alternate)
- How to Use crutches: 2, 3, 4 Points Gaits, Swing-To/Through, Stairs, Nursing NCLEX – Bing video
- **Transfer to crutches to seat:** hold armchair with strong hand, both crutches on weak side hand
- Amputation or not weight bearing: **swing to gait or swing through gait**
 - **Swing to gait** is when the patient advances the crutches and move both legs together till the same place of the crutches



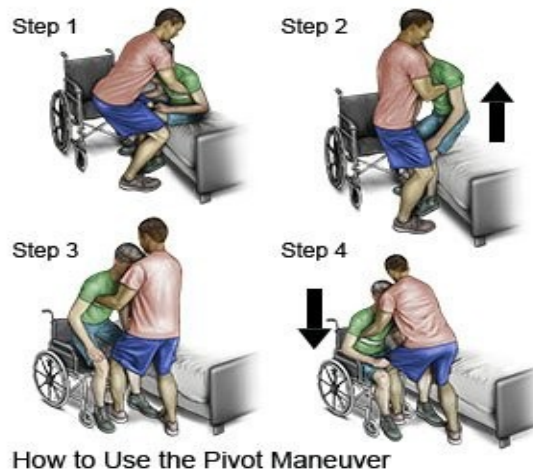
- **Swing through gait** is when the patient advances the crutches and move both legs together just in front of the place where he stop the crutches



- **Cane: “COAL cane opposite affected leg”**
 - Hold cane with **strong side’s hand**
 - Elbow flexed at 30 degrees, top at level of **great trochanter**
 - Advance cane and weak leg at the same time
- **Walker:**
 - Client must have strength on arms, elbows flexed at 30°
 - 1. Lift and advance walker: put all four points of walker on floor,
 - 2. Put weight on the hand pieces
 - 3 Then advance weak leg, after that advance the strong leg into the walker area

Walking a Client with Weak Side

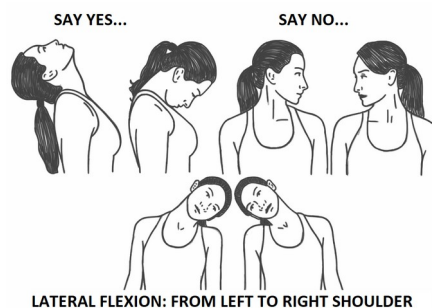
- Nurse stand on the **affected side**
- Grasp the security belt in the mid-spine
- Position free hand at the shoulder area so that the client may be pulled in the event of forward fall
- Client is instructed to **look up and outward** rather his feet. *If patient have Parkinson the nurse does not allow the patient look down*



How to Use the Pivot Maneuver

ROM: Passive or Active

Example: ROM for neck.



Consequences of Immobilization

- Atelectasis and pneumonia (NI: avoid with early ambulation and use incentive spirometer)
- Skin breakdown
- Muscle atrophy
- Bone demineralization, osteoporosis and osteopenia
- Urinary stasis and kidney stones
- Constipation
- Venous stasis and DVT **Venous Stasis are clot on the vein
- **Range of Motion:** help to keep normal joint mobility
 - Active
 - Passive

Class 5. IV and Blood. (Video 7)

Fluid Volume Excess (hypervolemia)

- Causes: *heart failure, renal failure, IV fluids*
- Assessment
 - Tachycardia (happen in Addisonian crisis)
 - Hypertension (bounding pulses)
 - Distended neck veins (it is JVD, nurse kept HOB at 45°, check vein)
 - Dyspnea, Orthopnea (with orthopnea the patient cannot be lie-down)
 - Moist crackles lung bases
 - Cough, **pink frothy sputum = Pulmonary edema**
 - Dependent edema (depending on position)
 - Elevated CVP
 - Weight gain: ≥ 3 lbs in 2 days (because patient have liquid retention, nurse needs weight daily, 1 L = 1 Kg = 2.2 lbs)
 - A new diaper, dry and clear the standard weight is 20 g.
 - Ex: a mother takes off the wet diaper for her baby and the weight is 65 g – 20 g = 45 g = 45 mL
- Interventions
 - Sodium restriction:
 - Avoid process food, canned, sauces, dressings, tomato juice, soups, dry fruits, sea food, dairy, pretzels, pickles, milk and dairy
 - Diuretics: potassium rich diet (except in CKD and spironolactone)

Fluid Volume Deficit (hypovolemia)

- Causes: *bleeding, vomits, diarrhea, polyuria, burns, sepsis, 3rd space*
- What is shock? Causes. Pathophysiology (**lactic acidosis**)
- What are insensible losses?
- Assessment
 - Hypotension: orthostatic dizziness
 - Tachycardia
 - Thirst, Oliguria + Increased specific gravity
 - Decreased skin turgor
 - Decreased CVP
 - Increased hematocrit

- Interventions

Oral rehydration preferred (better than IV, most in children)

IV hydration (severe): Isotonic fluids: NS, LR

IV Fluids

Type Solution	Osmolarity	Composition	Use
Normal Saline	Isotonic	0.9 g NaCl/100 mL H ₂ O	Fluid replacement, bleeding, burns
Lactated Ringer's	Isotonic	NS with K ⁺ , Ca ⁺ , lactate	Fluid replacement * Preferred for burns * No-no for dehydration in CKD
D5W	Isotonic in vitro, Hypotonic in vivo	5 g glucose/100 mL H ₂ O	Replacement of water or glucose
½ NS	Hypotonic	0.45 g NaCl/100 mL H ₂ O More water less salt	Replacement of water, hyponatremia, DI
3% NaCl	Hypertonic	3g NaCl/100 mL H ₂ O More salt less water	Hyponatremia or volume expansion
Dextrose + Saline (e.g., D ₅ NS)	Hypertonic	Glucose and NaCl in H ₂ O	Hypoglycemia or client cannot eat, need calories

Ringer Lactate: Each 100 mL of intravenous infusion contains: NaCl 600 mg, Sodium lactate 310 mg, KCl 30 mg, CaCl dihydrate 20 mg, Water for injection q.s.

Peripheral Venous Access Devices

Avoid:

- Dermatitis and cellulitis,
- Paralyzed arm
- Dominant arm
- Lymphedema or AV shunts (fistulas)
- Over joints e.g., wrist



INSERTING PERIPHERAL VENOUS ACCESS DEVICE (PVAD)

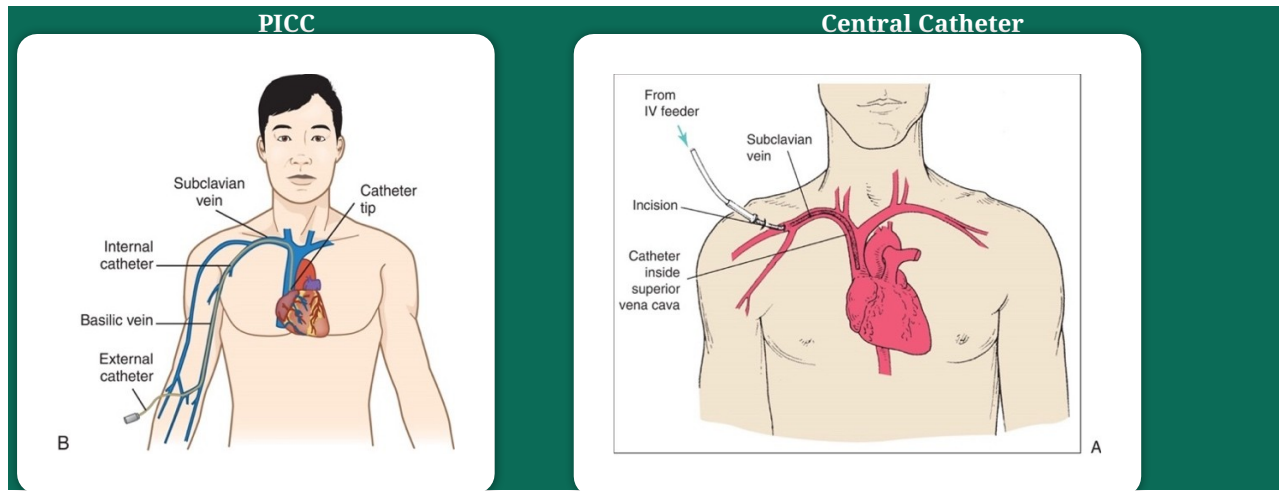
- Wash hand and don clean gloves
- Place tourniquet 3-5 inches proximal to insertion site
- Cleanse selected site
- Pull skin taut to anchor vein
- Insert needle bevel up until blood return on hub
- Advance catheter
- Remove stylet and activate safety device.
- Make pressure above insertion site and remove tourniquet

- Connect to infusion set or saline lock
- Apply transparent dressing and date

REMOVING PVAD (Remove IV)

- Check order, wash hands and don clean gloves
- Close clamp IV infusion
- Remove IV dressing
- Place clean gauze over IV site and apply pressure
- Pull catheter out slowly. Check for intactness
- Cover with transparent dressing or band-aid

Central Vein Access Device



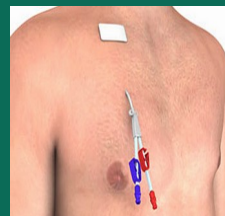
Positioning during insertion: Trendelenburg
X-ray after procedure (pneumothorax, cath tip)

Central Catheters Care

- **Signs of infection: Report ASAP**
- **Flush with syringe of 10 mL**
- **Dislodgement:**
 - Pain while flushing
 - Check length of catheter
- **Obstruction:**
 - Don't force flush
 - Label the obstructed lumen
 - Thrombolytic
- **Prevention air embolism: Valsalva or Trendelenburg with tube changes**



Port A-Cath



Tunneled

Both catheters the tip is placed at Superior Vena Cava. Tunneling place the port under the skin.

Which has the lowest risk of infection?

Air Embolism (Prevention)

- Related to central catheters
- Respiratory distress, hypotension
- Nursing interventions:
 - Clamp the tubing
 - **Trendelenburg position.** (If air embolism occurs, place the patient on left lateral with head down)
 - Oxygen
 - Notify MD or call Rapid response team
 - **Ask the patient to perform Valsalva maneuver**
 - **Instruct the patient NOT deep breath during insertion**

Central Venous Catheter Dressing Change

- Hand hygiene.
- Facial mask on nurse and client (Have patient wear a mask or turn his head away)
- Non-sterile gloves to remove old dressing
- Assess insertion site
- Remove gloves, wash hands and put on a pair of sterile gloves for site cleaning
- Clean around catheter with chlorhexidine swab, scrub for 30 seconds
- Allow to air dry for 30 seconds
- Clean the catheter in one motion from insertion site to the distal end, **never up to the insertion site**
- Do not use antibiotic ointment, except for dialysis catheters
- Apply transparent dressing over catheter
- Date, time, initial the dressing.
- Document

Interventions in CVC dressing change:

- Dressings needs to be transparent
- Change dressing sterile, if soft or dirty, do NOT do it every day
- Every time the RN touch the catheter, needs measure the outside part in centimeters and compare, if the result is more, call HCP
- Cover de dressing with plastic before shower
- Clean catheter away to the insertion site

PICC line remove

- Only can be removed by HCP
- If the catheter is broken when is pulled:
 - Apply a tourniquet to upper arm (used for lodged the catheter embolus)
 - Check the radial pulse to may sure the tourniquet is not tight
 - Call for emergency X-rays
 - Call HCP and rapid response team, and PICC insertion team
- Document

Draw blood in catheters

- The catheter has various lumen
- Check what kind of test is
- Waste first 3-5 ml of blood

- Draw in new syringe 3-5 ml of blood
- Flush with normal saline at start the test and at the end
- When flush notice resistance, tell the patient raise the arm and cough because maybe the top is close of the vein
- Do NOT flush the catheter with syringe less than 10 ml

IV Infiltration Vs. Thrombophlebitis



Elevate arm



Warm compresses

Assess IV

- Assess for signs of infiltration and phlebitis or infection
 - Skin turgor
 - Monitor lab test
 - Monitor In & out
 - Monitor daily weight
 - Auscultate lung sounds
- Use aseptic technique
- IV bolus admin into IV tubing injection port
 - Occlude IV tubing proximal above the injection port
- Explain procedure to the patient
- Time the medication administration with a watch

IV Bag Label

- Use a regular pen on a label or tape and place on the IV bag
- Do NOT use marker pen for label the IV bag (Ink of the marker in the bag can penetrate the plastic of the bag and diffuse into the solution)

Extravasation (infiltration) of Vesicants

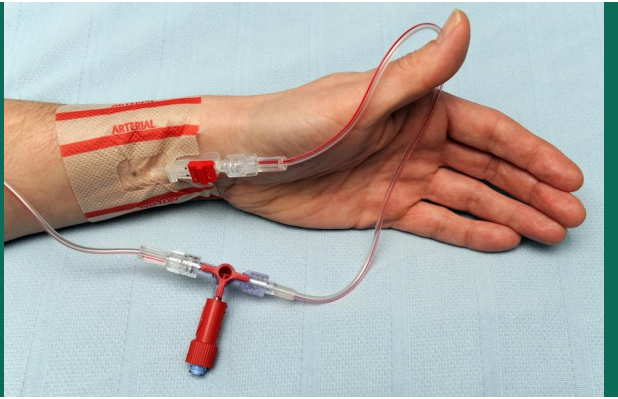
- Dopamine, chemotherapy, TPN



- Frequent assessment while infusing
- If infiltration occurs:
 - Remove IV fluids **without removing catheter**
 - Aspirate as much blood and fluid as possible before removing catheter
 - Follow physician instructions depending on type of IV fluid (e. g: phentolamine for dopamine)

Arterial Line

- Radial artery
- Purpose:
 - Blood pressure
 - ABGs
- Complications:
 - Bleeding
 - Thrombosis of artery
- Removal:
 - Pressure for 5 mins.



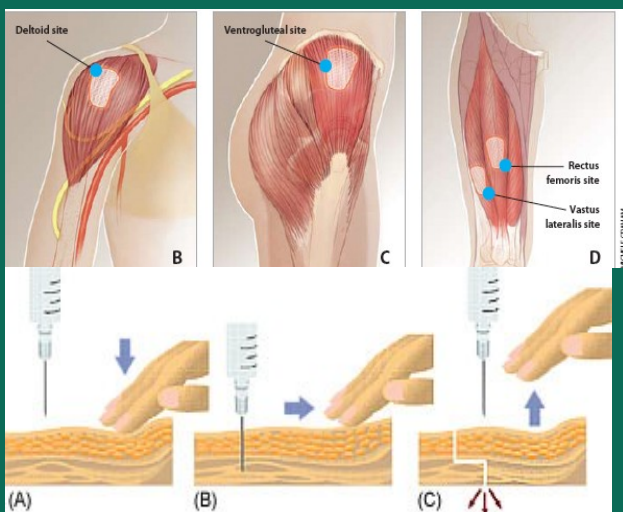
Intramuscular & SQ

IM:

- Needle adult: 21 Gauge, Length: 1-1½"
- Needle Infant: 23 or 25 Gauge, length: 5/8" (1.6 cm)
- Ventral gluteal
 - Major trochanter: do not use dorsal gluteal for IM.
 - Lateral position

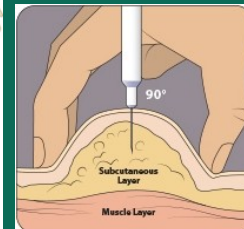
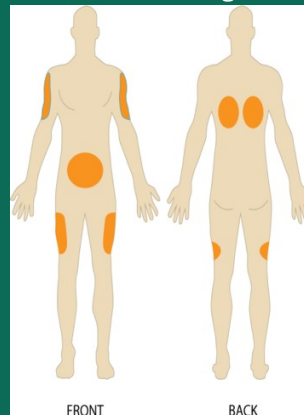


- Deltoid (2-3 fingers below the acromion), do not put more than 1 mL
- Vastus lateralis do not put more than 1 mL
- Z track



SQ:

- Needle 24-26 G, 3/8"
- Make a pinch
- 45 or 90 degrees
- 45° to thin or low BMI (<18)
- 90 degrees for fatty client (BMI>35)
- Don't remove air bubble
- Don't aspirate
- Don't massage



Z-track: for ampule needs to use filter needle. If you use the vial needs to introduce air first to draw the liquid. Change the needle. Not draw the air from the vial (IM or SQ). Use the nondominant hand to pull

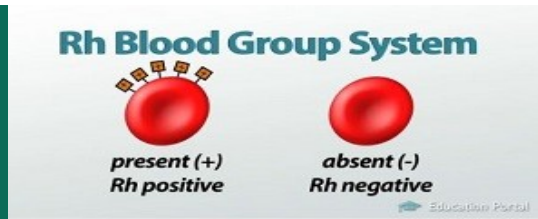
the skin backward. Without left the skin introduce the needle at 90 degrees, extract the needle and left the skin. Do not massage the skin.

TRALI (transfusion related acute lung injury)

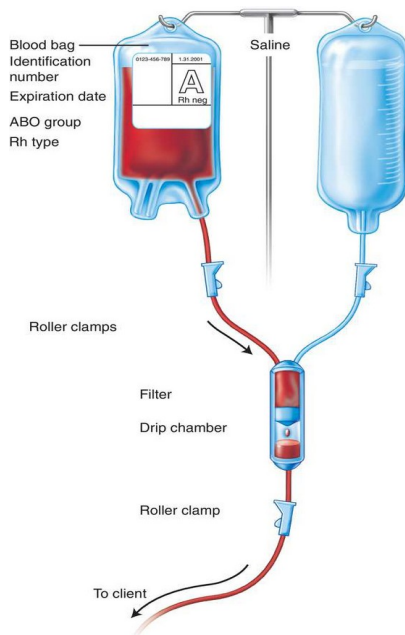
- Symptoms occur within 1-6 hours of transfusion
 - Fever, chills, hypotension, foamy oral secretions, tachypnea, frothy sputum, dyspnea, hypoxemia, respiratory failure
- Prevent TRALI requesting leukocytes reduced RBCs

Blood Products:

BLOOD GROUP COMPATIBILITY		
BLOOD GROUP	CAN DONATE TO	CAN RECEIVE FROM
O	O, A, B, AB	O
A	A, AB	O, A
B	B, AB	O, B
AB	AB	O, A, B, AB



NEGATIVE (-) only receives NEGATIVE (-)
POSITIVE (+) receives POSITIVE (+) and NEGATIVE (-)



Each bag PRBC increases: **1 gram hemoglobin**
3% hematocrit

Steps for Blood transfusion

- **Check vein: 18G catheter**
- Assess baseline vital signs
- Send errand/CNA for blood
- Y-set with filter primed with NS
- Verify blood with **other RN before administer**
- Spike blood and clamp NS
- Take vital signs

- Initiate blood **within 30' of received**
- **Clamp NS line and blood line open**
- Begin transfusion slow (acceptable 2 mL per min)
- Expect every bag of RBC increase Hg 1g and Hct 3 %
- Check VS **every 5 min for 15-30'**
- Complete within **less than 4 hours (Never more than 4 hours)**
 - Platelets need to pass faster
- **No medication** can be added
- Monitor for adverse reactions (elderly can develop cardiac insufficiency, anaphylactic shock, infection, **trans-fusional reaction**)

Transfusional Reaction Interventions

1. **STOP THE TRANSFUSION** and remove blood and tubing (**Do not remove the catheter**)
2. Keep IV catheter and start a new IV-line infusion with NS or 0.9% NaCl
3. Place the client in Fowler's position and administer O2 for ARDs
4. Notify the physician/RRT.
5. Obtain a urine specimen
6. Blood container, tubing, attached label, and transfusion record are saved and returned to the laboratory for analysis.

Incompatible reaction release cytokines causing shock, acute respiratory distress, DIC, and acute renal failure

Client restless, intense back pain, and dark urine (hemoglobinuria)

Note: Packed red blood cells or whole blood, white blood cells, and plasma requires ABO/Rh group compatibility.

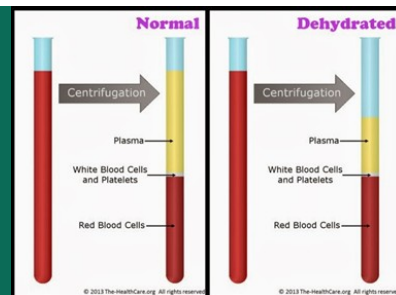
Platelets, albumin, gamma globulins, and cryoprecipitate of coagulation factors doesn't require ABO group compatibility (any group can be used)

Class 6. Labs and Procedures

Test requiring fasting: glucose and lipids (triglycerides). Fasting is not necessary for other blood or urine work

Some Lab Tests

- Serum Electrolytes:
 - Sodium (Na^+): 135-145 mEq/L
 - Potassium (K^+): 3.5 – 5 mEq/L
 - Calcium (Ca^{++}): 9 – 11 mg/dL
 - Magnesium (Mg^{++}): 1.5 – 2.5 mg/dL (preeclampsia until 7 is expected, 8 is toxic)
- Hemoglobin (12 to 16 g/dL. Critical < 8 g/dL) anemia (need transfusion), Except Pregnancy 11 to 16, newborn: 16 - 24
 - Hematocrit: $\text{Hgb} \times 3$ (when pt is dehydrated the Hct increase)
 - Related to hydration



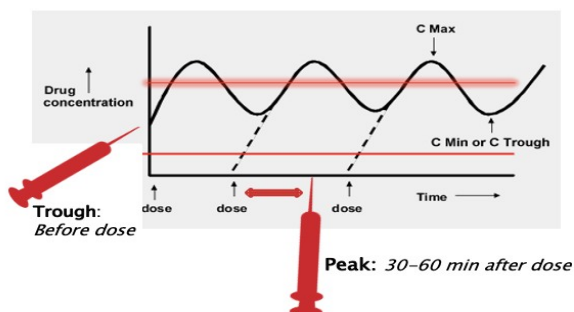
- **White Blood Cell Count** (5000 to 10,000 cells/mm³)
 - < 5000 is called **leukopenia**, > 10000 is called **leukocytosis possible infection**
 - Higher values are normal if **receiving steroids** or in **labor (prednisone, cortisone, glucocorticoids)**
 - **Teach the patient to report immediately sore-throat**
 - If the patient is receiving adalimumab (Humira) check for WBC elevated (infection)
 - Eosinophils (and **IgE**): allergies
 - Lymphocytes: viral infections, CD4
 - Neutropenia: severe when count <500 cells/mm³
 - Drug induced: clozapine, carbamazepine, PTU, chemotherapy (nadir)
 - Filgrastim (**increase WBC**)
- **Platelet count (150,000 to 450,000 cells/mm³)**
 - Critical thrombocytopenia:**
 - <20,000 (***** Medical Emergency**, possible head bleeding, platelet transfusion)
 - >20,000, bleeding precautions (prevent trauma)
 - Platelet < 150,000 is DIC (disseminated intravascular coagulation)
 - Platelet low in-patient using heparin, produce HIT (Heparin induced thrombocytopenia, it is **emergency, stop heparin, call HCP**) or ITP (Idiopathic thrombocytopenia purpura)
- **Activated partial thromboplastin time (aPTT):**
 - Normal 25 to 35 seconds**
 - Heparin therapeutic range: **aPTT at 1.5 to 2.5 times** control time
 - Memorize 75" as the limit of over-anticoagulation and bleeding risk
- **Prothrombin time (PT) and INR: 9.5 to 12 seconds**
 - Warfarin** therapeutic range: **INR 2.0 to 3.0** (more than 3 stop and call HCP)
 - 2.5 to 3.5 for high-dose warfarin therapy
- **D-dimer: fibrin degradation product (FDP) positive in thrombosis (confirm DVT, PE)**
 - Normal values for each local laboratory may vary. When activator reagents are used by the laboratory, the normal CLOTTING time is shortened. Common normal ranges are 20-30 sec in some laboratories and 30-40 sec in others. Therapeutic range for PE is 1.5-2.5 times the normal value (e.g., if normal is 20-30 sec, therapeutic range is 40-75 sec).
 - DVT: check arterial pulse
 - **Patient receiving heparin needs to have platelet normal**
- **Glycosylated hemoglobin A_{1C}**
 - DM diagnosis ≥ 6.5%.
 - A_{1C}: normal 5.6% or less.
 - GI: 5.7 to 6.4%.
 - DM: 6.5% or more.
 - DM well controlled: < 7%
- **BNP: Type B natriuretic peptide: Rule out heart failure if < 100 pg./mL**
 - **Used when the patient has cardiac insufficiency, cardiomegaly**
 - **BNP < 100 not heart failure (HF)**
 - **BNP > 100 yes HF**
- **Amylase & Lipase: Pancreatitis (enzymes of pancreas)**
- **Prostatic Specific Antigen (PSA):** Blood sample before digital rectal examination
 - **Digital exam can elevate PSA**
- **Liver Function Tests: LFT**

- Liver enzymes: AST, ALT (<40 IU/L): elevated in hepatocyte lesions, **caused by virus, preeclampsia**
- Bilirubin: *product of hemoglobin breakdown (hemolysis), excreted by liver*
- Albumin: 3.5 -5 g/dL
- Prothrombin time & INR (*normal < 1.2*)
- **Serum creatinine** (0.5 to 1.2 mg/dL)
- **Blood urea nitrogen (BUN):** 10 to 20 mg/dL, **elevated when kidney damage and dehydration are**
- Serum Enzymes and Cardiac Markers
 - Troponin I: remains elevated 10-14 days after MI, **not increase in angina, yes in MI, do first EKG**
 - Total Creatine kinase (CK-MB): skeletal muscle disease (rhabdomyolysis, dermatomyositis)
- **HIV and AIDS Testing**
 - Screening: **ELISA** (**can be false-positive, need confirmation**)
 - **If patient is high risk for HIV and Elisa is negative, repeat the test in 3 months)**
 - Confirmation: **Western blot**
 - CD4⁺ T-cell count (*Normal: >600 cells/mm³*)
CD4 <200/mm³: **AIDS**
 - **Viral load testing: PCR**
 - Med: Tenofovir, Emtricitabine
 - **Tx is effective if CD4 and PCR are normal results**
- **Syphilis Testing**
 - VDRL or Rapid Plasma Reagine (RPR): screening
 - Fluorescent Treponema Absorption Test (FTA-ABS): **confirmation**
- **UTI**
 - Leukocyturia
 - Bacteriuria
 - Positive nitrite and leukocytes esterase

Therapeutic Serum Medication Levels

- **Digoxin 0.5 to 2 ng/mL**
- **Lithium 0.5 to 1.2 mEq/L**
- **Phenytoin 10 to 20 mcg/mL**
- Theophylline 10 to 20 mcg/mL
- Valproic acid 50-100 mcg/mL

Peak and Trough



Trough can be drawn 30 minutes before dose, Peak 1 hour after administration in average

(Start Video 8)

ABG sampling (ABG blood test use radial artery ONLY)

- pH: 7.35-7.45
- $p\text{CO}_2$: 35 – 45 mmHg
- HCO_3 : 22 – 26 mEq/L
- $p\text{O}_2$: 80 – 100 mmHg

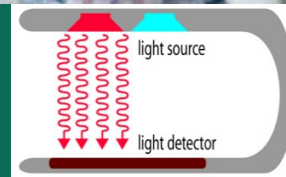


STEPS to demonstrate ulnar artery is able:

- ask patient close fist
- press both artery
- ask patient release hand and check hand blanching while RN compress both arteries
- release one artery, blood return to the hand

O₂ Saturation: Pulse Oximetry

- Below 95%: concern
- Below 90%: critical
- False readings:
 - Bright environment light
 - Severe anemia
 - Nail polish or artificial nails
 - Movement: shivering, seizures
 - Local ischemia:
 - Raynaud's,
 - Blood pressure cuff on site,
 - Shock, cardiac arrhythmia



GI Diagnostic Procedures

- **Occult blood: Guaiac**
 - Three days before avoiding:
 - False positives:
 - Read meat, meatballs, sausages
 - Broccoli,
 - Turnips,
 - Radishes
 - NSAIDs, ASA, and corticoids
 - False negative:
 - Ascorbic Acid (Vit. C), oranges, tomatoes

Diagnostic Procedures

- **Upper GI series or Barium swallow:** Esophagus, stomach, duodenum
 - NPO

- Barium sulfate
- X rays
- Stool softener, white feces
- Last test to practice
- **Barium enema:** colon and rectum
 - Bowel preparation:
 - NPO after midnight
 - Performed after US or colonoscopy
 - **Sim's position**
 - Stool softener, white feces

Endoscopic Diagnostic Procedures

- **Upper GI endoscopy (EED) (esophagus-gastro-duodenoscopy)**
 - NPO 6 - 8 hours
 - **Left lateral**
 - Topical anesthesia throat
 - NPO until gag reflex returns
 - Usually includes biopsy
 - Post-sedation and bleeding precautions
- **Bronchoscopy**
 - Same precautions/interventions as upper GI
 - **Bronchospasm** possible (albuterol, O₂)
- **Sigmoidoscopy/Colonoscopy**
 - Diet low fiber, no red liquids
 - Cleansing enemas (tap water) until clear or Polyethylene glycol *Golytely*®
 - Cleansing enema**
 - Sims left lateral, right knee flexed
 - Lubricated tip introduced **3 - 4 inches**
 - Bag 500-1000 ml warm solution
 - Elevate no more than **12 - 18 inches**
 - Lower or clamp if abdominal cramps
 - Administer in **10 - 15 minutes**
 - Hold liquid for a **1 - 2 minutes**
 - No more than 3 cleansing enemas
 - NPO
 - Sim's position (left lateral)
 - Conscious sedation: can't drive
 - Air is insufflated: bloating and cramps
 - Takes 30 – 60' plus recovery time
 - Monitor for bleeding or perforation (peritonitis)
- **Endoscopic retrograde-cholangio- pancreatography (ERCP)**
 - NPO until gag reflex return
 - Assess for return of gag reflex,**

- Iodine contrast administration (ask patient for allergic to iodine, shellfish is not problem for iodine) (If patient is taking metformin, needs STOP med day before of the test and reinitiate 2 days after test)

Monitor for pancreatitis, perforation

- **Trans-parietal-hepatic Cholangiography**

Liver puncture with needle

Iodine contrast injection

Risk for bleeding

- **Cystography**

- Void before
- Urinary catheter
- Iodine contrast local
- X ray
- Void again

- **Pyelography**

- IV catheter
- Iodine contrast IV (ask patient for allergic to iodine, shellfish is not problem for iodine) (If patient is taking metformin, needs STOP med day before of the test and reinitiate 2 days after test)
- X ray

- **Radioactive tests**

- **Nuclear test with isotopes:**
 - PET Scan** (radioactive glucose)
 - Nuclear stress tests (MUGA scan)
 - Bone scan**
 - Thyroid scan**
- Nursing care
 - Rule out pregnancy
 - Iodine is not used
 - IV injection
 - Fasting 6 hours, clear liquids allowed
 - No pain, no sedation, no rest, no bleeding
 - Force diuresis (liquids after tests)

- **Another test**

- Test using iodine and x-rays:
 - Angiographies** (arterial puncture)
 - CT scans**
 - IV pyelography
 - Cystography
 - Hysterosalpingography
- Nursing care:
 - Evaluate creatinine
 - Iodine allergy
 - Eliminate metformin
 - Fasting 6 hours, clear liquids
 - IV catheter

Force diuresis (liquids)
Care of arterial puncture

○ **MRI:**

No iodine (gadolinium), no x-rays
Report metallic objects/patches
Assess for claustrophobia

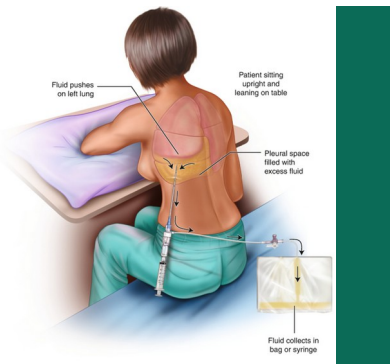
Paracentesis (Sterile procedure)

- Purpose: It is a palliative procedure to improve respiration (SOB). Effective when pt breath better
- Common in patient with fibrosis
- Abdominal girth and weight
- Have client void
- High-Fowler if dyspnea (high as tolerated)
- Post procedure monitor for:
 - Hypovolemia: Blood pressure, urinary output
 - Mental status changes
 - Check for leakage: is common, lateral opposite site
 - Monitor hypotension



Thoracentesis

- Bedside or ambulatory
- Local anesthesia (pressure)
- Position: upright resting arms on overbed table
- Stay still but can breathe normally
- X-rays before and after
- Rest on non-affected side with elevation of HOB 30°



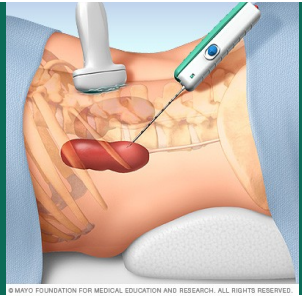
Liver Biopsy

- Pre procedure:
 - Position: supine with hand below neck
 - Coagulation tests
 - Teaching: Hold expiration while the needle is rapidly inserted and removed
- Post procedure:
 - Bed rest for 3-5 hours on right side (prevent bleeding)
 - Vital signs every 15 minutes
 - Dressing on site



Renal Biopsy

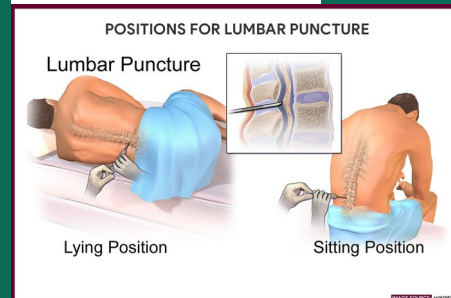
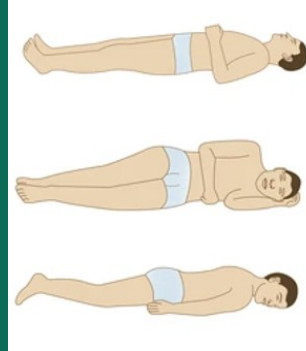
- Baseline coagulation
- Prone position during procedure
- Supine with support on biopsy site
- Bed rest for 8 hours
- Monitor BP and HR
- Mild hematuria is expected



Lumbar puncture

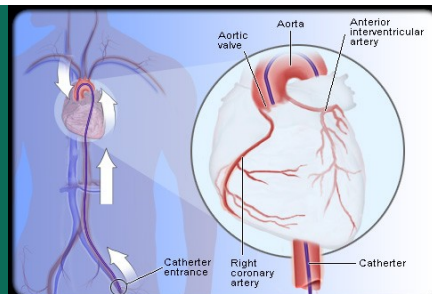
- ASAP before antibiotics
- No need for NPO
- Empty bladder
- Contraindicated in ICP
- **NO: ASA, clopidogrel, anticoagulant**
- Positioning
- Local anesthesia without sedation
- Post procedure:
 - **Flat** resting without pillow
 - Change position with log roll technique: supine, lateral, prone
- CSF in bacterial meningitis:
 - Increased neutrophils
 - Low glucose

POSITIONS POST PROCEDURE:

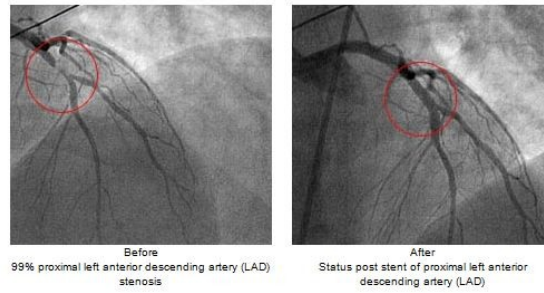


Catheterization for Angiographies

- Iodine: allergy
- Renal function
- Metformin (biguanide)
- NPO
- Most common femoral, made through femoral artery
- Post procedure interventions:
 - Monitor every 15 minutes:
 - Distal pulses
 - Signs of bleeding or hematoma
 - Nerve compression (numbness)



- Avoid hip flexion: extremity flat, Rest extremity flat, cannot sit for 4-6 hours (only 1 pillow below head)
- Elevate HOB no more than 30 °
- Encourage fluid intake
- Check below patient for bleeding



Tuberculin Skin Test, PPD, Mantoux

- Screening of TB
- Read induration at 48-72 hours
 - ≥ 15 mm: positive in any person (even without risk)
 - ≥ 10 mm: positive in any person at risk (who?)
 - ≥ 5 mm: positive if immunosuppressed or recent close contact with active TB
- Interpretation of PPD:
 - Exposition and **infection with or without active disease**
 - Positive test requires chest X ray and eventual sputum
 - Positive PPD **with normal X ray: latent disease**
 - Positive PPD, **symptoms + abnormal X-ray + sputum AFB: active disease**
 - BCG vaccine can make the test positive
 - A **negative PPD** can't rule out active disease

Groups at High Risk for Developing TB Disease

In general, people at high risk for developing TB disease once infected with *M. tuberculosis* include:

- People living with HIV
- Children younger than 5 years of age
- People recently infected with *M. tuberculosis* (within the last 2 years)
- People with a history of untreated or inadequately treated TB disease
- Persons who are receiving immunosuppressive therapy such as tumor necrosis factor-alpha (TNF) antagonists, systemic corticosteroids equivalent to/greater than 15 mg of prednisone per day, or immunosuppressive drug therapy following organ transplantation
- Persons with silicosis; chronic renal failure; leukemia; or cancer of the head, neck, or lung
- Persons with diabetes mellitus
- Persons who have had a gastrectomy or jejunioileal bypass
- Persons with low body weight (<90% of ideal body weight)
- People who use substances (such as injection drug use)
- Populations defined locally as having an increased incidence of disease due to *M. tuberculosis*, including medically underserved and low-income populations

Pre-Operative care

- **Assessment:**
 - Allergies: **latex**
 - History of allergy to tropical fruits: banana, avocado, kiwi
 - Previous multiple surgeries
 - Spina bifida
 - Personal or family history of **malignant hyperthermia**
- Informed consent: prior to pre-medication
- Teaching:
 - Incentive spirometry: 6-10 times every hour
 - Deep breath (pursed lips) and cough

- Surgical wound splinting
- Pain management: **PCA** (patient-controlled analgesia)

Tolerance: med is not effective to calm pain (requires more dose). Addiction: craving, no pain, before the medication is due

Surgeon: informed consent, alteration of coagulation tests,

Anesthesiologist: anesthesia reactions, allergies, muscle relaxants (aminoglycosides), SNC depressants,

- **Blood Pressure** medications usually are not held
- **ASA, clopidogrel and NSAIDs** withheld **7-10 days**. **Suspend before Sx**
- Glucocorticoids doses should be increased.
- **Oral hypoglycemic and NPH or gargline** are withheld (**No DM med before Sx**)
 - Order for regular insulin and sliding scale
 - **Frequent accu-check during the Sx**
- Remove dentures and bridgework
- **Keep eyeglasses (but not contact lenses) and hearing aids**
- Pre-medications:
 - Sedatives: **SIDE RAILS UP!**
 - Atropine **before Sx**
 - Prophylactic antibiotics: one hour before **Sx**
- Padding to prevent pressure ulcers developed during surgery

DVT Prevention

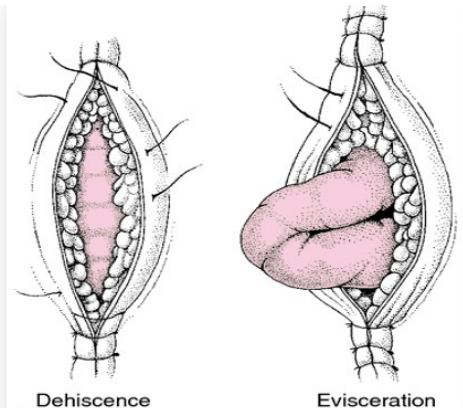
TED stocking	SCD
• Measure correct size • Apply before getting out of bed or make client rest for 30 minutes • Can prepare skin with prescribed lotion or talc • Roll and turn stocking inside out • Place on foot, the opening toward the sole, and pull up • Smooth to prevent wrinkles • Avoid the top rolling down • Assess for circulation and ability to move toes • Check skin at least once daily	• While on bed or sitting • 1-2 finger space between cuff and skin • Remove TID for skin check • Contraindicated in DVT

Perioperative Complications

- **Malignant hyperthermia:** Rigidity, muscles stiffness, fever. Dantrolene
- Hypoxemia, **atelectasis**, pneumonia: incentive spirometer
- Hemorrhage: check under the client
- Urinary retention (wait to void in 6-8 hours)
- Thrombophlebitis and PE:
 - Early ambulation, TED, SCD, LMW heparin
 - **No knee-gatch, no pillow under popliteal region**
- Paralytic ileus:
 - NPO until BS return (pass gas)
- Wound infection (late: 3 - 6 days)
- **Wound dehiscence and Wound evisceration:**
 - First: Resting in low Fowler and flexed knees, antiemetic or antitussive.

- o Cover wound with wet (NS) sterile dressing.
- o Prepare for OR

AVOID LIFTING MORE THAN 10 POUNDS FOR 6-8 WEEKS



Obesity: main risk factor
Other: Valsalva (cough)
First: lay down flexed knees
Second: cover wound



Dehiscence: separation of the edges of an incision

Evisceration: emergence of internal organs through the wound opening

STEPS:

1. Stay calm and stay with the patient
2. Place the patient in Semi-Fowler position with knees slightly flexed (reduce tension on the abdomen muscle and incision)
3. Check vital signs, especially BP and pulse
4. Ask the charge nurse to call the surgeon
5. Ask the UAP to get sterile supplies
6. Cover the intestines with sterile moistened gauze (keep wound moist and prevent)
7. Prepare the patient for Sx as ordered
- 8.

Class 7. ABG.

Sodium Alterations

Hyponatremia

- o Causes:
 - o Diuretics
 - o Water intoxication

Hypernatremia

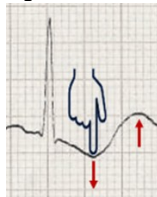
- o Causes:
 - o Diabetes Insipidus,
 - o Hypertonic solutions

- **IV hypotonic solutions,**
- **Irrigation** (enemas, bladder irrigation, gastric lavage)
- SIADH
- Assessment
 - **Critical: 120 mEq/dL** (produce neuron damage)
 - **Brain edema:** AMS, seizures
 - Muscle weakness, hyporeflexia
 - **Hyperactive peristalsis: cramps**
 - Na⁺ replacement:
 - NaCl 3%
 - **Water restriction**
- Lithium toxicity when Na⁺ is low

Potassium Alterations

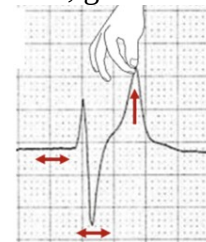
Hypokalemia

- Causes: **diuretics**, NG suction, vomits, diarrhea, glucocorticoids
- Assessment:
 - **Hypotonic** muscles: shallow respirations
 - **Muscle cramps**
 - **Hypo dynamic** peristalsis: **constipation**
 - **Bradycardia**
 - EKG: **U wave**, depressed ST
 - **Potential for digoxin toxicity**
- Nursing Interventions:
 - Assess **renal function** before K⁺ replacement
 - Monitor **cardiac rhythm**
 - IV K⁺ infusion rate <10 mEq/h
 - **High potassium diet**
- Ex: digoxin, and prednisone or furosemide



Hyperkalemia

- Causes:
 - **Renal failure**
 - K⁺ sparing diuretics
 - **ACE inhibitors**
 - Shift of intracellular K⁺ (burns, trauma)
- Assessment
 - **Cardiac** dysrhythmia
 - ECG: **Tall peaked T waves**
 - **Hyperreflexia**
- Nursing Interventions:
 - Monitor cardiac rhythm
 - **Ca⁺ gluconate**
 - **Dextrose + insulin**
 - **Sodium polystyrene** (Kayexalate)
 - Dialysis
 - Low potassium diet: apple, pears, peaches, berries, green beans



Calcium Alterations

Hypocalcemia

- Causes:
 - Hypoparathyroidism (thyroid surgery),

Hypercalcemia

- Causes:
 - **Hyperparathyroidism,**

- **Pancreatitis**
- Assessment (*hypoCa⁺ = everything up*)
 - **Paresthesia**: circumoral numbness
 - **Muscle spasms**: “tetany”
 - **Trousseau's** and **Chvostek's**
 - **Laryngeal spasm** and stridor
 - **Irritability SNC**: Seizures
 - **Increased bowel sounds (diarrhea)**
 - **Increased QT**: Cardiac dysrhythmias
- Nursing Interventions
 - Calcium gluconate or chloride
 - Cardiac monitor
 - Seizure precautions
 - Calcium rich diet
- **Metastatic cancer**
- Vitamin D intoxication
- Assessment (*hyperCa⁺ = everything down*)
 - **Constipation** and abdominal pain
 - **Dehydration** (polyuria)
 - **Renal failure**
 - Kidney stones
- Nursing Interventions
 - **Hydration**
 - Furosemide
 - **Calcitonin**

Magnesium Alterations 1.5 to 2.5 mg/dL

Hypomagnesemia

Causes:

- Malnutrition, **alcoholism**
- **Assessment**: (same as hypocalcemia)
- Paresthesia
- **Positive Trousseau's and Chvostek's**
- **Seizures**
- Prolonged QT
 - **Torsade de Points**

Nursing Interventions

Seizure precautions

Cardiac monitorization

Mg supplement,

Mg found in legumes, whole grains, nuts



• Hypermagnesemia

Causes:

- **CKD**
- **Mg laxatives or antacids**,
- MgSO₄ in eclampsia

• Assessment:

- **Hyporeflexia**: DTR < 2+
- **Respiratory depression**
- AMS and coma

• Nursing Interventions

- Calcium gluconate (antidote)
- Dialysis
- Avoid antacids containing magnesium

ARTERIAL BLOOD GASES

Acid-Base Balance

Normal Range

7.35.....7.4.....7.45

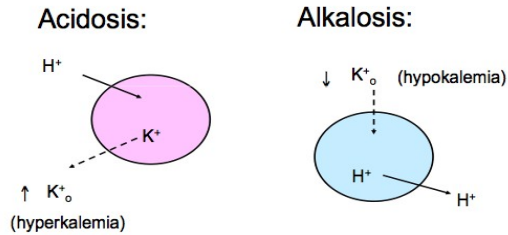
← “acidosis” “alkalosis” →

- **CO₂ is acid and respiratory**
- **HCO₃ is alkaline and metabolic**

- Lungs control CO_2 in minutes:
 - o Hyperventilation: $\downarrow \text{CO}_2$
 - o Hypoventilation: $\uparrow \text{CO}_2$
- Kidneys slowly control HCO_3
- Normal Values:
 - o pH: 7.35 – 7.45
 - o pCO_2 : 35 – 45 mmHg
 - o HCO_3 : 22 – 26 mEq/L

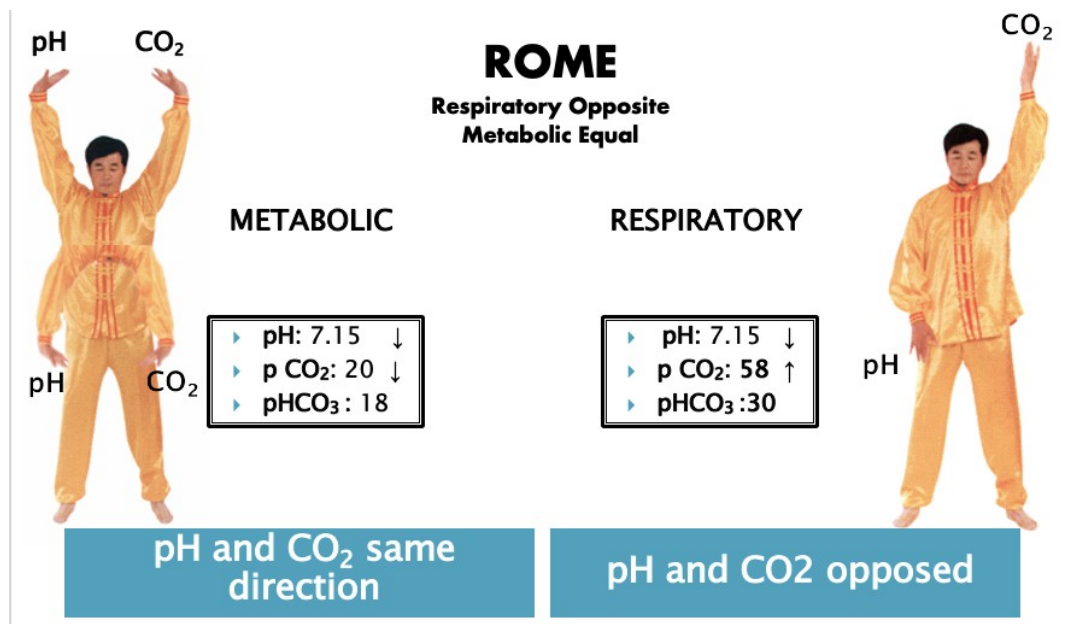
Potassium and pH:

Acid-base Balance And Internal K^+ Exchange



Relation of pH and CO_2

ROME
Respiratory Opposite
Metabolic Equal



ROME:

- RESPIRATORY
- OPPOSE
- METABOLIC
- EQUAL.

THE RELATION IS PH AND CO_2 , BUT ALSO PH AND HCO_3 , AT LEAST IN METABOLIC.

ACID BASE BALANCE

1. See pH: Is it an acidosis or alkalosis?

- pH: 7.35 – 7.45
Normal Range
7.35.....7.4.....7.45
"acidosis" "alkalosis"

2. Is it metabolic or respiratory?

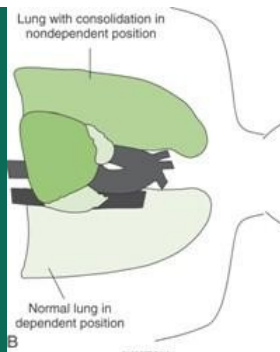
- Use ROME
- Acidosis = Too much CO_2
Too little HCO_3
- Alkalosis = Too little CO_2
Too much HCO_3
- Metabolic Acidosis:
 - Diabetic ketoacidosis
 - Renal failure
 - Lactic acidosis
 - Diarrhea
 - *MAC KUSSMAUL*: hyperventilation to compensate metabolic acidosis
- Respiratory Acidosis:
 - Hypoventilation
 - Chronic: COPD
 - Acute:
 - Opioid/CNS depressors overdose
 - Airway obstruction
 - Guillain Barre, myasthenic crisis
- Metabolic Alkalosis:
 - Vomits/NG suction
 - Antacids/Bicarb abuse
- Respiratory Alkalosis:
 - Hyperventilation:
 - Anxiety/panic attack
 - Mechanical ventilation induced

Class 8. RESPIRATORY DISORDERS

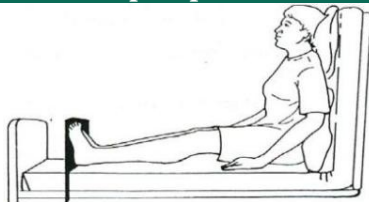
Positioning



Tripoid position

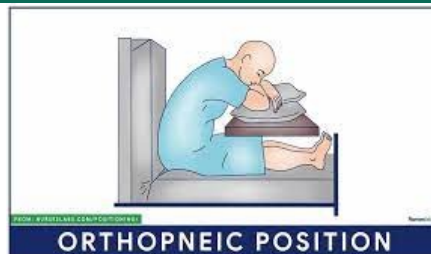


Good Lung Down:
favors perfusion to oxygenated lung (Pleural effusion)

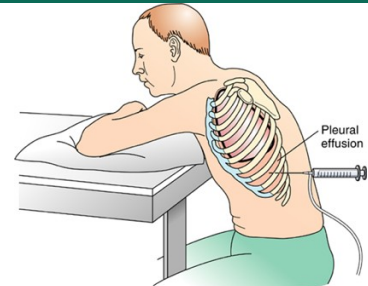


High Fowler

When the patient has very bad breathing



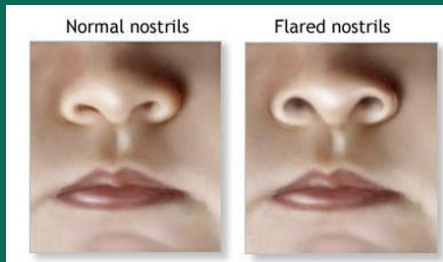
ORTHOPNEIC POSITION



Thoracentesis
Breathe normally
X-rays before/after

Infant respiratory distress: PREMATURITY

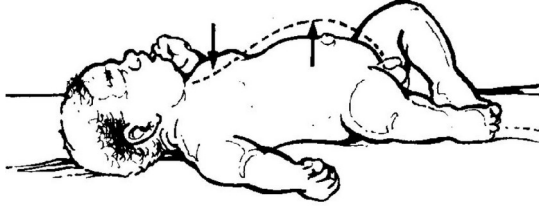
Nasal Flaring



Subcostal and intercostal retractions



Seesaw paradoxical respirations



Epiglottitis

Tripod positioning,

S/S: **Drizzling**, Stridor, “Sniffing” position

Absence of cough

Caused by Hemophile influenza B infection: Droplet precautions

Interventions

Auscultation and X-rays only

Don't visualize pharynx, culture or oral temperature (Nothing by mouth)

Do not force child to lie down, **allow presence of parents**

Intubation/tracheostomy available **at bed side**

Do not put the patient lie-down

Croup = laryngo-tracheo-bronchitis (include epiglottitis)

- Viral (parainfluenza, RSV (produce bronchiolitis), adenovirus: contact and droplets)
- **Barking cough**, stridor, wheezing
- Avoid cry (allow parents bounding the baby for exam)
- **Cool mist prevents the laryngospasm (humidifier, put the child front A/C unit)**

Bronchiolitis

- RSV (very contagious)
- Prevalent first 2 years of life
- Wheezes
- **Contact precautions:** Cohorting is frequent
- Management:
 - Cool humid O2,
 - Bronchodilators

- Ribavirin (nebulization)
- Palivizumab (Monoclonal antibody, administer monthly dose until the baby gain weight. It is used only for prevent RSV and is only administered in child with high risk, is expensive)
 - Prophylaxis in high-risk infants (means premature child and cystic fibrosis)

Asthma

- Allergic-inflammatory disease: Increased **E**osinophils and Ig-**E**
- Status asthmaticus (when the patient doesn't respond to medications)
 - Abolition of wheezes (patient can die, means wheezing are not listening)**
 - Lethargy**
 - Increased pCO₂**
- Medications
 - Rescue for attacks: albuterol and ipratropium** (prolonged use of nebulizer can cause Hypovolemia because has excessive amount of water vapor through the lung tissue)
 - Side effects: **Tachycardia**, tremor and **hyperglycemia**
 - Long term management:
 - Corticosteroids:** fluticasone, budesonide (after use wash your mouth)
 - Thrush.
 - Theophylline**
 - Toxicity: SNC: seizures. Cardiac arrhythmias. Nausea and vomits
 - Interaction with **caffeine**
 - Blood levels (10-20 mcg/dL)
 - Other prophylactics of asthma: **never for asthma attack**
 - Montelukast, zafirlukast (take with **empty stomach**)
 - Cromoglycate (exercise induced asthma)
 - Avoid ASA and NSAIDs
 - Environmental control: **dust mite**
 - Contraindicated with Cimetidine and Grapefruit

Dust Mite

- Change bed cloth weekly
- Wash and dry very hot

Peak flow meter:

Green >80%: *OK*

Yellow 50-80%: *rescue inhaler*

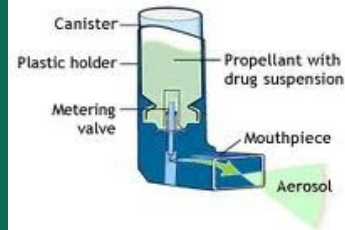
Red: < 50%: *rescue inhaler + 911*

USE OF PEAK FLOW METER

1. Move the marker to the bottom of the numbered scale
2. Stand up if you're able.
3. Take a deep breath, filling your lungs completely.
4. Place your lips tightly around the mouthpiece. Blow as hard and as fast as you can.
5. Note the final position of the marker.
6. Blow into the peak flow meter two more times.
7. Record the highest reading of the three blows.

Metered Dose Inhalers

- Shake the canister
- Prime if used for first time
- Sit or stand erect.
- Breath out
- Place mouthpiece 1inch away with open mouth
- Breath in as deep as possible, at the same time the MDI is triggered
- Hold respiration for 10 seconds
- Breath out through pursed lips
- Wait 1-5 minutes for a second puff
- Use Bronchodilator 1-3 minutes before steroid
- Rinse the mouth and throat if a steroid is used



Number of puffs to prime is variable. Follow the instructions in the insert, 1 to 4 puffs. Do it again if the mdi has not been used in 2 weeks.

COPD: susceptible to respiratory infections

Position: **Upright, tripod.**

Patients with COPD always have the CO₂ high

Emphysema	Chronic Bronchitis
• Pink puffer • Underweight • Smoking or α1anti-trypsin • Alveoli destruction • $\uparrow$ Residual Volume ○ Barrel chest • Expiratory dyspnea ○ Wheezing • $\downarrow$ pO₂, $\uparrow$ CO₂ later stage • Polycythemia, clubbing • Cor Pulmonale	• Blue bloater • Obese, edema • Smoking • Bronchi inflammation • Chronic cough (3 mo/2y), ○ $\uparrow$ Production of mucus • Expiratory dyspnea ○ Wheezing • $\downarrow$ pO₂, severe $\uparrow$ CO₂ • Polycythemia, clubbing • Cor Pulmonale

Cor Pulmonale: right side heart damage produced by lung. Right heart failure = right cardiac insufficiency

In home do not increase the Oxygen more than 2 L

Question: How is the Hg in COPD? High because is use for compensating (Polycythemia)

COPD: Management

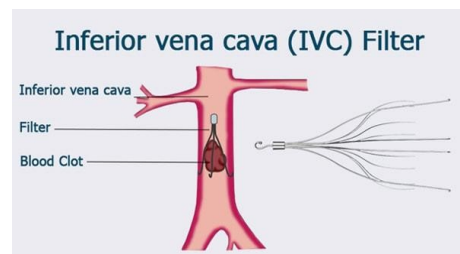
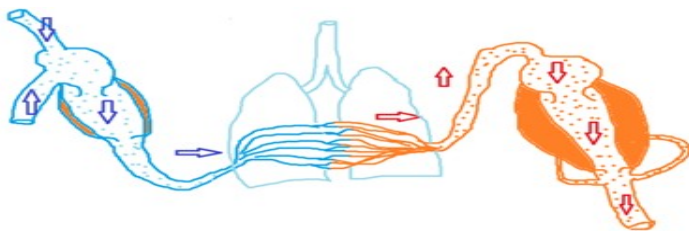
- Smoking cessation
- Pursed lip respiration
- Vaccines:
 - Influenza: annual
 - Pneumococcal: once in life.
- Nutrition:
 - High calorie and protein
 - Low carbohydrates (CO₂)

- Frequent small meals
- Avoid gas producing foods
- Avoid liquids with meals
- Food easy to chew
- Oxygen while eating
- DOC (**Drug of choice**): Anticholinergics
 - Ipratropium
Avoid if peanut allergy
 - Tiotropium
- Beta2-agonists: “**erol**”
 - Albut**erol**, Salmeter**erol**, Formoter**erol**, Terbutaline (parenteral)
- Inhaled glucocorticoids
 - Fluticasone
 - Budesonide
- Theophylline
- Low flow O2

Pursed Lips Respiration

- Breath in as if smelling a flower...
 - Through the nose...
- Breath out as if blowing a candle...
 - Through the mouth...
- Do not puff out cheeks
- Tuck lips to offer resistance exhaling
- Use abdominal muscles to breath out, place hand in abdomen
- Count until four when breathing out
- Benefits:
 - Reduction CO2 in blood
 - Increase O2 in blood

Pulmonary Embolism (clot in lungs).

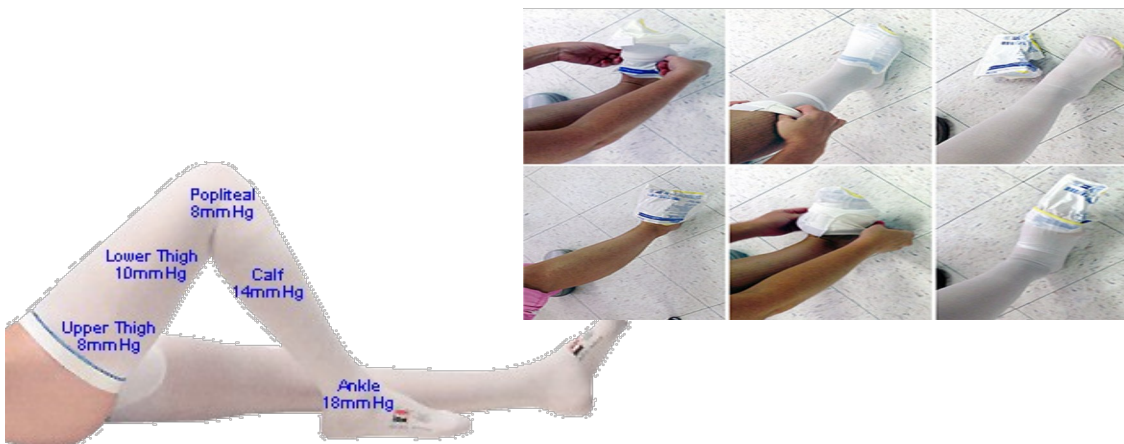


- Risk Factors:
 - DVT: has the potential of developing PE
 - **Immobility**: low extremities surgery, big fibroma, pregnancy & birth, obesity, long travels, hip surgery, trauma veins, estrogen therapy
 - **No No: PILLOWS UNDER KNEES** (popliteal) and **KNEE GATCH** (**elevation of the knee in hospital bed**)
 - Symptoms
 - Sudden: (**Dyspnea**, Collapse for hypotension, Chest pain), **tachypnea**, **hyperventilation**, **NOT vein pulses**, **DVT do NOT affect artery pulses**
 - **If necrosis on alveoli happens, the patient can vomit blood**

- ↓ O₂, ↓ CO₂, ↓ BP
- Diagnostic
 - Spiral CT Angio scan
 - V/Q scan
 - D-dimer
- Treatment:
 - Thrombolytic therapy (used for dissolve clot, med: alteplase)
 - Anticoagulants (fractionated heparin)
 - Vena Cava Filter (only apply for patients that cannot use anticoagulants, Ex: aneurysm)
 - New Tx with medications as (Dabigatran, Apixaban, Rivaroxaban) **Do NOT need check INR**, those med are taken oral. If bleeding occurs, No antidote, nurse can use Factor X (OTC)

DVT Prevention: TED stocking

- **Measure:** Correct size shouldn't harm skin, TED needs to be right size (Not big or small)
- Apply before getting out of bed, use all time only remove for shower
- Turn stocking inside out
- Place on foot, the opening toward the sole, and pull up
- Smooth to prevent wrinkles
- Assess for circulation and ability to move toes
- Check skin at least once daily (UAP can check TED after the nurse put them)



DVT PREVENTION

- Sequential Compression Device (SCD)
 - While on bed or sitting
 - 1-2 finger space between cuff and skin
 - Remove TID for skin check
 - Contraindicated in DVT
- Anticoagulation
- Unfractionated heparin
 - IV or SQ



- aPTT 1.5 - 2.5 control
- Protamine sulfate
- Heparin Induced Thrombocytopenia
 - Heparin induced antibodies
 - Thrombocytopenia
 - Thrombosis
 - Heparin replaced by Lepirudin or argatroban
- Enoxaparin
 - Only SQ
 - SQ technique

- Normal PTT for person without Heparin: around 30
- Normal PTT for person using Heparin ≥ 80 risks for bleeding, decrease heparin or STOP infusion
- If patient is receiving heparin and platelet is $< 150\,000$ call HCP
- LMWH (low molecule weight heparin): prevent clot
- Enoxaparin cannot be administered in IV only SQ
- Heparin antidote: **protamine**
- **(Start Heparin but at the same time administer warfarin and when the patient is stable stop heparin and continue with warfarin only)**

Acute Respiratory Distress Syndrome (ARDS)

- Secondary to sepsis, trauma, shock, Covid-19
- Hypoxemia refractory to 100 % O₂
- Non cardiogenic pulmonary edema
 - Increased permeability A-C membrane
- Management:
 - Artificial ventilation and PEEP
 - Prone position

SIRS (Systemic Inflammatory Response Syndrome) is produced by ARDS: lungs swelling and leakage



Cystic Fibrosis (affect 2 organs: lungs and pancreas)

- **Autosomal recessive:** both parent carriers, 25% chance
- **Meconial ileus** (first manifestation, newborn cannot have bowel movement in 24 hrs, stool is too hard)
- **“Salty child”:** **chloride in sweat** > 60 mEq/L (for stimulation of sweat during the test, the nurse can admin atropine)
- **Lung:**
 - Bronchiectasis (secretions produced by bronchi and the cilia cannot move and made cyst)
 - Recurrent respiratory infections. Antibiotic resistance.
 - Polycythemia**
 - Clubbing** (happens in tetralogy of fallout, COPD, cystic fibrosis)
- **Pancreas:**
 - Exocrine pancreas insufficiency: pancrelipase with meals
 - Malabsorption and **steatorrhea**
 - Malnutrition (ADEK) and growth retardation

◦ Interventions:

Hydration and acetylcysteine to lose bronchial mucus

Postural drainage

Cupping hand for help with secretions on babies (chest physiotherapy)

Do not do it over the skin put a t-shirt for baby

Position: head down, gravity help move secretions to trachea

Do therapy in the morning and before eating

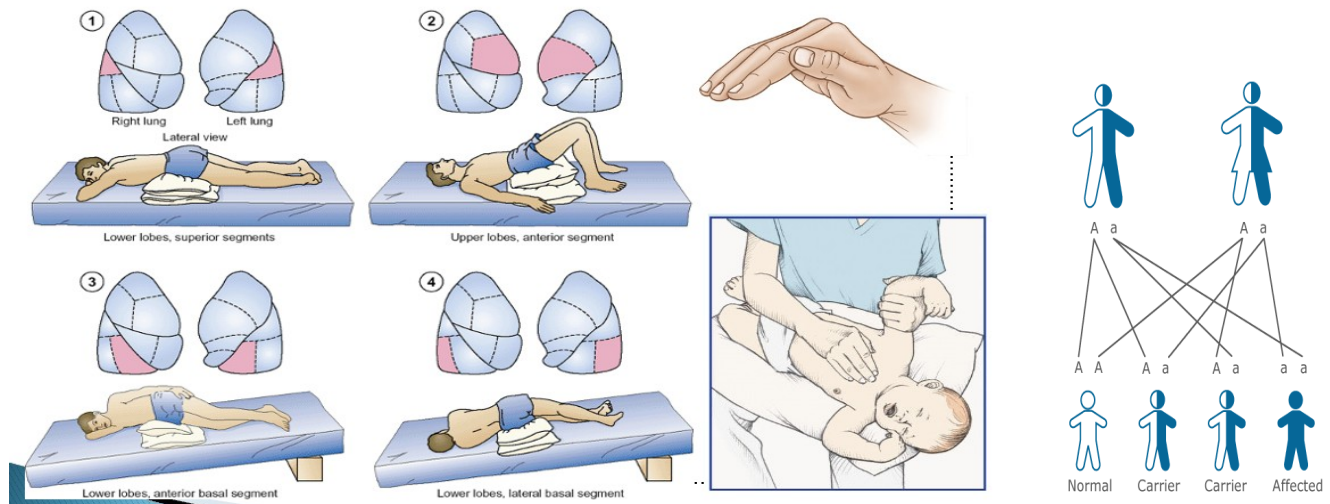
After therapy made oral hygiene

Diet: high calories, high protein, vitamins ADKE, **liberal salt** (salty snacks)

Pancreatic enzymes (**pancrelipase**) **IMMEDIATELY BEFORE** each meal or snack (in babies can mis with the food, capsules)

Verify high fat diet with enzymes, to compensate loss of fats and caloric deficiency: butter, whole dairy, half and half, etc.

Postural Drainage Positions



Pneumonia

- Risk factors:
- Assessment:
 - Cough, sputum (rusty, purulent), fever, chest pain, crackles, pleural effusion
 - Sputum: Gram stain, culture and sensitivity **before first dose antibiotics**
 - Most common is **Streptococcus pneumoniae**: droplets precautions
 - Other:

Droplets Precautions

Streptococcus **group A**

Mycoplasma (walking pneumonia) (no SOB, not fever)

H. influenza B (children) (has vaccine)

Pneumonic plague

Standard Precautions:

Streptococcus pneumoniae

Pneumocystis jirovecii: AIDS, **not contagious**. It is a fungal infection in lungs. Happens when the patient is severe immunosuppressed

Gram negative: Klebsiella pneumoniae

Legionnaire disease: **Contaminated water**.

Airborne and contact precautions

SARS 1or 2 (Coronavirus), MERS

- Treatment
 - Antibiotics, hydration, oxygen therapy
 - **Empyema:** thoracentesis / chest tube

Expected in elderly patients with bacterial pneumonia? Confusion and tachycardia

Risk for Acquiring Pneumonia

- Older adult or infants
- **No vaccination:** influenza and pneumococcal
- Aspiration (*enteral feeding, cough!*) (*Priority patient with TPN coughing frequent, dysphagia*)
- Endotracheal or nasogastric tubes (*risk for aspiration*)
- Mechanical ventilation
- Immunosuppression (*HIV, transplant, chemo...*)
- **Asplenia (not Spleen)**
- Malnutrition
- Chronic health problem: COPD, CHF, CKD (*needs vaccine, do not include asthma*)
- Use of tobacco or alcohol
- Altered level of consciousness

Ventilators

Ventilator Associated Pneumonia (VAP) Bundle	Ventilator Alarms
Prevention:	High pressure alarm: Obstruction:
◦ Hand washing ◦ HOB at 30 degrees ◦ Gastric residual check ◦ Oral care every 4 hours ◦ Chlorhexidine ◦ Endo-Tracheal Cuff pressure check ◦ Sedation holidays ◦ Daily evaluation for extubating readiness	◦ Kinking ◦ Mucus and secretions ◦ Bronchospasm ◦ Dislodgement ◦ Right bronchus intubation ◦ Biting tube
	Low pressure alarm: Leak of air:
	◦ Deflated cuff ◦ Misconnection

Tuberculosis (TB)

M. tuberculosis	TB drugs: RESPI (take 2 or 3 TB drugs together prevent develop of resistant organism)
◦ AFB (acid fast bacilli, confirmation test, technique color to detect TB)	
Mantoux, PPD, tuberculin skin test:	Rifampin
• Positive induration:	Discoloration of urine (Orange color), tears Hepatotoxicity (avoid alcohol)

◦ ≥ 15 mm: person without risk ◦ ≥ 10 mm: person at risk ◦ ≥ 5 mm: $\downarrow$ immunity or contacts ◦ Positive PPD: chest x-ray/sputum <i>Normal x-ray/sputum:</i> latent TB $\rightarrow$ chemoprophylaxis <i>Positive sputum:</i> active disease $\rightarrow$ combined tx. ◦ False positive PPD: Vaccine of Bacilli of Calmette Guerin (BCG) < 5 years ◦ False negative PPD: AIDS, steroid therapy, malnutrition	$\downarrow$ contraceptives (inactive them, risk of pregnancy) - Ethambutol Optic neuritis (blindness, annual eyes exam) - Streptomycin (teratogenic, no in pregnant) Nephrotoxicity & Ototoxicity - Pyrazinamide Arthralgia (increase uric acid) - Isoniazid (INH) Hepatotoxicity, neuropathy, Pyridoxine (B_6) Test: Check for Liver Enzymes
IGRTs (blood test):	Airborne precautions until three sputum come out negative
◦ Interferon Gamma Release Tests	DOT (direct observation treatment)
Symptoms of TB	
Latent: asymptomatic	
Active: low fever, sweating, loss weight, cough	

- 10 mm: Person with risk factor: immigrants (<5 years comers), healthcare workers, residents of nursing homes and jails, IV drug users, chronic disease and children
- 15 mm: Persons without any risk factor.
- All pt with TB should screen HIV and vice versa
- PPD and X-rays only do **NOT** confirm TB, the only one confirm TB is sputum
- TB can happen in meninges, lungs
- When the patient is discharge doesn't need any precautions

TB sputum collection

- The sputum is composed by purulence composed by white blood cells, debris, and mucus.
- Best collection time is in the morning
- **Sputum colors:** Dark green or yellow sputum: indicate respiratory infection

Pneumothorax

- Spontaneous (young male, thin)
- Traumatic
 - "Sucking wound" in inspiration
 - Dyspnea and pain
 - Hyper resonance to percussion
 - Tachypnea, vesicular sound
 - Pt can wait 12-24 hrs by care

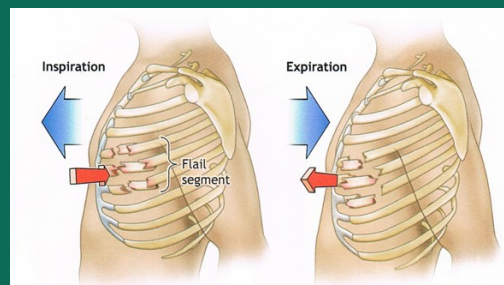


Tension pneumothorax:

- Trachea deviation
- Needle decompression

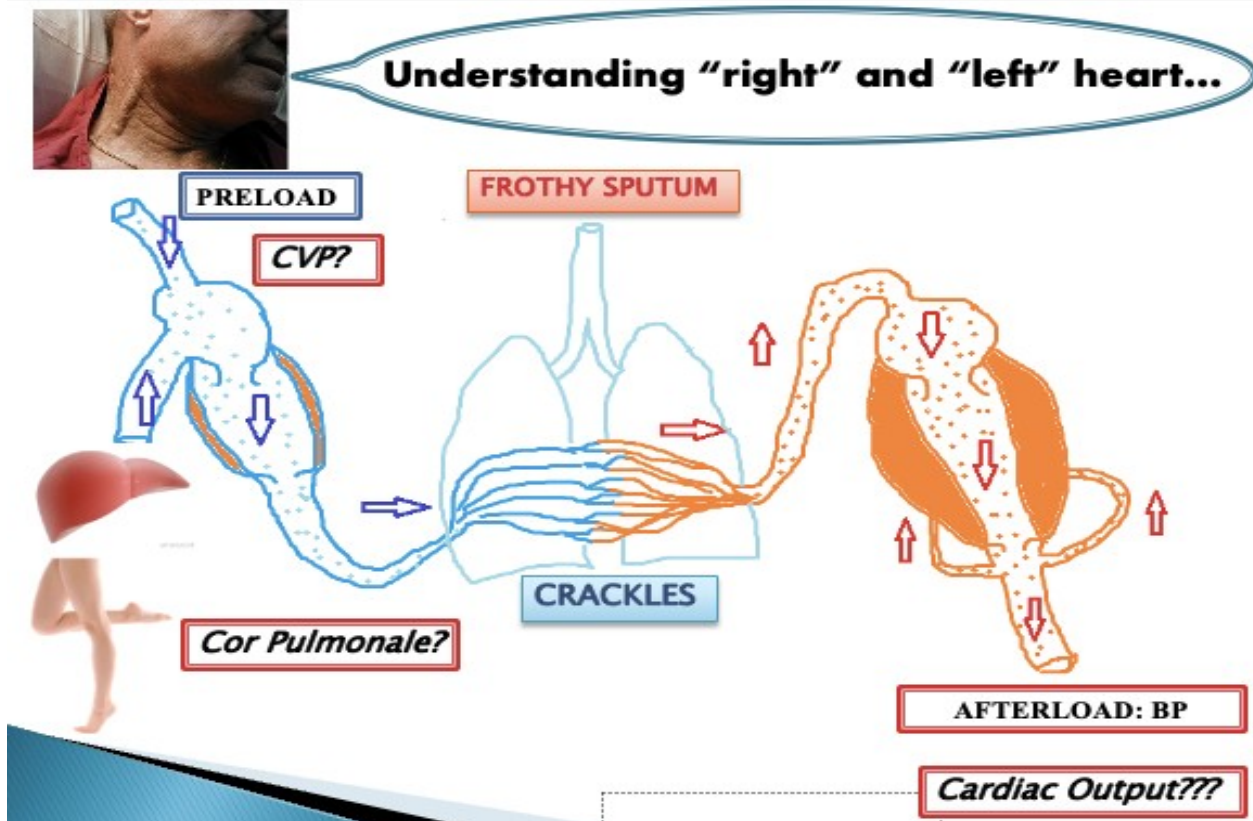
Flail chest **Med Emergency, need intubation

- Loose segment of chest wall
- Paradoxical movement of chest wall
- Ineffective ventilation
- ET intubation and mechanical ventilation



Glucocorticoids:

Ending in “one”: prednisone, triamcinolone, cortisone, hydrocortisone. These medications having risk for infections in patients immunosuppress.

CLASS 9: CARDIOVASCULAR

Cardiac output = stroke volume

Afterload: BP = ACE inhibitors

Cor Pulmonale = right side heart failure, ↑ volemia

CVP: 5 - 10 cm H₂O or 2 - 6 mm Hg, PCWP 6 - 12 mmHg

Heart Failure

RIGHT SIDE HEART FAILURE	LEFT SIDE HEART FAILURE
Dependent edema	Dyspnea, orthopnea, PND*
Hepatomegaly	Pulmonary edema, crackles
Ascites	Cough
Neck Vein Distension (at 45°)	Frothy, blood-tinged sputum
Weight gain	Fatigue, confusion
Anorexia, RUQ pain	S3 gallop
Nocturia	Decreased renal function

*PNP: paroxysmal nocturnal dyspnea

Right side Interventions:

- Check enemas in dependent zones (back, sacrum), elevate HOB and legs down
- Hepatomegaly is too painful
- Position patient head at 45 degrees
- S/S: Anorexia, RUQ pain, fullness, bloating
- Nocturia, orthopnea, hepatojugular reflex

Left side Interventions:

- Breathing in sitting position with pillows
- Crackles listen in the base of the lungs, Rales
- Decrease urinary output, liquid retention
- **BNP >100** = cardiac insufficiency and produce ventricular arrhythmias, elevated in heart failure
- Elevated **CVP** (5-10 cm, 2-6 mm Hg) and pulmonary capillary pressure
- Tx:
 - Raise HOB
 - Fluid restriction
 - Diuretic, ACE inhibitors, low dose Beta blockers, digoxin
 - Low Na⁺ (< 2 g/day), high K⁺ diet
 - Daily weight: 1 kg = 1 L
 - Blurred vision, halos (Digoxin toxicity)
 - Check apical pulse

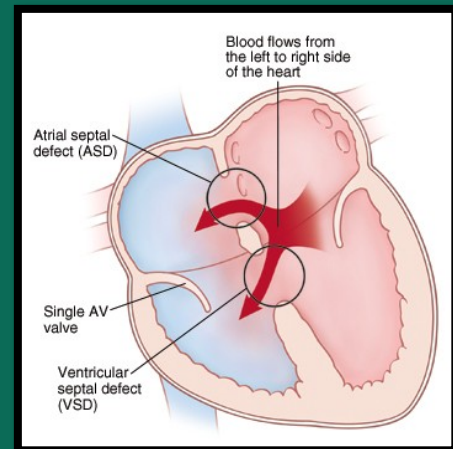
Question: Patient with anemia and low hemoglobin. What happen with cardiac output? **Increase**

PCWP in pulmonary edema > 20 mm Hg diagnostic criteria.

Diuretic given am for diuresis, digoxin pm when lab results available

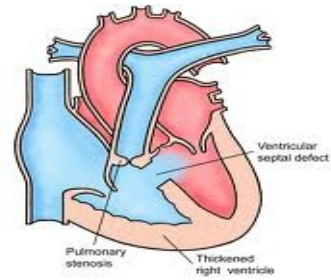
Acyanotic Heart Congenital Malformations

- **Acyanotic heart defects:**
 - Atrial septal defect
 - Ventricular septal defect
 - Persistent ductus arteriosus
- **Murmurs**
- **Fatigue: Crying, feeding, suctioning**
- **Interrupted feeding: Always hungry, expect low weight**
- **Forehead sweating**
- **Growth retardation**
- **Risk for endocarditis**
- **Infant easy to fatigue during feeding, interrupt the feeding and sleep. This baby eats frequently because doesn't feeding well.**



Cyanotic Heart Defect: Tetralogy of Fallot **

- Right to Left shunt
- Cyanotic spells when feeding, crying
 - Children presents squatting (it is a patient intervention to improve cyanosis)
 - Infants: Upright and knee chest position
- Polycythemia (too many RBCs)
- Clubbing (Chronic hypoxia)
- Growth retardation
- Baby born cyanotic and more blue by moments (feeding, crying, ...)

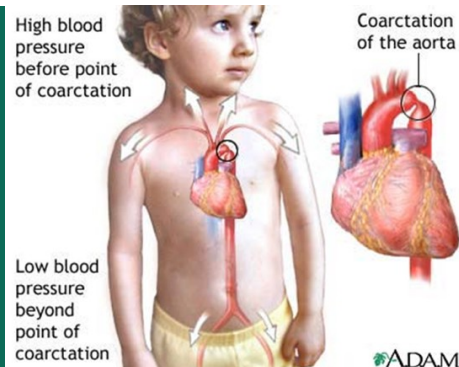


Question: When put oxygen to an infant? *When is crying, agitated, eating.*

Question: What to do when baby is crying? *Give to mom for bounding.*

Coarctation of the Aorta

- Strong pulses upper extremities
- Weak or absent pulses in lower extremities
- Ankle/brachial index < 1
 - Normal patient has ankle/brachial index higher in low extremities than upper extremities, Ex: arms 110 and legs 114, then index > 1
 - Patients' smokers or uncontrolled DM has upper extremities higher than low, then index < 1
- Left heart failure (Blood storage in left lungs)



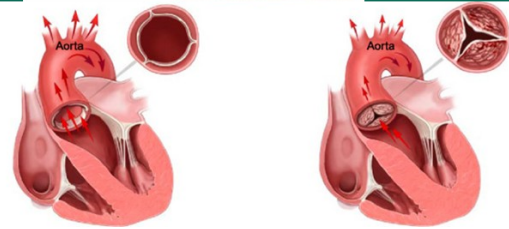
Kawasaki Disease:

- **Acute phase:** Fever, rash, conjunctivitis, strawberry tongue, swollen hands
- **Chronic phase:** coronary aneurisms, arrhythmias
- Treatment:
 - Acetylsalicylic acid (**aspirin**)
 - **IVIG (immunoglobulin): IV**



Rheumatic Fever (autoimmune disease):

- Post strep throat
- Fever
- Polyarthrititis
- Chorea movements
- Carditis
- ↑ Antistreptolysine O (ASO)
- Valvular damage (aorta, mitral)
- Treatment:
 - High doses ASA.
 - Prophylactic antibiotics



Bacterial Endocarditis

- History of
 - Valvular damage
 - IV drug users
- Fever
- Arterial Septic Embolus:
 - Petechiae
 - Purple nodules (Osler)
 - Stroke, brain abscesses
 - Renal, liver infarction
- Changing murmurs
- New onset heart failure
- Positive **blood cultures**
- **Antibiotics:** several weeks (most used Vancomycin)
- Valve replacements (mechanical valves, risk for clotting), Tx: heparin/warfarin for the rest of life)



Pericarditis.

- Etiology: Viral, uremia, post MI, lupus
- Pain: position **upright leaning forward**, sharp chest pain, scratching
- Fever
- **Pericardial rub**
- **Muffled sounds if pericardial effusion**, high pitch sound auscultated in precordium

Cardiac Tamponade (Medical Emergency, Life-threatening)** too much liquid store around the heart

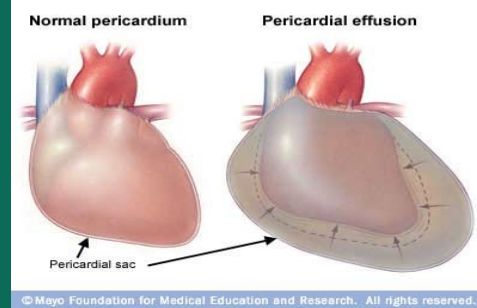
Compression of the heart caused by fluid collecting in the sac surrounding the heart. Do NOT allow the heart to relax and move the blood.

○ Assessment

- Muffled heart sounds (friction sounds between swollen walls of the pericardium)
- Jugular vein distention, pain, fever
- Hypotension, pulsus paradoxus (not audible in inspiration only in expiration)
- Clear lungs
- Pericardia effusion, respiratory distress

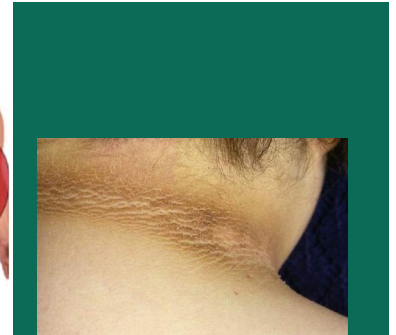
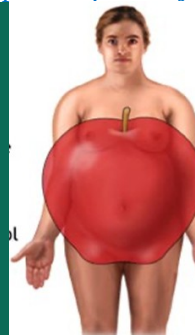
● Interventions

- IV fluids to force blood flow
- Pericardiocentesis (16–18-gauge needle, it is lifesaving intervention)
- Elevate HOB



Risk Factors CAD: Metabolic Syndrome (Syndrome X). Apple shape body form

- Central Obesity (waist > women 35 and men 40)
- High blood pressure, BP > 130/85
- High triglycerides: TG > 150
- Low HDL cholesterol: HDL < 50 women < 40 men
- Insulin resistance: BS > 100 mg/dL
- Pregnant woman + 300 cal/ day, Breastfeeding: + 500 cal/day
- Acanthosis Nigricans, atheroma's tumors (fat tumor)
- Risk factors: DM, HTN, obesity



• Anti-atherogenic Diet:

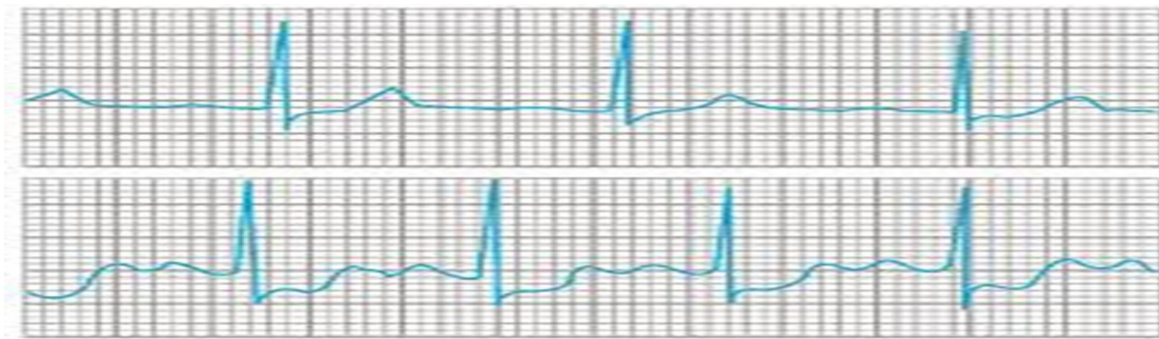
- Low saturated fats (animal)
- Low sugar and alcohol
- Low sodium, high potassium
- High vegetables, fruits, fish, poultry, grains, legumes, nuts, mono-poly unsaturated fats
- HDL-C: **“happy”** cholesterol
- LDL-C: **“lousy”** cholesterol

Coronary Artery Disease

• Risk Factors

- Non-Modifiable:
 - Genetic
 - Gender (male>female)
 - Aging
- Modifiable:
 - Hyperlipidemia: ↑LDL-C and TG, ↓HDL-C
 - Hypertension

- Diabetes mellitus
- Tobacco
- Sedentarism
- **Angina**
 - **Stable:** (flat plaque in the artery w/ischemia and exercises)
 - **Unstable:** (plaque is growing)
 - Variant Angina or Vasospastic (**Prinzmetal**):
 - Onset pain, vasospasm, no atheroma
 - Elevation of ST segment
 - Pain when the patient is resting, No Troponin able
 - Normal cardiac markers (Cardiac catheterization doesn't show coronary obstruction)
 - **Tx:** admin CCB (Calcium channel blockers)
- **Diagnostic studies**
 - **ECG:** reversible ST depression
 - **Stress test**
 - Cardiac enzyme levels **normal**
 - **MUGA Scan** (use radioactive substance)
 1. **Priority question: Ask female patient if she is pregnant?**
- **Treatment:**
 - Nitrates
 - Beta blockers: metoprolol
 - CCB: diltiazem
 - ASA, clopidogrel
 - Statins

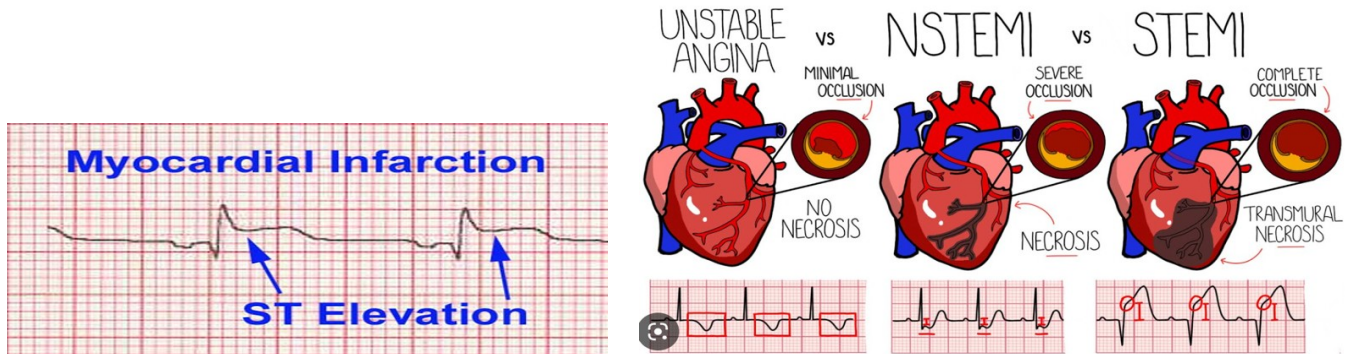


Angina

Myocardial Infarction (MI)

- **Atypical symptoms** in women, **DM**, elder adult
- Cardiac markers elevated
- ST segment elevation, Q wave
- **Increase Troponin levels** (heart cells death)
- **Treatment:**
 - **MONA (Morphine, Oxygen, Nitrates, Aspirin)**
 - Stool softener, bed side commode, **NI: Do not allow patient strain with MI**
 - **Fibrinolytic** (rTPA) = thrombolytic therapy
 - **Window period:** MI is around 6 hrs, stroke around 4 hrs
 - **Reperfusion** dysrhythmia (expect good therapy)
 - **Antiarrhythmic medications:** lidocaine, amiodarone
 - Risk for **hemorrhages**

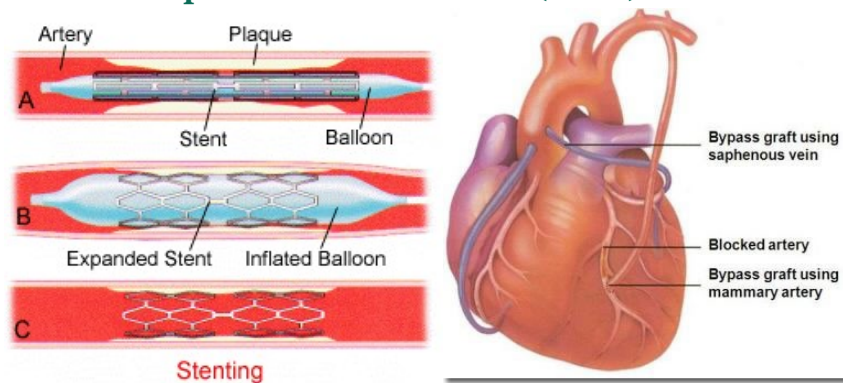
- **Alteplase:** RN needs to know a what time start the infarction. If less than 6 hrs., administer
- Surgical:
 - PTCA & Stent
 - CABG
- **Complications:**
 - **Ventricular dysrhythmias:** PVC & V-Tach & V-Fib
 - **Cardiac tamponade**
- **Rehabilitation:**
 - Sex? Consult HCP, two flights of stairs without pain or dyspnea



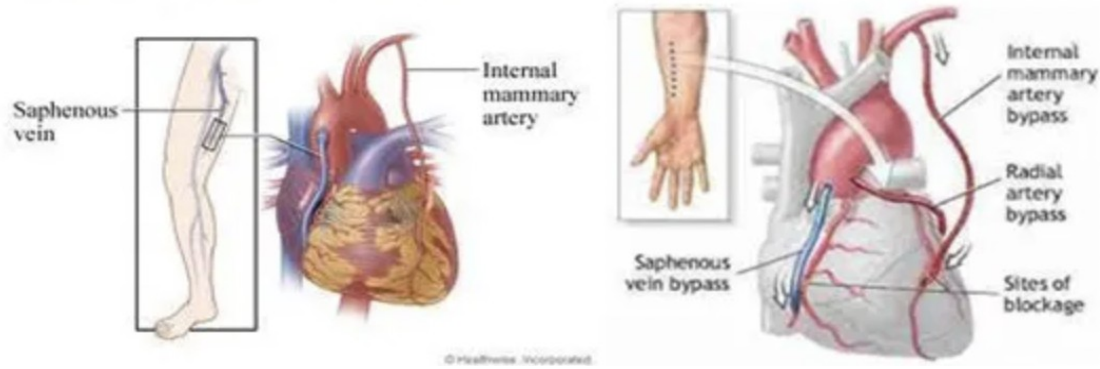
STEMI (means): ST segment, E elevation, MI myocardial infarction (non-stemi the ST segment still low, but in stemi the MI is happen now, so QRS is not completed because S going to T, heart attack is happen)

- **MI Interventions**
 - Morphine if pain doesn't improve with nitrate.
 - Oxygen: If O₂ sat is < 94%.
 - Hold nitrate if SBP < 90 mmHg or taken sildenafil.
 - Atypical symptoms: orthopnea, pain between shoulders, jaw, epigastrium, fatigue.
 - Bleeding precautions: GI, avoid IM and ABGs

Invasive Reperfusion Therapies "Catheterization" (Stent)

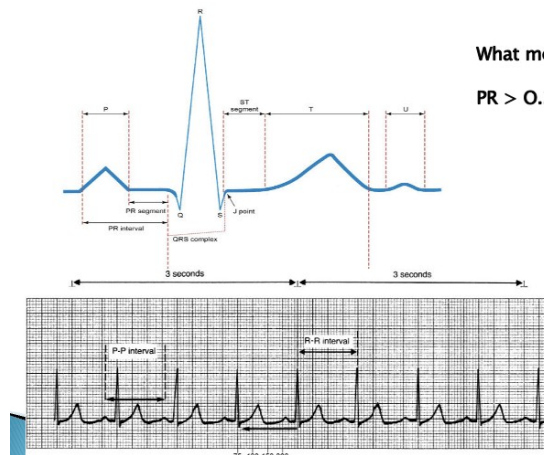


- **CABG (Coronary Artery Bypass Graph):** This procedure is performed when patient do not react with MONA and do NOT respond to fibrinolytic therapy. It is unstable patient.
- **CABG:** this procedure inserts a vein with an artery
- **CABG:** Saphenous Vein
 - Mammary Artery



EKG

NORMAL SINUS RHYTHM



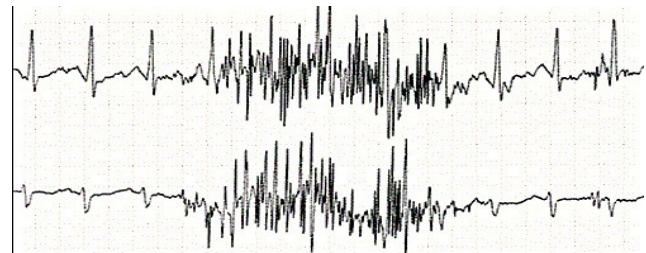
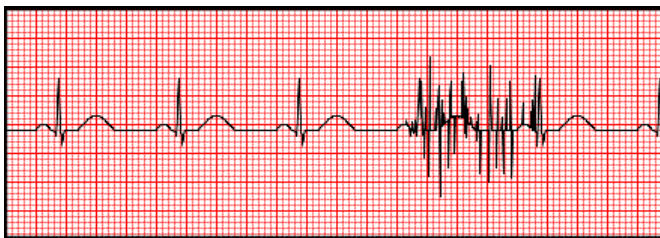
What means?

PR > 0.20 sec

P – atrium
depolarization
QRS – vent

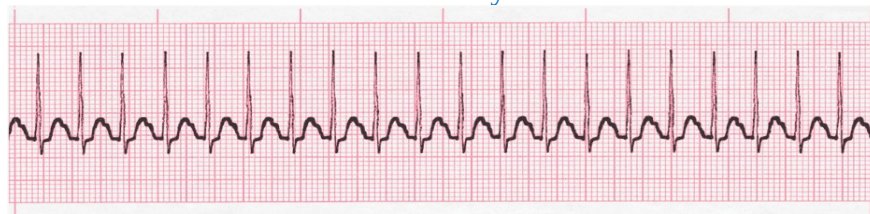
ECG (10 cables), ECG monitor (5 cables), and HOLTER

Normal ECG with Artifacts

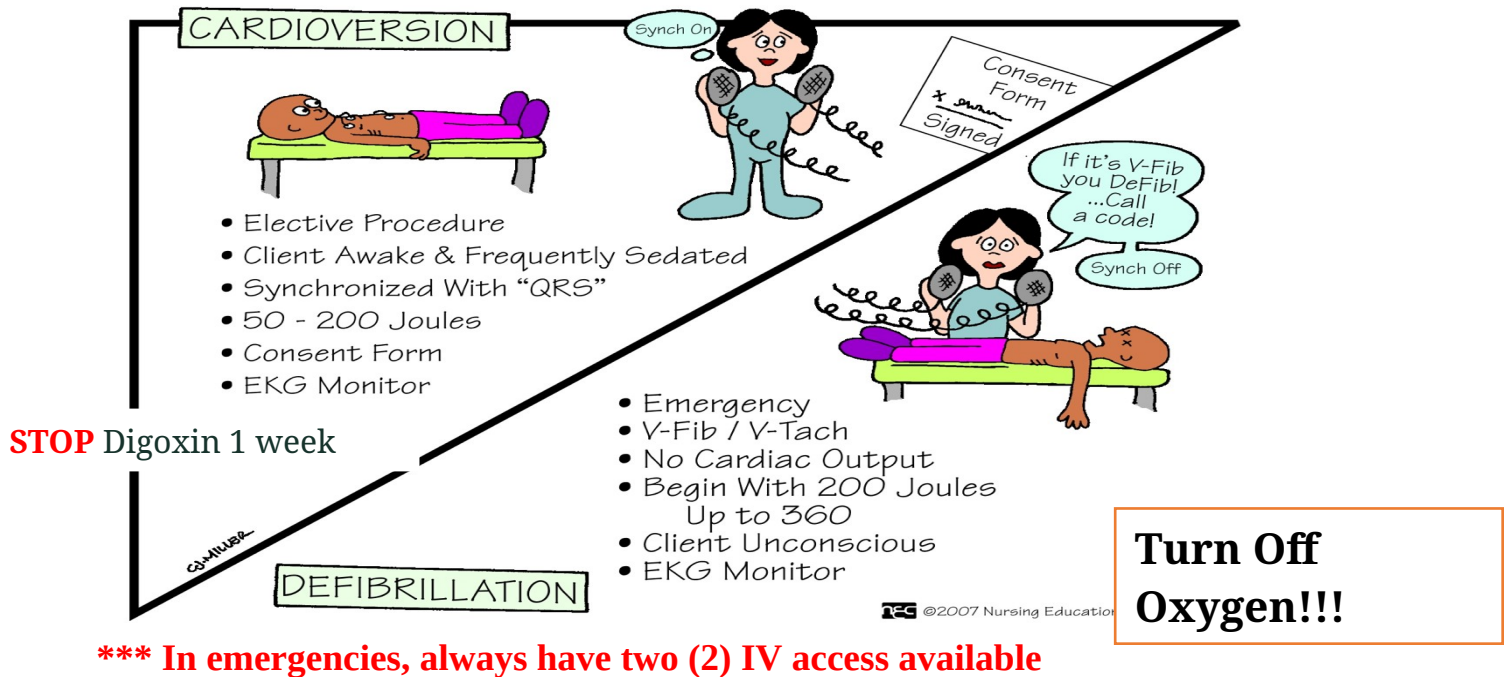


Supra Ventricular Tachycardia (SVT):

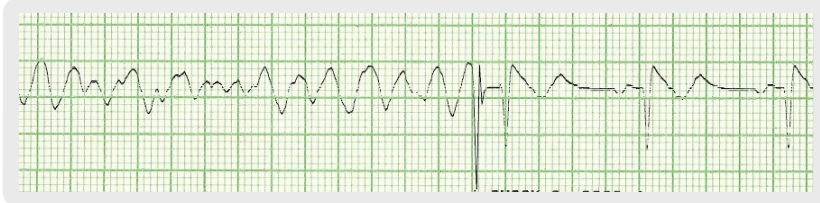
1. Vagal maneuvers (1st intervention): cough, Valsalva
2. If maneuvers don't work admin Adenosine
3. No "P" wave means arrhythmias
4. NI: Administers medications needs to be check by 2 RN



Cardioversion Vs. Defibrillation



*** In emergencies, always have two (2) IV access available

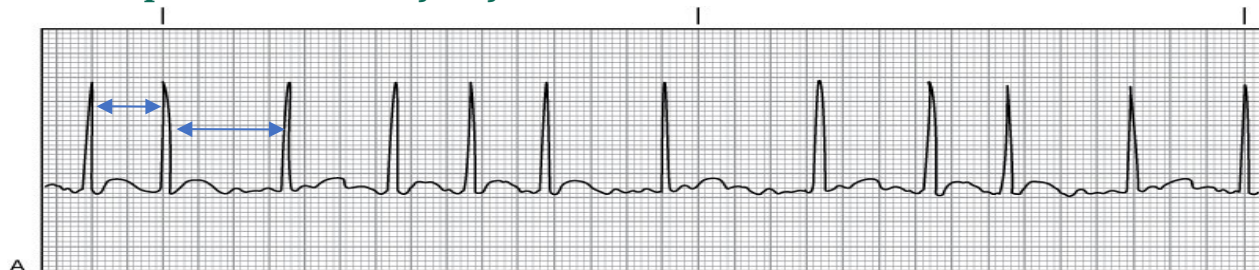


CPR +
Defibrillation

Question: What is the difference between cardioversion and defibrillation?

- cardioversion is synchronized with QRS
 - RN needs to make sure Synch button is **ON**
 - If button is **OFF**, you can kill the patient

Other Supra Ventricular Dysrhythmias



ATRIAL FIBRILLATION (R-R space is irregular)



ATRIAL FLUTTER (R-R space is regular, same distance), called saw tooth (diente de Tiburon) or saw cut (serrucho)

A-Fib and Atrial flutter

- **Chronic Atrial fibrillation/flutter:**
 - **Anticoagulant** (lifelong: warfarin, dabigatran)
 - **Chronotropic negative drugs**
 - CCB (diltiazem or verapamil) or beta blocker
- **New onset or hemodynamically unstable A. fibrillation/flutter:**
 - Anticoagulant, negative TEE (transesophageal echocardiogram) for clots, pt in NPO, sedation, throat
 - **Synchronized cardioversion** (electric) or **amiodarone** (chemical cardioversion), (for return the patient with A-Fib to normal rhythm)



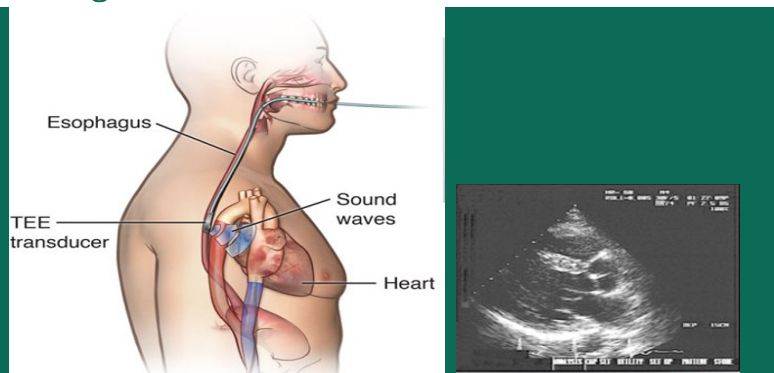
NOTES:

- A-Fib: blood inside the atrium will produce **clots in lung or brain**
 - If the clot is in right atrium, that cause **PE**
 - If the clot is in the left atrium, that cause **stroke**
- Patient with A-Fib/flutter take 2 pills for control of tachycardia
 - Warfarin or dabigatran
 - Diltiazem or beta blockers
- Chronic A-Fib/flutter is long life disease (**not reversible**), considered stable patient
- Acute A-Fib needs take Amiodarone for long-term therapy and produce **pulmonary fibrosis**. S/S: dry cough

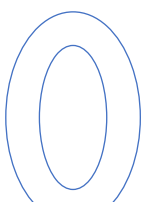
TEE (transesophageal echocardiogram)

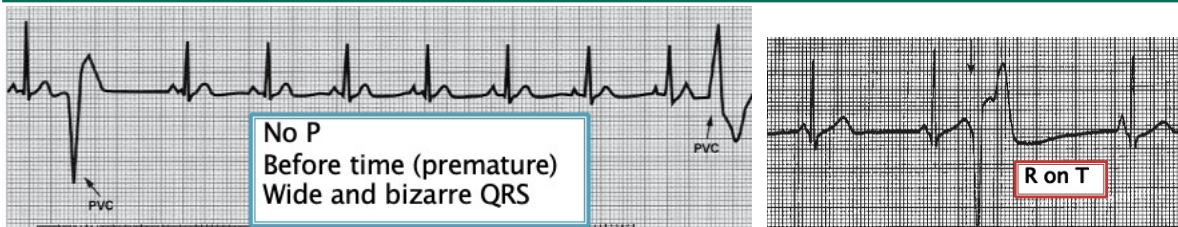
REQUIREMENTS:

- Before elective CV
- NPO
- Throat anesthesia
- Check gag reflex
- Sedation



Premature Ventricular Contractions





- **High Risk: PVC occurs near the T wave of precedent complex (R on T)**
- Interventions: EKG monitor, check **electrolytes**, report to MD
- Lidocaine and **amiodarone for ventricular dysrhythmia**
 - **Amiodarone:** pulmonary fibrosis (dry cough, dyspnea)
- Patients with $\downarrow$ Mg⁺, $\downarrow$ K⁺, $\downarrow$ Ca⁺, ischemia, $\uparrow$ Digoxin
- PVC can precede a V-Tac or V-Fib

Ventricular Tachycardia: CPR and defibrillation, “**Not call 911**”



Pulseless Ventricular Tachycardia: **Call a Code!**

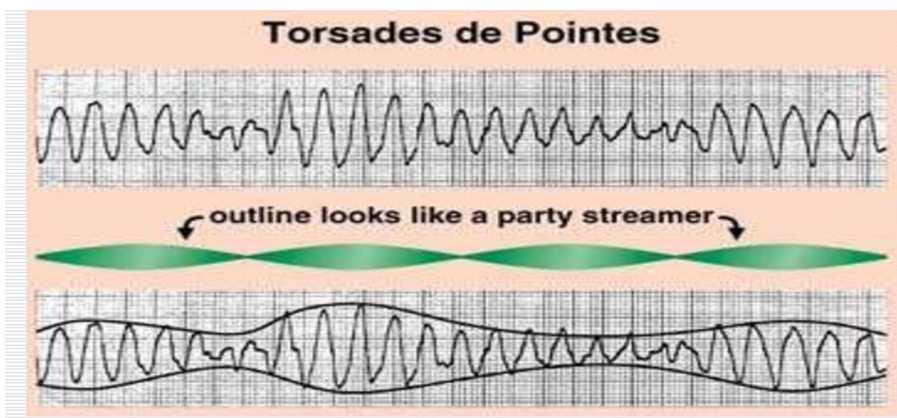
STEPS for CPR:

1. Check if patient is responsive, aware, ask “**Sr, Sr are you O.K?**”
2. Activate blue code
3. Check carotid pulse
4. If no pulse, start CPR

If the patient is gasping or EKG shows V-Tac, do CPR steps

Delay calls for help just 1 second, the RN kill the patient because pt can go to stop heart beat)

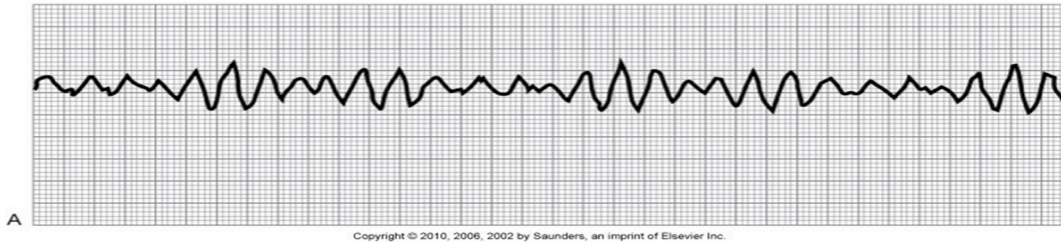
Question: What report for patient with acute MI? Troponin elevated, PVC presence, ST, elevated tachycardia



Ventricular-Fibrillation/Flutter (Call a Code!)

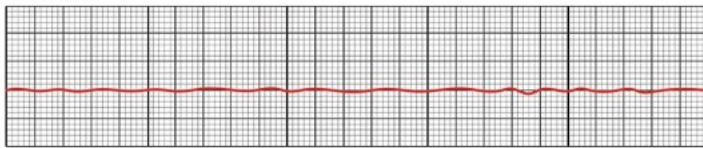
- Cardiac output: zero
- Never pulse
- CPR

- Defibrillation



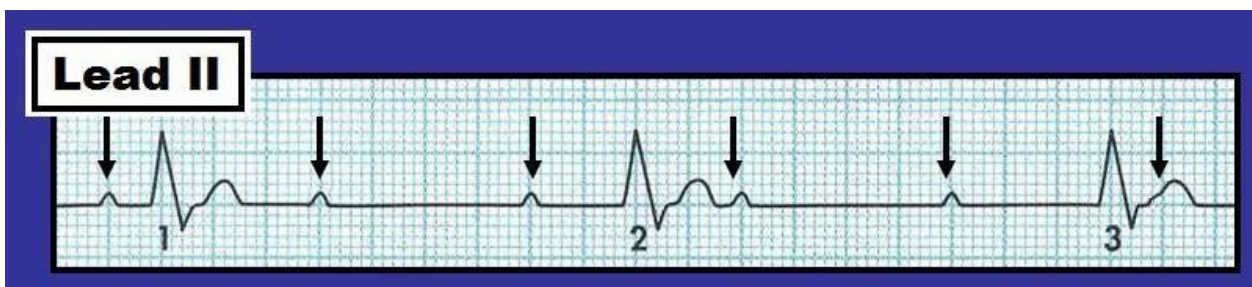
Asystole (Verify if it is Asystole)

- **FIRST:** Determine if it is a true Asystole.
 - Assess client and cables.
 - Move to other lead if on telemetry
 - Real asystole: cardiac output zero, no pulse
- **Code Blue**
- CPR: Chest compressions
 - ****No electrical shock**
- Anticipate: epinephrin & vasopressin IV



AV Block = Syncope (chichones) = Bradycardia

- A-V Block first degree
- A-V Block second degree:
 - Mobitz I
 - Mobitz II
- A-V Third degree



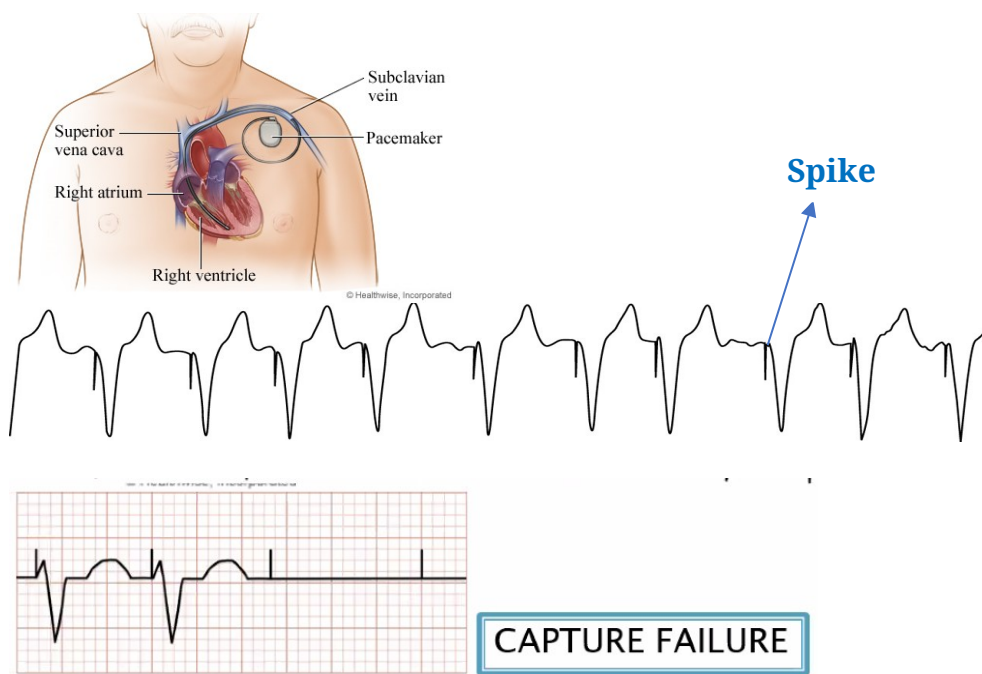
- ☺ **Interventions:** atropine, isoproterenol, pacemaker
- ☹ **Digoxin, Beta Blockers, CCB, lidocaine, amiodarone: NO!!!****
 - **Most important A-V III, HR: 30-40, do not use digoxin, beta blockers, Ca+ channels, they are decreasing BP and the patient is already in bradycardia**

RN put the patient lie-down and do not allow the patient stand up

Question: What means AV BLOCK? Intraventricular delay of conduction (hiss bundle branches)

Pacemaker and ICD (implantable cardioverter-defibrillator) care:

- Microwave or electronics don't affect the device
- Avoid direct hit or magnetic fields on generator (rifles, airport security, **MRI**)
- Avoid raising arm above shoulders (combing, tennis), weightlifting
- Teach radial pulse daily evaluation for malfunction
- Look medical attention if cardioverter/defibrillator is activated
- Monitor for signs of infection
- Teach the patient avoid exposure to electric magnet field (store metal bars outside)
- Do not use the cell phone in the pacemaker side
- **Pacemakers fail capture** happens when the electricity of the contact is not send to the right ventricle, the wire is too close
- Avoid thigh clothes (skin irritations)



CPR Guidelines

- Position client in supine or flat, hard surface
- If no pulse is felt (palpate carotid pulse for 5-10 seconds)
 - If client is not breathing start chest compression
 - Admin ratio: 30 compressions to 2 breaths (1 rescuer)
 - Admin ratio: 15 compressions to 2 breaths (2 rescuers)
 - Check pulse and change position every 2 minutes
- Use AED as soon is available

If the patient return to normal sinus rhythm, the nurse needs to continue the CPR for a couple of more cycle.
- Diagnose:
 - No monitor: Check response with sternal rub
 - Monitor: V Tach, V Fib, Asystole (**real**)
- Quick check CPR status (DNR wrist band)
- Activate code blue and **request AED**
- Check carotid pulse:
- **No pulse: initiate chest compression (CAB)**
 - A compression rate of at least 100/min

Compression depth:
 2 inches (5 cm) in adults and children
 1.5 inches (4 cm) in infants
 Allow complete chest recoil
 Compression-to-ventilation ratio of 30:2 or 15:1

- **Rapid defibrillation with AED ASAP**

ICD have extra precaution: patient needs avoid any hit over the pacemaker's zone because the pacemaker can understand that the patient needs a shock

Peripheral Vascular Disease	Venous Disease	Arterial Disease
History	Varicose veins, DVT	Intermittent claudication, DM, tobacco
Pain relief with legs	elevation	Hanging down
Capillary refill	< 3 seconds	> 3 seconds
ABI: ankle/brachial index (systolic BP)	Normal > 1	Abnormal < 1
Skin	Pigmented (hemosiderin), normal pulses, varicose veins and edema. Edema normally occurs in one leg only	Pale, cool, shiny, thin skin, loss of hair in toes , ↓ pedal, post tibial pulses. Metabolism change to anaerobic and accumulate lactic acid
Ulcer aspect	Irregular borders, beefy granulation, wet exudate	Punched with sharp borders, dry and pale bed

DM affects distal arteries: pedal, posterior tibial. Atherosclerosis affects femoro popliteal arteries

Varicose ulcers are typical in ankles, skin change to dark color, blood stagnation, iron accumulation on ulcer site.

Buerger's disease (Vasculitis)

- Strong tobacco use
- Young adults (<45)
- Endothelial inflammation and thrombosis of arteries and **veins**
- Tobacco abstinence is the only treatment
- **Rehabilitation (walking until limit of pain)**
- **Pentoxifylline (trental)**



Buerger's: HLA B5 and 2A9: genetic predisposition.
 Sympathetic hyperactivity: cool but sweating feet.

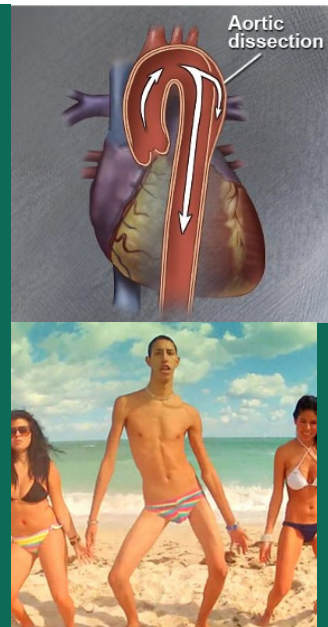
Raynaud's Disease

- **Etiology:**
 - Idiopathic
 - Occupational
 - Hand-arm vibration syndrome
 - Connective tissue disease (Scleroderma)
- **Triggers of vasospasm:**
 - Cold
 - Stress
 - Beta blockers
 - Caffeine and tobacco
- **Treatment:**
 - Avoidance of cold, stress, tobacco
 - CCB: *nifedipine, amlodipine, verapamil, diltiazem*



Aortic Aneurysms, ** Medical Emergency

- Occurs between 1-12 hrs before rupture
- Risk factors: old male, hypertension, tobacco, Marfan syndrome, atherosclerosis
- Pulsatile mass in abdomen
 - Bruit in skin
 - Decreased pulses
- Dissection:
 - Tearing pain lower back (AAA) and groin (aorta is close to the spinal and that is why pain on back)
 - Abdominal girth enlargement
 - Hematoma groin, scrotum and perineum
 - Rupture: tachycardia, hypotension, shock
- Interventions:
 - Maintain BP within normal limits
 - Emergency surgery
 - Post-Surgery care:
 - Monitor LE distal pulses
 - Monitor renal function
 - Check urine output
 - Check creatinine



Some Cardiovascular Drugs

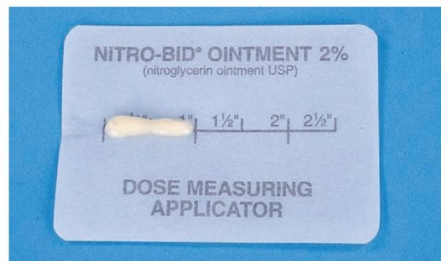
Nitrates

- **Nitroglycerin:** Use for chest pain, decrease BP for short time
 - **STEPS:**
 - Administer nitroglycerin sublingual
 - Call 911
 - Administer 2nd doses
 - Sublingual for emergency treatment of angina
 - Potent vasodilator: reduce oxygen needs
 - HYPOTENSION (use gloves): Monitor blood pressure. **Do not leave patient standing**
 - Orthostatic changes
 - **Lethal interaction with sildenafil, tadalafil, vardenafil**
 - Keep in opaque glass container

- Avoid exposition to warm temperatures
- Expect headaches, [migraines](#)
- Call ambulance if pain doesn't disappear after first dose
- Patches to prevent angina (around the clock, 12 hours off and 12 hrs on)
 - [Do not watch the patch](#)
 - [Rotate the side](#)
 - [Apply in hairless skin, do not shave](#)
 - [Clean are before putting a new one](#)

Nitroglycerin is not indicated or may even be contraindicated if the patient is suffering from:

- Low systolic blood pressure of < 90 mm/Hg
- Right-sided ventricular infarction
- Severe bradycardia of < 50 bpm
- Tachycardia > 100 bpm in the absence of heart failure
- When the patient is taking a phosphodiesterase type 5 inhibitor (a class of medications including Sildenafil) within the past 24 hours, nitroglycerin could cause severe hypotensive side effects. It is important to gather a thorough medication list from the patient to avoid these serious contraindications. Use of tadalafil (known as brand names Cialis) within the past 48 hours



- **Angiotensin Converting Enzyme (ACE) Inhibitors: pril**
- **Angiotensin II Receptor Blockers (ARB): sartan**
 - Indications: HTN, heart failure, diabetic nephropathy
 - Side effects
 - **Angioedema** ("funny voice") [most common in ARBs](#)
 - Persistent dry cough, [call HCP for change med](#)
 - Orthostatic hypotension: dizziness, avoid **hot showers**
 - **Hyperkalemia** (No: salt substitute, renal failure)
 - Teratogenic
 - Only **IV drug** of this group: **enalaprilat**
- **Beta-Adrenergic Blockers**
 - **Beta1 selective:** metoprolol, atenolol, acebutolol, bisoprolol, esmolol
 - **Non-selective:** rest of betablockers: propranolol, nadolol, carvedilol, etc.
 - **Bronchospasm:** Beta 2 block (**non-selective** contraindicated in asthma/COPD)
 - **Beta 1 selective** (metoprolol) *could be used in asthma/COPD*
 - **Rebound HTN and angina: taper when discontinued**
 - May mask symptoms of **hypoglycemia** ([tachycardia](#), [tremors](#), [anxiety](#))
 - [Tx: orange juice](#)
 - [Frequent accu check](#)
 - Avoid in symptomatic bradycardia and AV Block, Raynaud, Prinzmetal angina
- **Calcium channel blockers**

- Diltiazem, verapamil (**CHRONO NEGATIVE**), nifedipine, amlodipine
- Avoid grapefruit juice
- **Constipation**
- **Beta blockers Cardio selective:** less vasoconstriction: Metoprolol, Atenolol, Acebutolol (most use in asthma and COPD)
- **Beta 2 agonist: erol:** albuterol, salmeterol, formoterol. **Azole:** ketoconazole, clotrimazole. Alcohol either.

Cardiac Glycosides: Digoxin Range: 0.6 to 1.2 ng/mL

- Mechanism of Action:
 - Increase concentration of intracellular calcium
 - **Positive Inotropism**
 - **Negative chronotropism**
- Toxic effects
 - Visual disturbances: yellow-green colored halos
 - Gastrointestinal (GI) disturbances (**anorexia, nausea, vomiting**)
 - Cardiac disturbances: *late symptoms*
- Bradycardia (<60 bpm) or ventricular cardiac dysrhythmia, **children < 90 bpm**
- Interventions
 - Toxicity; digoxin level above 2 ng/mL
 - Monitor potassium level for **hypokalemia**
 - Monitor apical pulse; if less than 60/min, hold medication, notify MD
 - Antidote: digoxin FAB (Digibind®)

Antiplatelet:

- ASA, clopidogrel, prasugrel, ticagrelor, ticlopidine, dipyridamole
- Glycoprotein IIb/IIIa inhibitors: **abciximab**
- **Discontinue** before elective surgery (5 days)
- **Contraindicated Lumbar test**

Anticoagulants:

- **Heparin sodium**
 - Activated partial thromboplastin time (aPTT)
 - Prolongation Outcome: 1.5 to 2.5 times normal control: ↑ to ± 45-75", **> 75 is risk for bleeding**
 - LMW heparin: Enoxaparin (Lovenox): no need of aPTT testing, **only in emergency admin SQ**
 - **Heparin Induced Thrombocytopenia (HIT):** CBC ☹
 - Antidote: protamine sulfate
 - **Administered during hemodialysis**
 - **NEVER administer IM**
- **Warfarin sodium, PO**
 - **Delayed** beginning of action (inhibit vitamin K depending on factors)
 - Lab: PT and INR. Desired INR 2-3
 - Antidote: vitamin K [**phytonadione**]:
 - Constant moderate intake of green vegetables, oils, egg yolk, liver

- **Interaction:** ASA, NSAIDs, Garlic, Ginseng, Ginger, Ginkgo, St. John Wort, feverfew, Vitamin E, levothyroxine, licorice, green tea
- **Nobel oral anticoagulants:** dabigatran, apixaban, rivaroxaban, Xarelto, pradaxa: **vitamin K is not an antidote, neither needed INR nor diet**

Diuretics

- **Thiazides:** HCTZ, chlortalidone, indapamide (**Do not admin to a patient allergic to Sulfa**)
- **Loop diuretics:** furosemide, bumetanide, torsemide, ethacrynic acid (**can be ototoxic when inject too fast, < 2 min**)
- **Potassium sparing diuretics:** Spironolactone, eplerenone, triamterene, amiloride
 - **No use in patients using K+**
 - **Produce male impotence**
 - **Gynecomastia on male**
- **AC inhibitors:** acetazolamide
- **Osmotic diuretics:** mannitol (**used in ER, acute glaucoma, pulmonary edema**)

Antilipemic Medications

- o **Statins**
 - **Hepatotoxicity:** Elevated serum liver enzyme
 - **Myopathy:** Muscle pain and elevated CK (**grapefruit**)
 - **Contraindicated in pregnancy** (fetal myelin)
 - **Avoid alcohol, check AST, ALT, jaundice, decrease appetite, dark urine**
- **Gemfibrozil:** potentiates adverse effects of statins:
 - **Rhabdomyolysis** (**muscle damage, dark urine**)
 - **Red urine indicated in CK total test, means appear myoglobinuria, protein in urine**
- **Niacin:** flushing relieved by ASA, **hepatotoxicity**
- **Cholestyramine:** with meals. Interact with absorption of other drugs and fat-soluble vitamins ADEK

NOTES:

Medication cannot stop abruptly:

- Beta blockers (produce angina)
- Clonidine (sudden increase HTN)
- Cortisone, prednisone, TPN
- Benzodiazepines (produce seizures)
- Morphine, opioids

Class 10. NEUROLOGIC DISORDERS

Positioning

Increased Intracranial Pressure:

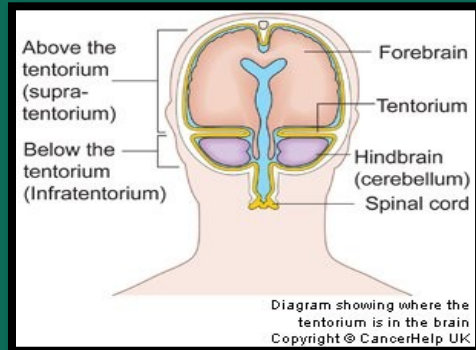
- Head elevated, neck in neutral position,
- Avoid elevation of legs or flexion of the hips (not pillows)

Supratentorial neurosurgery: Elevate HOB

Infratentorial neurosurgery: Flat position

Spinal cord injuries & procedures:

- Log Roll technique (need > 1 person and roll sheet)
- Flat (no pillow)



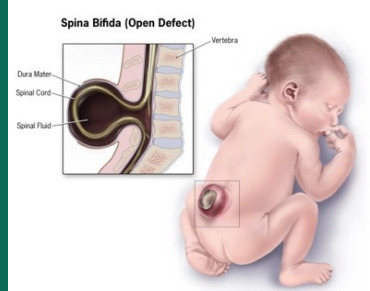
Neural Tube Defects

Types:

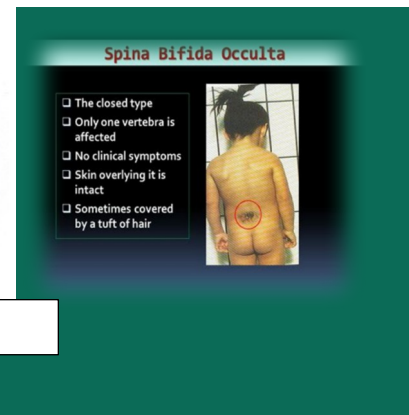
- Spina bifida occulta
- Meningocele
- Myelomeningocele

Interventions:

- Prone or lateral position
- Moist, sterile dressing
- Avoid latex
- Risk for infection
- R/O hydrocephalus



Lumbar tap between



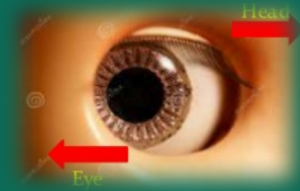
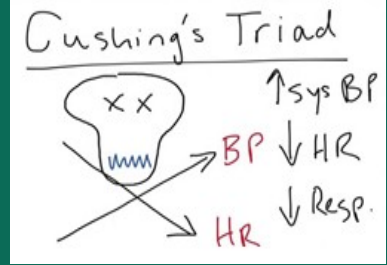
Diseases of the Cranial Nerves

- **Trigeminal Neuralgia:** (swollen nerve, unknown reason)
 - V cranial nerve territory **stabbing** bouts of pain
 - Avoid triggers (touch, brush **teeth too strong**, wind, cold **air in that side of face**, vibration, **chewing food too cold or hot**)
 - Drugs: carbamazepine, gabapentin, pregabalin, amitriptyline. Surgery?
 - **Check CBC** because ↓WBC (1st symptoms is sore throat)
- **Bell's Palsy:**
 - VII cranial nerve paralysis (**one side of the face**)
 - Inability to raise eyebrows, frown, smile, close eyelids, or puff out cheeks
 - **Protect cornea from dryness.** How?
 - Rehabilitation with exercises, heat, electric stimulation and **therapeutic massage**. Steroids
- **Meniere's Disease:**
 - VIII cranial nerve dysfunction (usually one ear) in recurrent attacks
 - Tinnitus**, loss of hearing, **ear fullness**
 - Vertigo (equilibrium): Falls prevention** (when changing position)
 - Pathophysiology: Increased endolymph:
 - Low sodium diet**, **diuretics**, meclizine, lorazepam
- **Acoustic Neuroma:**

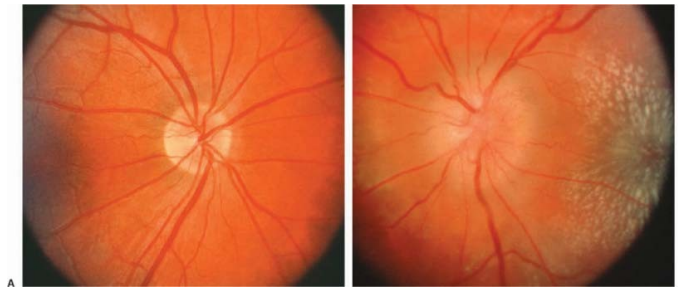
- **VIII** cranial nerve **unilateral** benign tumor (vestibular branch) causing tinnitus, **unilateral** loss of hearing (no deafness), facial numbness, and severe **loss of equilibrium** (vertigo). Requires microsurgery.

Increased Intracranial Pressure (ICP)

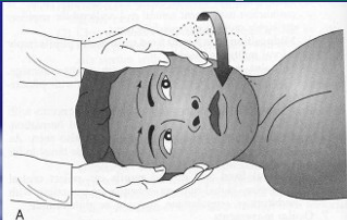
- ICP: normal 7-15 mmHg.
- Early signs increased ICP:
 - Mild changes on LOC: mood changes, confusion, agitation
 - Headache more intense while lay down (at awakening)
 - Diplopia
 - Projectile vomiting
 - Bulging fontanel
- Late signs increased ICP:
 - Low Glasgow scale (less than 8)
 - Cushing's triad
 - ↑ Systolic BP, bradycardia, widening of pulse pressure
 - Decortications/decerebration posture
 - Asymmetric, fixed and dilated pupils
 - Negative oculo-cephalic reflex



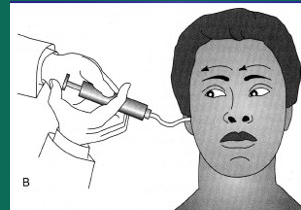
PAPILLEDEMA = ICP



Doll's eye maneuver (Oculocephalic reflex)

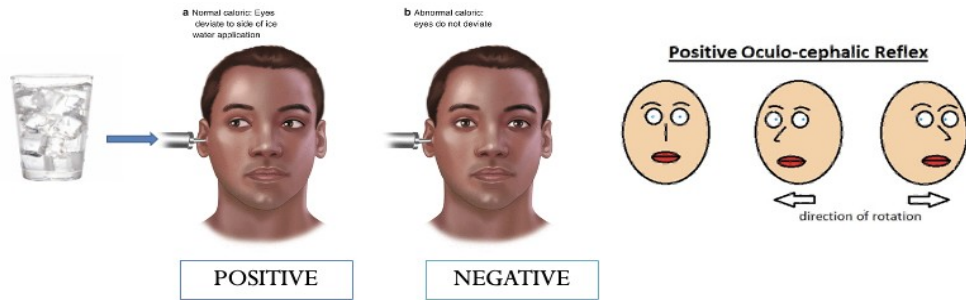


Cold caloric test (Oculovestibular reflex)



Evidence of Brain Death

- Coma: No evidence of responsiveness..
- Absence of brain stem reflexes:
 - Absence of pupillary response to bright light
 - Absence of ocular movements using oculocephalic testing (negative doll's eyes) or caloric test (cold water: negative)
 - Absence of corneal reflexes.
 - Absence of pharyngeal (gag) and tracheal (cough) reflexes.
- Apnea test is the final step in the determination of brain death



Head Injury and Increased Intracranial Pressure

- Airway: NPO, Jaw thrust (cervical injury)
- Immobilize neck
- Elevate HOB
- Prevent hip/neck flexion
- Avoid:
 - Straining, coughing, fever, vomiting, suctioning, hypoventilation, opioids
- Contraindicated lumbar tap
- Collaborative:
 - **Steroids: dexamethasone**
 - Mannitol: in-line filter.
 - 3% NaCl solution
 - Acetazolamide diuretic
 - Forced hyperventilation
 - Regular neuro check

Basilar Skull Fracture and Leakage of CSF

Basilar Skull fracture:

- Battle sign
- Raccoon's sign
- Hemotympanum: (hearing loss)
- Trigeminal lesion: (numbness)

CSF leaking:

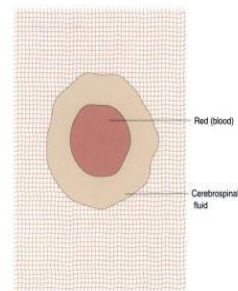
- Otorrhea, rinorrhea
- Clear CSF: Glucose test:
- Serosanguinous CSF: Halo sign test:

Battle sign



Clear Glucose

Raccoon's sign



Halo test

CSF Leak

- Don't suction CSF
- Don't pack ear/nose
- Elevate HOB

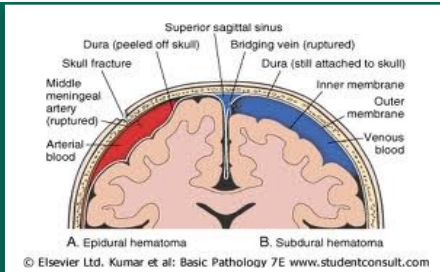
Epidural Hematoma

- Severe trauma
- Neurologic emergency
- Meningeal Artery bleeding
- Above Dura

Momentary unconsciousness followed by lucid interval and then rapid deterioration of LOC

Subdural Hematoma

- Venous bleeding
- Beneath Dura
- Subacute or chronic
- Older and alcoholics



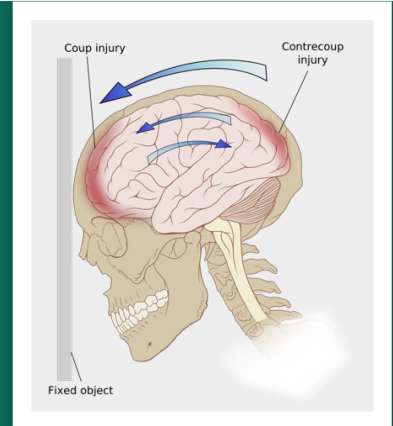
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Concussion Vs. Contusion**Concussion:**

- Temporary unconsciousness, amnesia: self-limited
- CT is normal
- Observation, can sleep, wake up only for neuro check, normal diet
- Post-concussion syndrome

Contusion

- Bruise/laceration of the brain
- Coup and contra-coup
- CT abnormal
- More severe periods of unconsciousness and amnesia

**Spinal cord injuries****Spinal shock: (neurogenic shock)**

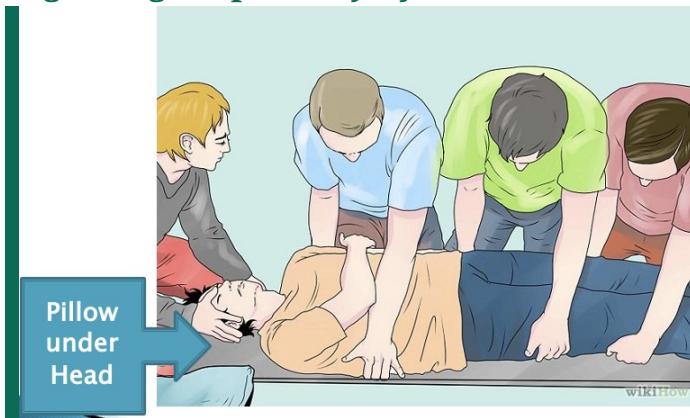
- Respiratory arrest (C3-C5), (C4 start nerve go to diaphragm, paralyzed respiration)
- Bradycardia
- Hypotension
- FLACCID paralysis, no DTR
- Quadriplegia or paraplegia (low extremity)
- Loss of sensation below
- Neurogenic bladder
- Hyperthermia (bad prognosis)
- When spinal shock resolves:
 - Hyperreflexia
 - Spasticity paralysis (leg extended, crossed, tightness)
 - Baclofen (relax muscle, produce epilepsy)

Autonomic dysreflexia: (happen after spinal shock)

- T6 or higher (paralysis)
- Severe hypertension (HTN crisis) (patient can die)
- Headache (strong)
- Nasal congestion
- Profuse sweating and flushing
- Bradycardia
- Interventions:
 - Elevate the HOB (decrease BP)
 - Check for urinary retention (palpate ABD, if Foley, check for kink)
 - Remove TED stocking (return blood to lower extremities)
 - Other triggers:
 - fecal impaction, skin sores, bed sore

Intermittent bladder catheterization. Baclofen and diazepam

Logrolling in Spinal Injury



1 person only for the head
Polytrauma patient, use Jaw thrust maneuver
Use logrolling technique for move the patient or transfer
Avoid twist the spinal

Halo Fixation

- Scanning when ambulate
- Avoid bending over/back
- Risk for infection (pins)
- Don't adjust the screws
- Skin care on pressure areas
- Don't hold the bars or halo
- Driving prohibited
- Risk of choking with food
- Keep wrench at hand (used for adjusting the screw bolts by HCP)
- Cervical fractures



Meningitis

- Bacterial Vs. Viral
- Assessment
 - Fever
 - Photophobia
 - Bulging anterior fontanel
 - Purpuric rashes
 - Seizures
 - Disorientation
 - Opisthotonos
 - Nuchal rigidity
 - Kernig's signs
 - Brudzinski's
- Lumbar tap before starting antibiotics: CSF test
- Dim light
- Seizures precautions
- IV cephalosporins (Abx)
- Droplets until 24 h on antibiotics, Bacterial Meningitis:
 - *Neisseria meningitidis* (meningococcus) worst
 - *H. influenza B.* (put mask on patient)
 - Pneumococcal
 - Bacterial meningitis produce vasculitis, amputation, hearing loss
- Aseptic meningitis (Viral), TB, gram (-), streptococcus pneumoniae (pneumococcus): no isolation (patient go home)

Nuchal rigidity = Stiff neck



Brudzinski's neck sign



ADAM

Kernig's sign



Brudzinski's: Bend the Brain.
Kernig's: Knee elevation



Opisthotonos



Meningococemia Rash

Seizure Disorders

Classification:

- Generalized (grand mal)
- Absence seizure (petit mal)

Seizure precautions

- Decrease stimuli
- Padded side rails
- Avoid pillows (prevent asphyxia)
- Suction available
- Avoid swimming, tub baths, cooking, driving
- Have IV access if needed

- Ensure airway patency
- Ease patient to floor
- Prevent head trauma
- Remove pillows if in bed
- Do not restrain
- Turn patient on his/her side: after movements stop, "NEVER during seizure"
- Do not put anything in mouth
- Decrease stimuli
- Observe and document the characteristics of the seizure
- If the seizure is > 2 min, admin benzodiazepine

Anticonvulsants (require Blood test, do not use Oral contraceptive)

- Encourage compliance
- Monitor blood levels
- Drug interactions (OC)
- Avoid change of brands
- Stevens-Johnson syndrome
- Phenytoin (10-20 Ug/mL):
 - Gingival hyperplasia (soft toothbrush)
 - Toxic: Ataxia, nystagmus
 - Teratogenic
- Carbamazepine (4-12 Ug/mL)
 - Sedation (admin at night)
 - Bone marrow depression (↓WBC)
 - Hyponatremia
 - Oxcarbazepine
- Valproic acid (50-100 Ug/mL):
 - Hepatotoxic
 - Mood stabilizer
- Levetiracetam
 - Increase risk of suicide
 - Psychosis
- Lamotrigine
 - Stevens-Johnson (rash)
 - Suicidality
- Benzodiazepines:
 - DOC for status epilepticus (admin parenteral)
 - Antidote: flumazenil

NOTE: Teach the patient do not change the brand of the medication, if that happen, notify HCP and need a new blood test.

Brain Attack (stroke): Hemorrhagic Vs. Ischemic

	HEMORRHAGIC	ISCHEMIC
PRODROMAL:	"Worst headache..."	TIA/palpitations
ONSET:	Sudden	Progressive or sudden
Level of Consciousness:	Progression to comma, vasospasm, rebleed	Can be conscious
CSF:	Hemorrhagic	Clear
RISK FACTORS:	HTN, anticoagulants Aneurysms	Female, black, HTN, cholesterol, smoking, estrogens, DM, Atrial Fibrillation
THROMBOLITIC (TPA)	CONTRAINDICATED	INDICATED BEFORE 4.5 HOURS

TIA (partial obstruction)

S/S:

Headache
Hydrocephalus
Rebleeding



NI: soft tool, avoid strain, rest, avoid cough, not vomit

Question: Who has more risk for hemorrhagic attack? Patients with HTN

Question: What test is used for detecting hemorrhagic attack? CT Scan

Dominant Vs. Non-Dominant

Dominant Hemisphere (Left):

- Usually left hemisphere
- Right hemiplegia
- Aphasia (speech), Alexia (reading), and Agraphia (writing)
 - Expressive or motor aphasia: Broca's area
 - Receptive or sensorial aphasia: Wernicke's area
- Awareness of deficits: Emotionally affected

Non-Dominant Hemisphere (Right):

- Usually, right hemisphere
- Left hemiplegia
- Left neglect syndrome:
 - ANOSOGNOSIA: Unawareness of deficits, patient doesn't know he is paralyzed
 - Hemianopsia (only can see a middle visual field)
 - Impulsiveness, poor judgment
 - Confabulation
- Risk for injury: Scan while ambulating

PRONATOR DRIFT



HEMIANOPSIA AND NEGLECT SYNDROME



Myasthenia Gravis

- Autoimmune: ↓ receptors of acetyl choline
 - Muscle weakness as day advances. (Weakness is worse in face than the rest of the body)
 - Ptosis, diplopia, difficulty chewing and
- Edrophonium test: Positive (means too Low Ach (paralysis), then patient needs more Cholinergic med for improves)

swallowing (dysphagia) • Myasthenic crisis: ↓ breathing • Interventions ◦ Concentrate ADLs in morning ◦ Avoid hot shower, tub bath ◦ Cholinergic medications: Pyridostigmine, neostigmine ◦ Immunosuppressive therapy ◦ Thymectomy ◦ Avoid muscle relax medications	Myasthenic crisis ◦ Acute exacerbation of M.G. ◦ Needing more cholinergic medication ◦ Edrophonium test improvement Cholinergic crisis ◦ Too much cholinergic drugs ◦ <i>Edrophonium</i> test: Negative (worsening): (means too much Ach, then patient needs reduce Cholinergic med) ◦ Admin Atropine (anticholinergic)
--	---

NOTE: M.G crisis and Cholinergic crisis both of them producing patient paralysis. How to know which of them is? **Do the Edrophonium injection test. This injection has effect and result in 2 minutes.**

Question: When Myasthenia Gravis treatment is effective? **When patient exhibits a fast and observable increase in muscle strength.**

Question: What is the best time to take the Myasthenia Gravis medication? **1 hour before meals, help with swallowing food and move intestine.**

Question: What is the priority in Myasthenia Gravis? **Stress (psychologic)**

Question: What ABG produce Myasthenia Gravis? **Respiratory Acidosis**

Cholinergic medications secreting saliva, move the intestines and produce urine.

Guillain-Barré Syndrome

- **Autoimmune:** self-limited peripheral nerves demyelination
- **Post-infectious:**
 - Upper respiratory/diarrhea/vaccination (flu)
- **Assessment**
 - Ascending weakness of lower extremities
 - Dysphagia, and possible progression to respiratory failure
 - S/S: Paresthesia and **acute pain**, hyperventilation, **can occur partial bell-palsy**
 - Absence of deep tendon reflexes (**DTR**)
 - Autonomic instability: labile BP, HR, and dysrhythmias, **increase, CO₂, decrease O₂**
- **Interventions**
 - Respiratory support (**as soon as you note the patient start to use respiratory accessories muscles for breathing**). Patients starting to be anxious and hyperventilated, increase CO₂, RN instruct the patient to blow air deeply with force in paper bag. If patient cannot blow needs intubation.
 - **Communication code**
 - **Cardiac monitor: cardiac dysrrhythmia**
 - **Tx:** IV immunoglobulins, plasmapheresis
 - Prolonged rehabilitation, **start to do exercises for remyelination.**

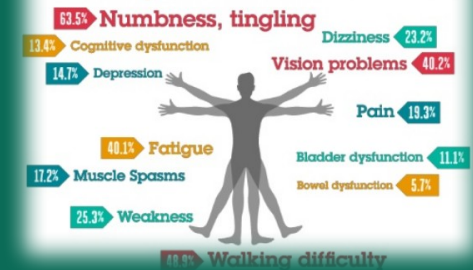
Question: What we need to ask the patient before administer Flu vaccine? **Do you have history of Guillain-Barre?**

Question: What to do for help the patient with G.B communication? **Made a eyes code**

Multiple Sclerosis

- Autoimmune disease
- Demyelination of any part of nervous system
- Relapsing-remitting or progressive
- Assessment:
 - Paralysis
 - Spasticity
 - Nystagmus
 - Blindness (optic neuritis)
 - Neurogenic bladder
 - MRI: demyelination plaques. (screening test)
- Management:
 - Avoid fatigue, pregnancy and extreme temperatures
 - Avoid infections
 - Immunosuppressors: Interferon beta, steroids, adalimumab

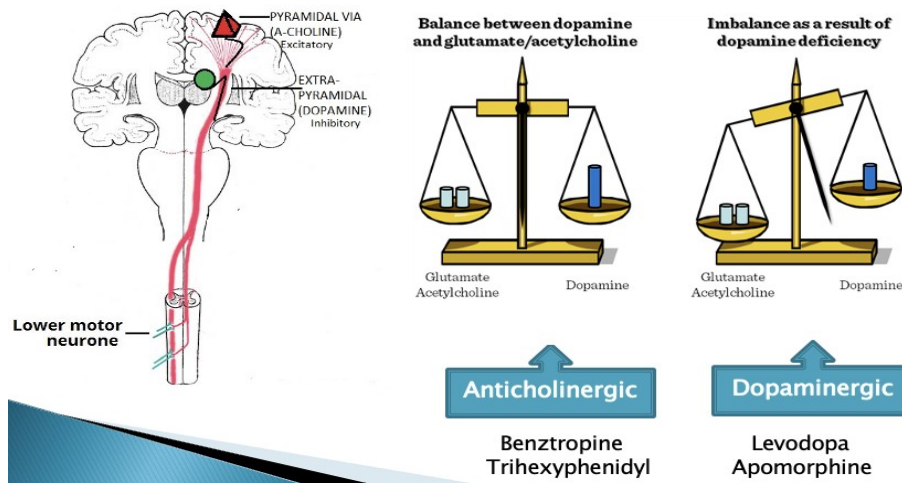
Multiple Sclerosis



Question: What can trigger MS attack? [Stress, infection, UTI, pregnancy, hot temp.](#)

Question: What is the risk for a patient with MS taking immunosuppressors? [Increase the risk to have severe infection, monitor for sore throat, UTI and notify HCP](#)

Parkinson's (more frequent in elderly patients)



- Degenerative and progressive
- Early:
 - Unilateral tremor, bradykinesia
- Imbalance dopamine/A. Choline
 - Rigidity (cogwheel, lead pipe)
 - Bradykinesia (slow movement)
 - Resting tremor (pill rolling),
 - Stooped gait
 - Shuffling,
 - Dysphagia (risk for aspiration)
 - Mask face, drooling
 - Risk for fall:
 - Sleep in prone position (no pillow)
 - Cognitive changes
 - Can develop dementia

Anti-parkinsonian Medications

- Dopamine medications
 - Levodopa
 - Abnormal movements (dyskinesia)
 - Hallucinations
 - Dark urine and sweat
 - Sexual and gambling urge
 - Inhibited by B6
 - Other:
 - Apomorphine, bromocriptine, pramipexole, ropinirole
- Anticholinergics: Benztropine, Trihexyphenidyl
- MAOi: Selegiline: *hypertensive crisis*

- Spasticity

Question: What is the Parkinson's reason? [Ach deficiency](#)

Question: What to expect in Parkinson's diet? [Mechanical modified, increase calories](#)

Migraine

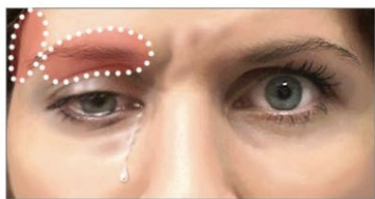
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| <ul style="list-style-type: none"> • Triggers: <ul style="list-style-type: none"> ○ Food Tyramine, MSG • Symptoms <ul style="list-style-type: none"> ○ Aura: visual (scotomas) ○ Attack: <ul style="list-style-type: none"> ▪ Unilateral throbbing pain ▪ Photophobia: dark ▪ Phonophobia: quiet ▪ Nausea and vomit • <i>POUND (pulsatile, one day 4-72 hrs, unilateral, nausea, disabling)</i> | <ul style="list-style-type: none"> • Treatment: <ul style="list-style-type: none"> • Quiet, silent environment • Cold compresses • Abortive: <ul style="list-style-type: none"> ○ Ergotamine and caffeine. ○ Triptans (sumatriptan): do not take > 2 sumatriptan per day, left 1 hour between the pills <ul style="list-style-type: none"> ▪ almotriptan, rizatriptan ○ Vasoconstriction ○ Take ASAP (aura) ○ Cardiovascular side effects: Myocardial ischemia |
|--|--|

Question: When is it not migraine? *“Worst headache of my life”*

Question: What cause the pain? vasodilation

Cluster Headache

- Men > women
- Clustered attacks of pain, 30' to 2 hrs
- Unilateral,
- No throbbing,
- Periocular and forehead
- Ptosis, and myosis
- Tearing,
- Rhinorrhea
- Improves inhaling O₂ 100 %



Cluster headaches may involve pain around one eye, along with drooping of the lid, tearing and congestion on the same side as the pain

Temporal Arteritis

- Severe headache in older
- Tenderness in temporal area
- Fever, weight loss, arthralgia
- ESR very elevated (vasculitis)
- Risk for **blindness** and stroke
- Treatment: Prednisone



Cluster = aggrupation

Amyotrophic Lateral Sclerosis (ALS)

- *Lou Gehrig's disease*
- Muscle atrophy, fasciculation
- No known cure
- Intact cognitive functions
- Risk for aspiration & apnea
- Suicidal
- Respiratory support
- Hospice
- Without indication for rehab

Huntington's Chorea

- Hereditary (autosomal dominant)
 - Male and females: 50%
- Appears 30-50-year age
- Dementia
- Ataxia, chorea
- Risk for aspiration
- Suicidal
- No treatment available
- Abnormal movements

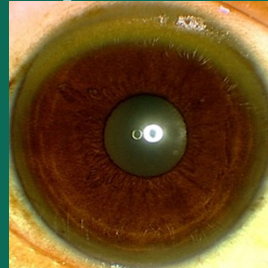
- Person does not speak
- Rehabilitation doesn't work
- Slow develop (in years)

TAY SACHS Disease

- Hereditary (autosomal recessive, both parents)
- Exaggerated startle (Moro) reaction to loud noises
- Seizures
- Paralysis
- Psychomotor retardation
- Cherry-red spot-on retina

**Wilson Disease**

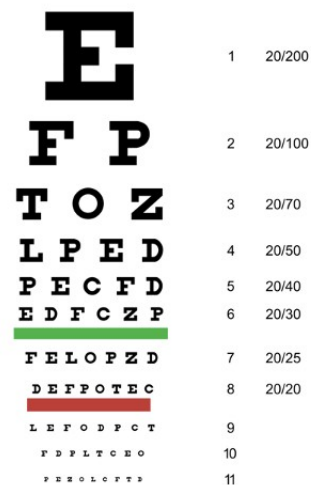
- Hereditary (autosomal recessive)
- High blood levels of copper
- Extrapyrimalidal syndrome and dementia
- Liver damage: cirrhosis
- Keyser Fleischer ring
- Avoid organ meats, shellfish, nuts, mushrooms, chocolate
- Treatment: d-penicillamine

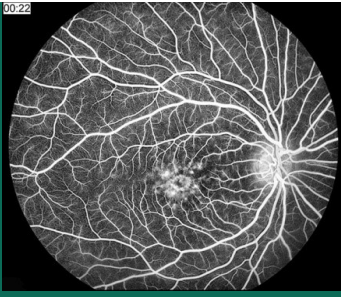
**Screening test: Electroencephalogram (EEG)**

- Recording brain spontaneous electrical activity. Multiple electrodes place on the scalp
- Used for evaluating seizures, cerebral disease, CNS effects, brain injury, brain death
- Patients do not feel any sensation during the test
- Patient do not need to be on NPO
- Explain the procedure
- Instruct wash hair night before the test and avoid sprays or gel
- Instruct refrain from consuming caffeine 24-48 hrs prior test

Class 11. Eyes and ENT**DIAGNOSTIC TESTS: One Eye Each Time****Snellen Chart:**

- One eye is covered while the other is examined
- 20/20: Normal vision
 - Numerator: the client at 20 feet
 - Denominator: the higher, the worst vision
- Legal Blindness: 20/200 or less
 - Fluorescein Angiography (not iodine)

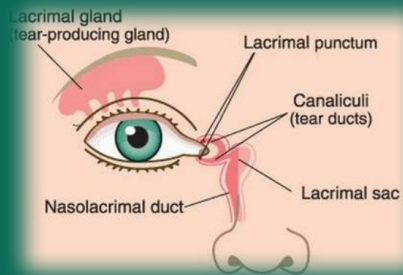




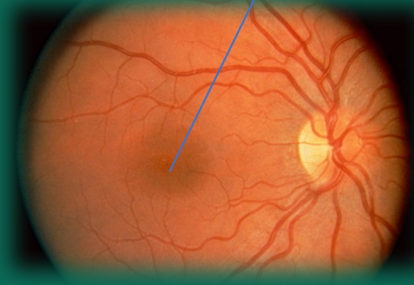
Tonometry: IOP 12 – 21 mmHg

Anatomy of the Eye

Lacrimal System



Retina



Macula: place where more precision vision and colors is

After putting the eye drop, make pressure in the nasolacrimal duct for 1 min to absorb the drop inside the eye

Blindness

- Alert client with sound when approaching (knock the door, speak identifying yourself, etc.)
- Orient to the room and **don't move objects**
- Use clock placement of foods
- Ambulate client one step behind grasping your arm which is kept close to body, to detect changes of direction
- **Promote independence**

Strabismus

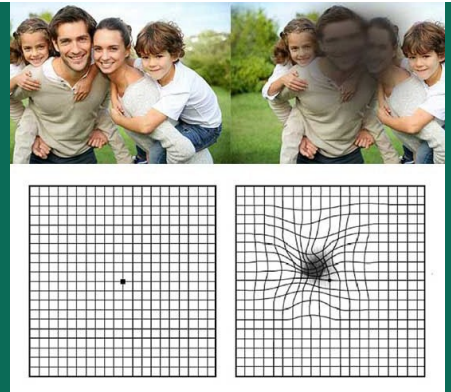
- First 3-month child has blurred nearsightedness
- Strabismus is normal until 3-4 months age
- Child tilts the head or closes one eye.
- Diplopia: double vision
 - Asymmetric corneal light reflex
 - **Unaffected** eye is patched

Question: What eye do you cover? *The good eye for exercise the bad eye*

Disorders of the Eyes

- **Presbyopia:**
 - Inability to accommodate
 - Farsightedness
 - Aging related
- **Macular degeneration:**

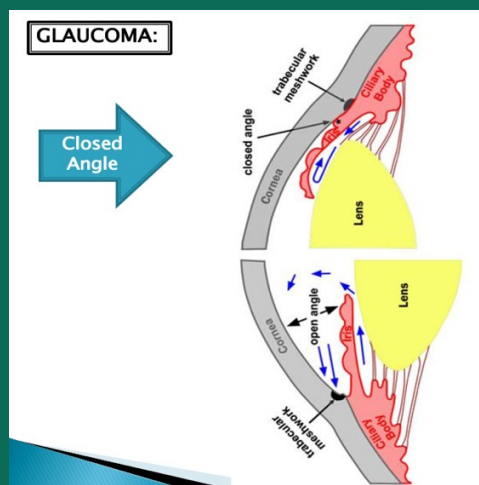
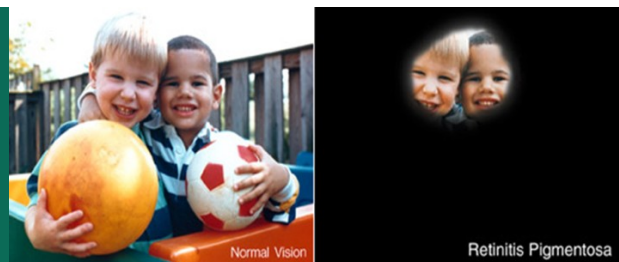
- Loss of central vision (most common in white people)
 - Dry: age related: 85-90% of all cases
- Difficulty with focused vision:
 - Sewing, needlepoint, reading
 - Straight lines appear crooked or wavy
 - Loss of central vision
- Nursing care:
 - Safety
 - Approach from sides
 - Enhanced illumination,
 - Magnifiers, large print books and audio books
 - Board games, gardening, bowling
 - Amsler grid for self-monitoring
 - Not treatment



Photodynamic therapy: light activated drug injected IV and circulates to retinal blood vessels, laser light shone to retina activating the drug which eventually occludes leaking blood vessels

Glaucoma:

- Risk factors:
 - Family history,
 - African American,
 - Corticosteroids
- Open angle: Loss of peripheral vision (years duration, slow development)
- Angle-closure:
 - Acute loss of vision
 - Intense pain
 - Vomits,
 - Red eye
 - Avoid pupil dilation: atropine
- Treatment:
 - Lifelong
 - Pilocarpine: miosis
 - Betablockers:
 - Bradycardia, bronchospasm
 - Latanoprost
 - Acetazolamide
 - Mannitol



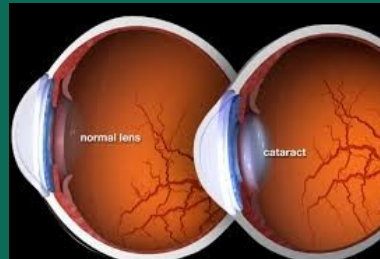
Question: What do you report ASAP to HCP in one patient with Glaucoma? [Patient reports start seeing halos](#)

Cataracts

- Loss of *red reflex*
- Blurred vision, diplopia, and glare

Care following cataract surgery: (ambulatory, not patch)

- HOB 30-45 degrees
- Supine or lateral non operative side
- Eye shield at bedtime
- Dark glasses for day light
- Avoid cough and straining:
 - Stool softeners, tight collars, bending over
- Report decreased vision, eye pain, severe redness or discharge (pus), Call HCP
- Scratching sensation expected in upper lid
- Sleep with head elevated at night or from not Sx side



Eye Emergencies

Retina detachment:

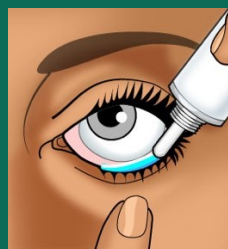
- Flashing scotomas and curtain of blindness on visual field (loss vision upper-down)
- *Painless*
- Physical and vision rest: bed rest
- Vitrectomy:
 - Keep detached area dependent
 - Sleep in prone position

Eye Trauma: evaluate visual acuity

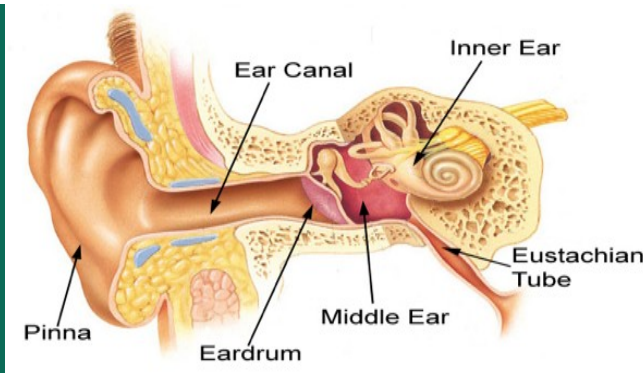
- Contusions
 - Ice or cold compresses on eye
- Penetrating objects
 - Do not remove.
 - Cover both eyes with eye shield
- Chemical burns
 - Flush eyes with water for 15 to 20 minutes until pH is neutral

Ophthalmic Drops and Ointments

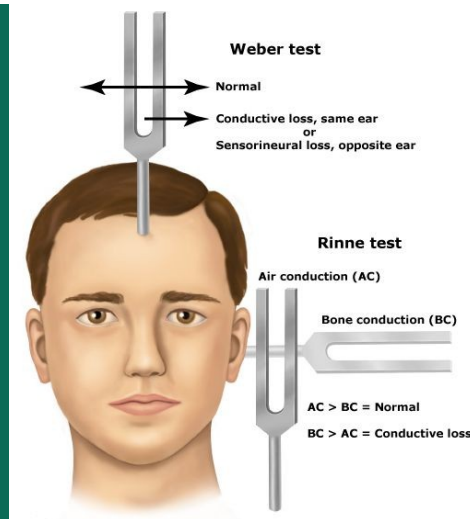
- Wear gloves
- Clean secretions
- Remove contact lenses
- Client sits up, tilt head back and look up
 - If eyes are dirty, clean inner canthus to external canthus
- Pull lower lid to create a pocket
- Dominant hand resting on forehead or cheek
- Hold eyedropper ½ inch above eyeball
- Administer ordered drops on pocket
- Client gently close eyelids (don't squeeze)
- Instruct patient move eyes side to side
- Press nasal bone for 60"
- Wait 3-5 minutes between different drops
- Apply ointments along inner side of lower eyelid, after drops
 - Vision may be blurred for a while
- *Topical anesthetics* can't be used at home



Ear



Auditive Nerve VIII



Conductive vibration: Damage inside the ear produce conductive deafness

NOTE: PVC-13 is administered in children and elderly, also, used for patients with chlochear implant

Communication with a Hearing-Impaired Client

- Approach within client's line of vision or **tap on shoulder**
- Visual clues, illumination
- Reduce background noise: TV off or another noise background
- Avoid covering mouth
- Face client
- Speak slowly with normal tone of voice (low pitched)
- **Use nonverbal cues, gesticulations, mimics, and written messages**
- Evaluate understanding asking client to repeat back

Hearing Deficiency

Presbycusis: expected in elderly

- Loss high-frequency sounds
- Normal tone of voice
- Face audient
- Pantomime and write
- Check hearing aid is on

Sensorineural Loss of Hearing

- Noise
- Presbycusis
- Ototoxicity:
 - Loop diuretics
 - Aminoglycosides and vancomycin
 - ASA
 - Cisplatin

Otosclerosis:

- Hereditary ossification stapes
- Surgery: stapedectomy
- Post stapedectomy care:
 - Anticipate vertigo,
 - Loss of hearing until edema subsides
 - Avoid swimming, cough, sneeze, use of straw, working out

Post-op stapedectomy

STAPES: STEIPIS

- Inform hearing will be muffled
- Avoid airplanes and physical activity for one week
- Change cotton ball daily
- Keep ear dry for 6 weeks
- Assess facial nerves
- No shampoo for 1 week

- Blow nose gently: one nostril each time, with open mouth
- Do not use straw
- Low Fowler's position
- Anti-motion meds

Removing and Inserting hearing aid

- **Removing:**
 - Lower volume of HA and turn it off
 - Remove HA from ear
 - Separate and wash the HA mold with warm water and soap, and dry.
 - Open battery door and store HA in storage bag
 - Clean client's ear with a wet towel and dry
- **Insertion:**
 - Turn volume down and turn power off
 - Check battery and close battery door
 - Pull ear lobule **down**
 - Bring mold into the ear and insert snug
 - Turn on and adjust volume
 - Wash hands and document

Ear infections

- Otitis Externa
 - Risk factors: Swimming
 - Topical antibiotics
- Otitis Media:
 - Risk factors:
 - Secondhand smoking,
 - Horizontal bottle feeding.
 - Breastfeeding is protective
 - Myringotomy (procedure realized by HCP to drain the pus on the middle ear, make an orifice)
 - Affected ear down to drain
 - Earplugs, diving not allowed
 - Avoid straws, blowing nose
 - Tubes come out spontaneously
 - Avoid cotton tips inside the ear, produce ear infarction. Patient needs ear irrigation with warm water or tepid water. Direct the water to the top, not the center. No use cold water, produce vertigo.

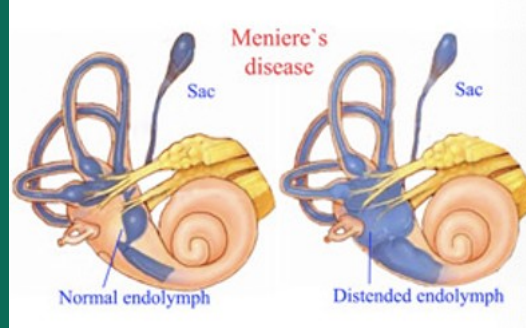
Mastoiditis

- Infection of mucosa of mastoid process
- Complication otitis media:
 - Meningitis and abscess
 - Bell palsy
- Treatment: Antibiotics, Surgery



Meniere's Disease:

- Excessive endolymph
- Chronic, recurrent vertigo
- Tinnitus
- Unilateral loss of hearing
- Ear fullness
- Treatment:
 - Na⁺ restriction, Diuretics (apply for avoid recurrent vertigo)
 - Meclizine: anticholinergic/sedation (Dramamine, scopolamine)
 - Safety precautions
- Interventions: dark room, slow change positions, semi-fowler position. Scopolamine most used in patch by 3 days



Ear Drops

4 year or older



3 years or younger



STEPS to administer Eardrops.

1. Wash hands
2. **Warm the drops** in your hands
3. Gently shake the bottle
4. Lie down or tilt your head to have the ear upside
5. Gently pull and hold the ear up and back (adult) or down and back (child)
6. Squeeze the upside-down bottle to drop the correct number of drops without touching the ear
7. Keep the head tilted for two to three minutes. Can place a clean cotton ball gently just inside your ear
8. Wipe any excess of liquid
9. Document

Tonsillectomy and Adenoidectomy

Postoperative interventions

- Monitor for signs of bleeding:
 - Frequent swallowing
- Avoid coughing, retching, Valsalva, [suctioning](#), [not allow straw](#)
- Ice collar
- Avoid red liquids ([cherry](#), [cranberry](#), [strawberry](#), [gelatin](#))
- Avoid acid liquids (**orange**, lemonade)
- Avoid hot liquids

- Provide soft food, soups mashed potatoes, pasta and clear non-citrus, cool liquids like water, apple or grape juice, apple sauce, gelatin, ice cream, yogurt, pudding, popsicles

Epistaxis

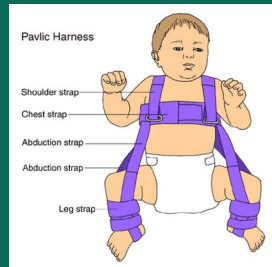
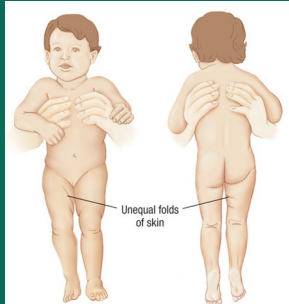
- Posterior nasal bleeding is an emergency!
 - Frequent swallowing
- Management:
 - Seat **upward and leaning forward**,
 - Do not blow nose
 - Lateral pressure **by 5-10 min**
 - Local cold compresses
 - Packing
 - Cauterization

Class 12. MUSCULOSKELETAL DISORDERS

Congenital Diseases

Developmental Dysplasia of the Hip (DDH)

- Asymmetrical gluteal folds
- Positive Ortolani's
- Pavlik harness 24 hours (adjust only by HCP)
- Skin care
- Confirm disease with X-rays



Idiopathic Scoliosis

- Screening: girls school age
- Brace or body cast 24 hrs/day
 - Cast syndrome
 - Vomits, pain
 - NI: soft straps



Muscular Dystrophy (Duchene's)

- Hereditary: X linked (by mother)
 - Child walk tiptoes
 - Gower's sign
 - No Cure
 - Multidisciplinary team:
 - Physical therapy
 - Occupational therapy
 - Geneticist



PAGET'S DISEASE

- Genetic influence
- Adults > 40
- Rapid bone turnover, osteosarcoma
- Bone deformities: legs, skull
- Elevated blood calcium and alkaline phosphatase
- Bone pain, fractures, tumors, inflammation
- Deafness, frequent headache
- Heart failure
- Kidney stones
- Symptomatic treatment

Multidisciplinary team:

- Pain management
- Cardiology
- Nephrology
- Physical therapy

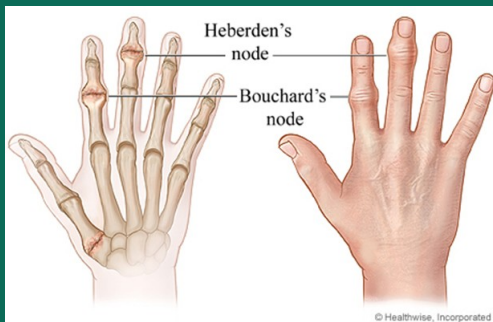


Chondromalacia patella (knee pain): is the softening and breakdown of the tissue (cartilage) on the underside of the kneecap (patella). Pain results when the knee and the thigh bone (femur) rub together. Dull, aching pain and/or a feeling of grinding when the knee is flexed may occur.

	Rheumatoid Arthritis	Osteoarthritis (DJD)
Definition	Inflammatory	Non-inflammatory
Etiologies	Autoimmune	Degenerative Obesity , overuse (athletes)
Stiffness	Lasting > one hour, worse at morning, rigidity	Short lasting
Joints	Symmetric polyarthritis: Severe inflammation Severe deformities	Asymmetric oligoarthritis Little inflammation Crepitation
Systemic symptoms	Low grade fever Anemia	None
Diagnostic Test	Elevated ESR, Rheumatoid factor	Normal ESR (no inflammation)
Treatment	ASA DMARDs: IMMUNOSUPPRESSORS <i>methotrexate</i> <i>leflunomide</i> <i>Adalimumab, etanercept</i> Occupational and physical therapy Labs: ALT, AST, CBC	Acetaminophen Not more than 3 g for older adult

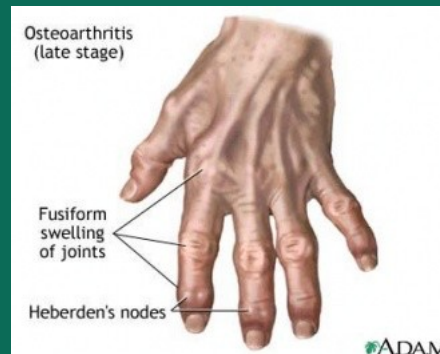
If **RA** complains of fatigue: look for anemia

Osteoarthritis



- Avoid walking, running
- Damage in knee, hip
- Good exercises: swimming

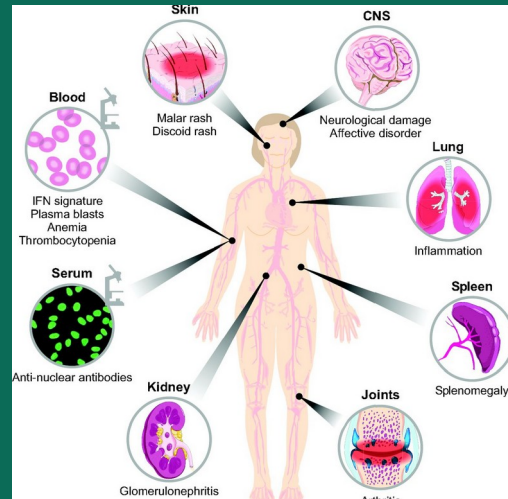
Rheumatoid Arthritis



- Hot improve morning stiffness
- Flu vaccine, check for infections
- Antidote: methotrexate, folic acid (prevent toxicity)

Systemic Lupus Erythematosus (SLE)

- Autoimmune
- Women childbearing age
- Findings
 - Fatigue, fever, weight loss
 - Alopecia
 - Malar rash (erythema, butterfly)
 - Mouth ulcers
 - Photosensitivity
 - Pericarditis and pleuritis, cardio tamponade
 - Autoimmune hepatitis
 - Renal: Chronic Glomerulonephritis: hematuria, CKD
 - Arthritis
 - Anemia (hemolytic), leucopenia, thrombocytopenia (platelet low)
 - Positive ANA
- Management:
 - Avoid sunlight, estrogens, stress
OC contraindicated: barrier methods
 - Immunosuppressors
Steroids, cyclosporine, azathioprine (these med are so strong and only admin if the patient is hospitalized)
Hydroxychloroquine (**affect retina)
Risk for infections



NOTE: Lupus do **NOT** affect Thyroid and Intestines. Thyroids have Hashimoto disease with similar symptoms to lupus, and intestines have Chron's disease.

SCLERODERMA

- Autoimmune
- Skin:
 - Swelling and later sclerosis
 - Raynaud's
 - Ulcers in the top of the fingers
 - Sclerodactyly
 - Telangiectasis (vessels on skin dilated, vascular spiders)
- Esophagus:
 - Dysphagia (cannot open well the mouth, loss weight)

MNEMONIC: CREST

Calcinosis
Raynaud
Esophagus
Sclerodactyly
Telangiectasis

- Kidneys:
 - Nephrosclerosis
 - HTN

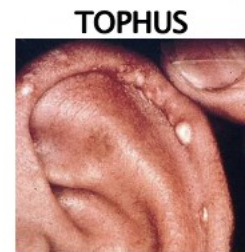
SJÖGREN SYNDROME

- Autoimmune
- Sicca syndrome
- Eye dryness: Corneal ulcers
- Mouth dryness: Tooth decay
- Interventions:
 - Artificial Tears
 - Artificial Saliva
 - Vaginal lubrication
 - Diet with gravy sauce



Gout (too much acid uric in blood)

- Arthritis: *PODAGRA* (red, swollen foot)
 - Urate crystals in synovial fluid
 - Avoid Allopurinol, admin Colchicine
 - NI: Use bed cradle
 - Complication: hepatic encephalopathy
- Tophi (crystal tumors in ears border)
- Renal calculi (risk factor: dehydration)
 - Interventions
 - Fluids
 - Low purine diet: organ meats, seafood, wine
 - Avoid thiazide diuretics & ASA
 - Rapid loss of weight can trigger gout attack
 - Drug Treatment:
 - Acute arthritis:
 - Colchicine (diarrhea)
 - NSAIDs: Indomethacin
 - Hyperuricemia:
 - Allopurinol (avoid in Podagra)
 - Hydration



Osteoporosis (Priority: fractures)

- Risk factors:
 - Menopause
 - Asian and Caucasian
 - Thin body frame
 - Tobacco and alcoholism
 - Family history
- Management
 - Diet rich in calcium and Vit D
 - Weight bearing exercise
 - Best exercise is stand up (yoga, tai-chi, running, walking)
 - Bisphosphonates: take w/empty stomach and

- Low consume of calcium
 - Vitamin D deficiency
 - Sedentary life
 - Drugs:
 - Glucocorticoids
 - Omeprazole
 - Progesterone use (contraceptive)
 - Anticonvulsants
 - Loss of height
 - Hip, wrist and spine:
 - Pathologic fractures
- water, best time in the morning. After take med stay 30 minutes sitting
- Alendronate,
Ibandronate,
Risedronate
- Esophagitis
 - Jaw necrosis
 - Raloxifene
Estrogen receptor modulator
 - DVT
 - Screen test: Bone density

Complications of Fractures

Compartment syndrome: (any extremity, ME)

- Neurovascular compression
- Severe refractory pain (not relief w/analgesic)
- Paresthesia's, numbness, tingling
- Paleness, coolness
- Decreased pulses
- 6 P's: Pain, Poikilothermy, Pallor, Paresthesia, Paralysis, Pulselessness
- Nursing interventions:
 - call HCP
 - RN needs to compare pulses with the other extremities
 - Check capillary refill
 - Only elevate the extremity before the compartment syndrome
- Emergency fasciotomy/cast cut

Fat embolism:

- Long bone fractures (femur)
- Symptoms:
 - Hypoxemia and SOB
 - Altered mental status
 - Petechiae upper body
- ICU: Life support



Affect brain and skin vs thromboembolism only affect lungs

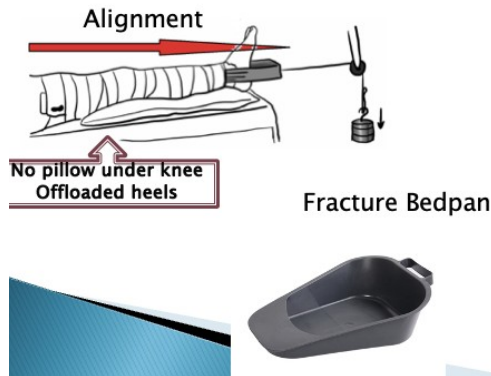
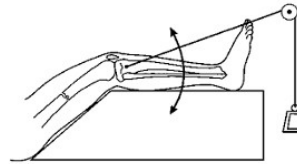
TRACTIONS: REDUCTION OF FRACTURE

SKIN TRACTION: BUCK'S



← Skin: 5-10#
Skeletal: 10-20# →

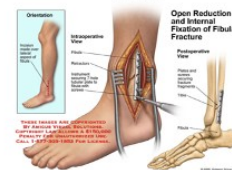
SKELETAL TRACTION



EXTERNAL FIXATION



ORIF



Comminute Fracture

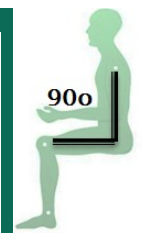


Greenstick Fracture (common in kids)



Hip Fracture

- Hip fracture:
 - Shortening and external rotation
 - Avoid, NO, NO, NO
 - Adduction (use abductor wedge, UAP DO NOT remove it)
 - Flexion less 90 degrees before 12 wk
 - Lateral lying over surgical site
 - Rotation, Crossing legs
 - Stoop, kneel or squat
 - Sitting low chair/toilette
 - Bending to tie shoes or pulling socks
 - Sex before 6 weeks (top or down)
 - Drive before 6 weeks
 - Monitor for infection.
 - Before visit the dentist needs Abx
 - Prophylaxis of DVT:
 - LMH or warfarin until 35 days.

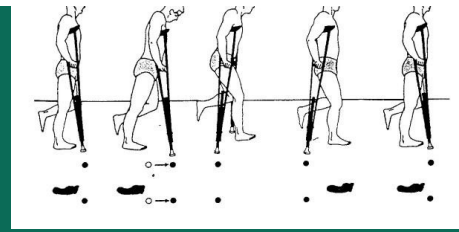


Complications: DVT, PE

Lower Extremity Amputation

Rehabilitation after amputation

- Risk for bleeding (keep tourniquet at bedside)
- Avoid pillow under stump after first 24 h.
- Avoid sitting with flexion of hip or knee
- ROM to prevent flexion contractures
- Trapeze and overhead frame
- Firm mattress
- Prone position every 3 to 4 hours
- Wrapping/cast of the stump to promote remodeling of shape
- 3-point-gait ambulation with crutches



3 point-gait

Phantom limb pain and sensation

- Recognize that this pain is **real**
- **Opioids are not the drug of choice** for phantom pain
- Other drugs
 - Antiepileptic drugs (gabapentin, carbamazepine),
 - IV infusion of **calcitonin**
 - Beta blockers.

Muscle Relaxants

- **Baclofen:**
 - Central relaxant
 - Sedation
 - Seizures (Do not use in epilepsy)
- **Cyclobenzaprine:**
 - Sedation (do not drive)
 - Anticholinergic effect

Aspirin and NSAIDs

- **NSAIDs:**
 - Ketorolac, ibuprofen, naproxen, meloxicam, diclofenac, piroxicam, celecoxib, indomethacin
 - GI and renal toxicity
- **ASA:**
 - salicylic acid/salicylate
 - Reye's syndrome (w/Flu, fever or varicella)
 - GI bleeding
 - Nephrotoxicity

Aspirin Overdose: produce

- Mix imbalance (respiratory alkalosis + metabolic acidosis)
- GI Bleeding
- Pulmonary Edema

AKA (above knee amputation)

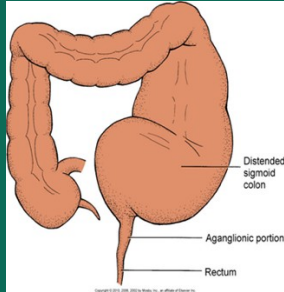
- Before dressing the amputation site, maintain compression to minimize edema, prevent infection

Class 13. Gastrointestinal System

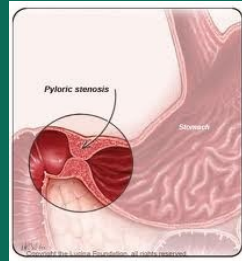
Congenital Diseases

- **Hirschsprung's Disease (Megacolon)**
- **Hypertrophic Pyloric Stenosis**

Obstruction distal colon
 Delayed passage of meconium
 Distended abdomen
 Ribbon-like stools (feces)
 Priority Complication is chronic enterocolitis: fever, bleeding, and explosive watery diarrhea.
 Surgical Tx: Pull-through or ostomy
 S/S: diarrhea with mucus, blood, and septic status
 Risk of sepsis



Congenital disease
 Projectile vomits without bile
 Child is always hungry (child can eat 30 min after vomit)
 Risk for aspiration, HOB elevated, position side lying
 Visible peristaltic waves from left to right
 Olive-shaped mass in RUQ, palpable
 Dehydration, metabolic alkalosis, tenting in skin produce skin turgor
 Surgical treatment



Cleft Lip and Palate (congenital disease)

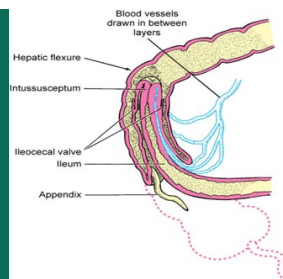
- Feeding:
 - Upright
 - Bulk elongated nipple.
 - Burp frequently
- Surgery
 - Cheiloplasty: nose-lips-mouth done between 3 to 6 months
 - Clean with NS incisional site



- Postop care, NI:
 - No oral temperature
 - No straws, pacifiers, toothbrushes, spoons for 7-10 days
 - Use elbow restrains
 - Supine position
 - Risk for aspiration
 - Suction equipment at bedside

Intussusception (Acquired disease)

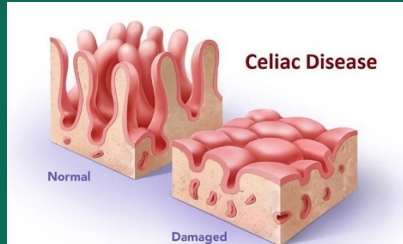
- Telescoping of bowel
- Sudden colicky pain, fetal position w/pain
- Passage of Currant jelly-like content with mucus and blood but not feces
- Vomits and abdominal distension
- Sausage shaped mass palpable
- Passage of brown stools translates resolution and spontaneous reduction
- Risk for recurrence
- Treatment: Air or hydrostatic enema reduction
 - Outcome: No pain and passage of brown stools



Celiac Disease (Non-Tropical Sprue)

Autoimmune, Life disease
 Intestinal villous atrophy by gluten
 Diarrhea, steatorrhea
 Tx: Avoid gluten
 Malnutrition. ADEK deficiency
 < 50 % percentile, avoid cookies,
 meat balls, bread crumbs
 Celiac Crisis Tx: IV fluids

BROW (Mnemonic)
 Barley (Cebada)
 Rye (Centeno)
 Oats (Avena)
 Wheat (Trigo)

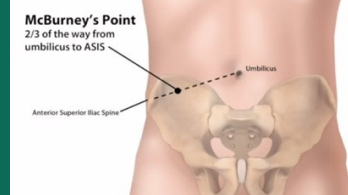


AVOID	CONSUME
Breads	Corn
Pasta	Rice
Noodles	Quinoa
Bread crumbs	Tapioca
Oatmeal	Vegetables
Beer	Meats
Cookies	Fruits

Appendicitis (most common in young adult)

Pain: start umbilical and radiate to McBurney's point
 Anorexia
 Tender McBurney's point.
 Positive Psoas and Obturator signs.
 Elevated WBC (expected around 15000)

McBurney's Point (Appendicitis)

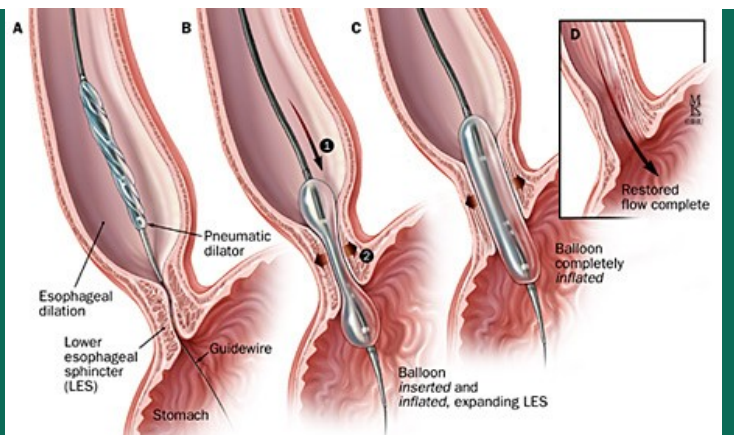


Interventions:

- NPO.
- Semi-Fowler on right side-lying (to prevent infected fluids going to left side)
- **No heat or laxatives, not palpate RLQ**
- Surgery in less than 24 hrs, quickly recovery
- Do no cohort with patients immunosuppress
- **Priority: Sudden relief of pain or don't feel pain, suggests perforation. Risk of peritonitis**

Achalasia

Failure of Lower Esophageal Sphincter (LES) to relax
 Dilation of proximal esophagus
 Symptoms:
 ◦ Dysphagia to solids and liquids
 ◦ Chest pain
 ◦ Regurgitation of undigested food
 ◦ Cough and broncho aspiration
 Treatment: Endoscopic dilations (sedation)

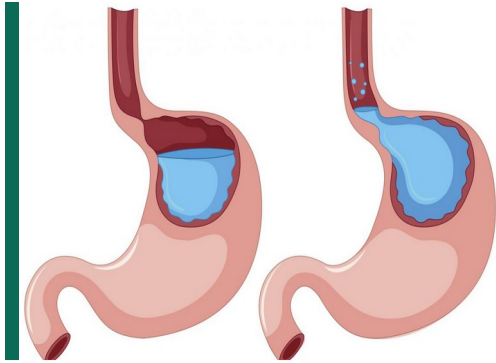


NI:

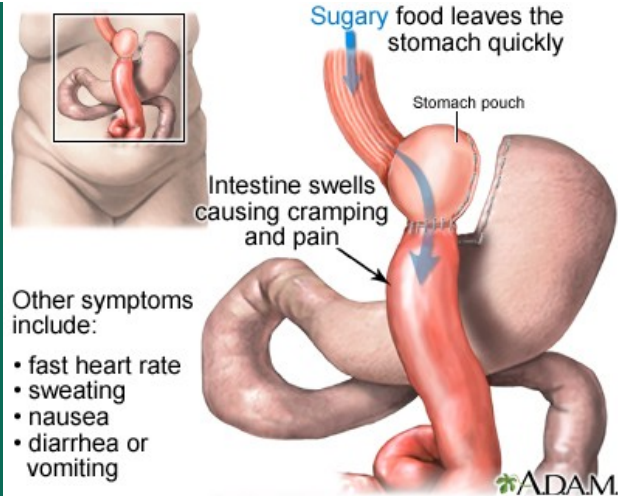
- Teach the patient that is > 1 dilation before can eat
- Risk for aspiration

- Instruct patient do not drive after procedure
- The procedure requires several dilations, different days
- Complications: mediastinal rupture by perforation, fever, pain. **Call HCP**

GERD AND DUMPING SYNDROME



- Prostaglandins protect the stomach walls from the stomach acid
- Avoid NSAIDs, produce gastric ulcer



GERD	DUMPING SYNDROME
Goal: accelerate gastric emptying	Goal: retard gastric emptying
Upright after eating, elevate HOB	Lie down after meals
Lose weight, avoid tight clothing, bending Avoid: alcohol, tobacco, coffee, peppermint, fats, milk (gastritis), spicy foods Expected in pregnancy	Eat fats and proteins, carbohydrates complex No: sugars
Small frequent meals	Small frequent meals
No liquids with meals	No liquid with meals
Tx: Omeprazole, Metoclopramide, Famotidine S/S: heartburn, severe esophagitis, nocturnal cough, nocturnal asthma, recurrent pneumonia right lung	Tx: Anticholinergics Early symptoms: 15-30 min: tachycardia, vertigo, sweating, pallor, palpitations, desire to lie down: fluid shift due to hypertonic chyme, hypovolemia and GI distension, Late symptoms: 2-3 hrs: cramping pain, N/V, diarrhea, high pitched BS, plus symptoms of hypoglycemia Billroth I: gastroduodenal anastomosis hypoglycemia, ↓ BP, stomach pain
Complications: Barrett's esophagus (Pre-cancer)	
How to Elevate the Head of the Bed	Dumping Syndrome after Gastric Bypass Surgery

Do not drink liquid with food, do it between meals

NOTE: Barrett's Esophagus: chronic burn of the esophagus, produce by heartburn and can develop esophagus cancer

Peptic Ulcer Disease (PUD)

- Etiology: NSAIDs and H. pylori bacteria, alcohol and tobacco use
- PUD vomit means:
 - If vomit is coffee ground, means was bleeding before
 - If have melena means was bleeding hours ago
 - **If it is red and now, it is priority Volemic**
- Gastric ulcers
 - Pain soon after eating
 - Association with gastric cancer, more frequent in stomach or duodenum
- Duodenal ulcers
 - Pain 2 to 4 hours after eating
 - Relieves eating: gain weight
- Complications:
 - Hemorrhage:
 - Hematemesis, melena
 - **Hypovolemia (bleeding now): orthostatic hypotension**
 - Perforation and peritonitis:
 - Sudden pain and rigid abdomen, reborn, tenderness
 - Tx: surgery
- Drugs Used in PUD
 - Antacids: AL OH with Mg OH (Mylanta®) and Calcium carbonate:
 - Concomitant drugs need to be given one hour before antacids
 - Magnesium contraindicated in renal failure (**Mg+ and K+ NEVER admin in renal failure**)
 - Alkalosis (metabolic alkalosis if patient takes too many antacids)
 - H2 Receptor Antagonists: Famotidine, Cimetidine (acts the same as grapefruit, ↑ strength of medication)
 - Inhibit secretion of gastric acid
 - Cimetidine: increase concentration of drugs (toxicity). Confusion
 - Proton Pump Inhibitors: omeprazole, esomeprazole, lansoprazole
 - Suppress production of gastric acid: **inhibit calcium absorption**
 - Lansoprazole: **granules** can be given in PEG or dysphagia
 - Best with empty stomach (**30' before breakfast**)
 - PPI use for long time produce osteoporosis
 - Misoprostol
 - **ABORTIVE**
 - Sucralfate
 - Needs to take with empty stomach
 - May impede absorption of other drugs

Viral Hepatitis

	HEPATITIS A	HEPATITIS B	HEPATITIS C
TRANSMISSION	FECAL-ORAL (Shellfish) STANDARD PRECAUTIONS (continent)	Body fluids, sexual , perinatal, dialysis?? STANDARD PRECAUTIONS	Blood Transfusion , IV drugs, sexual, HIV, dialysis?? STANDARD PRECAUTIONS
LONG TERM EFFECTS	Cure.	Chronicity, Cirrhosis , Cancer	Chronicity, Cirrhosis , Cancer
PROPHYLAXIS	HAV vaccine immunoglobulin	HBV vaccine immunoglobulin	Vaccine: not available

- **Hepatitis D: co-infection with hepatitis B**
 - Patient needs to have Hep B before getting Hep D
 - 2 virus affecting liver
 - If mom has Hep B when baby born, admin immunoglobulin for kill the virus, the baby can breastfeed
- **Hepatitis E: fecal-oral in underdeveloped countries (Not in USA)**
- **Toxic hepatitis:**
 - Alcohol, medications, toxins
 - Not contagious

NOTE: When the patient has diarrhea and is incontinent, then apply contact precautions. Ex. C-difficile, rotavirus (kids), norovirus (diarrhea in cruises)

Liver Functions and Portal Circulation

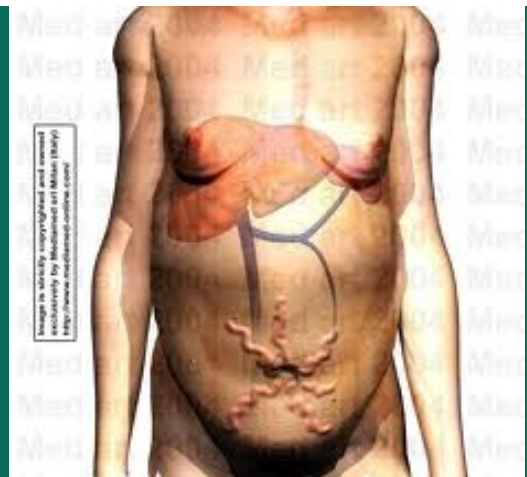
- There is a relation between Liver-INR-Vitamin K
- Liver functions
- Protein synthesis albumin
- Excretion of bile
- Synthesis coagulation factors
- Detoxification of ammonia
- Drugs metabolism
- Store of glycogen
- Immunity
- Splenomegaly

S/S: Caput medusa, jaundice, ascites, palmar erythema, spider angioma, splenomegaly

Tx: Paracentesis (improve respiration).

- Do not stand the patient after the procedure
- Monitor BP, hypotension
- Monitor hemorrhage

NI: Palliative treatment for ascites
Position client on semi-fowler

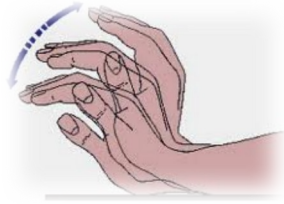


Caput Medusa

Cirrhosis

- Complications
 - **Portal hypertension:** Ascites (produced by decreased albumin) and esophageal varices

- **Coagulation defects:** Prolonged PT & INR
- **Portal systemic encephalopathy:** Asterixis (flapping hands) and AMS (produced by increase BUN)



Asterixis

Question: How determine early encephalopathy?

- Ask the patient to write some sentence and compare with past calligraphy
- Ask patient to extend both hands and check for asterixis (hand don't stop to moving)

- Interventions
 - Diet:
 - **Restriction protein, fats, liquids and sodium.**
 - Increased **carbohydrates** and calories
 - Monitor coagulation: PT & INR
 - Patients have normal liver and normal vitamin K, the INR will be 1.5
 - Patients have normal vitamin K, is taking coumadin and INR is elevated, means **liver damage**
 - Patients using coumadin or warfarin, have celiac disease and have liver damage, then the INR will be elevated
 - Reinforce Vitamin K
 - **Lactulose** (cause loose stool 3-5 times but not diarrhea, decrease ammonium and improve AMS)
 - Avoid opioids, **sedatives** (can produce coma and risk for death)

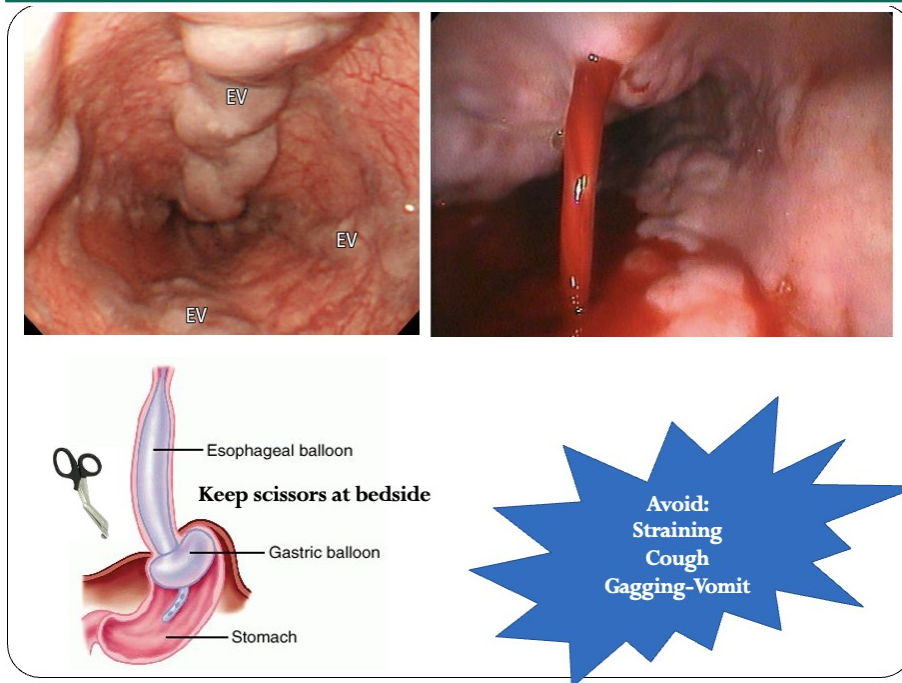
Laboratory in cirrhosis:

- ↑AST & ALT,
- ↑ **Prothrombin time & INR**
- ↑Ammonia,
- ↓Albumin,
- ↓**Glucose**,
- Pancytopenia (decrease RBCs)



Esophageal varices Bleeding:

Tx: beta blockers,
vasopressin IV
(vasoconstriction, stop
bleeding, ADH)



**** Medical Emergency:** patient with cirrhosis and **esophageal varices** who is vomiting and have nausea (Means vein rupture).

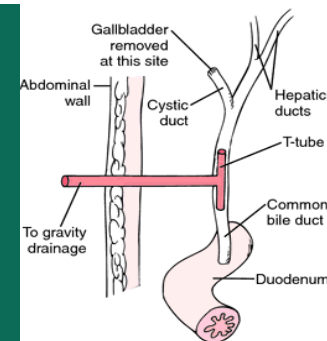
- Tx: beta blockers, vasopressin IV (vasoconstriction, stop bleeding, ADH)

NOTE:

- If the patient has liver damage and is using coumadin or warfarin or has celiac disease (block vitamin K), then the INR will be elevated.
- If the patient has liver normal and has vitamin K normal in the body, INR= 1.5
- Patient not using coumadin or warfarin, has vitamin K and INR elevated, means liver damage

Cholelithiasis (gallstones)

- Risk factors: female, obese, multipara
- Biliary colic: after a fatty meal, RUQ radiating to right scapula or shoulder
- Positive Murphy's sign (patient stops inspiration because it is painful and RN can palpate when deep breath)
- Jaundice. Clay color stools, jaundice eye.
- Fever (infection), red urine
- Can cause hemorrhagic Pancreatitis
- Treatment:
 - Diet: fat free
 - Laparoscopy surgery
 - Open surgery: T tube
- T-tube Care:
 - Expect >500 ml/day of bile first day
 - Do not clamp or irrigate
 - Keep below level of liver
 - Remove when feces become brown and no bile in bag



Question: When is effective the Tx? **When no bile is found in the bag, feces are brown, not red urine, and stone was destroyed**

Pancreatitis

- **Acute pancreatitis**
 - Pain radiating to back
 - Etiology: **gallstones, alcohol, hypertriglyceridemia > 500 mg/dL**
 - **Priorities:** Fluid volume deficit, ileus paralytic, pain
 - **Cullen's sign, Turner's sign** (means pancreas is bleeding)
 - **Hypocalcemia:** Chvostek, Trousseau's (Torsade de points, tingling)
 - **Diagnosis:**
 - ↑Amylase and lipase
 - ↑ WBC and glucose
 - ↓Calcium
 - **Abdominal CT**
 - Interventions: **NPO status, NG suction for 24-48 hrs**, IV fluids, side lying position, fetal position (NEVER supine)

Chronic pancreatitis

- Alcoholism
- Chronic pain
- Decrease bowel sounds
- Isotonic fluids
- Endo and exocrine pancreatic insufficiency
 - Insulin (DM 1)
 - Pancreatic enzymes with meals (Capsules)
- Tx is effective when steatorrhea disappear
- Do not use Meperidine, cause seizure

Pancreatic Cancer

- More common in African Americans men
- Risk Factors: smoke, older age, alcohol use, diabetes, obesity, Hx pancreatitis, consumption high-fat-diet, previous abdominal irradiation

Cullen Sign



Turner's Sign



Inflammatory Bowel Disease (IBD), Autoimmune.

	ULCERATIVE COLITIS	CROHN'S DISEASE
Affected Organs	Colon & Rectum	All GI (distal ileum), mouth, anus
Malnutrition	Mild	Severe (inactive enzyme pancreatic) (absorb water and left only fiber creating solid fecal bolus), affect joints, eyes, skin
Blood in stools	Frequent	Absent
Fistulas	Absent	Present
Risk colorectal cancer	Increased	Increased
Colectomy	Curative (diarrhea with mucus/blood)	Not indicated (diarrhea with mucus/blood)
Extra intestinal symptoms	Less frequent	Frequent

- Treatment:
 - **Diet: low residue, high protein and calories, reduced fat and no lactose**
 - **Sulfasalazine:** anti-inflammatory (both diseases decrease pain, decrease diarrhea)
 - Corticosteroids and Immunosuppressors: adalimumab, infliximab
 - Surgery:
 - **UC:** colectomy in **toxic megacolon**
 - **CD:** obstructions, abscesses, fistulas

Toxic Megacolon: avoid antidiarrheal and opioids in clients with **UC or C. diff diarrhea**

Ileostomy ● Total resection of colon ● Terminal ileum open to skin ● Effluent is liquid ● Digestive enzymes damage skin (enzymes can infect the skin around the stoma and produce infection) ● Risk for fluid and electrolytes imbalance ● Avoid use of enteric coated pills ● Feces are liquid ● Metabolic acidosis, ↓K⁺, ↓COH₃	Colostomy ● Colon fecal diversion ● RN measures and assesses stoma ● Report if stoma becomes pale or cyanotic ● Proper fit of appliance: 1/8" around stoma ● Delay irrigation until healing ● Role of UAP limited to emptying bag ● Feces are solid ● Most frequent left side (descending colon or rectum) ● ER stoma color (white, black, pale pink, purple, brown), means necrosis, call HCP
	

Diverticulosis and Diverticulitis (most common in older adult w/Hx of constipation)

- Out pouching of mucosa
 - **Diverticulosis:** "culosis" means asymptomatic, no pain, most located in sigmoid colon
 - Asymptomatic. Chronic constipation

- Finding colonoscopy/barium enema (routine after 45years-old)
- **Diverticulitis:**
 - **Left lower quadrant** abdominal pain, fever
 - Palpable, tender mass
 - Constipation, lower fever, increase WBC
 - Melena or rectorrhagia
 - CT scan. Colonoscopy contraindicated (produce peritonitis)

Management:

- Diverticulitis: No roughage diet (NPO or clear liquid), Tx: Abx
- Diverticulosis: (not cure)
 - High bulk (fiber) diet
 - **Avoid foods containing seeds**, nuts: raspberry, figs, corn, grapes, blueberry, cucumber, okra (quimbombo), popcorn, tomato, peanuts

Constipation. Irritable Bowel Syndrome-C

- Nursing Interventions:
 - Rule out cancer, hypothyroidism, medications (opioids, iron,CCB, calcium-Tums, antidepressants)
 - Increase fiber, liquids and exercise.
- Bulk forming agents: **Psyllium**. OTC. 2 cups water
- **Stool softeners: Docusate**
- Mineral oil (fleet) and glycerin suppositories
- Osmotics:
 - o Milk of magnesium
 - Contraindicated in renal failure
 - o Lactulose
 - o Polyethylene glycol
- Cathartics stimulants: Bisacodyl, Senna
- Lubiprostone: Constipation caused by opioids
- Linaclotide: IBS-C treatment.

Bowel Sounds:

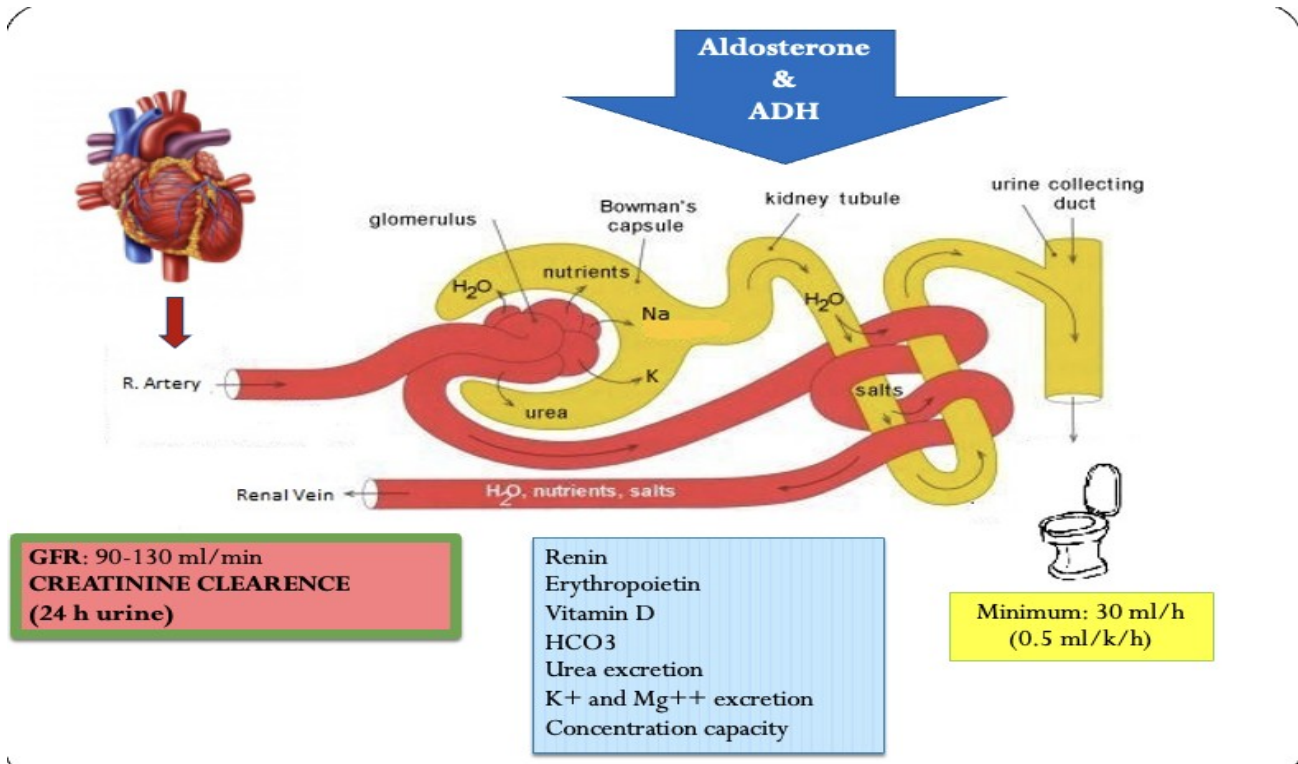
Adult average: 5 to 15 seconds

Absent sound: 3 to 5 minutes (auscultate every quadrant and call HCP), possible bowel obstruction, ileus or ischemia

Hyperactive: IBD, gastroenteritis, GI bleeding, use of laxative, high pitched > 30

Hypoactive: normal when sleep

Class 14. RENAL AND URINARY DISORDERS

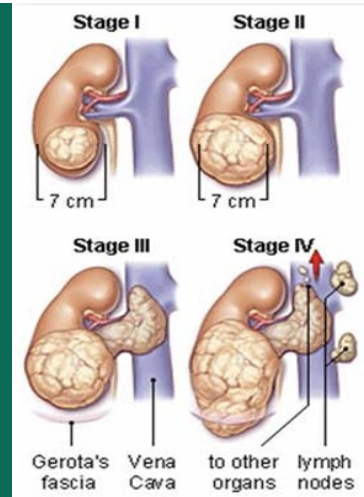


Sequences for blood circulation relate kidneys:

1. **Before kidneys:** before blood go inside kidneys needs leave from the heart, but if the heart has left ventricular failure, dehydration, hemorrhage or volemia then going to the kidneys is not enough.
 2. **Inside Kidneys:** Glomerulus filter the blood retaining protein and red globules, capacity for filter 6 L of plasma per min.
 3. **Post-kidneys:** damage in the glomerulus produce prostate regressive, kidney stones, ect.
- Patient with renal insufficiency does not release HCO₃ and produce **metabolic acidosis**
 - **Creatinine Clearance** is the best test, more precise for diagnosis kidneys problem
 - **Creatinine levels: 0.6 to 1.2 mg/dL**
 - **BUN: 7 to 20 mg/dL** (urea levels)
 - If creatinine level is normal but BUN is elevated, the kidneys are good, but the patient is dehydrated. **NI:** Admin IV fluids
 - If creatinine level >2 and BUN >20, it is **kidney damage**
 - Creatinine Clearance Collection STEPS:
 - Collect urine for 24 hrs
 - Drop the first urine
 - All urine needs to be collected in the same container and keep refrigerated
 - Weight and height the patient before starting the test and draw blood sample for calculate the Body surface area.
 - Last urine collection needs to be taken it at the same time the collection start the day before, the patient needs to urinate no matter what
 - HCP determine damage of the kidney and the stage
 - For the last 2 stages the patient needs dialysis 3 times per week.

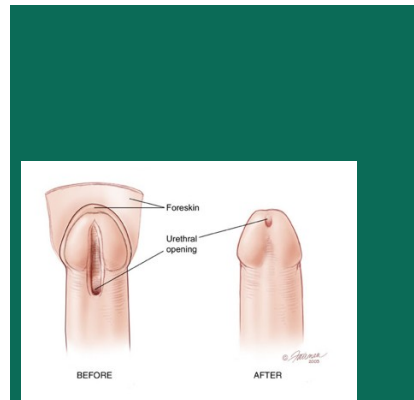
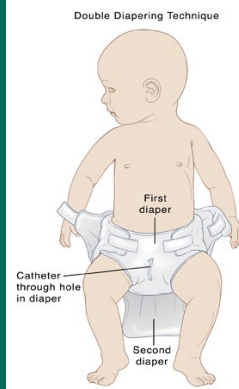
Wilms Tumor (nephroblastoma), most common in children, **NEVER** can be touch or palpate kidney area (Cancer).

- Presentation:
 - Abdominal mass (DO NOT PALPATE), rapid onset.
 - Pain, hematuria, fever
 - Renal failure
- Easy rupture:
 - Hemorrhage
 - Peritoneal dissemination
- Sign at bedside: **"DO NOT PALPATE ABDOMEN"**
- Treatment: Immediate surgery, Chemo and radiotherapy



Hypospadias

- Abnormal opening of urethra on ventral surface of penis
- Corrected around 18 months
- No circumcision (use the skin for fix)
- Double diaper technique after hypospadias surgery: urine is derivate through a catheter the first day after surgery; this catch passes through a hole made in first diaper and the second diaper collect the urine and is changed every 4 hours. The inner diaper is changed only when passes stools.



Testicular Pathology

Testicular Torsion (** Medical Emergency)

- Call HCP
- Acute pain, one testicle is red, vomiting
- Absent cremasteric reflex
- Surgical emergency

Cryptorchidism

- Failure to testicular descend (prematurity)
- 80% descend spontaneously
- HCP wait between 2-3 months to see if the testicle descent spontaneous. If not occur admin HCG.
- Untreated: risk for testicular cancer

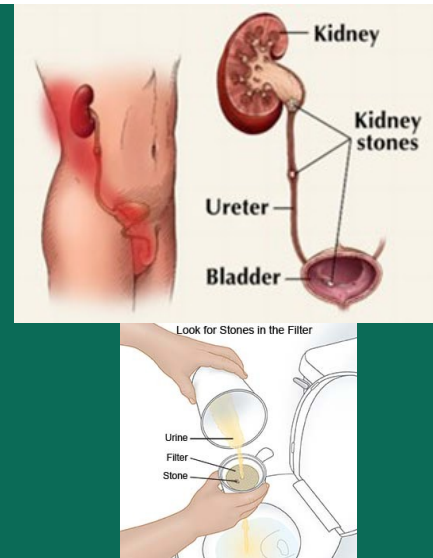
Testicular Cancer

- Young men or history of cryptorchidism
- Slow growing painless lump
- Elevation on AFP (alfa feto protein) and HCG, they are tumoral markers



Kidney Stones:

- Pain radiated to groin and genitals
- Hematuria
- Hydronephrosis
- Management:
 - Opioids
 - Hydration (instruct the patient to drink plenty fluids)
 - Strain urine (can use kitchen strainer or a tea strainer to catch the stone)
 - Procedures: Nephrolithotomy, Transurethral endoscopy and stent
- Ca oxalate: Avoid cocoa, tea, cola soda, beans, rhubarb, green vegetables, spinach
- Urate: Avoid organ meats, sardines, herring, salmon, goose, venison



Acid ash: cranberries, plums, grapes, prunes, tomatoes

UTI: (Urine infected with entero bacteria's Gram-Neg from colon).

- **Cystitis:** Frequency, urgency, dysuria, hematuria
- **Pyelonephritis** (produce urosepsis)
 - Costo-vertebral angle tenderness
 - High Fever, chills, leukocytosis
 - Elderly: confusion, AMS
- **Urine culture:** >100,000 colonies/cc (needs to be done before Rx an antibiotic Tx for identify bacteria)

Treatment:

- **Phenazopyridine:** No antibiotic. **Orange urine.** Do not use for >3 days is **hepatotoxic**. This medication is AZO + Pyridine and is acquired OTC is an anesthetic.
- **Antibiotics** (before start Abx the HCP needs order a culture test for identify bacteria)
 - Sulfonamides: (take full glass of water with the pill)
 - crystalluria, allergy, photosensitivity
 - Quinolones: tendonitis
 - Nitrofurantoin: nausea (can use in pregnancy).

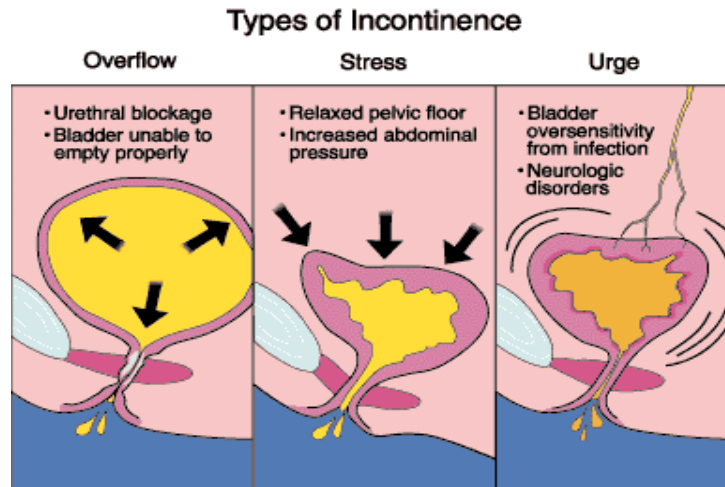
Urinary Incontinence (more in women than men)

1. **Stress incontinence**
 - Leak with increased abdominal pressure (lifting weight, sneezing, coughing, obese, multipara)
 - Kegel exercises
 - Pessaries
2. **Urge incontinence**
 - **Strong urge to void:** Overactive bladder reflex
 - Anticholinergics: oxybutynin
3. **Overflow incontinence**
 - **Overflow:** neurogenic bladder, spinal cord injury, DM₂
 - Continuous dribble
 - Crede's and Valsalva's maneuvers
 - Intermittent catheterization, **DO NOT** use indwelling catheter

- Down sterile gloves
- Patient can do it in home:
 - Clean gloves
 - Clean perineal area with soap and water
 - Catheter is reusable
 - Lubricate catheter before insertion

4. Functional incontinence

- Unable to control (dementia, forgetful)
- *Schedule BR travels every 2 hrs*



Acute Glomerulonephritis (Post-infectious) most common in children

- Produce by Streptococcal infection: ASO+
 - Strep throat
 - Scarlet fever
 - Impetigo
 These infections occurring at least 1 week before
- Facial edema
- Hematuria (reddish color urine)
- Hypertension
- Complications:
 - Acute renal failure (needs dialysis)
 - Hypertensive encephalopathy
 - Seizures precautions
 - Heart failure, pulmonary edema
- Management:
 - Strict bed rest in acute phase
 - Diet: Na restriction
 - Antihypertensive and antibiotics

NOTE: Acute Glomerulonephritis have HTN and Hematuria and Nephrotic syndrome not, the rest are the same

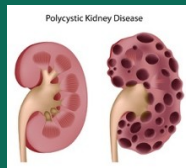
Minimal Change Nephrotic Syndrome, most common in children

- Autoimmune
- Symptoms:
 - Serum Proteinuria,
 - Hypoalbuminemia,
 - Edema:
 - Weight gain,
 - Facial edema (more in mornings)
 - Hyperlipidemia
- Diet:
 - Normal protein or high protein
 - Na⁺ and cholesterol restriction
- Corticosteroids
- Test: Dip stick



Polycystic Kidney Disease

- Hereditary (most frequent Dominant)
- Enlarged kidneys
- Flank pain
- Hematuria,
- Hypertension
- Renal failure (last stage)
- Increase creatinine

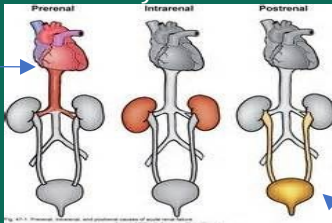


Hemolytic-Uremic Syndrome

- Small children
- After bloody diarrhea
 - (E. coli O157:H7), this bacterium is most common located in animals
 - Raw vegetables, fruits, travel
- Priority question: patient *travel other countries?*
- Hemolytic anemia
- Acute renal failure
- Thrombocytopenia
- Hydration, transfusions, dialysis

Acute Renal Failure

- Etiology: patient can recover in 3-4 weeks
 - Pre-renal (before kidney)
 - Renal (inside kidney)
 - Post-renal
- Oliguric phase
 - Urinary output < 30 mL/h
 - Restriction of fluids, Na, K, ↓ protein
- Polyuric phase
 - Hourly fluid replacement ml/ml
- ARF; Short term dialysis



Cause
hypovolemic
shock

Nephrotoxic:
vancomycin, acute
glomerulonephritis

Cannot
urinate

Chronic Renal Failure (develop by years)

- Etiology: DM, HTN (worse in black people), glomerulonephritis
- Hyperkalemia and hypermagnesemia
- Hypocalcemia and hyperphosphatemia (admin phosphate binder with each meal for inhibit P⁺)
- Metabolic acidosis
- Osteomalacia (↓ Ca⁺, ↓ vitamin D)
- Anemia (no erythropoietin). The kidney stops to produce Erythropoietin hormone, then needs to receive Epoetin alfa injection, it is a synthetic hormone replacement.
- Renal osteodystrophy (risk for injury)
- Diet: ↓ protein, Na⁺, K⁺, P⁺
 - Fluid urinary output + insensible losses: 500-700 ml
 - Epoetin alfa injection can be SQ or IV, administer after dialysis and stop when the patient acquire the Hgb level required like 12 or more.
 - Monitor test for Hgb and Hct
 - Monitor complications for DVT
- Hyperkalemia: Diet, Na polystyrene
- Hyperphosphatemia: Al OH, sevelamer, Ca

acetate: with meals

- Avoid NSAIDs, metformin (cause lactic acidosis), iodine (nephrotoxic)
- Dialysis and renal transplant
- Test: clearance creatinine

NOTE: If the patient doesn't have one kidney, then the erythropoietin hormone is no segregate, patient will have severe anemia, needs Epoetin alfa injection.

Glomerulus = Filter = left protein and red globules (6 L of plasma)

Kidneys need 2 doses of vitamin D: 1st for skin through the sun, and 2nd for kidneys: if the patient has dialysis, needs aluminum hydroxide

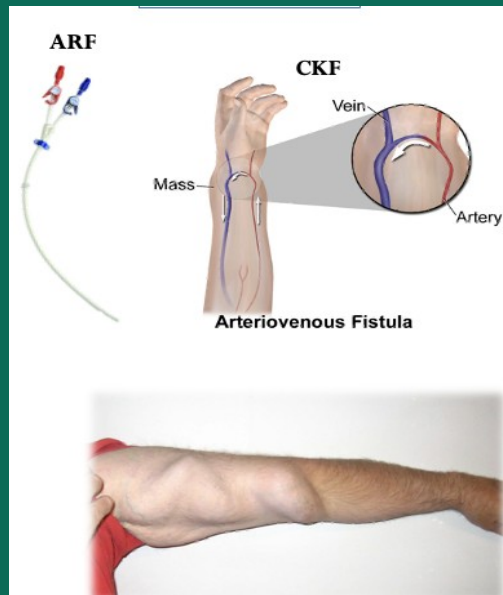
Acute Vs. Chronic Renal Disease

	ACUTE KIDNEY INJURY	CHRONIC KIDNEY DISEASE
ONSET	SUDDEN	GRADUAL
NEPHRON LOSS	≠ 50%	>90 %
REVERSIBLE	YES	NO
ETHIOLOGY	HYPOVOLEMIA, TOXICS, SEPSIS	HTN, DM, CHRONIC GLOMERULOPATHIES
PATHOPHYSIOLOGY	AZOTEMIA, FLUID & LYTES, A.B. BALANCE	SAME + ANEMIA, OSTEODYSTROPHY
TREATMENT	LIMITED DIALYSIS	SAME plus: Epoetin α, PO4 BINDERS, KIDNEY TRANSPLANT

Hemodialysis

- **Fistula:** (only for CKD), catheter for AKI
 - **Bruit and thrill per shift**
 - **Bruit:** auscultate with stethoscope, it is turbulence, vibration
 - **Thrill:** palpable vibration.
 - If not appears, means block with cloth and pt needs thrombolytic therapy
 - Six weeks to mature
 - **Thrombosis:** thrombolytic therapy (TPA, Alteplase)
 - No IV, BP, trauma
 - **Arterial steal syndrome (ASS):**
 - Pale and cool hand distal to fistula
- **Hemodialysis care:** (hemodialysis machine warm the blood)
 - Weight and VS before and after
 - Withhold BP medications, because drawn blood during dialysis
 - Hypotension: most common
 - Anticoagulation, patient is heparinized during the procedure (protamine sulfate at side of the pt)
 - Cramps, nausea: expected
 - **Disequilibrium syndrome:** (post-dialysis)
 - Brain edema,
 - Headache, nausea and vomit
 - AMS, slurred speech and seizures.

Vascular Accesses



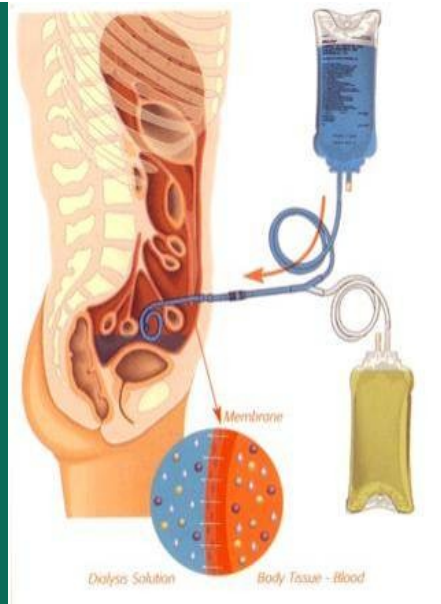
Vein hypertrophy

- Elevate HOB, report to MD

NOTE: Risk factor for long-term dialysis is Hepatitis C

Peritoneal Dialysis or CAPD (Continuous Ambulatory Peritoneal Dialysis), sterile procedure
For long-term treatment, 3 times a day

- **Complications**
 - **Infection:** cloudy effluent. Take sample and send to the LAB, call HCP, expect Rx Abx
 - Blood tinged is OK if client is menstruating
 - **Obstructions:** avoid constipation, change position
 - **Pain:** warm dialysate in a warmer, do not use microwave
 - **Monitor blood sugar** for hyperglycemia
 - **Weight gain** (calories from glucose)
 - **Increased needs of proteins** (loss during lavage)
 - **Hernias**
- **Advantages:** Freedom and better quality of life
- **I & O calculations**
 - If dialysate bag has 1000 mL + 200 mL in body = 1200 mL out
 - **Deficit of effluent** (out less than in or <1200) is added to Intake, = 1000 mL
 - **Excess of effluent** (out more than in) is added to Output, = 200 mL, means possible kicking, obstruction, change position a little bit
- **Interventions:**
 - Teach patient do not take BP in affected arm, not IV
 - Teach patient do not carry any weight in the affect arm (purse)
 - Teach the patient do not sleep for the affected side, not accu check or drawn blood
 - Position patient in modified Trendelenburg if hypotension occurs during dialysis



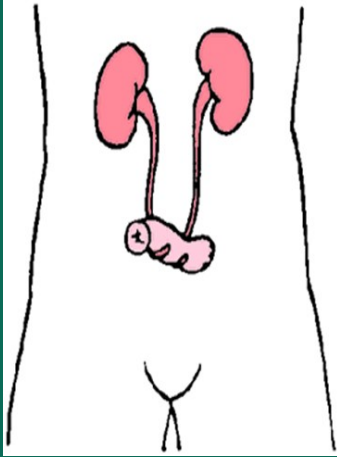
Renal Transplant

- **Immunosuppression:** **Lifelong treatment**, Neutropenic precautions, **high risk infection**
- **Report** urinary output **less 100 ml/hr**
 - Replace lost volume ml by ml
 - Expect mild hematuria
- **Hyper acute rejection:** First hours (wrong matching), **happens as soon as the surgery is done**, **high creatinine**
- **Acute rejection:** **most frequent**
 - Days to months. **Around 2 weeks**
 - Fever, decreased output, increase BUN, **↑creatinine**
 - Requires increase dose of immunosuppressors
- **Chronic rejection:**
 - Months to years. **Around 2 years**
 - Gradual increase of BUN & creatinine.
 - Requires return to hemodialysis and new transplant

ILEAL CONDUIT

Cancer on bladder, need artificial bladder connected to the ureters.

Complications: constipation caused by AL OH



Expect urine in the bag (no blood or feces)
Mucus is present in the urine

Some Renal Medications

- Diuretics:
 - **Loop diuretics:** excrete Ca^{+}
 - *Furosemide, bumetanide, torsemide, ethacrynic acid*
 - Hypokalemia (**muscle cramps**)
 - Hyponatremia, hyperuricemia
 - Dehydration
 - **Ototoxic:** slow IV push
 - Cross allergy with sulfas
 - **Thiazide:** retention Ca^{+}
 - *Hydrochlorothiazide (HCTZ), chlortalidone*
 - Hypokalemia (**muscle cramps**)
 - **Cross allergy with sulfas, means allergy for both sulfa and thiazide**
 - **Photosensitivity**
 - Increase glucose, uric acid, lipids
 - **Potassium sparing diuretics:** $\uparrow \text{K}^{+}$, $\downarrow \text{H}_2\text{O}$, $\downarrow \text{Na}^{+}$
 - *Spironolactone, amiloride, eplerenone, triamterene*
 - Potential hyperkalemia
 - **Antiandrogen: Gynecomastia, impotence and irregular menses**
 - **Osmotic diuretics:**
 - *Mannitol:* IV bolus with **inline filter**
 - Brain edema and acute glaucoma
 - Contraindicated in heart failure
 - **Carbonic anhydrase inhibitor**
 - *Acetazolamide*
 - Treatment of glaucoma
 - **Cross allergy with sulfas**

Acute Urinary Retention

- **Straight Catheterization:** Insert catheter and obtain around 500 ml of urine, clamp the catheter and wait between 15 to 30 minutes and then reassume draining.

BPH

Tx: TURP

S/S: nocturia, hypotension, terminal dripping, dropping urine, weak urine flow, urgency

Rx: alpha blockers (relax prostate)

- ending in "zosin": Terrazosin, Tamsulosin (admin the first 2 doses lie-down)
- finasteride, dutasteride

Prostatitis

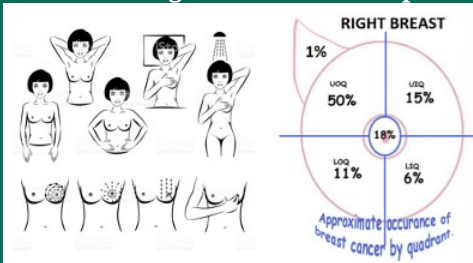
Pain in the rectum

Test: PSA (do not do any rectal tact before the PSA test, increase PSA level)

Class 15. Oncology

Breast Cancer

- **Risk Factors for Breast Cancer:**
 - Age (> 50)
 - Family history in first-degree: Mother, sister, daughter.
 - BRCA-1, BRCA-2 (breast-ovary-colon)
 - Early menarche (< 12)
 - Late menopause (> 55)
 - Black cohort used for hot flashes
 - Hormonal replacement (HRT)
 - Obesity.
 - Nulliparity or first child after 30
 - Breastfeeding is protective
 - Alcohol and tobacco
 - Dense breasts
 - Chest radiation
 - History of breast cancer.
 - Pregnancy and breastfeeding are protective
 - Breast fibrocystic disease is not risk factor
- **Screening:**
 - Mammography (don't use talc and deodorant, better after menstruation)
 - BSE and mammogram. 1 week after period
- Mastectomy and lymphedema (ganglia in axilla)
- Elevation and compression
- Prevention of trauma and infections
- Exercise: ROM (crawl wall with fingers)
- NI: Elevate arm with pillows. Avoid skin breakdown (cut nails), drawn blood there can cause lymphangitis
- Tamoxifen and raloxifene (S/S: hot flashes, dry vaginal, DVT, calf tenderness)
 - *Estrogen receptor modulators*



Tamoxifen (Novaldex): Antagonist of estrogen receptors, reduce breast cancer cell growth, induce hot flashes, increase risk of developing clots (DVT, MI, stroke). **Increase risk of endometrium cancer.**

Peau d'orange (Lymphatic edema)



Nipple retraction



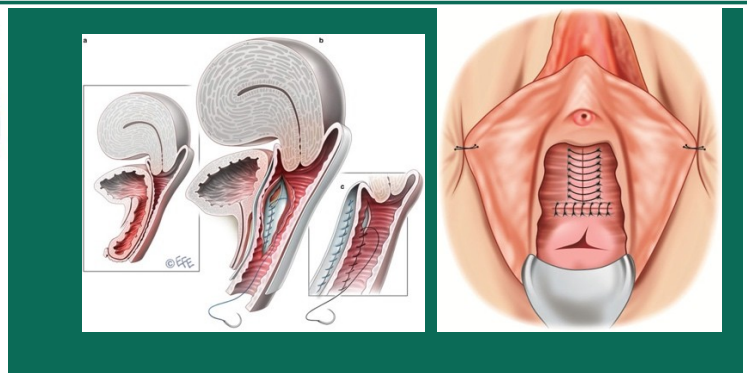
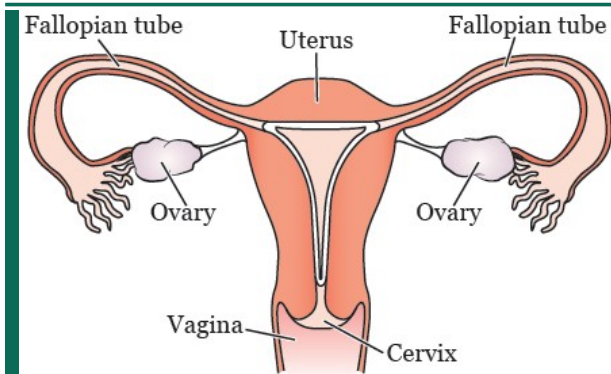
NOTE: Dimple on breast suggest breast cancer

Other Cancers. Risk Factors

- **Ovarian:**
 - Age >50
 - Family history
 - Hormonal replacement therapy
 - Nulliparity
 - Obesity
 - Oral Contraceptives are protective! (Prevent ovarian Cancer)
- **Endometrial:**
 - Age > 50 and family history
 - Obesity and type 2 DM
 - Endometrial hyperplasia
 - Early menarche, late menopause, HRT
 - Nullipara
 - Tamoxifen
 - IUD and oral contraceptives protective!
- **Cervical: (young women) associated with virus**
 - HPV, HIV,
 - Early sexual activity,
 - Multiple partners without protection
 - Smoking
- **Colon:**
 - Family history
 - Polyps (develop in years, adenomatous)
 - IBD (UC/Crohn's) colonoscopy every 3 years, ↑ risk
 - Diet low in fiber, high in fat, red meat
 - Alcohol, tobacco, and obesity
 - No relation with: Peptic ulcers, diverticulosis, laxatives, parasites, etc.
 - Screening:
 - Sigmoidoscopy every 5 years plus annual occult blood (FOBT) or Colonoscopy every 10 years
- **Liver:**
 - Hepatitis B and C,
 - Alcohol (cirrhosis)
 - Obesity and fatty liver
 - Smoking
 - Not related to acetaminophen, amphetamines
- **Prostate:**
 - Age > 50
 - Family history
 - African American

Hysterosalpingo - oophorectomy. Anticipate estrogenic insufficiency

Colporrhaphy.
Avoid Valsalva and intercourse



Leukemia: (Blood cancer) Most common in test for children

- Acute or chronic, lymphocyte or myelogenous
- Assessment
 - Weight loss, bruises, gum bleeding
 - Bone pain, cancer in bone marrow, painful
 - Headache, vomits (SNC involvement)
 - Pallor and fatigue: anemia
 - Bleeding, bruises: thrombocytopenia
 - Fever, night sweats: infections
 - Lymph nodes, liver and spleen enlargement
 - CBC: Immature WBCs, ↓ RBC and platelets
 - **Bone marrow biopsy:** diagnostic (local anesthesia) in iliac bone, position: side lying
- Interventions:
 - Chemo and radiotherapy,
 - **Neutropenic and bleeding precautions**
 - BM transplant: bed side transfusion (it is similar to blood transfusion, BM liquid, is clear in bag)
 - Isolation with Positive Pressure
 - Not fruits, salads, sushi

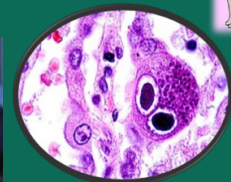
Lymphoma

Hodgkin's Lymphoma (young patients)

- Malignancy of lymph nodes
- Firm, painless progressively growing lymph nodes
- Fever, night sweats,
- Loss of weight
- Infiltration of liver, spleen and bone marrow
- Presence of Reed-Sternberg cells, test: biopsy

Non-Hodgkin lymphoma (older adult)

- Older patients,
- Disseminated
- Worst prognosis



Oncological Emergencies

● Hypercalcemia (** Medical Emergency)

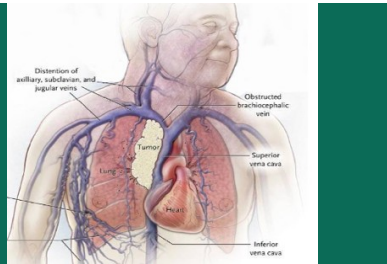
- Metastasis (Ca⁺ move from bone to the blood)
- Hypercalciuria and renal failure
- Oral and/or parenteral fluids are prescribed (admin IV with NS)
- Calcitonin, mithramycin or bisphosphonates, **Foxamax**

● Tumor lysis syndrome (** Medical Emergency)

- Large numbers of tumor cells destroyed rapidly by effective chemotherapy
- ↑K⁺, ↑Uric Acid, ↑PO₄↓Calcium
- Gout, azotemia and renal failure
- **Hydration**; diuretics, urine alkalization, allopurinol
- **NI**: Administer Ca⁺ sulfonate, Kayexalate, Allopurinol

● Superior vena cava syndrome (** Medical Emergency)

- SVC is compressed by tumor growth:
- Edema on arms, head and neck.
- Brain edema
- Arms and neck vein distension
- Radiation therapy for decompression



Chemotherapy

- **Cyclophosphamide**:
 - Hemorrhagic Cystitis
 - Hydration, Mesna
- < V **Vincristine**: Peripheral neuropathy
- < C **Cisplatin**:
 - Ototoxicity (tinnitus)
 - Nephrotoxicity
- T **Trastuzumab**: Cardiotoxic (breast cancer)
- D **Doxorubicin**: Cardiotoxic (ECG/EKG daily)
- B **Bleomycin**: Pulmonary toxicity
- Nausea and vomits:
 - Ondansetron/Granisetron + Dexamethasone:
 - Combined with dexamethasone & aprepitant
 - Side Effects: Sedation, EPS (oculogyric crisis), prolongation QT and torsade d' points
- Nadir of low WBC: Filgrastim
- Chemotherapy produce Bone marrow depression:
 - Anemia: fatigue, decrease energy, activities with rest periods
 - Transfusion
 - Neutropenic precautions: not sick visitors, not allow visitor with vaccine life virus
 - Low platelets: electric razor, avoid strain



Manipulation



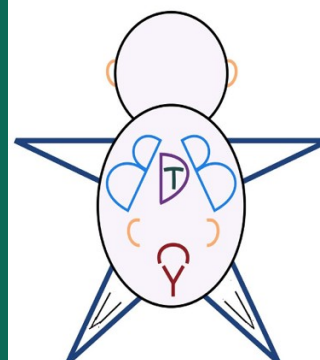
Stomatitis



BM depression



Vesicant extravasations



- **Cisplatin**
 - ototoxicity
 - nephrotoxicity
- **Vincristine**
 - peripheral neuropathy
- **Bleomycin**
 - pulmonary fibrosis
- **Trastuzumab and Doxorubicin**
 - cardiotoxicity
- **Cyclophosphamide**
 - hemorrhagic cystitis

Stomatitis:

- Frequent mouth care with normal saline after meals and bedtime
- Use soft-bristle toothbrush
- Use viscous lidocaine before meals to alleviate pain
- Use water-soluble lubricant to moisten mouth
- Avoid hot, acid, spicy or scraping food
- Avoid alcohol-based mouthwash, peroxide, and glycerin in mouth

Certain Chemotoxicities

- Cisplatin
 - Ototoxicity
 - Nephrotoxicity
- Vincristine: peripheral neuropathy
- Bleomycin: pulmonary fibrosis
- Trastuzumab and Doxorubicin: cardiotoxicity
- Cyclophosphamide: hemorrhagic cystitis

Beam Radiotherapy. (Teletherapy Radiation)

- Pregnant nurse can take care of this patient it is not radioactive
- S/S: loss hair, severe diarrhea, chronic fatigue
- Radiation dermatitis:
 - Avoid sunlight
 - Avoid heat pads or cold pads
 - Avoid lotions, creams, oils (hydrocortisone)
 - Wash with lukewarm water and soft soaps
 - Pat dry, don't rub
 - Don't erase the radiation markings
 - Use loose cotton clothing, soft clothes

Brachytherapy: radioactive implants (Pregnant nurse cannot care for this patient)



Internal Radiation Implant Precautions

- Private room
- Keep on bed (48 to 72 hours)
- Rotate nursing assignments: no pregnant nurse (post-menopausal best option)
- Limit exposure to **30-60 minutes/shift**
- Nurse wears a dosimeter film badge
- Wearing a lead shield, face patient, don't turn back
- Limit visitors to 30 minutes per day, no pregnant women or children
- Visitors should stay at least **6 feet** from the source
- Keep all bed linens and dressings client's room until the implant is removed
- Keep a lead container and long-handled forceps in the room
- Secretions are not radioactive with implants

Note: urine is radioactive with I131 (radioactive iodine) but not in implants

Class 16. HEMATOLOGICAL DISORDERS

Iron Deficiency Anemia (most frequent)

- S/S: weakness, low energy, pale skin, pica, tissue hypoxia, polys, ulcer, hemorrhage, severe period
- Explore cause: occult blood
- Tachycardia: ↑ cardiac output, ↑ HR
- Spoon like deformity of nails
- Lab:
 - Low Hgb, normal WBC/platelets
 - RBC: microcytic (small cell), hypochromic (pale)
 - Low serum iron and ferritin, ↓ B₁₂, ↓ protein
 - Elevated TIBC, ↓ folic acid, ↓ Fe

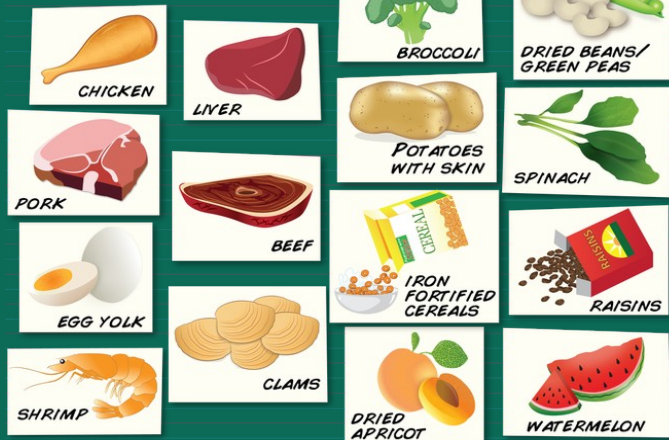
Interventions

- Iron rich food
- Administer 2 hrs before or 4 hrs after taking antacids
- Oral iron: empty stomach (better absorption), acid liquid (best vitamin C)
- Liquid iron: straw to avoid teeth stain.
- “Milk anemia”: limit milk to 3 cups/day
- Ferrous sulfate: dark stools expected
 - Admin with vitamin C for double absorption of iron
- Intramuscular: Z track



Spoon like deformity of nails

IRON RICH FOOD:



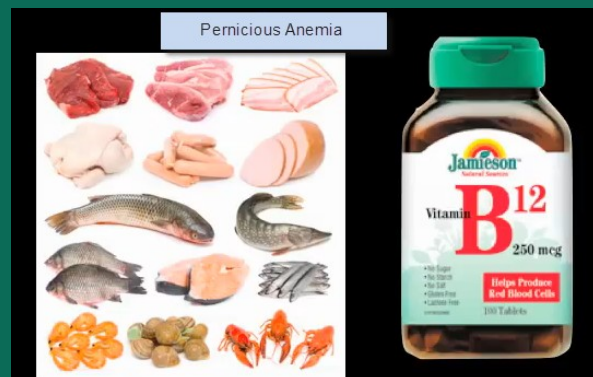
NOTE: Orange juice and beet don't have Iron. Tomatoes juice ↑ vitamin C

Macrocytic Anemia (Folic acid or B₁₂ deficiency)

- Folate deficiency:
 - Alcoholism (B₁)
 - Sulfonamides (sulfasalazine used for Chron's disease and UC long-term), methotrexate (used for rheumatoid arthritis)
- B₁₂ deficiency = Pernicious Anemia
 - Gastrectomy, bypass, atrophic gastritis, vegans
 - Intrinsic factor deficiency
 - Neurologic deficiency (myelin):
 - Paresthesia, impaired gait, memory loss, numbness, tingling
- Lab:
 - Macrocytic anemia
 - Low folate or B₁₂
 - Gastric secretion: achlorhydria
 - Schilling test: ↓ absorption B₁₂
- Interventions:
 - Vegans: B₁₂ supplement (pills)
 - Parenteral B₁₂ for pernicious anemia
 - Folate only if confirmed:
 - E.g., alcoholic or sulfas.



Red, beefy tongue



Question: Risk for Pernicious Anemia? Patient with stomach Sx, remove or bypass

Sickle Cell Anemia

- Autosomal recessive: 25% (Heredity)
- African descendent
- **Vaso-occlusive crisis**
 - Triggers:
 - Dehydration (NI: IV fluids)
 - Hypoxia (NI: O₂)
 - **Cold temperatures** (NI: warm room temperature)
 - Infections (most common pneumococcal, admin vaccine)
 - Pain = microinfarction (opioids, morphine)
 - Hemolysis: anemia and **jaundice** (↑bilirubin)
 - **Iron overload**: no iron supplement (NI: no Fe⁺ deficiency, no need B₁₂ injection)
- **Interventions**
 - Hydration, Oxygen, Pain (HOP)
 - **Meperidine Contraindicated** (NO NO, produce seizures)
 - **Priority Pain vs Hydration**
 - **Warm** environment
 - Hydroxyurea
 - Pneumococcal vaccination
- Hypotonic solutions: D5W, 1/2 NS in D5W, intracellular shift reduces sickling. Sickle c. patients have decreased renal concentration capacity and
- Loss of free water. Hemosiderosis: deferoxamine. Crisis: hemolytic, vaso-occlusive, sequestration and aplastic (parvovirus B19)
- **Spleen remove abnormal cells from the body and old cell**

Thalassemia

- Hemolytic anemia like sickle cell
- Hereditary:
 - Autosomal recessive
 - Alpha: Asian descend
 - Beta: Mediterranean descend



- Splenomegaly
 - Splenectomy
 -



- Iron overload (transfusions)

Aplastic Anemia (Bone marrow)

- Pancytopenia:
 - ↓ RBC, WBC & platelets
 - Pallor, bleeding and infections
 - Drugs, radiations
- Diagnostic: bone marrow biopsy
- Treatment
 - Transfusions
 - Epoetin alfa
 - Filgrastim

Polycythemia Vera

- Premalignant condition
- Elevation of RBC, WBC and platelets
- Excessive blood viscosity
- Risk for thrombosis
- Intense itchiness (hot baths)
- Splenomegaly
- Risk factors: COPD, cystic fibrosis
- Complications: stroke, MI, headache
- Treatment:
 - Phlebotomy (draw blood intentional, excess)

- Bone marrow transplant
- blood)
 - Anti-platelets (aspirin)
 - Antihistamines (itching)
 - Chemotherapy (hydroxyurea) bone marrow depression
 - Avoid Iron and vitamin supplements

Hemophilia

- Hereditary X-linked: Only male children
- Factors produce in Liver:
 - Hemophilia A: VIII factor
 - Hemophilia B: IX factor
 - (IX = Christmas factor).
- Abnormal bleeding
- Neck hematoma: **emergency**
- Laboratory:
 - Normal platelets (150 000 – 400 000)
 - Prolonged aPTT (> 30 sec)
- Treatment: Concentrate of factor VIII or IX

Idiopathic Thrombocytopenic Purpura (ITP)

- Autoimmune
- Thrombocytopenia:
 - Antibody-platelet destroyed in spleen (Platelets < 150 000)
 - Bruising, petechiae, bleeding



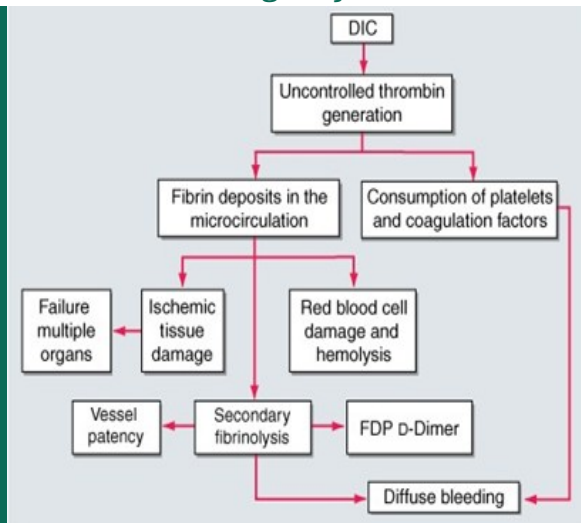
Management IPT:

- Count:
 - > 20,000: bleeding precautions
 - < 20,000: critical bleeding (** Medical emergency), needs transfusion
 - Platelet transfusion:
 - RN verify consent before transfusion and verify blood product with another RN before start
 - Platelets are infused over 15-30 min. Check vital signs after transfusion
 - Not requires ABO compatible
- Immunosuppressor: prednisone
- Avoid contact sports, use electric razor
- Avoid ASA
- Avoid IM shots
- Splenectomy (needs pneumococcal vaccine)

DIC: Disseminated Intravascular Coagulation (** Medical emergency)

- Activation of intravascular coagulation:
 - Sepsis most common cause, amniotic fluid embolism.
- Clot formation & destruction (fibrinolysis):
 - Fibrin degradation products (FDP):
 - D-dimer
- All coagulation tests are abnormal:
 - aPTT, PT, bleeding time, platelets
- “Oozing blood” in venipunctures
- Treatment:
 - Transfusions
 - Fresh frozen plasma
 - Platelets

NOTE: this patient bleeding for anything



Bleeding Precautions

- Handle gently, use lift sheet
- Avoid IM shots
- Test urine and stool for occult blood
- Avoid rectal trauma (enema, thermometer)
- Use electric shaver rather than razor
- Soft-bristled toothbrush, no flossing
- Don't blow nose forcefully or pick in it
- No ASA or NSAIDs
- No contact sports
- Stool softener, no straining
- Avoid playing wind instruments

Neutropenic Precautions

- Room reverses isolation
- Positive air pressure, close door
- Dedicate instruments
- Standard precautions, staff wear mask and gown
- Client wears mask when out
- Visitors screened for diseases
- No raw or rare cooked food
- No standing water, no flowers
- No animals, pets
- Avoid constipation, rectal trauma
- No live virus vaccines
- Clean toothbrush (bleach, dishwasher)
- Report fever, sore throat, cloudy urine

Some hematologic drugs

- **Epoetin Alpha**
 - Do not shake, don't freeze (keep refrigerated (stored at 36 to 46 °F))
 - SQ or IV
 - Adverse effects:
 - **Hypertension:** stop when Hgb 12 g/dL
 - **Thrombosis** (caused when do not Stop med after Hg result was normal)
 - Depletion of iron reserves: Fe supplement
- **Filgrastim**
 - Treatment of neutropenia
 - Chemotherapy **nadir** (Admin 24 hrs after receive chemo)
 - SQ or IV
 - Don't shake or freeze
 - Secondary effects: Bone and joint pain (admin Tylenol)
- **Antiplatelets:**
 - ASA, clopidogrel, prasugrel, ticagrelor, ticlopidine, dipyridamole, Abciximab (IV)

Class 17. Endocrine

Pathology of the ADH (Posterior Pituitary)

Pituitary gland = Hypophysis

	Diabetes Insipidus (DI)	SIADH
ADH	Low	High
Urinary output	High (pee a lot)	Low (No pee)
Urine Specific Gravity	Low < 1.005 (clear, diluted)	High > 1.030 (concentrated)
Plasma osmolarity	High (Concentrated)	Low
Plasma Sodium	High (Concentrated)	Low (hyponatremia dilutional)
Hematocrit & BUN	High (Concentrated)	Low
Symptoms	Polyuria, thirst, polydipsia	Oliguria, brain edema , AMS headache, weight gain
Causes	Lesion of pituitary gland, hypophysectomy, trauma	Lung cancer with small cells, trauma, infection, Desmopressin toxicity, TB
Treatment	Vasopressin/Desmopressin Free water/Hypotonic fluids Weight every day at same time	Demeclocycline, conivaptan, Tolvaptan , lithium Water restriction, IV NaCl 3% (for replace Na+)

NOTES:

ADH: Hormone do NOT pee (Only affect H₂O no Na⁺)

Too much desmopressin in DI produce SIADH

Desmopressin most used as nasal spray, blow nose before dose.

Vasopressin: Injectable, inhalation (it is vasoconstriction, stop bleeding, CPR)

Hyponatremia < 120 produce cerebral damage

Many nursing questions include administration of sodium and hypertonic sodium solutions.

Transsphenoidal Surgery

COMPLICATIONS

CSF leaking: avoid Valsalva maneuver

Diabetes insipidus: monitor urinary output

Disruption of under lip suture (no toothbrush)

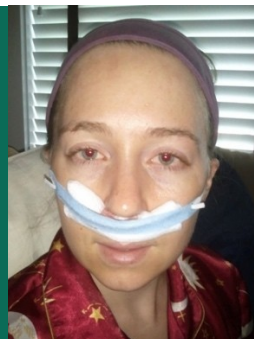
Priority: Clear liquid on nasal drainage (glucose), means CSF leakage

Risk Factors:

- Acromegaly (hands/feet/jaw)
- Elevated prolactin (Galactorrhea)
- Pituitary tumor
- Cranial frangioma

NI: No cough, no strain, no blow nose

Tx: Ondansetron to avoid nausea and vomits



SELLA TURCICA. MOUSTACHE DRESSING

Hypothyroidism (most common in middle adult women)

- Hypothyroidism: ↓ T3, ↓ T4: TSH ↑ (early Dx)
 - Hashimoto thyroiditis (autoimmune Dx)
- ↓ Basal metabolic rate:
 - Weight gain
 - Cold intolerance: Provide warmth (warm blanket)
 - Bradycardia, ↓ RR, ↓ HR
 - Low cardiac output,
 - Constipation,
 - Hyperlipidemia
 - Myxedema: palpebral edema
- Myxedema coma: (long-term hypothyroidism)
 - Brought on by infections or surgery
 - Coma
 - Hypothermia
 - Respiratory insufficiency
 - Provide warmth
 - Hypoventilation (CO₂ retention, not O₂)
 - Voice change, sac on lower eyelids
 - IV levothyroxine (Synthroid) for increase metabolism
 - Hydrocortisone



- Treatment:
 - T₃, Levothyroxine (pill every day, fasting, wait 1 hour before breakfast)
 - T₄, Liothyronine
 - Empty stomach
 - Annually checkup visit and check TSH is normal
 - Monitor:
 - Toxicity: hyperthyroidism
 - Angina
- Diet: low calories, low cholesterol, high fiber, ↓ Ca⁺, ↑ P⁺
- Improve:
 - Void frequent (no constipation)
 - ↑ energy
 - Loss weight

Hyperthyroidism: Graves' Disease (produce heat, energy)

- Hyperthyroidism: ↑ T3, T4, TSH ↓
- ↑ Basal metabolic rate:
 - Tachycardia
 - Heat intolerance, sweating
 - Insomnia, nervousness, tremor
 - Weight loss
 - Exophthalmos
 - High cardiac output heart failure, produce fast arrhythmias, most common A-Fib
 - Propranolol (Beta Blockers)
- Thyroid storm = thyrotoxicosis:
 - Tachyarrhythmia: Atrial fibrillation
 - Hyperthermia: no aspirin (can have Rhabdomyolysis, fever)
 - First: IV propranolol (rapid action, rapid block hormone effect)
 - Second: Oral PTU/Methimazole (report onset of sore throat or fever to HCP, depress bone marrow), slow effect



Care of exophthalmos (cannot close eyelids, dryness)

- Sleep supine, not prone, not side
- Semi-Fowler position
- Use tape to close eyes at night
- Ice compress, steroids
- Artificial eyes drop
- Exophthalmos do not disappear after thyroid removal

- Cool, quiet environment
- High-calorie diet, no caffeine, high fats
- Treatment
- Antithyroid drugs:
 - Propylthiouracil (PTU): neutropenia (CBC)
 - Methimazole (Teratogenic)
 - Prednisone: decrease GI absorption of Ca and decrease Ca level in hyperthyroidism
- Radioactive Iodine I131 ablation:
 - Low iodine diet, discontinue PTU/tapazole
 - It takes 6-8 wks. to observe improvement
 - Lifelong levothyroxine after ablation
 - Precautions:
 - Vomit is radioactive for 6-8 hours
 - Urine and saliva are radioactive for 24-48 h
 - Hydration and frequent urination: Wash hands, flush toilette twice
 - Can trigger thyroid storm

Causes of hyperthyroidism:

- Grave disease (autoimmune), ↑ stimuli T3, T4 and produce Goiter (enlarge thyroid gland)
- Exophthalmos (edema o eyes)

Interventions:

- Far from nurse station
- Decrease hyperthermia using ice, cooling blanket
- Sleep alone
- Recommended drink plenty fluid to help flush iodine from the body
- Diet: ↑ Ca⁺, ↓ P⁺

Question: What is the secondary effect to the patient hyperthyroidism taking iodine radioactive? Will convert in hypothyroidism.

Thyroidectomy

Preoperative: Iodine solution (Lugol)

- Admin 2-3 days before
- Decrease vascularization of thyroid
- Prevent hemorrhage

Postoperatively:

- Semi Fowler position
- Tracheostomy set at bedside:
- Check behind neck for bleeding
- Hypocalcemia:
 - Paresthesia, Trousseau's, Chvostek, tingling (lips/fingers), muscle spasm, torsade de points
 - Calcium gluconate available
- Assess vocal cord's function
- Protect cornea in exophthalmos
- Thyroid storm can appear
- Risk for bleeding, aspiration



Complications: obstruction airway, damage vocal cords nerve. Parathyroid damage produce hypoparathyroidism.

Adrenal Gland: Addison Disease (Autoimmune)

“Vallin says: Na⁺ and water for me, K⁺ for. Toilet”

↓ Cortisol and Aldosterone (patient can die without cortisol and aldosterone)
 ↓ Weight
 Stress intolerance
 Volemia (↓ Sodium and water, hypovolemia)
 ↓ Blood Pressure (tachycardia)
 ↓ Blood Sugar
 ↑ Potassium
 ↑ Pigmentation (dark skin because pituitary produce Acth with melanina)
 ↑ Heart rate (tachycardia compensatory by decrease BP)



● Adrenal /Addisonian crisis / Adrenal shock

- Brought on by physiologic stress (infection, sepsis)
- Hyponatremia, hypovolemia
- **Shock:** ↓BP (severe hypotension, blood doesn't move, need fluid urgent)
- **Tachycardia**, Weakness
- **Hypoglycemia**
- **Hyperkalemia**
- AMS, fever
- Management:
 - **First: IV fluids:** NS and glucose (2 veins, 2 IVs)
 - **Second: IV hydrocortisone**
 - Limit number of visitors

Prevention Adrenal Crisis

- Taper steroids when D/C
- Increase steroids in physical stress
- Increase Prednisone dose

Causes of Addison Disease:

- Long-term use steroids (prednisone, cortisol), stopped abruptly
- Patient has Sx and the doses wasn't increased
- Cushing patient who has removed adrenal gland

NOTE:

Aldosterone control Na^+ and H_2O , no need K^+ .
 Cortisol control Glucose ↑ glucose, ↑ Na^+ , ↓ K^+
 Immunosuppress ↓ WBC

Cushing Disease: pituitary tumours ACTH, ↑ secretion of Acth; syndrome adrenal hyperplasia. CHF could be a complication of Cushing. Short treatments of steroids (<3-4 days) could be stopped suddenly

Cushing's Disease/ Syndrome

- ↑ Cortisol and Aldosterone
- ↑ Sodium and water: CHF
- ↑ Blood Pressure
- ↑ Blood Sugar
- ↑ Weight:
 - Central obesity, buffalo hump
- ↑ Hirsutism.
- ↓ Immunity
- ↓ Potassium
- ↓ Bone density (osteoporosis), falls precautions

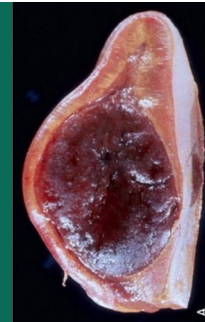


- **Diet:** ↑ protein (minimize muscle wasting), ↓ carbohydrates (risk DM₂), ↓ calories
- **Interventions:**
 - No hold patient by the armpit when move up, use draw sheet (2 RN)
 - Isolation by depress immunity

Note: Cushing's is related to **psychiatric problems** (psychosis)

Adrenal Medulla: PHEOCHROMOCYTOMA

- Tumor of adrenal medulla: Epinephrine (adrenaline)
- Hypertensive crisis
- Tachycardia, HTN, headache, hyperglycemia
- Interventions:
 - Do not touch or palpate because the pressure can produce HTN crisis, not diuretics.
- Treatment:
 - Sodium nitroprusside IV
 - Alpha 1 blockers: Phentolamine, Phenoxybenzamine, terazosin
 - Surgery



DIABETES MELLITUS

Normal A1c 5.6 or less.

5.7 to 6.4 **Prediabetes.**

6.5 or higher: diabetes

Good Control < 7

Enuresis (urine on bed), check for UTI, check glucose in urine

DM type 1	DM type 2
5 – 10 % (most common in children, young people)	90 – 95 %
Insulin dependent: Absent insulin	Resistance to insulin
Ketosis: Ketoacidosis < 500 mg/dL, Medical Emergency Abdominal pain, ketones bodies in urine, low pH, Kussmaul respirations, fruit odor breath	No Ketones: HHNK , > 500 mg/dL, Medical Emergency Normal pH, higher glucose, AMS
DKA and HHNK priorities:	

- **Hypovolemia: IV NS FIRST**
- **Hypokalemia: IV KCL**
- **IV Regular insulin** (when the patient get the expected number, STOP insulin to prevent cerebral edema)
 - until BS is 200 mg/dL → Sliding scale, SQ
 - D5W when BS < 250 mg/dL
 - Avoid hypoglycemia

Younger age (nocturia, enuresis), stomach pain on peritoneal zone	Middle and older age
No obesity	Obesity
Autoimmune disease	Strong genetic
Treatment: insulin (NEVER stop insulin), Metabolic acidosis, ↓pH, ↓CO ₂	Lifestyle modifications Oral hypoglycemic. Insulin, Metformin



- **Diabetes: Good control: A1c < 7% (for 3 months)**
 - Osmotic diuresis and dehydration (admin NS 0.9 % and KCl + Regular insulin IV)
 - Accelerated atherosclerosis
 - Target organs:
 - Heart: CAD (risk MI, silent, not pain)
 - Peripheral arterial disease (aorta aneurysm)
 - Retina: Diabetic retinopathy (aneurysm)
 - Kidneys: Proteinuria (No Tx ending in Renal failure)
 - Microalbuminuria
 - Role of ACE inhibitors (protect kidneys)
 - Peripheral neuropathy:
 - Autonomic neuropathy
 - Impotence
 - Water heater at 120 °F
 - Foot care
 - Check daily for injuries
 - Never walk barefoot
 - Cut nail filled straight and file it
 - Use mirror, white cotton sock
 - Not put feet in hot water
 - Check temp with thermometer
 - Do not apply lotion between the toes
 - Use supportive shoes w/closed toes
- **Hypoglycemia: (patients always know when the glucose is low), BS < 70**
 - Sympathetic arousal: tachycardia, paleness, cold sweating, tremor, anxiety, hunger
 - 15/15: Conscious, give 15 g of refined carbohydrate (6 oz OJ, 8 oz skim milk, candies, 1 tbs honey) every 15 minutes until BS > 70 mg/dL
Nothing with fat (chocolate), fat inhibit glucose absorption
 - Unconscious: glucagon IM or D₅₀W 20 mL IV, BS < 70, (Nausea/Vomit),
 - All DM medications decrease glucose, except Metformin (frequent Accu-check)
- **Hyperglycemia Recognition:**
 - Polyuria, thirst, polydipsia
 - Abdominal pain
 - Ketonuria, fruity breath

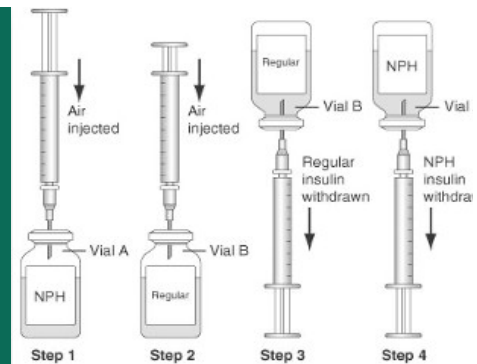
Question: If the patient arrives dehydrated and RN administer 0.9 % NS + KCl + Insulin regular IV. When the HCP stop the insulin? When the patient gets around 200-250 level of glucose, the HCP change the IV by shot the insulin.

Question: Which patient the RN see first, DM₁ or DM₂? DM₁ because the patient can decompensate very fast and die, DM₂ is slow.

Question: What the RN gives the patient with hypoglycemia? If the patient is conscious (saccharose, 6 candies, honey 1 tbs, lactose (fat milk skin), orange juice 6 oz. If the glucose still < 70 repeat until glucose is > 70. If the patient is unconscious, administer Glucagon.

Medications for Diabetes Mellitus: Multidisciplinary Team

- **Insulin Peaks of action:**
 - **Clear Insulin:** admin insulin first and food after
 - Aspart, lispro, glulisine: fast, food present (onset 15 min)
 - Regular: few hours. Best for IV
 - **Cloudy Insulin:** NPH (also called Hayeman or Isophane): 6 - 10 hours (memorize 8 hrs)
 - Resuspend rolling, NO SHAKING
 - Never give in IV
 - Can be mixed with regular, aspart, lispro
 - Storage: Left syringe with needle up
 - Glargine, detemir: don't mix, no peak
- **Sick day:** test frequent, keep insulin / sliding scale
- **Exercise:** extra snack, avoid thigh and arms shots
Put the shot on abdomen, not thigh, not arms
- **Rotation of shots:** (Lipodystrophy)
 - Same anatomic area until completed, same area but separate the shot at least 1 cm
- **Hyperglycemic drugs:** corticosteroids, albuterol
- **Mask hypoglycemia:** β -adrenergic blockers
- **Vials:**
 - Once open needs to be room temperature, expiration date in 28 days after open
 - Do not inject insulin that is cold
 - Unopen vial keep refrigerate to maintain patency

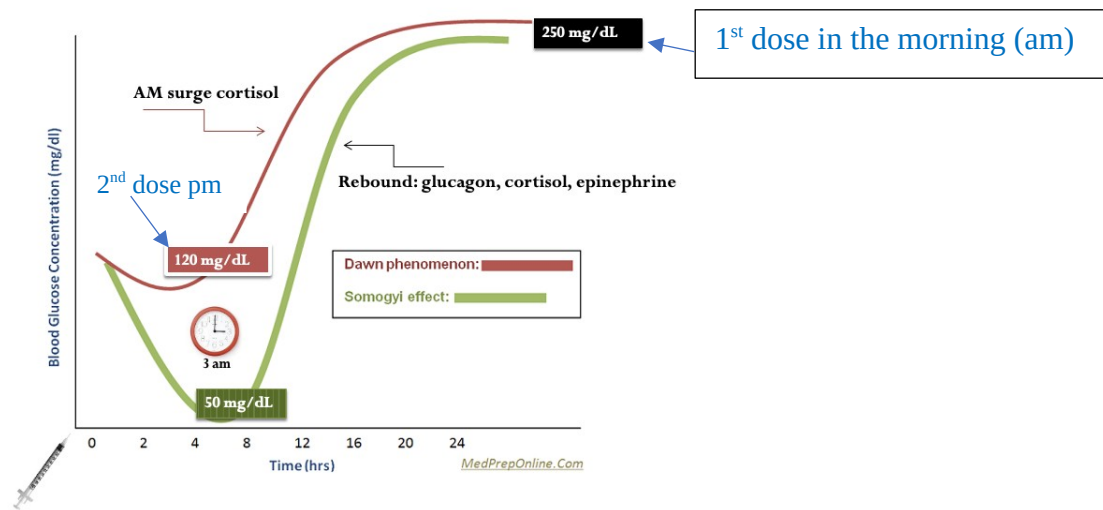


Insulin

Onset, Peak, & Duration Times

Type	Generic Name	Onset	Peak	Duration
Rapid-Acting Used to manage blood sugar during snacks/meals and to correct high blood glucose levels	Insulin aspart	15 min	1-3 hrs	3-5 hrs
	Insulin lispro	15 min	30-90 min	5 hrs
Short-Acting Used to manage blood sugar during snacks/meals and to correct high blood glucose levels	Regular	30 min	2-4 hrs	5-12 hrs
Intermediate-Acting Used to control blood sugar between meals, while fasting, and overnight	NPH	1-4 hrs	4-12 hrs	10-24+
Long-Acting Used to control blood sugar between meals, while fasting, and overnight	Glargine	2-4 hrs	None	24 hrs

Somogyi Phenomenon (produce by NPH insulin) = Hypoglycemia (more danger at night when the patient is sleeping)



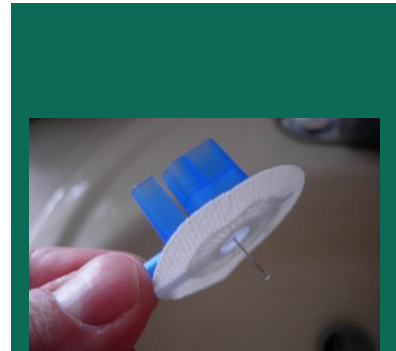
Interventions:

- Decrease insulin doses
 - **Somogyi:** give the second dose pm or recommend the patient eat before to go to the bed or move the second dose more late to have the peak early in the morning
 - **Dawn phenomenon:** patient has stable glucose during the day and only increase in the morning. Needs increase insulin at morning

Question: How do you know which phenomenon is? [Check glucose at peak](#)

Insulin Pump (for DM₁, DM₂, Pregnant)

- Catheter and needle inserted in sub-Q.
- Delivers rapid (aspart, lispro, glulisine) or regular insulin two ways:
 - Continuous (basal)
 - Bolus triggered by own patient before meals or when BS is high (sliding scale)
- Still requires frequent finger stick glucose monitoring
- Expensive: Labile forms DM (type I or II)
- Advantages:
 - Less shots to deliver insulin
 - Improvement A1c
 - Less episodes of hypoglycemia
- Risk: Dislodgement of needle: DKA



Oral Hypoglycemic Medications: Type 2 DM

- **Biguanides:** Metformin or Metformin XR (**don't crush**) (do not use metformin in patient with kidneys problems, not renal failure)
 - Lactic acidosis in **CKD** renal disease (↑ BUN, ↑ creatinine) or when combined with Iodine IV contrast. For screening test, suspend metformin 2 days before and resume 2 days after
 - Usually does not cause hypoglycemia
 - Help to lose weight
- **Sulfonyl ureas:**
 - Glipizide, glibenclamide, glyburide, glimepiride, tolbutamide (medications name that start with "Gli" or "Gly" make pancreas produce too much insulin, risk for hypoglycemia)

- **Hypoglycemia**, potentiated by **Ginseng, oolong tea, and alcohol**
- Disulfiram reaction with alcohol (**when alcoholic patient arrives, put IV with glucose + B1**)
- Cross allergy with sulfas (**Metronidazole, Flagyl**)
- **Incretins:**
 - Mimetics: **exenatide, liraglutide, dulaglutide (injections)**, decrease secretion of glucagon
 - **Risk for pancreatitis** (abdominal pain)
 - DP4 inhibitors: **sitagliptin, saxagliptin**
 - **Risk for heart failure**
 - **Januvia:** stimuli production insulin, improve insulin resistance, help reduce weight because produce satiety and fullness
- Thiazolidinedione: **Pioglitazone**
 - **Hepatotoxicity**
 - Fluid retention: **heart failure**
- Acarbose:
 - Alpha glucosidase inhibitor: inhibit the absorption of glucose.
 - First bite of a meal. **Do not cause** Hypoglycemia, bloating. **Reduce intestinal uptake of sugar**

Sitagliptin (Januvia), saxagliptin (Onglyza) exenatide (Byetta, SQ), liraglutide (Victoza, SQ), dulaglutide (Trulicity, SQ)

NOTES:

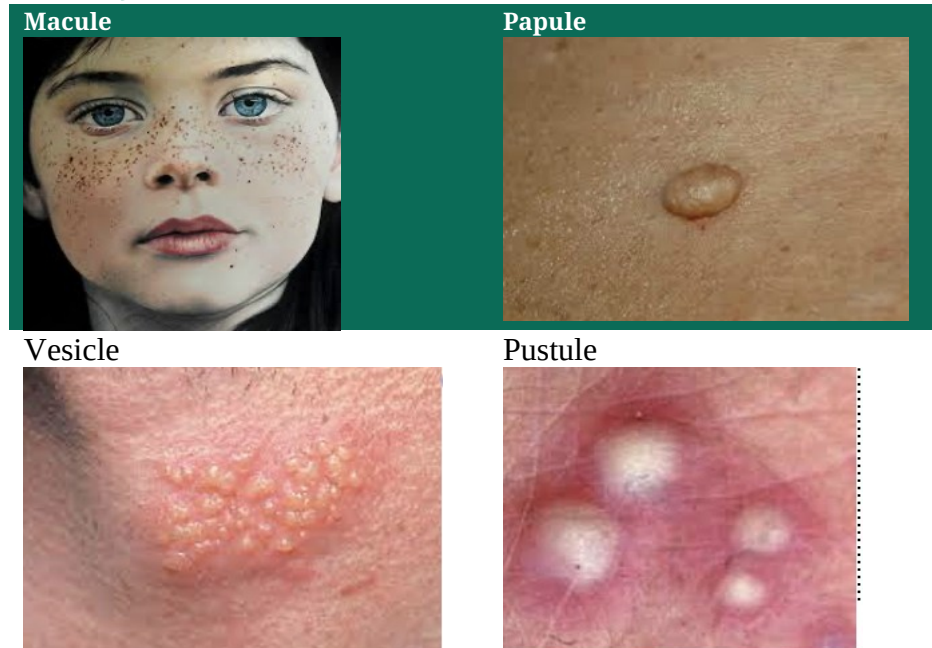
- DM multidisciplinary team:
 - Cardiology
 - Nephrology
 - Dietitian
 - OT
 - Case manager
 - Social worker

Question: What happen if the patient with NPH insulin sleep lunch 6-10 hrs after receive the insulin?
Hypoglycemia occurs

•

Class 18. Integumentary

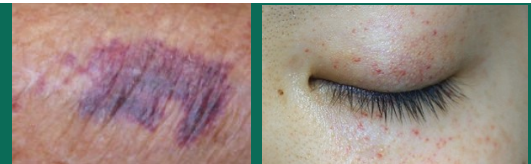
Primary Skin Lesions



Aging and Skin

- **Lentigo Senile:**
 - Age spots, liver spots
 - Dorsal hands, face
 - No cancerous
 - Just document
- Seborrheic Keratosis
- Purpura Senile (don't assume abuse, assess medication)
 - Most common in arms
 - Ask patient if is taken aspirin or Plavix?

• **Purpura: Petechiae and Ecchymosis**
Reddish-purple, don't blanch
IMPLICATIONS: bleeding in GI, brain.
 Monitor for melena



Atopic Dermatitis (Eczema)

- Allergic background
 - Stress increases symptoms
- Pruritus → excoriations → infections
- Treatment:
 - Avoid heat and hot baths
 - Keep skin hydrated and lubricated
 - Avoid bubble baths, wool or latex
 - Avoid hydrocortisone
 - Prevent secondary infections (scratching)
 - Fingernails short, cotton gloves



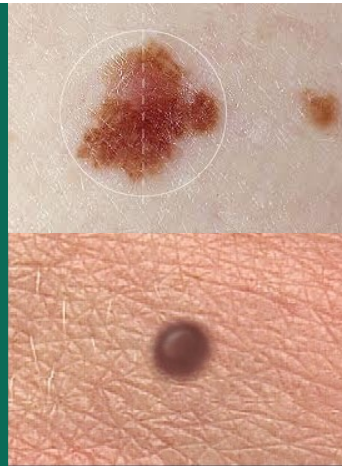
Psoriasis: “Silvery scales”

- Autoimmune, not contagious
- Stress exacerbated
- Disturbed body image
- Psoriatic Arthritis
- Social Isolation
- Treatment
 - Sunlight improves the lesions
 - UV therapy (PUVA)
 - Topical steroids (no systemic),
 - Immunosuppress: Methotrexate, Adalimumab, Etanercept



Malignant Melanoma and Dysplastic Moles (Nevus)

- ABCDE rule
 - A—asymmetry
 - B—border
 - C—color
 - D—diameter > 6 mm
 - E—elevation and enlargement
- Risk factors:
 - Family history
 - Sunlight exposition: SPF > 15
 - Dysplastic nevi (ABCDE)
 - Biopsy



Acne

- Comedone (black heads), papules and pustules
- Diet has not influence in acne
- Treatment:
 - Topical: benzoyl peroxide, azelaic acid, topical antibiotics
 - Systemic antibiotics: tetracycline
 - Isotretinoin: (for acne and wards)
 - Teratogenic
 - Hypertriglyceridemia
 - Avoid vitamin A supplements

Neurofibromatosis

- Hereditary
- Café au lait spots
- Neurofibromas

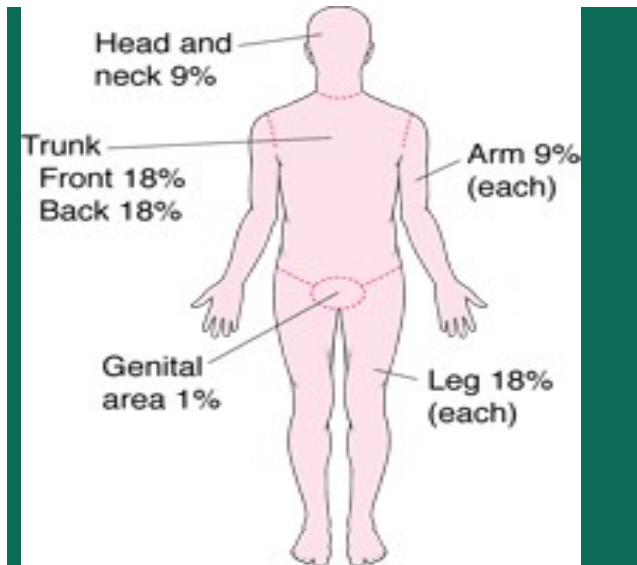


Candidiasis

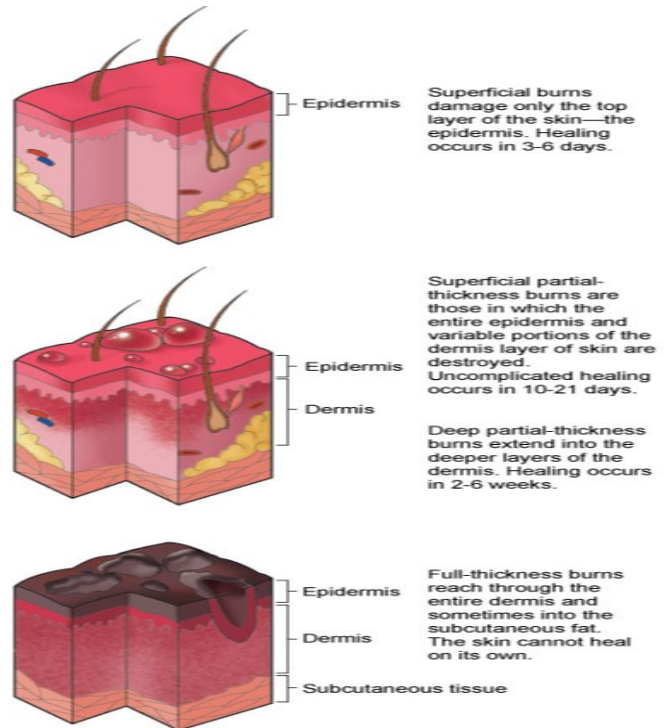
- Risk factors:
 - Decompensated diabetes mellitus
 - Immunosuppression (AIDS, steroids)
 - Broad spectrum antibiotics

- Mucosa: Vaginal, Thrush, Esophageal candidiasis
- **Nystatin:**
 - Swish (bucches) and swallow (need swallow because the esophagus also has infection)
 - Don't drink water 1 hour
 - Vaginal: recumbent 20 min
- Fluconazole: One dose PO

Burns



Effective Tx: patient urine > 30



Complications in burns: K^+ ↑, Myoglobinuria

First Aid: ABCs, cover burns

Emergent Phase:

- **Fluid shift:**
 - Capillary leak syndrome: first 12-24 hr
 - Compartment syndrome:
 - Remove jewelry and clothing
 - **Escharotomy**
- **Hyperkalemia**
- **Myoglobinuria** (dark brown urine)
- **Acute renal failure**
- Curling stress gastric ulcers
- **Fluid replacement:**
 - **Lactated Ringer**
 - Parkland formula: $4 \times \text{Kg} \times \text{TBSA}$
 - $\frac{1}{2}$ first 8 hours
 - $\frac{1}{2}$ remaining 16 hours
 - Keep urinary output > **30 ml/h**
- **Electrical Burns:** Cardiac dysrhythmias (need monitor troponin, need Hospitalization)
- **Carbon Monoxide:**
 - Cherry red color, normal O₂ Sat.

Thickness: redness

Blister: partial burns, pain, curable

Dry, hard skin: full burns, painless

Chemical Burns Steps:

1. STOP burning process, remove any unknown substance from the body by decontamination.
2. Remove burned clothing and jewelry
3. Assess ABC (airway, breathing and circulation)
4. Cover the wound with clean cloth

- Give oxygen 100%

Burns of the Respiratory System

- Burns in head and face
- Singed nasal hair or eyelashes
- Burned beard or eyebrows
- Blisters on mouth
- Hoarseness or brassy cough,
- Airway edema and stridor
- Wheezes, bronchospasm
- Drooling

Priority: Airway



FROSBITE (Hypothermia, patient can die), most frequent in fingers, nose, ears

- Interventions:
 - Rewarm rapidly with a warm water 37-39°C (98.6-102.2°F).
 - Manage hypothermia:
 - Warmer oxygen
 - Warmer IV
 - Warmer blankets
 - NPO
 - Give opioid analgesics
 - Cover with bulky dressing
 - Monitor for compartment syndrome
 - Prophylaxis tetanus
 - Prepare for surgical debridement

Drug Induced Steven Johnson Syndrome

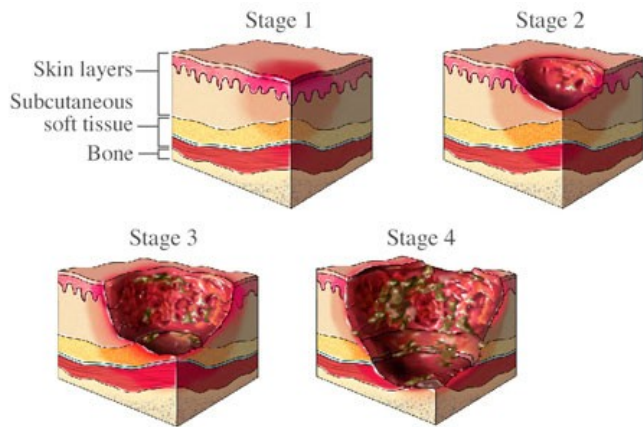
- Antibiotics (Sulfas)
- Anticonvulsants (Phenytoin)
- Allopurinol

Rash: "Target" lesions, Blisters

Discontinuation of the associated drug is critically important.

Pressure Ulcers

- Causes: pressure, friction, shear, maceration, malnutrition (albumin)
- Staging
- Nutrition: protein, vitamin C, Zinc
- Healing through second intention (granulation tissue)
- Dressings:
 - Don't leave ulcer open to air
 - Dressings are changed prn: 3-7 days
 - No gauze or wet to dry
 - Transparent films: protection (stage I)
 - Hydrogels: provide moisture (dry wounds)
 - Hydrocolloid: protect, absorb, moisture
 - Alginate & foams: absorption/maceration
 - Wound-Vac: foam must be shrunk



Question: When prevent pressure injury? **When the skin is blanching erythema**
Shear, friction, maceration and donuts

Shear and



NO-NO



Maceratio



DON'T MASSAGE SKIN!!!

Braden Score: the lower score, the higher risk

Braden Risk Assessment Scale					
Sensory/ Mental	Moisture	Activity	Mobility	Nutrition	Friction/ Shear
1. Totally limited	1. Constantly moist	1. Bedfast	1. 100% immobile	1. Very poor	1. Frequent sliding
2. Very limited	2. Very moist	2. Chairfast	2. Very limited	2. < ½ daily portion	2. Feeble corrections
3. Slightly limited	3. Occasionally moist	3. Walks w/ assistance	3. Slightly limited	3. Most of portion	3. Independent corrections
4. No impairment	4. Dry	4. Walks w/out assistance	4. Full mobility	4. Eats everything	
Total Braden Score _____ 15-16 Mild Risk 12-14 Moderate Risk <12 High Risk 15-18 is considered Mild Risk for those > 75 years					

HEEL PRESSURE ULCER: DO NOT REMOVE STABLE SCHAR



Unstageable

Tx: schar debridement by Surgery in others areas, no on heel

NOTES:

Integumentary changes in older adult skin include:

- Transparency and fragility
- Thinning of subcutaneous layer (risk for hypothermia/pressure injury)
- Loss of tone and elasticity
- Decrease sensation, hair thinning, changes in hair color
- Decrease perspiration and dry skin
- Skin dryness and delayed wound healing
- Changes in pigmentation, age spots, liver spots
- Overgrowth of capillary (benign cherry hemangiomas)
- Thickening of nails
- Complications: potential infection, allergic reaction or injury

Class 19. WOMEN'S HEALTH, MATERNITY AND NEONATE

Changes during Pregnancy

- Increase vaginal discharge, breast size
- Oxygen consumption increase 20 % and RR
- Increase circulation volume
- Pressure of the bladder cause urinary leakage

Signs of Pregnancy

- Presumptive: Amenorrhea, fatigue, nausea & vomiting, breast changes, quickening, urinary frequency.
- Probable: Goodell's sign, Chadwick's sign, Hegar's signs, ballottement, positive pregnancy test (serum urine), Braxton hicks ctx.
- Positive: hearing fetal tones, visualization of the fetus, and palpating fetal movement, visualizing fetal movements.

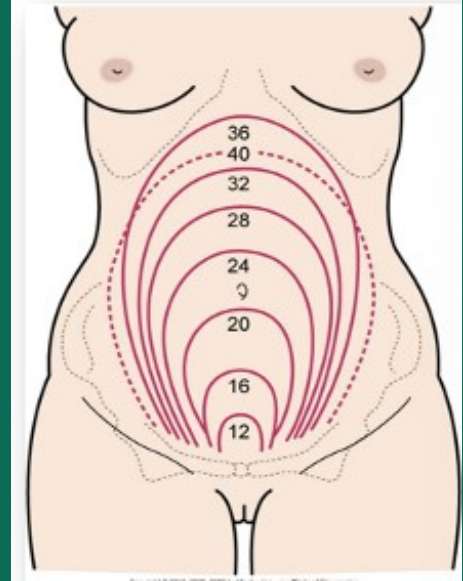
Chadwick sign is a bluish discoloration of the cervix, vagina and vulva

Goodell's sign is a significant softening of cervix.

Hegar's sign is compressibility and softening of the cervical isthmus (segment between uterus and vaginal cervix)

NORMAL PREGNANCY:

- **Fundal height:**
 - At 20th week: umbilicus.
 - Fundal height in centimeters equals fetal age in weeks $\pm$ 2 cm
- **Quickening: Fetal movements: 16-18 weeks**
 - If not kicks, FHR means baby, is death. NI: RN needs to auscultate FHR.
 - Kick count: Report when < 10 kick/2 hours. Expected: 10 kicks in 2 hours or less
Unexpected: 3 kicks in 1 hour, the RN needs to do non-stress test, call HCP
- **Nägele's rule: Time to reach 40 weeks of gestation (LMP - 3 months) + 7 days**
 - LMP: May 23rd: Date of confinement?
- **Cardiovascular system:**
 - Increased volemia and cardiac output
 - Pulse or HR may increase 10-15 beats/min.
- **Amniocentesis:**
 - Before 20 weeks (malformations): full bladder
 - After 20 weeks (lung maturity): empty bladder
- **Weight gain 25-35 # (300-500 Kcal extra)**



After 38 weeks the uterus height decreases because the baby change position to head down

GERD: Morning sickness

- NI: RN can administer thumbs in pregnancy for heartburn.

On a trimester basis in a woman with normal pre-pregnancy weight:

- First trimester: 1 - 4.5 pounds
- Second trimester: 1 - 2 pounds per week
- Third trimester: 1 - 2 pounds per week

Ultrasound:

- Gestational age
- Location (ectopic)
- Placental insertion
- Fetal wellness
- Amniotic Fluid
- Malformations

Leopold Maneuvers

* Start off by looking at the belly from the feet - the fuller side is often the side with the legs



1. Fundus

Facing the head
Figure out if it's the head or butt



2. Sides

Facing the head
What side are the legs and arms?



3. Presenting Part

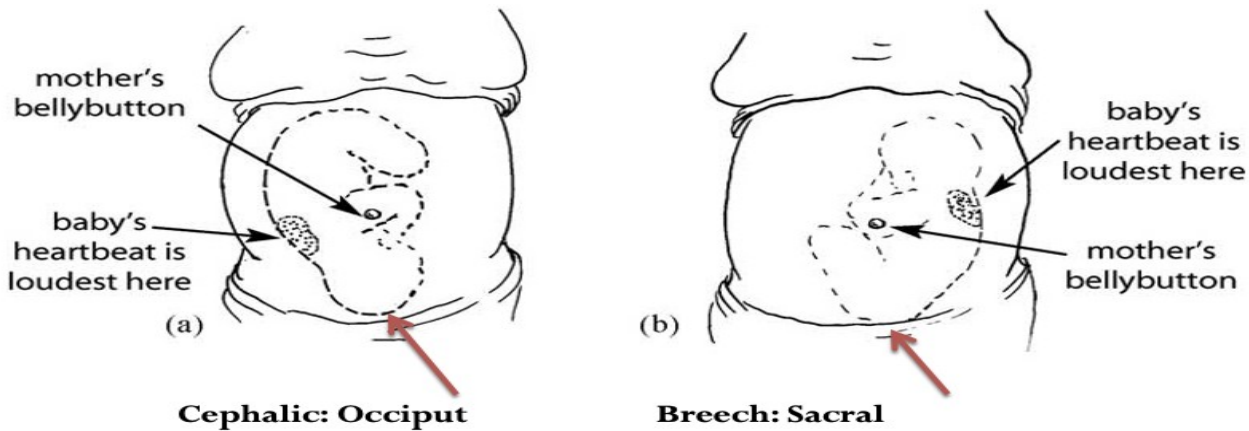
Facing the head
Breach vs cephalic?



4. Pubic Symphysis

Facing the feet
Assess descent into pelvis

Localization of FHR: Cephalic Vs. Breech



Baby HR can be listening on the back

Labor

- Stages:
 - First stage:** first contraction until complete dilation: *less than 24 hours*
 - Latent Phase: until 3 cm
 - Active Phase: **Opioids or epidural** could be started, **Keep bed rest lateral position.**
 - Transition Phase: Contractions every 2-3 min lasting 60-90 sec. *After 90 seconds STOP Pitocin, baby distress*

Latent phase: 0-3 cm	Contraction: 10-30 sec	Long: 5-30 min
Active phase: 4-7 cm	Contraction: 40-60 sec	Long: 3-5 min
Transition phase: 8-10 cm	Contraction: 60-90 sec	Long: 1 ½ - 2 min

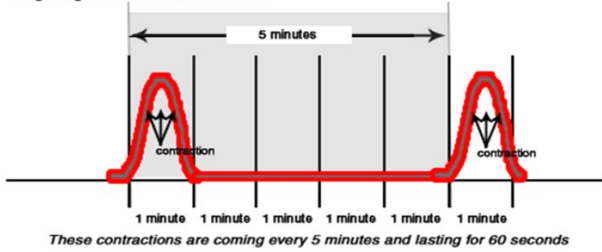
- **Encourage to drink fluids, void every 1-2 hours, avoid supine/sitting position (lateral)**
- Second Stage:** full dilation (10 cm) to birth, **Time to PUSH!** (30 minutes to 1 hour)
- Third Stage:** delivery of placenta: *less than 30 minutes*
 - Signs of placenta detachment: cord descends, uterus globular, gush of dark blood
- Fourth Stage:** 2-4 h postpartum. **Bleeding can be an issue**

In the early stages of labor, narcotics are known to decrease the strength of contractions and slow dilatation of the cervix. If given too late, they can depress baby's breathing. The best time to administer narcotic pain medication during labor is when your labor is very active (6-8 centimeters), just before you enter transition

During labor:

- Sutures and fontanels allow baby pass through the pelvis
- Fetal attitude is normally one of flexion
- Fetal occiput is in the left lower quadrant of the mother pelvis

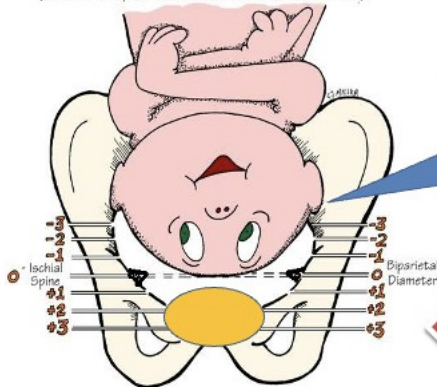
DURATION: beginning to end of one contraction
FREQUENCY: beginning of one contraction to the beginning of the next contraction.



Meconial fluid

NOTE: If meconial fluid is dark, notify HCP, means baby is not good

FETAL STATION (Relationship of Fetal Head to Mother's Pelvis)



Station 0 = "engaged"
 Station + 4 = "crowning"

Distended Bladder

I'm At Zero... From Here It's All Positive... I'm On My Way Out!!!

Effacement - the gradual thinning, shortening and drawing up of the cervix measured in percentages from 0 to 100%.

Borramiento - el adelgazamiento, acortamiento y encogimiento gradual del cervix medido en porcentajes del 0 al 100 %.



Dilation - the gradual opening of the cervix measured in centimeters from 0 to 10 cms.

Cervix before/after birth:

Dimple like os.

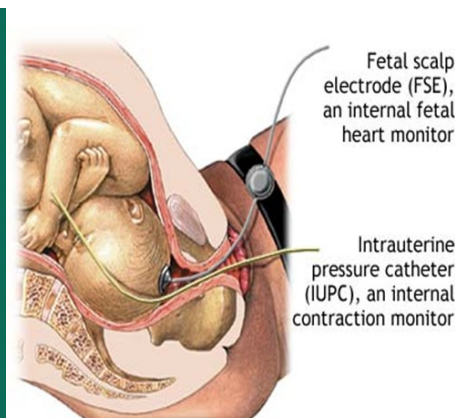
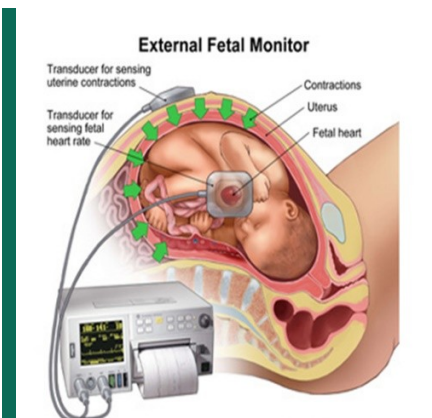
Slit like os



Never vaginal

Vaginal

Contraction Test: External vs Internal Fetal Monitoring



Use when mom is not in labor

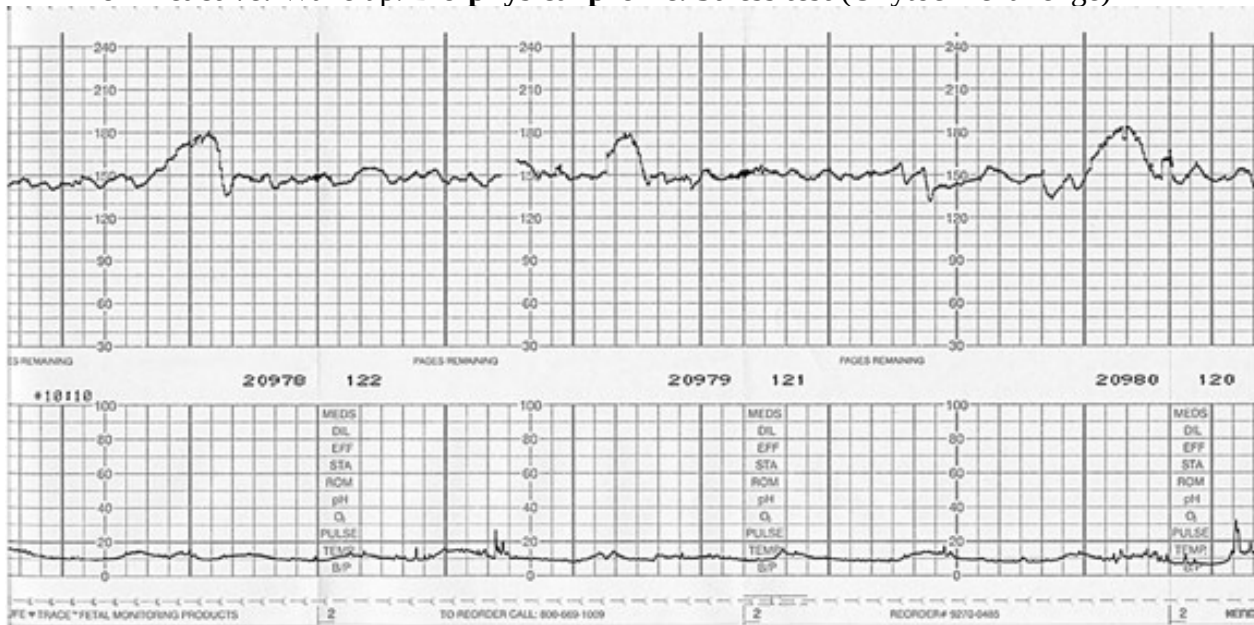
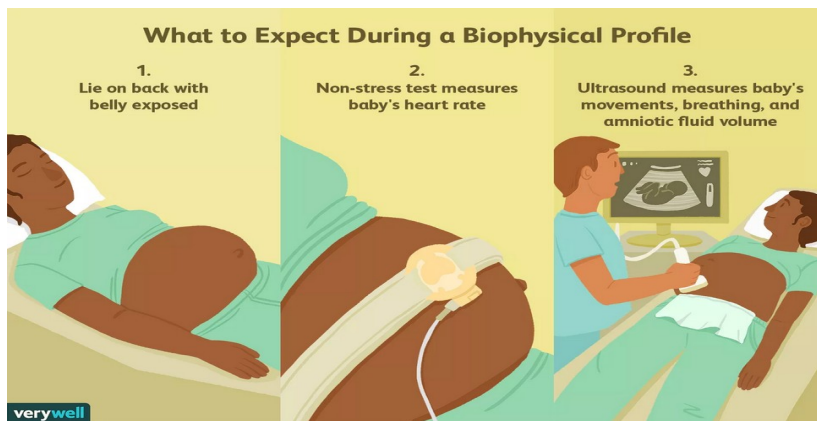
Internal fetal monitor (Do not use in mom w/HIV), requires dilation 2-3 cm. Used with or without labor

How are the membranes in internal fetal monitoring?

Variable = Cord compression or Cord prolapse
 Early = Head compression
 Acceleration = O.K.
 Late = Placenta Insufficiency

NON-STRESS TEST: Accelerations (2/20/15/15)**REACTIVE: GOOD. NON-REACTIVE: MORE TEST**

- Normal fetal response to movement or contractions. Ambulatory. No contractions. External monitorization
- Non-Stress Test: Desired findings: **Variability and Accelerations = Reactive = Good!**
- **Non-Reactive:** Wake up. **Bio-physical profile. Stress test (Oxytocin challenge)**

**Oligohydramnios:** can be big uterus or malformation

Biophysical profile		
BPP score	Interpretation	Management
8-10	No fetal asphyxia	Repeat testing at weekly interval
6	Chronic asphyxia	If >36 weeks- deliver
4	Chronic asphyxia	If ≥36 weeks deliver, if <32 weeks repeat testing in 4-6 hours.
0-2	Certain asphyxia	Test for 120 min—Persistent score ≤4—deliver regardless of gestational age.

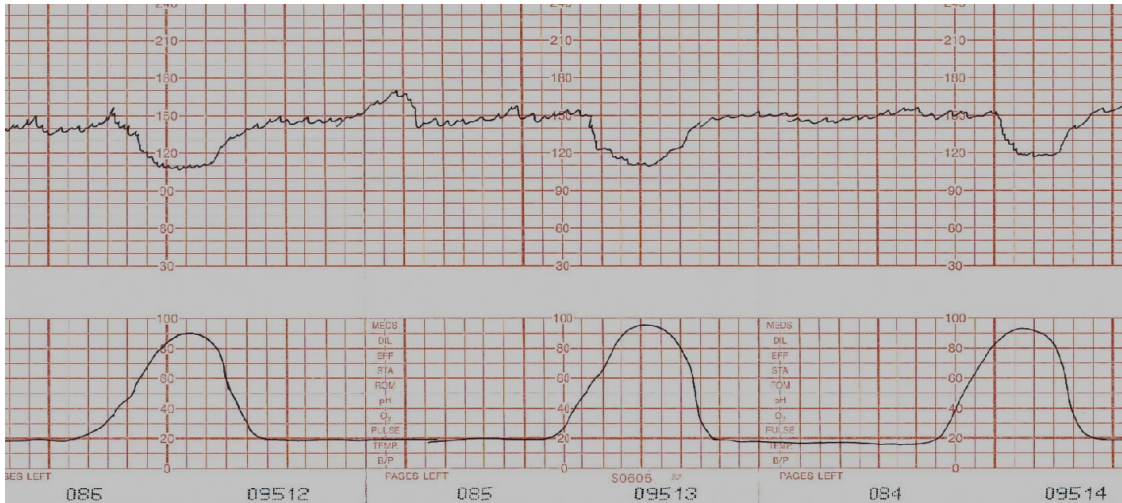
STRESS TEST

- **Negative: good. Positive: concern** (placental insufficiency)

- High risk pregnancy, usually done in hospital, after week 32nd, supine tilted to left, external monitor, IV oxytocin
- **Negative:** no late deceleration. Variability and Accelerations are OK (negative)
- **Positive:** late decelerations (placental insufficiency, induction or C section ASAP)

Monitorization in Labor: Early Deceleration

- Nadir coincides with the peak of the contraction
- Due to compression of the fetal head by the cervix, eliciting a vagal reflex
- FHR returns to baseline at same time the contraction ends
- Intervention: continue monitorization

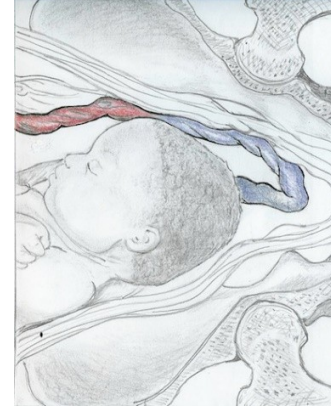
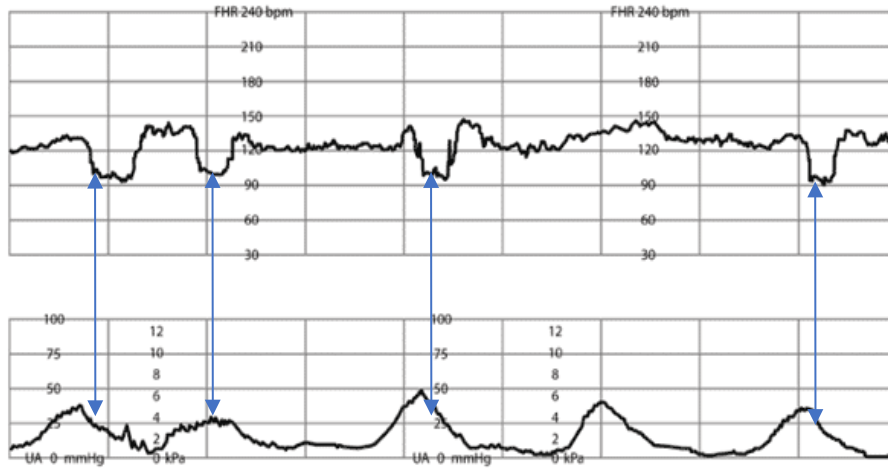


Monitorization in Labor: Late Deceleration

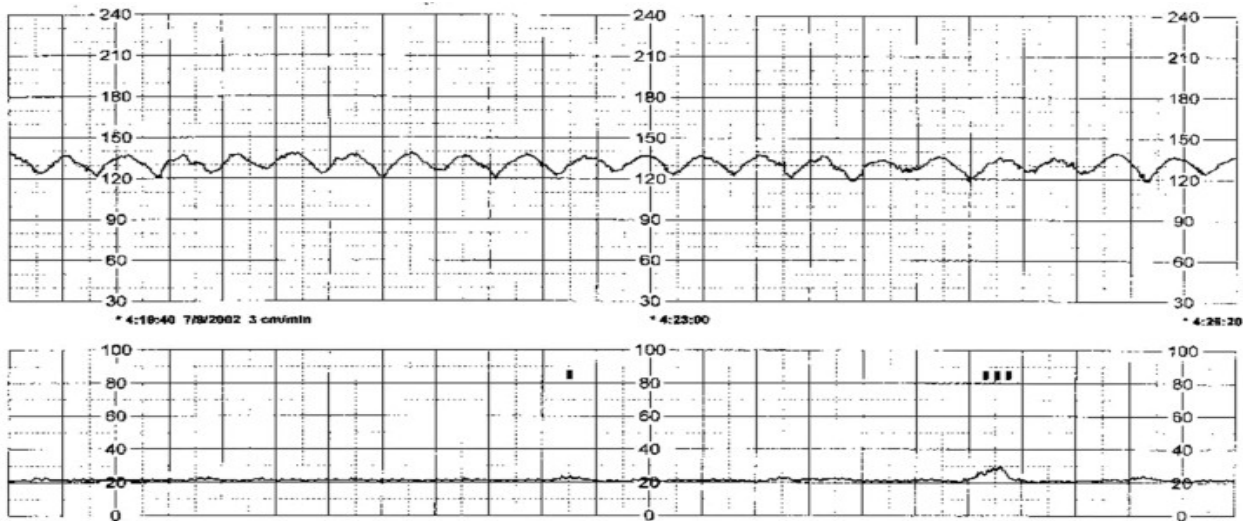
- Nadir occurs after the peak of the contraction.
- FHR returns to baseline **AFTER** the contraction ends.
- Is usually due to utero-placental insufficiency and fetal hypoxia.
- Intervention:
 - **Stop oxytocin**
 - Left lateral
 - Oxygen (10 lpm)
 - Inform MD (**STOP**)

Monitorization in Labor: Variable Decelerations: happens FHR around 120 or less

- Notice “V” shape
 - Don’t confuse with “variability”
- Umbilical Cord Compression
 - Don’t confuse with cord prolapse but it must rule out
- Interventions:
 - Change mother positions (recommended left side or chest knee)
 - Stop oxytocin if severe or persisting
 - Administer oxygen
- Inform MD if persists



SINUSOIDAL FETAL HEART RATE PATTERN



SHR is a rare occurrence. A true SHR is an ominous sign of fetal jeopardy needing immediate intervention. The correct diagnosis of true SHR pattern should also include fetal biophysical profile

Fetal Monitoring: Summary

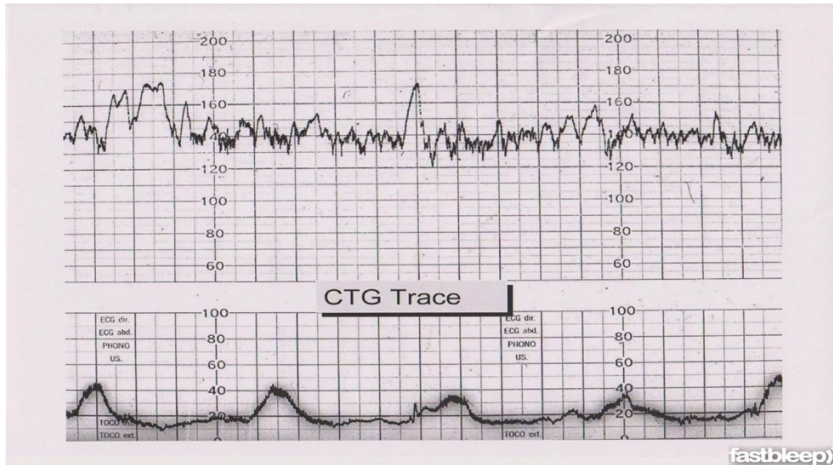
- **Non stress (no contractions) test:**
 - **Reactive** is reassuring (good). **Non-reactive (?)**: more tests needed
 - **Reactive:** Accelerations 2/20/15/15: good! **Non-reactive:** no accelerations
- **Contraction (stress) test:**
 - **Negative:** good, no late decelerations. **Positive:** bad, late decelerations
 - **Early decelerations:** fetal head is pressed against woman's pelvis
 - **No intervention:** continue monitorization
- **Late decelerations:** utero-placental insufficiency: **bad, bad!**
- **Interventions: STOP**
 - Stop oxytocin, **Turn** mother to left side, **Oxygen** by mask (10 lpm), and call the **Physician**
- **Variable decelerations:** umbilical cord compression inside uterus:
 - **Change maternal positioning**
 - *Do not confuse with prolapse of umbilical cord*

2/20/15/15: at least 2 accelerations in 20 minutes of at least 15 beats/min, lasting at least 15 seconds.
Nursing Interventions: The usual priority is as follows:

1. Discontinue oxytocin if infusing.

2. Assist woman to lateral (side-lying) position.
3. Administer oxygen at 10 L/min by nonrebreather face mask.
4. Correct maternal hypotension by elevating legs.
5. Increase rate of maintenance intravenous solution.
6. Palpate uterus to assess for tachysystole.
7. Notify physician or nurse-midwife.
8. Consider internal monitoring for more accurate fetal and uterine assessment.
9. Assist with birth (vaginal assisted or cesarean) if pattern cannot be corrected.

Which intervention should take the nurse in this case? **Variability and accelerations. Continue monitorization and document.**



Rh Compatibility

Coombs = antibodies

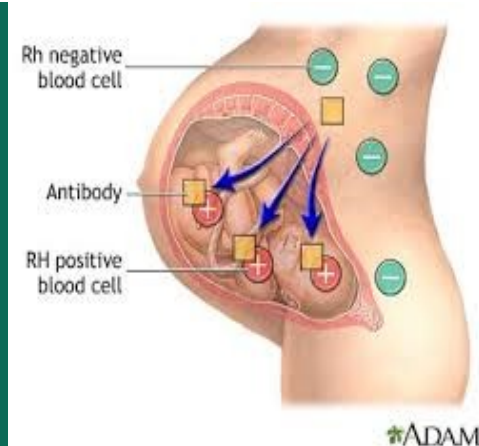
- Indirect: mother's blood
- Direct: newborn's blood

All Rh- pregnancy, Coomb's - test requires Rhogam shot in each gestation:

- 23-26 week of gestation
- After amniocentesis, chorionic villi sampling (CVS) or trauma
- Within 72 hours of blood exposition (IM) and birth

Coomb's + (make this test after 1st pregnancy)

- COOMB'S TEST + : NO RHOGAN (Injection)
- Fetal anemia and jaundice
- Unconjugated bilirubin can produce Kernicterus:
 - Phototherapy
 - Blood exchange
 - Albumin
 - Phenytoin



Test to the mom

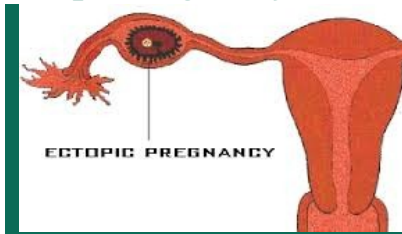
Rhogam admin to mom Rh⁻ and Comb's

Rhogam shot (injection) of Rho (D) immune globulin is administered for all Rh(-) mothers after delivery to prevent Rh Hemolytic disease in future pregnancy. Also needs to be administer to women after:

- Ectopic pregnancy

- Therapeutic abortion
- Miscarriage
- Amniocentesis
- Uterine bleeding

Ectopic Pregnancy

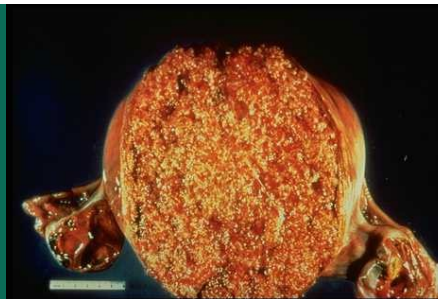


- Main risk factor: PID
- PID = Chlamydia (caused)
- S/S: sudden pelvis pain, hypovolemic shock, sharp ABD pain and shoulder pain, unilateral LQ pain, after 4-6 weeks, rigid and tender ABD.
- Tx: methotrexate
- Hysterosalpingography: check fallopian tubes

Trophoblastic Disease: Hydatidiform Mole and Choriocarcinoma

Assessment

- Non-detectable fetal heart rate
- Vaginal bleeding
- Fundal height greater than expected
- Snowstorm pattern on ultra-sonogram



Interventions

- Curettage
- Monitor hCG for one year or more
- Risk for Choriocarcinoma
- Avoid pregnancy for one year (monitoring if cancer growth)



DM in Pregnancy and Gestational Diabetes

- Pregnant previously diabetic mother
 - Only insulin is safe for fetus.
 - During first trimester, maternal insulin needs **decrease**
 - During second and third trimesters, maternal insulin needs **increase**
 - While **breastfeeding glucose decreases (risk hypoglycemia)**, a snack should be added.
 - Macrosomia and congenital malformations **if glucose is not controlled**
 - Newborn of diabetic is at risk for **hypoglycemia, hyperbilirubinemia, seizures**, and **hypocalcemia**. Risk for resp. insufficiency because immaturity lungs.

Mother Diabetic

Insulin (decreased)

II, III trimester, Insulin Increase (Hyperglycemia)

I trimester
(hypoglycemia)

Delivery, Insulin decrease, risk for baby hypoglycemia after cut the cord.
Mom needs eat before breastfeeding

- **Gestational diabetes**

- Appears in second trimester of pregnancy and disappear after birth
- Risk factors:
 - Older 25
 - Family history DM
 - Obesity: BMI > 30
 - Previous fetus weighing > **9 pounds (macrosomia)**
- Two steps screening: first glucose challenge test at **24-28 week**
 - Doesn't require fasting (normal diet prior test)**
 - Glucose: 50 g. (drink with water and wait 1 hour)
 - If > 140 mg in one hour: second step: glucose tolerance test 3 hours (fasting, drink 70 g and drawn blood every hour)
- Most gestational diabetes is controlled with diet (**dietitian**).
- Only insulin can be used if hypoglycemic drug is required

Preeclampsia and Eclampsia

• Hypertension (after 20th week), proteinuria, and severe edema • Risk Factors: ◦ Previous chronic hypertension or renal disease, lupus, sickle cell ◦ Nulliparas or >10 years since last birth ◦ Age > 40 years ◦ Obesity and diabetes mellitus ◦ Gestational diabetes ◦ Multiple Gestation ◦ Family history ◦ African American ◦ In-Vitro fertilization • Red flag for eclampsia: ◦ Epigastric pain RUQ, retina detach, rupture membrane, blood in liver ◦ Severe edema face and hands ◦ Eclampsia can appear after birth	• HELLP: coexistent severe preeclampsia plus: ◦ Hemolysis: anemia and jaundice ◦ Elevated Liver enzymes: ALT or AST ◦ Low Platelets (< 100,000/mm³), possibly DIC ◦ Magnesium sulfate to prevent seizures: ● <i>Mg increases risk for uterine atony and fetal respiratory depression</i> ● <i>Flushing and sedation are expected</i> ● <i>Monitor patellar reflex (reflex hammer)</i> ● <i>Monitor urinary output, respiration, and LOC</i> ◦ Stop infusion of magnesium if: ● Urinary output < 30 ml/h ● DTR: patellar reflex < 2+ ● Mg levels > 8 mg/dl ● Respiratory rate < 12 rpm (calcium gluconate) ● Have available calcium gluconate antidote
--	---

	BP	Proteinuria	Seizure	Hyperreflexia	Other
Mild preeclampsia	> 140/90	Yes >300	No	No	Facial edema
Severe preeclampsia	> 160/110	Yes >500	No	Yes	RUQ pain, HELLP
Eclampsia (ME)	> 160/110	Marked	Yes	No	Renal failure

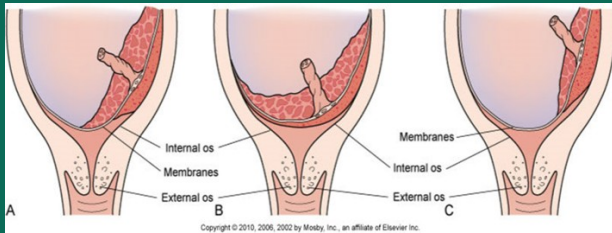
NOTE: Eclampsia is a **** Medical emergency**, and the RN needs to ensure patent airway, O₂ support, DIC management, delivery of fetus.

DTR: Normal (0, 1+, 2+), Preeclampsia (3+, 4+) no receiving Mg

Third Trimester Bleeding: Pain or no Pain

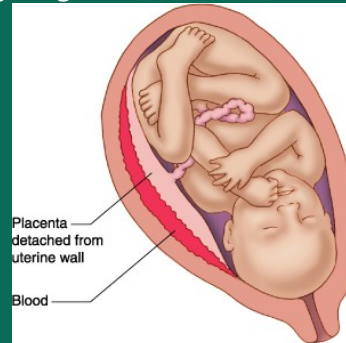
Placenta Previa

- Painless
- Bright red vaginal bleeding
- NO VAGINAL EXAM
- Diagnostic: Ultrasound
- No labor and dilation allowed
- I mom cannot delivery, C-section
- Never touch with finger the cervix (malpractice'negligence)
- Risk for death mom and baby



Abruptio Placentae (associated to preeclampsia)

- Painful (extreme)
- Uterine rigidity: uterine *tetany* (hard)
- Concealed hematoma or dark red vaginal bleeding
- Fetus is in severe distress
- DIC (d-dimer) +
- Risk for amniotic liquid pneumonia (placenta is baby lung)



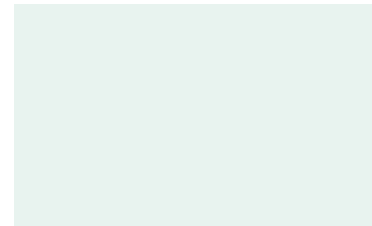
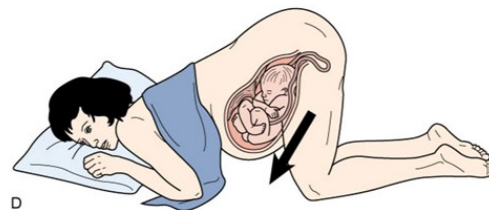
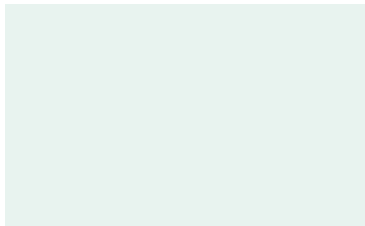
Cord Prolapse

- Occurs when membranes break
- Check FHR after membranes break and check if cord is out
- Do not try to reintroduce the cord
- Protect cord from desiccation
- Release pressure on cord
- Risk for baby death

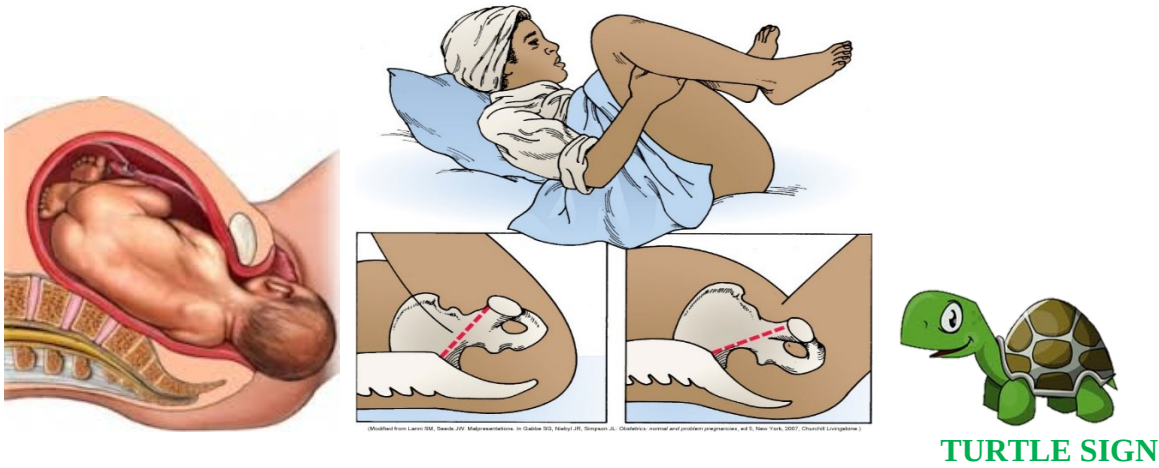


Umbilical Cord Prolapse

Knee-chest position



Shoulder Dystocia: McRoberts maneuver

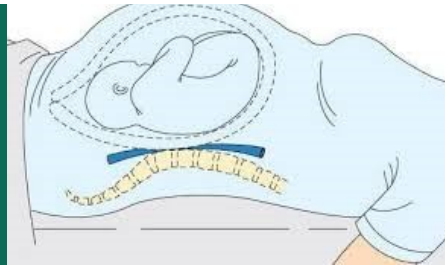


TURTLE SIGN

Hyperreflexia of Hips is recommended to open the way for baby delivery

Supine Hypotensive Syndrome (Vena Cava Syndrome)

- Compression of vena cava
- Reduced cardiac return:
 - Faintness, lightheadedness
 - Decrease placenta perfusion and fetal distress
- Interventions
 - Lateral recumbent position
 - Wedge under right hip during:
 - CPR
 - Surgery, C-section
 - Ultrasound or NST



Drugs in OB

Oxytocin: utero stimulant

Hold if frequency < 2 minutes (**tachysystole**) or duration > 90"

Hold presence of **fetal distress** (late decelerations)

Risk of uterine rupture: **previous C section**

Contraindicated in dystocia presentations (transverse, shoulder)

Water intoxication risk: brain edema (AMS)

Blood pressure elevation (preeclampsia)

Oxytocin in bolus can produce hypertension, prolonged infusion hypertension

- **Carboprost and misoprostol:** utero stimulant prostaglandins
 - Used for utero contraction (post-partum hemorrhage)
 - Nausea, vomits and diarrhea
 - Bronchospasm (asthma)
- Methyl ergonovine: utero-stimulant. Can cause hypertension
- Tocolytics:
 - Magnesium sulfate
 - Nifedipine: calcium channel blockers
 - Terbutaline
 - Beta 2 agonist used as tocolytic: utero inhibitor
 - Adverse effects: **Tachycardia**, dysrhythmia, hypertension, **hyperglycemia**
- Betamethasone
 - Induce fetal lung maturity

- Used premature labor, premature rupture of membranes, etc.

Contraindicated in pregnancy: NSAIDs (3rd trimester), statins, methotrexate, misoprostol, isotretinoin, tetracyclines, sulfonamides, aminoglycosides, ciprofloxacin, warfarin, benzodiazepines, finasteride, etc.

Post-partum

Assessment Post-partum Mnemonic: BUBBLE

B – Breast

U – Uterus

B – Bladder

B – Bowel

L – Lochia

E - Episiotomy

● Breasts

- Engorgement occurs in 48 to 72 hours in non-breastfeeding
- Breast-feeding will relieve engorgement
 - **NI:** Encourage the client to **wear a supportive bra**
 - **NI:** Instruct to express some milk to relieve the distension
 - **NI:** Apply ice pads between feedings
 - **NI:** Increase the frequency of breastfeeding
- Warmth if breastfeeding, cold if not
- **Uterine cramping expected during feeding**
 - **NI:** Administer pain medication

● Vital signs

- Temperature elevation until 100.4° F common in first 24 hours
- Bradycardia as low as 50 bpm is common in first week
- Orthostatic hypotension: risk for injury
- Leukocytosis as high as 25000 cell/mm³ during labor and even days postpartum, so WBC will be elevated.

● Uterus: post-partum: Firm fundus, 1-2 cm **above umbilicus, midline**. Then decreases 1 cm/day

- Administer oxytocic medication and analgesics for afterpains
- If uterus is boggy means the gallbladder is full or distended
 - **NI:** Instruct the client to **void and massage the uterus**
 - If the massage in the uterus is too strong can produce Uterine Inversion.
- Report unresponsive uterus to the HCP

● Lochia: no more 1 pad every 1-2 hours after birth,

- Lochia rubra for 2-3 days, then serosa and alba until a couple of weeks
- Lochia cannot smell
- If client saturated 1 perineal pad in 1 hour, Call HCP.

● Perineum: local ice for the first 24 hours.

- After 24 hrs instruct the client to use **sitz bath** (bano de agua tibia)

● Vaginal hematoma: painful lump in vaginal wall. Most occur when used forceps for delivery and it is very painful, call HCP.

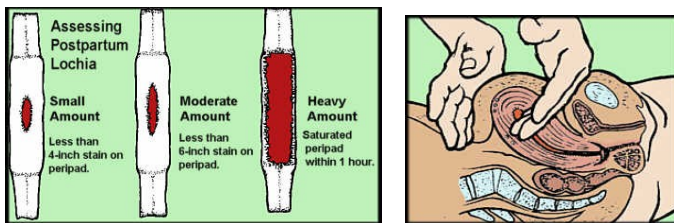
● Complications Postpartum:

- Bleeding

- Post-partum onset preeclampsia: MgSO₄ continues after birth
- DVT and PE
 - **NI:** will be check and monitor for DVT

Post-Partum Bleeding

- Abnormal bleeding: soaked perineal pad in one hour or less (> 500 mL vaginal birth, > 1000 mL C-section)
 - **Full bladders cause postpartum hemorrhage (uterus displacement)**
 - It can displace the uterus from the midline, usually to the right.
 - **Risk factors:** Overdistended uterus (macrosomia, multigestation, hydramnios), grand multiparity, drugs (MgSO₄, oxytocin use), prolonged labor, tissue retention, **infection, abruptio, preeclampsia.**
- Interventions for hypotonic uterus
 - Massage fundus, but no over massage
 - Assess for bladder distension, empty bladder first
 - Utero-stimulants: Oxytocin, methylergoline, carboprost
 - Water intoxication, hypertension, bronchospasm



Post-partum sexual activity:

- Abstain for intercourse until episiotomy is healed and lochia has ceased (3-4 weeks)
- Breastfeeding do not give adequate protection
- Oral contraceptive should not use during breastfeeding

Breastfeeding

- **Advantages:** passive immunity. **NCLEX support breastfeeding**
- **Frequency:** Breastfeeding newborn every 1.5 to 3 hrs.
- Hormones: Prolactin and Oxytocin
 - Estrogens inhibit production of milk (OC)
 - **NI: woman cannot take anti-conceptive pill during breastfeed, that cause STOP milk production.**
- **Rooting reflex: reflex to latching**
 - Rooting reflex is when the baby is stimulated touching close the lips and looking for the nipple
 - **Latching:** the baby needs to have inside his mouth the nipple, and 1/3 of the breast, cannot have dimpling when suction
- **Diet:** 500 Kcal extra, fluids
- Correct position: football
- **Nipple care:** Clean with warm water, no soap, dry to air
 - Let nipple air dry
- Mouth filled with areola (no dimpling)
- **Break suction before removing**
 - **Mom needs to put one finger inside the baby mouth to stop the suction.**
- 10' first breast, second until satisfied

- Begin lactation with **the last breast**. Mother needs to remember which the last breast was suctioned.
- **Contraindications:** HIV, drug addiction
- **Hepatitis B or C is not contraindication** to breastfeed unless nipples are wounded (blood)
- **Engorgement:** warmth & suction
- **Inverted nipples:** breast shields
- **Newborn Feces:** gold yellow and seedy, smell like yogurt
- How I know enough milk? 6-10 changes. If this occurs in a day **means mother is well hydrate**.



Mastitis

- Inflammation of breast frequently related to Staphylococcus infection
- Interventions
 - Good **hand washing**
 - **Maintain lactation** or use of breast pump every 4 hours
 - Good breastfeeding technique
 - **Supportive bra**

Newborn Priorities

- **Airway and oxygenation**
 - First: suction mouth, open airway
 - Second: after suction, dry baby and swallow
- **Heat loss:** dry, wrap, and use radiant heat (97.5 - 98.6 °F)
 - Newborn can't shiver.
 - Loss of heat: *evaporation* (wet), *radiation* (naked), *conduction* (cold walls), *convection* (drafts)
 - **Mottling, acrocyanosis, lazy blue: cold**
 - Dry and place skin to skin with mom as soon as stable
- Prevention of **hypoglycemia**, **Normal glycemia in newborn: 40-70**
 - Hypoglycemia: **<40**
 - Premature, post-term, LBW (**low birth weight**), child of diabetic mother
 - **Jitteriness** or tremors, poor feeding, lethargy, decreased muscle tone
- Nutrition
 - Early feeding
 - Adequate latching
 - Correct breastfeeding

Infant abduction prevention: electronic alarms, double check bracelets, parent education to identify staff authorized to transport infant, locked area, staff authorized with identification, only transport infant in bassinet, only give infant to mother, visitors identification in an out, check big packages

- Authorized Staff Badges: teach parents to recognize the unique badges for staff authorized to transport infants.
- Matching bracelets: parents wear matching bracelets with infants. Double check
- Transport infant within Hospital in bassinets, never arm carried
- Keep infant in direct line of sight all time even going to bathroom or taking naps
- Have a full front face photo taken of the baby
- Footprints: keep a copy
- Require identification to any home health care authorized
- Be careful when disclosing announces online of the birth and address

CARE OF THE NEWBORN

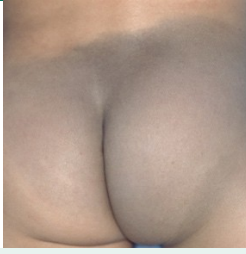
- Assessment
 - Monitor for nasal flaring, grunting, retractions, abdominal respirations
 - Monitor vital signs, signs of hypothermia or hypoglycemia
 - **Infant physical Assessment:**
 - Auscultate heart and lungs
 - Record heart and respirations rates
 - Palpate and percuss the abdomen
 - Examine eyes, ears, and mouth
 - Test for moro reflex
- Interventions
 - **Suction mouth first**, nose after the mouth
 - Dry newborn, maintain temperature stability: wrap and cap
 - Prevention of **neonatorum ophthalmopathy** (Chlamydia-gonorrhea) erythromycin ointment
 - Vitamin K IM within first hour of birth. Hemorrhage prevention
 - Hepatitis B vaccine
 - Keep newborn with mother to facilitate bonding
 - **Abduction prevention.** Ensure newborn's proper identification
- Normal Vital signs:
 - HR: 110 – 160 bpm (sleeping 100, crying 180 bpm)
 - RR: 30 – 60 rpm: **IRREGULAR**
 - T: 97.5 – 98.6° F (36-38° C)
 - **BS: 45 – 70 mg/dL. If BS <40 feed and recheck. Symptomatic or <35 → IV glucose.**
 - Weight at term: > 2500-4000 g (2500-3800 g): 6-9 pounds
 - Expect loss of until 10% of birth weight during the first week of life
 - Head circumference **32-37 cm (average 35 cm)**
 - Bilirubin: normal <6 mg/dL at birth. Physiologic jaundice usually <12-14 mg/dL (after 24 h.)
 - Hgb: 12-24 g/dL. WBC: 9,000-30,000 cell/mm³
 - **normal Hgb for newborn: 14-24**
 - **normal WBC count for newborn: 9,000-30,000**
 - **normal calcium: 7.4-14**
 - **normal BP for newborn: 80-60/45-40 at birth**

- normal length range: 48-52cm
- normal chest circumference: 30-35 cm

APGAR SCORING

INDICATOR	0	1	2
HR	Absent	<100	>100
RR	Absent	Slow, irregular weak cry	Good vigorous cry
MUSCLE TONE	Flaccid, limp	Some flexion of extremities	Good flexion, active motion
REFLEX IRRITABILITY	NR	Weak cry and grimace	Vigorous cry, cough, sneeze
SKIN COLOR	Blue	Acrocyanosis	Pink

- **Apgar:** 1 and 5 minutes (after birth, made this test twice: 1st 1 min after birth and 2nd at 5 min after birth)
- 8 to 10 No intervention required except to support infant's spontaneous efforts
- 4 to 7 Gently stimulate; rub infant's back; administer oxygen (by mask) to infant
- 0 to 3 Infant requires resuscitation

Umbilical Cord Blood Vessels	Millia (fat points)	Mongolian Spot
 Umbilical arteries Umbilical vein AVA (2 arteries, 1 vein)		
Lanugo 	Gynecomastia 	Molding 

Mongolian Spot = sacral melanocytosis (happens in native American and Asian)

NORMAL AND BENIGN FINDING

Epstein Pearls: White, shiny, round epithelial cysts on gums and palate
Normal finding

Harlequin sign: disappear soon, it is vasoconstriction

STRAWBERRY ANGIOMAS



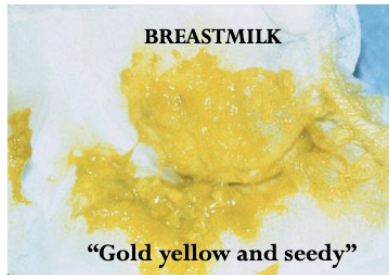
NORMAL FINDINGS



Erythema Toxicum Neonatorum (Baby Acne) and Telangiectatic Nevi



Vernix caseosa



BREASTMILK

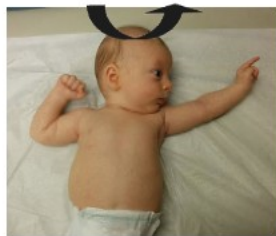
"Gold yellow and seedy"

TRANSIENT FLAT HEMANGIOMAS INDUCED BY MATERNAL ESTROGENS

NORMAL PRIMITIVE REFLEXES (Disappear between 4-6 months, except Babinski at year)



MORO REFLEX



TONIC NECK REFLEX



GALANT REFLEX



GRASP REFLEX



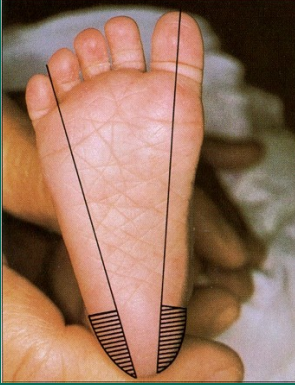
Step reflex



BABINSKI



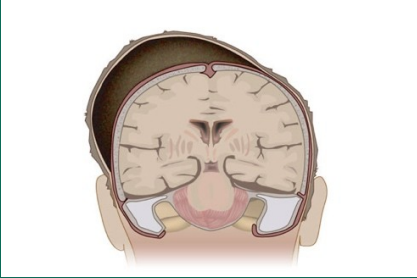
Parachute Reflex: **Appears at 6-8 months**

HEEL STICK: Lateral Areas

Never pock the baby on the center of the feet

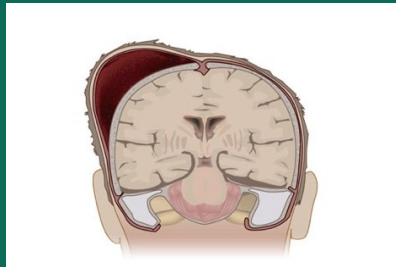
Neonatal Screening

- Deafness
- Congenital Hypothyroidism
- PKU
- Galactosemia
- Sickle Cell Disease
- Homocystinuria
- Maple Syrup Disease
- MCAD (Medium chain acyl-CoA dehydrogenase deficiency)
- Congenital Adrenal Hyperplasia
- Congenital Heart Disease

Caput Succedaneum: may cross over the suture

Parents need to be explained these lesions are benign and self-resolved.

Caput succedaneum disappears in days

Cephalohematoma: does not cross over the suture

Between periosteum and bone

Not painful

No related to intracranial hemorrhage

Assure parent of resolution in 4-6 wks

Jaundice: Physiologic Vs. Pathologic

- Diagnosed blanching skin
- **Pathologic:** ABO and Rh groups incompatibility occurs in the first 24 hours and is more severe: **Kernicterus**
 - Present if mother is Rh- and fetus is Rh+
 - Mother is group O, and fetus is A, B, or AB
- **Physiologic:** appears after 24 hours and is benign
 - Usually less than 15 mg/dL in preterm, 12 mg/dL in full-term newborns
 - Improvement with hydration (breast milk) and phototherapy
- **Breast milk jaundice:**
 - Variation of late onset physiologic jaundice (1st to 6th week of birth) in an otherwise healthy, breast-fed baby.
 - Recommendation: Breastfeed about 10 to 12 times each day to well hydrate.

Phototherapy (when bilirubin is > 15)

- Expose as much of newborn's skin as possible, except for eyes and genitals
- Monitor skin temperature frequently (axillary, aural, skin)
- Increase fluid intake breastfeeding more frequent (every 2-3 hours)
- Educate parents that stools and urine may be green
- Expect loose stools and need for frequent changes
- Feed and remove eye cover every 2 hours
- Don't apply oils or lotions in skin

- Blanket phototherapy doesn't require eye protection
- Sunlight therapy: 15-30 minutes of **indirect** sunlight before 10 am or after 4 pm
- Phototherapy is maintained until bilirubin is below 12-14 mg/dL
- Use pacifier for prevent SID



Parent Teaching

● Cord care

- Keep cord clean and dry
- Clamp removed at 24 h.
- Wash skin surrounding cord with soap and water (sponge) and let cord dry to air. Don't cover. No alcohol.
- Keep diaper folded below cord
- Usually falls after 10 days birth
Little bleeding could happen



● Circumcision care

- **Anesthesia: Pacifier with Sucrose**
- The yellow crust is normal
- Protect with **petroleum gauze** until healing
- Monitor for edema
- Call HCP if the baby does not have urinary output within 6-8 hrs after circumcision



Question: How the RN decrease the pain in a newborn? Administer glucose solution (not need Rx)

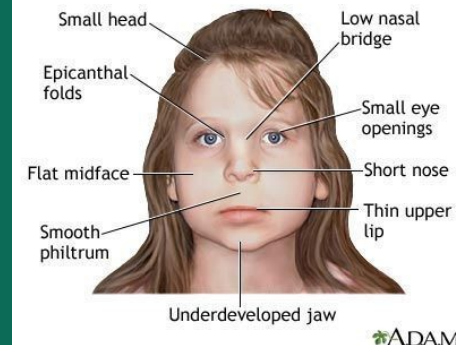
The Addicted Newborn

- **Description:** Newborn passively addicted to drugs that have passed through placenta
Addicting drugs: Heroin, Methadone, Cocaine
- **Assessment**
 - Opioid withdrawal is observed **12-24 hours after birth**
 - Irritability
 - Tremors
 - High-pitched cry
 - Extreme sucking of fists but poor feeding
 - Seizures
- **Interventions**
 - Monitor respiratory and cardiac status at the **time of withdrawal**

- Priority: no feeding: dehydration, hypoglycemia, etc
- Provide small, frequent feedings
- **Reduce environmental stimulation**

Fetal alcohol syndrome

- Low birth weight
- Small palpebral opening, epicanthic fold
- Short nose, low nasal bridge
- Very thin upper lip
- Small teeth, defective enamel
- Low implantation ears,
- Congenital heart defects
- Poor wake and sleep patterns.
- Tremors
- Seizures
- Interventions:
 - Vitamin deficiencies (thiamine)
 - Cuddling (give stimulation)

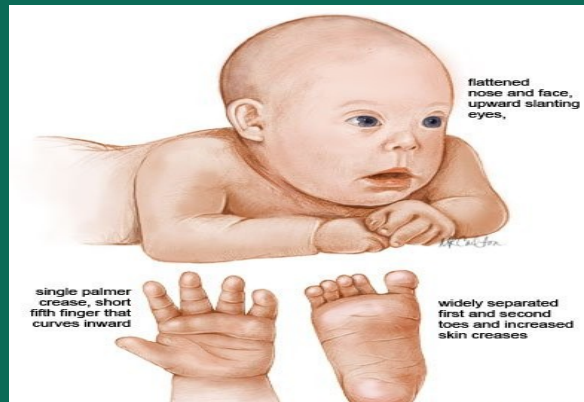


Phenylketonuria (PKU)

- Autosomal recessive (Caucasians)
- Deficiency of phenylalanine hydroxylase:
 - Increased blood levels of **phenylalanine** (amino acid from proteins) in blood and urine (urine musty odor)
 - Irritability, seizures and brain damage if not treated with diet
- Screening (Guthrie blood test or Guthrie bacterial inhibition assay):
 - **After 24-48 h of normal feedings**
 - Delayed if infant weighs less than 5 pounds
 - Could be done in urine 6 weeks after born
- Purpose of treatment: control intake of phenylalanine (dietary protein) along first 6 years and during pregnancy in adult women with PKU
- Diet:
 - Recommend breast milk (low in phenylalanine)
 - Use of mature breast milk or PA-free formula: Lafenalac®
 - Avoid cow milk and dairy, reduce intake of breads, beef, fish, poultry, cheeses, legumes, nuts, eggs
 - Calories provided by carbohydrates and healthy fats
 - Total restriction of Aspartame

Trisomy 21 (Down's Syndrome)

- Congenital, maternal age linked
- Decreased AFP
- Cardiac malformations: VSD
- Hypotonia
- Protruding tongue
- Epicanthic fold, slanted eyes
- Brushfield spots on iris
- Simian creases on palms
- Mental retardation
- Brachycephaly, dry skin
- Screening: AFP (alfa fetal protein): predict neural tube defects, done between 16-18 weeks, ↓ AFP suggest possible Down's syndrome



Also, redundant skin on back of neck and fat pat which is translucent to US

NOTES:

- Stepping reflex disappears at 4 weeks
- Begins drooling around 4 months
- Respond to own name 6-8 months
- Takes deliberate steps when standing 9-10 months
- Picks up bite size pieces of cereal 10 months

Anatomic changes for newborn as soon as they birth. It is happen when they left the mother thumb.

- ↓ pulmonary vascular resistance
- Closure the foramen ovale
- Closure of ductus arteriosus
- Closure of ductus venosus

Feces color in newborn:

- Dark green without odor if pass meconium
- If breastfeeding:
 - Orange – yellow stool each day soft
 - Gold – yellow and seedy and smell like yogurt
 - Expect 3-6 small soft stool each day

Newborn first bath:

- Waxy vernix carnosus protect newborn
- First bath is simple remove blood from birth process
- Parents can or not be in the first bath
- RN use pH neutral soap for cleansing to minimize skin injury

Toilet training: (18 months – 3 years)

- Bladder sphincter control (wake up dry from a nap)
- Able to adjust or remove own clothing
- Able to sit and wait for period of time (5-10 min)
- Parents able for child elimination process (weeks off, vacation, PTO)

Class 20. Psychiatric

MENTAL HEALTH Definition: A state of well-being in which every individual realizes his or her own potential, can cope with the normal stresses of life, can work productively, and fruitfully and is able to contribute to her or his community.

Arab Americans

- Don't assume, ask the client about his/her culture
- Large majorities are Islamic (Muslim)
- Child circumcised by Rabi before 8 days
- **Position bed facing toward Mecca (East).** They are praying 5 times a day on the floor
- Ramadan: fasting from sunrise to sunset. Administer medications at night
- Left hand used for toileting. Do not given nothing in the left hand (medication)
- Alcohol forbidden (cough syrups).
- **Pork is not allowed**
- Concordant sex healthcare worker (nurse/UAP)
- Traditional families: male member takes decisions
- Post-mortem
 - Members of the same sex of the death patient preparing the cadaver
 - Not allow cremation

Judaism

- Kosher food:
 - Food and utensils not touched or mixed
 - **Pork not allowed (gelatin)**
 - **Shellfish (crab, lobster) not allowed:** Fish with scales and fins are allowed
 - Dairy and meat are not eaten at the same time.
- Infant circumcised on the eighth day of life
- **Pain: Allow pain management, IV medications for pain is appropriate.**
- End of Life Issues:
 - Rabbi's presence.
 - A dying person or cadaver should not be left alone.
 - All organs need to be buried with the body. They do not donate organs
 - Prohibits cremation or autopsy
 - Life support, Do not take off

Asian Americans

- Illness: imbalance between **yin** (cold, internal, female, moon) and **yang** (hot, external, men, sun)
 - Yin (cold) illness treated with hot temperatures and cold food and vice versa
- Nodding or smiling doesn't always mean agreement or **understanding**
- Eye or physical contact could be inappropriate, it is a form to demonstrate respect
- Head shouldn't be touched without permission. Feet consider dirty
- Alternative medicine: acupuncture, cupping, coining, herbs and teas

Abuse: Report, and include referral of abuser for treatment (Mandatory by Law for children < 18 years old and elderly > 65 years old). Include the abuse on the treatment plan.

Child Abuse

- It is deliberately or intentionally inflicting physical injury
- Physical abuse is not the most common
- Abusive head trauma is usually an isolate event
- Malnutrition, dehydration
- **Indicators of Abuse: Physical, Emotional & Sexual**
 - **Any injury not consistent with history provided by the adult or child**
 - Overwhelmed parents, unplanned pregnancy, poverty, substance abuse, animal cruelty
 - **Pattern contact burns:** cigarette shapes or scalds from hot water
 - Multiple fractures in different stage of healing or fractured teeth
 - Bruises or bite marks in different stage of healing
 - Withdrawn or fearful from parents, absence of crying with procedures/strangers
 - Genital trauma or infection, STDs or pregnancy
 - **Shaken Baby:** Subdural hemorrhage, retinal hemorrhage, bulging fontanel, AMS, seizures, ICP
 - Recommend leaving crying child on crib for a while before holding him/her.

Elderly abuse/neglect: physical, psychologic and ECONOMIC

- If suspected: evaluate patient without parent/caregiver present, **report, and include referral of abuser for treatment**
- Always assess for domestic violence, refer victims to support groups, and develop a plan of help during crisis, include abuser in referrals

Abuse: Munchausen's Syndrome by Proxy

- Caregiver fabricates, exaggerates, or induces health problems in those who are in their care.
- Caregiver gains benefits and attention with it
- Mortality rate between 6% and 10%
- Child improve when caregiver is absent
- Death mostly from apnea (a common result of smothering) or starvation
- Suspected in a child with multiple unusual medical problems that don't respond to treatment

Don't break Defense Mechanisms until better coping strategies are learned

Denial: not accepting reality because is too painful, inconvenient	• I can stop drinking when I want
Projection: attributing one's thoughts or impulse to others	• You blame your children because you can't concentrate to study
Displacement: transferring feelings toward one person to another less threatening.	• You argue with your boss and then scream to your partner
Regression: returning to an earlier developmental stage	• A child bed wetting after being continent
Rationalization: attempt to justify unacceptable feelings	• I continue smoking to feel less anxiety, or I failed the test because I didn't have a good night sleep
Sublimation: replacement of unacceptable thoughts with other socially acceptable.	• Intense anger redirected to boxing, or a woman disappointed about not having children and is now volunteering at a

	children's center
Repression: unconscious block of unacceptable thoughts. Suppression: (is on purpose)	• A rape victim doesn't remember and is acting like nothing happen
Reactive formation: behavior exaggeratedly opposite of what one really wants to do (inappropriately)	• Cold attitude when you are really angry, or hate turned into love

Suicide

- Highest Risks:
 - **White, male, old, widow**
 - Depression
 - Previous attempt
 - Method lethality: Firearms at home
 - **Warning signs:** sudden improvement on mood, having a plan, giving away belongings, refusal to sign a non-harm contract.
- Eliminate any potential weapon or instrument to commit suicide
- Place with roommate
- **Place patient in constant observation 24 hours**
- All staff need to be informed and aware of suicidal risk.

Cognitive Therapy: based and focuses on how a person perceives or interpret an experience as a determinant for behavior.

NI: Encourage examine ways to reach the goals.

End of Life Care Problems

- **Dyspnea and pain:** elevate HOB, oxygen (mask poor tolerated), a blowing fan, morphine (for pain and dyspnea)
- Mouth of care, water-base lubricant for lips (don't use petroleum)
- Oral and lung secretions: rattle death sounds
 - Anticholinergic: atropine, scopolamine patches
 - Lateral position
- IV hydration, TPN, and enteral nutrition are not comfort measures (not needed)
- **Signs of impending death:** restlessness, Cheyne-stokes respirations, death rattle, mottled skin and cyanosis, decreased urinary output, hypotension, tachycardia, AMS, illusions, delirium, either insomnia or sleepiness
- **Spiritual needs:**
 - Catholics (**anointing of the sick**)
 - Islam (**facing Mecca, same gender handles the corpse**)
 - Judaism (**dying person should not be left alone. A family or near friend must stay with the cadaver from death until burial**)

Postmortem Care of Bodies

- **Close eyes and place dentures in mouth**
- Clean the body and place incontinent pads
- Remove external tubing and drains (**not in a legal case, and do not do if the patient death is a child the family can see what the team did for save his life**)
- Align the body but keep a cushion under head
- Pull a sheet up to neck to cover the body for viewing
- Allow family to see cadaver before autopsy. **Ask the family if they want to see the cadaver?**

- Give client's belongings to family
- Keep wrist band on place and attach two more labels to toe and outside the shroud
- Warn in the label if special biohazard care needs to be taken
- Document time of death, deposition of the body, and belongings
- Provide support and resources to the staff involved (if patient death is a child)
- Care in case of **stillbirths**

Grief

- Natural response to loss.
- Stages of grief, mnemonic: **DABDA**
 - **Denial, Anger, Bargaining, Depression, Acceptance.**
- Types of grief:
 - Normal grief
 - Complicated grief:
 - Delayed (denial)
 - Anticipatory grief
 - Traumatic (unexpected)
 - Disenfranchised grief (socially unacceptable)
- Nursing Interventions:
 - Assess stage of grief
 - Show empathy
 - Help identify and express feelings like anger or guilt
 - Identify pathological mechanisms of defense or complicated grief

Anxiety Disorders.

- Pathological anxiety disorders are present > 6 months: **Generalized Anxiety Disorder**
- **Panic attack:**
 - Perception grossly distorted, unable to problem solve.
 - Personality disorganized: inappropriate reactions
 - Risk for themselves and others
 - ↑HR, ↑BP, chest pain, hyperventilation
 - Respiratory alkalosis
 - Rule out organic condition (could be something more than nervousness)
 - Recurrent panic attack: **PTSD, phobias, and OCD clients**
 - **Nursing priority: client's safety**
 - Offer self, don't abandon: *"I will stay with you to make sure you're safe"*
 - Manage respiratory alkalosis (re-breathing)
 - Low impact physical activity decrease anxiety
 - Administer benzos
 - Provide privacy
 - Write down important information
- Mild: driving. Moderate: preparing for test. Severe: fighting

Post-Traumatic Stress Disorder (PTSD)

- History of traumatic event: war, rape, catastrophes
 - **Flashbacks:** intrusive unwanted memories
 - Panic attacks
 - **Defense mechanisms:** Denial, suppression, and repression
 - Depression. Suicidality

- Interventions:
 - Provide nonthreatening environment: *offer self*
 - *Therapeutic communication: restating, reflecting*

Anxiety Disorders

- Phobias

- An irrational fear of an object or situation → fear and avoidance → **panic** level anxiety
- Agoraphobia:
 - Fear of being alone in public places where it is difficult to exit or obtain help: open spaces, crowds, stores, bridges, public transports, most are outside home alone
- **Myophobia or microphobia:**
 - Pathological fear to germs
- Confrontation with phobic object is incorrect. Instead:
- **Systematic desensitization**

- Obsessive-Compulsive Disorder (OCD)

- **Obsessions:** persistent intrusive thoughts and ideas
- **Compulsions:** performance of rituals or repetitive behaviors
- Anxiety (panic level) develops if compulsion is blocked
- Allow performance of compulsion, **but limiting time to do it**
- Teaching is more effective immediate **after the compulsive behavior**

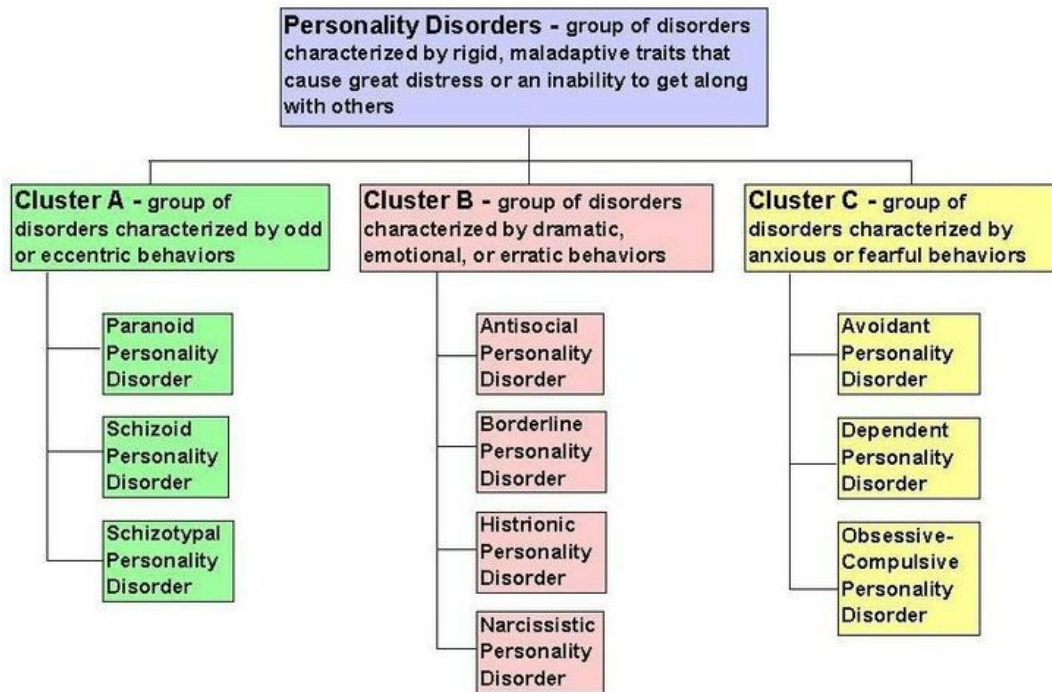
The following specifiers are used, according to the phobic stimulus present:

- Animal - Fear of dogs (cynophobia), cats (ailurophobia), bees (apiphobia), spiders (arachnophobia), snakes (ophidiophobia), or other animals
- Natural environment – Fear of heights (acrophobia), water (hydrophobia), or thunderstorms (astraphobia)
- Blood-injection-injury - Fear of needles or invasive medical procedures
- Situational - Fear of flying, elevators, or enclosed spaces
- Other - Fear of situations that may lead to choking or vomiting, in children, loud sounds or costumed characters

Anxiety: Somatoform disorders

- o **Somatization:**
 - Real symptoms that arise with emotional distress
 - Tension headache, IBS, fibromyalgia.
- o **Hypochondriasis:** Exaggerated worry with symptoms: multiple tests, doctor visits, and treatments
- o **Body dysmorphic disorder:** Exaggerated concern for minor defects in appearance: cosmetic surgeries
- o **Conversion:**
 - Neurologic symptoms like loss of movement, sensation, speech, or vision; or pain
 - Without organic basis: normal objective findings
 - Client doesn't look concerned: **La Belle Indifference**
 - **Primary gain:** decreased anxiety: E.g., hand paralysis the day before taking NCLEX exam...
- o **Dissociative Disorders**
 - Transient loss of memory, awareness, identity, or perception
 - Dissociative Amnesia:
 - o Selective memory loss for important personal information during disturbing events:
 - E.g., Survivor of a MVA who doesn't remember the accident until days after

- Dissociative Fugues
 - Client leaves usual place with no memory for his/her past or identity
 - E.g., An overwhelmed nurse travels to unusual place without memory of the way



Personality Disorders

- **Antisocial:** disregard for right of others, manipulative, **delinquents**, no remorse
 - **Conduct disorder in childhood:** disregard for others' rights, animal cruelty, property destruction, truancy, substance abuse, evolution to antisocial disorder, run away
 - **Oppositional Defiant Disorder:** child is negativistic, hostile, argumentative, annoying, but respects other's rights, is not cruel and not associated to antisocial personality disorder
 - Intermittent Aggressive Disorder:
 - Explosive outburst of anger and violence in which one reacts out of the proportion of the situation, with sympathetic discharge, road rage, domestic abuse, throwing or breaking objects, or other temper tantrums. Homicidal/suicidal risk.
- **Borderline:** very unstable personality, emotional lability, low self-worth, **impulsivity**, **self-mutilation**, **divisive** (good/bad), frequent bulimia, suicidal
- **Histrionic:** drama queen, emotional, **attention seeking**, inappropriate seductive behavior
- **Narcissistic:** feeling superior (grandiosity), need for admiration, lack of empathy, put others down, arrogant

Antisocial, borderline, maniac: frequency require confrontation (set limits), no bargaining.

Mood Disorders

- **Bipolar disorder:** episodes of mania and/or depression.
- **Major Depression:**
 - Inhibited mental and physical:
 - Self-neglect, hygiene, nutrition,
 - Anhedonia
 - Psychomotor retardation, inaction
 - Over-sleeping

- Low self-esteem: **suicide**
- **Interventions for Depression:**
 - Assess for suicidal ideation, safety
 - Never leave the patient alone is at risk of suicide
 - Assist with ADLs and attend to physiological needs (E.g, nutrition, rest)
 - Avoid presenting complex decision-making situations to client
 - Insist on participation in activities “It is time to go to the gym”
 - Provide structured environment and activities in which client can achieve success
 - **Onset change to depress for happy, the patient plan to suicide**
 - **NI: 1 to 1 observation, close nurse station**

o **Mania:**

- Elevated mood and **grandiosity delusions**
- Flight of ideas, talkative and physical restlessness
- Increased needs for rest, and high calories food
- Insomnia
- Increased risk for violence directed at others
 - Impulsivity,
 - External opposition or competence make them argumentative
- **Interventions for Mania:**
 - Implement limits setting and avoid manipulation/bargaining
 - Provide high-calorie finger foods, fluids
 - Provide physical, gross motor activities, outlets for tension
 - Avoid competitive games and concentration games (reduce span of attention)
 - Supervise self-administration of medication

Mood stabilizers: Lithium, Divalproate (Depakote), Carbamazepine (Tegretol)

Tx: Electro-convulsive therapy (ECT)

- **Contraindications:** Pregnancy, brain tumors, brain aneurysm, stroke
- Call HCP if the patient is taken **Alendronate**, this medication is for osteoporosis and cause muscle contractions.

Schizophrenia

o **Positive symptoms**

- Delusions (abnormal thought content)
 - **Paranoia:** unrealistic threat idea. “The doctors wants poisoning me with this pill”
 - The devil is forcing me to think on killing my child ‘ (thoughts insertion)
 - “An alien performed surgery and replaced all my blood with Kool-Aid without leaving a scar”
- **Hallucinations** (abnormal perception of **something that doesn’t exist**)
 - “I hear demons talking in my head (without any stimulus)
- **Illusion: wrong interpretations** of something really existing
 - “I hear demons talking outside (real nurses are talking in the hall)
- **Language disturbances:**
 - **Tangentiality:** wandering from the topic and never returning to it
 - o “Where are you from? “...”from England. They have fish and chips there. Fish have gills.”
 - **Clang association:** Rhyme: “I heard the bell. Well, hell, then I fell.”
 - **Echolalia:** Hello! hello, hello, hello...
 - **Flight of ideas:** Rapidly jumping from one topic to another nonrelated topic
 - **Word salad:** Looseness of association

- **Neologism:** made up words

o Negative symptoms

- Inappropriate affect: **flat affect (blunted) affect**
- Neglecting physical needs like **hygiene**
- **Anhedonia**, anergia, avolition (lack motivation), alogia (poverty speech)
- Social withdrawal

Thought insertion: others insert thoughts in their mind.

Thought broadcasting: other can read their thoughts

Interventions in Schizophrenia

- Active hallucinations
 - Ensure safety: command hallucinations.
 - **Acknowledge feelings** of fear or anger associated with hallucinations
 - **Don't reinforce hallucinations** (what is demon saying), **but don't challenge it** (demons don't exist)
 - Provide **reality orientation**
 - "I don't hear any voice, but I like to know what you are feeling"
 - "I don't see any blood coming out from the faucet, but it should feel scaring to you..."
 - Set limits (safety, end conversations conducting to loss of contact with reality)
- **Delusions**
 - Establish trust
 - Acknowledge feelings, but do not share, confront, or argue with the delusion
 - Manage paranoia delusions for medications and food
 - Avoid touching or staring to the client
 - Set limits (end conversations conducting to loss of contact with reality)
- **Catatonic Schizophrenia:** Basic needs (food, water, hygiene) must be covered.

Delirium Vs. Dementia

DELIRIUM	DEMENTIA
Acute, sudden (hours to days)	Chronic, progressive (years, for ever)
Reversible	Irreversible
Specific etiology: hypoxia, infection, ↓Na, ↓BS, alcohol, drugs, etc.	Unknown etiology: Alzheimer's, vascular
Altered level of consciousness: lethargy, confusion	Impaired memory with normal level of consciousness
Tx: underlying cause	Tx: donepezil, memantine?

Note: Sundown Syndrome: well lit room, bed alarm, frequent reorientation, company
(Confusion and disorientation that worsen at the end of the day, can present in delirium and dementia.)

Depression in the elderly can mimic dementia, with memory and attention changes

Alzheimer's disease

- **Classified mil, moderate (longest stage), and severe**
- **Interventions**
 - Basic reorientation, but don't expect learning, memorization or reality orientation
 - **Instead redirect to activities they can achieve**

- Provide calendar and clock, familiar possessions and pictures
- Use **distractions** when the client becomes agitated
- Provide consistent routines, simple instructions, closed ended questions, short sentences
- **Prevent unsafe wandering (door locks, alarms, don't leave alone in garden)**
- Respite care for caregivers
- Encourage **reminiscence** (groups, pictures, memories, movies)
- Promote daytime exercise to promote sleep at night
- Drugs used in Alzheimer's (cholinergic)
 - o Donepezil
 - o Rivastigmine
 - o Galantamine
 - o Memantine

Home safety old age/Alzheimer's

- Smoke and carbon monoxide detectors in kitchen and bedrooms
- Emergency phone number near all phones
- Safety knobs on the stove
- Childproof plugs for unused electrical outlets
- Childproof latches on cabinet to lock medicines, alcohol, cleaning products, gums, knives, scissors, ect...
- Reduce furniture and eliminate clutter
- Put carpet on stairs, or mark the edges of steps with brightly colored tape
- Remove small throw rugs, use rugs with nonskid backing instead
- Make sure cords to electrical outlets are out of the way or tacked to baseboards
- Clean up spills right away
- Non-skin shoes
- Make floors and walls different brilliant colors (no pastel, dark colors)
- Limit the size and number of mirrors
- Signs or simple pictures to label the bathroom, bedroom and kitchen
- Take care with pets and falls
- Reset your water heater to 120 0F
- **Prevent wander.** Safety locks on doors and identification tags

Eating Disorders

o Anorexia nervosa

- Develop as early as 10 years, food rituals, peer pressure
- Distorted body image and disturbed self-concept.
- Fear of gaining weight, frequent weighing, baggy clothing
- Starving, ketosis, **metabolic acidosis**
- Lanugo, amenorrhea, low vital signs
- Over-exercises
- Suicidality and electrolyte imbalances

o Bulimia nervosa

- More frequent than anorexia nervosa: normal weight
- Client indulges in eating binges followed by purging behaviors
- Anxiety, depression, low self esteem
- Tooth decay, enlarged parotids, esophageal tears,
- Russell's sign: calluses on knuckles of dominant hand
- **Metabolic alkalosis, hypokalemia**

o Interventions for Eating disorders:

- Assess risk for suicide,
- Monitor AB balance and electrolyte misbalances (**K+**)
- Supervise meals (*giving away food*) and after meals (*purge*)
- Set limits for physical activities
- Monitor vital signs vs food/fluid intake
- Daily weight, same time, same clothing
- Observe before-during and after eating
- Avoid discuss about food
- Refer family for therapy

Alcohol Abuse

- Screening: **CAGE** (Cut, Annoy, Guilt, Eye Opener)
- Wernicke's encephalopathy: **thiamine (B1)** deficiency
- **Codependency:** When the significant other derives positive self-esteem from doing things for or making decisions for the abuser:
 - Hiding alcohol
 - Covering for the addicted: calling sick, justifying the behavior
 - Experiencing shame.
 - Support groups: Al-Anon
- **Alcohol Withdrawal:**
 - **Early Stage I (6-24 hours):** hyper alertness, nausea, tremors, tachycardia, hypertension, agitation, headache, anxiety
 - **Late Stage II (24-72 hours):** hyperthermia, disorientation, seizures, **Delirium Tremens** (hallucinations: visual and kinesthetics, prickling or craving bugs), fever, HTN, sweating
 - **CIWA-AR scale** to evaluate alcohol withdrawal. More 8-10 benzodiazepine
 - Death
- **Treatment:**
 - Benzodiazepines (chlordiazepoxide, lorazepam)
 - Thiamine B1 and glucose
 - **Disulfiram is not used! NEVER given to alcoholic client**
- **Alcohol and disulfiram:**
 - Avoid OTC (cough syrups), elixirs, vanilla, mouth wash, vinegar, wine sauces

Drug Abuse

- **Opioids: CNS depressors**, heroin, morphine, oxycodone, methadone
 - o **Toxicity:** coma, respiratory depression, **pinpoint pupils**, constipation
 - **Naloxone** (before than oxygen) or naltrexone
 - o **Withdrawal (6-12 hours):**
 - Early: **Diaphoresis**, cravings, generalized pain, runny nose, dilated pupils,
 - Late: mydriasis, goose bumps, high BP, abdominal cramps and diarrhea,
 - Methadone
- **Cocaine, PCP, LSD, and methamphetamines:**
 - o CNS stimulants
 - o **Toxicity:** Paranoia, mydriasis, tachycardia, hypertension, hyperventilation, nasal septum perforation, seizures
 - o **Withdrawal:** cravings, depression, apathy, fatigue, sleepiness, hunger

Sometimes: paranoia, hallucinations, and agitation, seizures

- **Benzodiazepines (DOWNERS):** lorazepam, alprazolam, clonazepam
 - Toxicity: CNS depression, coma, apnea and death. Potentiating with **alcohol**
 - **Flumazenil** antagonist of benzodiazepine
 - **Withdrawal:** agitations, tremor, hallucinations, nausea, and vomits, seizures

Ecstasy: MDMA (3, 4-methylenedioxymethamphetamine). The drug sold on the streets rarely contains MDMA; it is made up of ketamine, caffeine and other narcotics and stimulants. Ecstasy release large amounts of dopamine and serotonin, which is why people feel euphoric, “You have an increased heart rate, increased blood pressure, you lose thermal regulation, and it makes your body more prone to experience fatal cardiac arrhythmia

PCP phencyclidine, related to anesthetic ketamine, produces hallucination, delirium, and mania.

Antagonist NMDA receptors (inhibitory), psychotic state like schizophrenia

<http://www.drugs.com/illicit/pcp.html>

Drugs Abuse

- **Addiction in Health Care professionals:**
 - Drug wasted without witness
 - Giving maximum doses when other nurses don't
 - Clients complains of unrelieved pain
 - Volunteering to carry narcotic keys, or passing drugs for other nurses
 - Coming too early, leaving too late, choosing shifts with less supervision
 - Frequent breaks, decrease alertness and mistakes
 - Obligation to report to chain of command or DOH **without confronting**

SSRIs:

Mnemonic: Ola Cita Para Ser Flu: **O**lanzapine, **C**italopram, **P**aroxetine, **S**ertraline, **F**luoxetine

- Monitor suicidal client: take 1-3 weeks to become effective
- Sexual dysfunction
- Serotonin syndrome (St John Wort, tramadol, linezolid, triptans, etc.)
- **Hyponatremia**
- Fluoxetine (Prozac) early in **morning**, with food

Tricyclic Antidepressants: amitriptyline, imipramine

- **Anticholinergic:** contraindication in glaucoma and BPH
 - Cardiac arrhythmias
 - Orthostatic hypotension: risk for fall
 - Sedative: administer **at bedtime**

Tetracyclic Antidepressant: trazadone, mirtazapine

- Sedative, useful for **insomnia**, give **at bedtime**

Bupropion and Varenicline:

- Atypical antidepressant **used to quit smoking**
- Suicidal ideation
- Decrease seizures threshold: avoid in epilepsy

Monoamine Oxidase Inhibitors (MAOIs)

- *Phenelzine, Isocarboxazid, Tranylcypromine, Selegiline*
- **Hypertensive crisis!**

- Interaction with SSRIs, opioids, ephedra, cocaine, and many OTCs
- Avoid tyramine-containing foods: wine, beer, aged cheese (not cottage)
- If the HCP STOP the MAOI, client needs to wait 2-3 weeks to start another medication.

Lithium

- Mood stabilizer (bipolar)
- Blood levels: **0.5 to 1.2 mEq/L**; toxicity > **1.5 mEq/L**
- Early toxicity: coarse tremor, polyuria, **nausea and vomits, diarrheas, coarse tremor, confusion**
 - Adequate **sodium intake** to avoid toxicity
 - Avoid **diuretics, low salt diet**
 - Monitor for dehydration (**vomits or diarrhea**)
 - Monitor **renal function (creatinine)**: lithium is excreted by kidneys
- Inhibit ADH: diabetes insipidus and polyuria
 - Encourage 2-3 L/day of fluids intake to compensate fluid loss
- Monitor thyroid function: lithium can cause **hypothyroidism**
- **Weight gain** is expected

Food rich in tyramine:

Avocados	Caffeine products, e.g., coffee, tea, chocolate
Bananas	Aged cheeses, no cottage cheese
Beef/chicken liver	Figs, including Fig Newton's® cookies
Brewer's yeast	Meat tenderizers
Papaya	Overripe fruit
Pickled herring	Sour cream
Raisins	Soy sauce
Red wine, sherry, beer	Yogurt
Sausage, bologna, pepperoni, salami	Liver

Antipsychotic Medications (neuroleptics)

- **Typical: haloperidol, chlorpromazine, trifluoperazine, fluphenazine**
 - Mechanism of action: **antagonist of dopamine**
 - Side effects more frequent than atypical
- **Atypical: clozapine, risperidone, quetiapine, olanzapine, ziprasidone, aripiprazole**
 - **Clozapine** produces agranulocytopenia
 - **Metabolic syndrome**, ↑ weight and glucose
- **Side effects in common t typical (+) and atypical (-) antipsychotics**
 - **Extrapyramidal effects** (Parkinson, akathisia, tardive dyskinesia, **oculogyric crisis**)
 - Antidote: **Benztropine**, diphenhydramine (anticholinergics)
 - Doses of antipsychotics must be reduced (discontinued in tardive dyskinesia)
 - **Neuroleptic Malignant Syndrome**: **rigidity, fever, sweat, tachycardia, AMS, seizures**
 - ICU life support and management of hyperthermia.
 - Treatment with dopaminergic drugs: Bromocriptine
 - QT prolongation (Torsade de points)
 - **Anticholinergic effects**: dry mouth, blurred vision, risk for heat stroke
 - **Photosensitivity**
 - Haloperidol: ↑ prolactin: **Gynecomastia, galactorrhea, amenorrhea**
 - **Antipsychotic drugs are not addictive, not controlled by DEA**

Extra NOTES:

KCl 10 mEq/L it is a correct dose and is safe, > 10 is not correct

Glucoside Cardio tonic = Digoxin, used for A-Fib and heart failure

- Check apical pulse 1 full min
 < 60 hold and call HCP
 >100 or between 60 – 100 give the dose
 < 90 in baby hold
- Check K⁺ levels and < 3.0 hold the dose
- Check nausea and vomit, it is toxic, Hold the dose

How many times the nurse needs to check the medication before to administer?

- First when check HCP order of the medication
- Second when prepare medication
- Third (right dose, right time, right patient, right medication)

**** 3 times ****